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(54) **Title:** NUCLEIC ACID-ASSOCIATED PROTEINS

(57) **Abstract:** Various embodiments of the invention provide human nucleic acid-associated proteins (NAAP) and polynucleotides which identify and encode NAAP. Embodiments of the invention also provide expression vectors, host cells, antibodies, agonists, and antagonists. Other embodiments provide methods for diagnosing, treating, or preventing disorders associated with aberrant expression of NAAP.

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NUCLEIC ACID-ASSOCIATED PROTEINS

TECHNICAL FIELD

The invention relates to novel nucleic acids, nucleic acid-associated proteins encoded by these nucleic acids, and to the use of these nucleic acids and proteins in the diagnosis, treatment, and prevention of cell proliferative, neurological, developmental, autoimmune/inflammatory, and lipid metabolism disorders, and infections. The invention also relates to the assessment of the effects of exogenous compounds on the expression of nucleic acids and nucleic acid-associated proteins.

BACKGROUND OF THE INVENTION

Multicellular organisms are comprised of diverse cell types that differ dramatically both in structure and function. The identity of a cell is determined by its characteristic pattern of gene expression, and different cell types express overlapping but distinctive sets of genes throughout development. Spatial and temporal regulation of gene expression is critical for the control of cell proliferation, cell differentiation, apoptosis, and other processes that contribute to organismal development. Furthermore, gene expression is regulated in response to extracellular signals that mediate cell-cell communication and coordinate the activities of different cell types. Appropriate gene regulation also ensures that cells function efficiently by expressing only those genes whose functions are required at a given time.

Transcription Factors

Transcriptional regulatory proteins are essential for the control of gene expression. Some of these proteins function as transcription factors that initiate, activate, repress, or terminate gene transcription. Transcription factors generally bind to the promoter, enhancer, and upstream regulatory regions of a gene in a sequence-specific manner, although some factors bind regulatory elements within or downstream of a gene coding region. Transcription factors may bind to a specific region of DNA singly or as a complex with other accessory factors. (Reviewed in Lewin, B. (1990) Genes IV, Oxford University Press, New York, NY, and Cell Press, Cambridge, MA, pp. 554-570.)

The double helix structure and repeated sequences of DNA create topological and chemical features which can be recognized by transcription factors. These features are hydrogen bond donor and acceptor groups, hydrophobic patches, major and minor grooves, and regular, repeated stretches of sequence which induce distinct bends in the helix. Typically, transcription factors recognize specific DNA sequence motifs of about 20 nucleotides in length. Multiple, adjacent transcription factor-binding motifs may be required for gene regulation.

Many transcription factors incorporate DNA-binding structural motifs which comprise either a helices or ß sheets that bind to the major groove of DNA. Four well-characterized structural motifs are helix-turn-helix, zinc finger, leucine zipper, and helix-loop-helix. Proteins containing these motifs may act alone as monomers, or they may form homo- or heterodimers that interact with DNA.

The helix-turn-helix motif consists of two α helices connected at a fixed angle by a short chain of amino acids. One of the helices binds to the major groove. Helix-turn-helix motifs are exemplified by the homeobox motif which is present in homeodomain proteins. These proteins are critical for specifying the anterior-posterior body axis during development and are conserved throughout the animal kingdom. The Antennapedia and Ultrabithorax proteins of *Drosophila melanogaster* are prototypical homeodomain proteins. (Pabo, C.O. and R.T. Sauer (1992) Annu. Rev. Biochem. 61:1053-1095.)

The zinc finger motif, which binds zinc ions, generally contains tandem repeats of about 30 amino acids consisting of periodically spaced cysteine and histidine residues. Examples of this sequence pattern, designated C2H2 and C3HC4 ("RING" finger), have been described. (Lewin, *supra.*) Zinc finger proteins each contain an α helix and an antiparallel ß sheet whose proximity and conformation are maintained by the zinc ion. Contact with DNA is made by the arginine preceding the α helix and by the second, third, and sixth residues of the α helix. Variants of the zinc finger motif include poorly defined cysteine-rich motifs which bind zinc or other metal ions. These motifs may not contain histidine residues and are generally nonrepetitive. The zinc finger motif may be repeated in a tandem array within a protein, such that the α helix of each zinc finger in the protein makes contact with the major groove of the DNA double helix. This repeated contact between the protein and the DNA produces a strong and specific DNA-protein interaction. The strength and specificity of the interaction can be regulated by the number of zinc finger motifs within the protein. Though originally identified in DNA-binding proteins as regions that interact directly with DNA, zinc fingers occur in a variety of proteins that do not bind DNA (Lodish, H. et al. (1995) Molecular Cell Biology, Scientific American Books, New York, NY, pp. 447-451). For example, Galcheva-Gargova, Z. et al. (1996) Science 272:1797-1802) have identified zinc finger proteins that interact with various cytokine receptors.

The C2H2-type zinc finger signature motif contains a 28 amino acid sequence, including 2 conserved Cys and 2 conserved His residues in a C-2-C-12-H-3-H type motif. The motif generally occurs in multiple tandem repeats. A cysteine-rich domain including the motif Asp-His-His-Cys (DHHC-CRD) has been identified as a distinct subgroup of zinc finger proteins. The DHHC-CRD region has been implicated in growth and development. One DHHC-CRD mutant shows defective function of Ras, a small membrane-associated GTP-binding protein that regulates cell growth and differentiation, while other DHHC-CRD proteins probably function in pathways not involving Ras (Bartels, D.J. et al. (1999) Mol. Cell Biol. 19:6775-6787).

Zinc-finger transcription factors are often accompanied by modular sequence motifs such as the Kruppel-associated box (KRAB) and the SCAN domain. For example, the hypoalphalipoproteinemia susceptibility gene ZNF202 encodes a SCAN box and a KRAB domain followed by eight C2H2 zinc-finger motifs (Honer, C. et al. (2001) Biochim. Biophys. Acta 1517:441-448). The SCAN domain is a highly conserved, leucine-rich motif of approximately 60 amino acids found at the amino-terminal end of zinc finger transcription factors. SCAN domains are most often linked to C2H2 zinc finger motifs through their carboxyl-terminal end. Biochemical binding studies have established the SCAN domain as a selective hetero- and homotypic oligomerization domain. SCAN domain-mediated protein complexes may function to modulate the biological function of transcription factors (Schumacher, C. et al. (2000) J. Biol. Chem. 275:17173-17179).

The KRAB (Kruppel-associated box) domain is a conserved amino acid sequence spanning approximately 75 amino acids and is found in almost one-third of the 300 to 700 genes encoding C2H2 zinc fingers. The KRAB domain is found N-terminally with respect to the finger repeats. The KRAB domain is generally encoded by two exons; the KRAB-A region or box is encoded by one exon and the KRAB-B region or box is encoded by a second exon. The function of the KRAB domain is the repression of transcription. Transcription repression is accomplished by recruitment of either the KRAB-associated protein-1, a transcriptional corepressor, or the KRAB-A interacting protein. Proteins containing the KRAB domain are likely to play a regulatory role during development (Williams, A.J. et al. (1999) Mol. Cell Biol. 19:8526-8535). A subgroup of highly related human KRAB zinc finger proteins detectable in all human tissues is highly expressed in human T lymphoid cells (Bellefroid, E.J. et al. (1993) EMBO J. 12:1363-1374). The ZNF85 KRAB zinc finger gene, a member of the human ZNF91 family, is highly expressed in normal adult testis, in seminomas, and in the NT2/D1 teratocarcinoma cell line (Poncelet, D.A. et al. (1998) DNA Cell Biol.17:931-943).

The C4 motif is found in hormone-regulated proteins. The C4 motif generally includes only 2 repeats. A number of eukaryotic and viral proteins contain a conserved cysteine-rich domain of 40 to 60 residues (called C3HC4 zinc-finger or RING finger) that binds two atoms of zinc, and is probably involved in mediating protein-protein interactions. The 3D "cross-brace" structure of the zinc ligation system is unique to the RING domain. The spacing of the cysteines in such a domain is C-x(2)-C-x(9 to 39)-C-x(1 to 3)-H-x(2 to3)-C-x(2)-C-x(4 to 48)-C-x(2)-C. The PHD finger is a C4HC3 zinc-finger-like motif found in nuclear proteins thought to be involved in chromatin-mediated transcriptional regulation.

GATA-type transcription factors contain one or two zinc finger domains which bind specifically to a region of DNA that contains the consecutive nucleotide sequence GATA. NMR studies indicate that the zinc finger comprises two irregular anti-parallel β sheets and an α helix, followed by a long loop to the C-terminal end of the finger (Ominchinski, J.G. (1993) Science 261:438-446). The helix and the loop connecting the two β-sheets contact the major groove of the DNA, while the C-terminal part, which determines the specificity of binding, wraps around into the minor groove.

The LIM motif consists of about 60 amino acid residues and contains seven conserved cysteine residues and a histidine within a consensus sequence (Schmeichel, K.L. and Beckerle, M.C. (1994) Cell 79:211-219). The LIM family includes transcription factors and cytoskeletal proteins which may be involved in development, differentiation, and cell growth. One example is actin-binding LIM protein, which may play roles in regulation of the cytoskeleton and cellular morphogenesis (Roof, D.J. et al. (1997) J. Cell Biol. 138:575-588). The N-terminal domain of actin-binding LIM protein has four double zinc finger motifs with the LIM consensus sequence. The C-terminal domain of actin-binding LIM protein shows sequence similarity to known actin-binding proteins such as dematin and villin. Actin-binding LIM protein binds to F-actin through its dematin-like C-terminal domain. The LIM domain may mediate protein-protein interactions with other LIM-binding proteins.

Myeloid cell development is controlled by tissue-specific transcription factors. Myeloid zinc finger proteins (MZF) include MZF-1 and MZF-2. MZF-1 functions in regulation of the development of neutrophilic granulocytes. A murine homolog MZF-2 is expressed in myeloid cells, particularly in the cells committed to the neutrophilic lineage. MZF-2 is down-regulated by G-CSF and appears to have a unique function in neutrophil development (Murai, K. et al. (1997) Genes Cells 2:581-591).

The leucine zipper motif comprises a stretch of amino acids rich in leucine which can form an amphipathic α helix. This structure provides the basis for dimerization of two leucine zipper proteins. The region adjacent to the leucine zipper is usually basic, and upon protein dimerization, is optimally positioned for binding to the major groove. Proteins containing such motifs are generally referred to as bZIP transcription factors. The leucine zipper motif is found in the proto-oncogenes Fos and Jun, which comprise the heterodimeric transcription factor AP1 involved in cell growth and the determination of cell lineage (Papavassiliou, A.G. (1995) N. Engl. J. Med. 332:45-47).

The helix-loop-helix motif (HLH) consists of a short α helix connected by a loop to a longer α helix. The loop is flexible and allows the two helices to fold back against each other and to bind to DNA. The transcription factor Myc contains a prototypical HLH motif.

The NF-kappa-B/Rel signature defines a family of eukaryotic transcription factors involved in oncogenesis, embryonic development, differentiation and immune response. Most transcription factors containing the Rel homology domain (RHD) bind as dimers to a consensus DNA sequence motif termed kappa-B. Members of the Rel family share a highly conserved 300 amino acid domain termed the Rel homology domain. The characteristic Rel C-terminal domain is involved in gene activation and cytoplasmic anchoring functions. Proteins known to contain the RHD domain include vertebrate nuclear factor NF-kappa-B, which is a heterodimer of a DNA-binding subunit and the transcription factor p65, mammalian transcription factor RelB, and vertebrate proto-oncogene c-rel, a protein associated with differentiation and lymphopoiesis (Kabrun, N. and Enrietto, P.J. (1994) Semin. Cancer Biol. 5:103-112).

A DNA binding motif termed ARID (AT-rich interactive domain) distinguishes an evolutionarily conserved family of proteins. The approximately 100-residue ARID sequence is present in a series of proteins strongly implicated in the regulation of cell growth, development, and tissue-specific gene expression. ARID proteins include Bright (a regulator of B-cell-specific gene expression), dead ringer (involved in development), and MRF-2 (which represses expression from the cytomegalovirus enhancer) (Dallas, P.B. et al. (2000) Mol. Cell Biol. 20:3137-3146).

The ELM2 (Egl-27 and MTA1 homology 2) domain is found in metastasis-associated protein MTA1 and protein ER1. The *Caenorhabditis elegans* gene egl-27 is required for embryonic patterning MTA1, a human gene with elevated expression in metastatic carcinomas, is a component of a protein complex with histone deacetylase and nucleosome remodelling activities (Solari, F. et al. (1999) Development 126:2483-2494). The ELM2 domain is usually found to the N terminus of a myb-like DNA binding domain. ELM2 is also found associated with an ARID DNA.

The Iroquois (Irx) family of genes are found in nematodes, insects and vertebrates. Irx genes usually occur in one or two genomic clusters of three genes each and encode transcriptional controllers that possess a characteristic homeodomain. The Irx genes function early in development to specify the identity of diverse territories of the body. Later in development in both *Drosophila* and vertebrates, the Irx genes function again to subdivide those territories into smaller domains. (For a review of Iroquois genes, see Cavodeassi, F. et al. (2001) Development 128:2847-2855.) For example, mouse and human Irx4 proteins are 83% conserved and their 63-aa homeodomain is more than 93% identical to that of the *Drosophila* Iroquois patterning genes. Irx4 transcripts are predominantly expressed in the cardiac ventricles. The homeobox gene Irx4 mediates ventricular differentiation during cardiac development (Bruneau, B.G. et al. (2000) Dev. Biol. 217:266-77).

Histidine triad (HIT) proteins share residues in distinctive dimeric, 10-stranded half-barrel structures that form two identical purine nucleotide-binding sites. Hint (histidine triad nucleotide-binding protein)-related proteins, found in all forms of life, and fragile histidine triad (Fhit)-related proteins, found in animals and fungi, represent the two main branches of the HIT superfamily. Fhit homologs bind and cleave diadenosine polyphosphates. Fhit-Ap(n)A complexes appear to function in a proapoptotic tumor suppression pathway in epithelial tissues (Brenner C. et al. (1999) J. Cell Physiol.181:179-187).

Most transcription factors contain characteristic DNA binding motifs, and variations on the above motifs and new motifs have been and are currently being characterized. (Faisst, S. and S. Meyer (1992) Nucleic Acids Res. 20:3-26.)

Chromatin Associated Proteins

In the nucleus, DNA is packaged into chromatin, the compact organization of which limits the accessibility of DNA to transcription factors and plays a key role in gene regulation. (Lewin, *supra*, pp. 409-410.) The compact structure of chromatin is determined and influenced by chromatin-associated proteins such as the histones, the high mobility group (HMG) proteins, and the chromodomain proteins. There are five classes of histones, H1, H2A, H2B, H3, and H4, all of which are highly basic, low molecular weight proteins. The fundamental unit of chromatin, the nucleosome, consists of 200 base pairs of DNA associated with two copies each of H2A, H2B, H3, and H4. H1 links adjacent nucleosomes. HMG proteins are low molecular weight, non-histone proteins that may play a role in unwinding DNA and stabilizing single-stranded DNA. Chromodomain proteins play a key role in the formation of highly compacted heterochromatin, which is transcriptionally silent.

Diseases and Disorders Related to Gene Regulation

Many neoplastic disorders in humans can be attributed to inappropriate gene expression. Malignant cell growth may result from either excessive expression of tumor promoting genes or insufficient expression of tumor suppressor genes. (Cleary, M.L. (1992) Cancer Surv. 15:89-104.) The zinc finger-type transcriptional regulator WT1 is a tumor-suppressor protein that is inactivated in children with Wilm's tumor. The oncogene bcl-6, which plays an important role in large-cell lymphoma, is also a zinc-finger protein (Papavassiliou, A.G. (1995) N. Engl. J. Med. 332:45-47). Chromosomal translocations may also produce chimeric loci that fuse the coding sequence of one gene with the regulatory regions of a second unrelated gene. Such an arrangement likely results in inappropriate gene transcription, potentially contributing to malignancy. In Burkitt's lymphoma, for example, the transcription factor Myc is translocated to the immunoglobulin heavy chain locus, greatly enhancing Myc expression and resulting in rapid cell growth leading to leukemia (Latchman, D.S. (1996) N. Engl. J. Med. 334:28-33).

In addition, the immune system responds to infection or trauma by activating a cascade of events that coordinate the progressive selection, amplification, and mobilization of cellular defense mechanisms. A complex and balanced program of gene activation and repression is involved in this process. However, hyperactivity of the immune system as a result of improper or insufficient regulation of gene expression may result in considerable tissue or organ damage. This damage is well-documented in immunological responses associated with arthritis, allergens, heart attack, stroke, and infections. (Isselbacher et al. Harrison's Principles of Internal Medicine, 13/e, McGraw Hill, Inc. and Teton Data Systems Software, 1996.) The causative gene for autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED) was recently isolated and found to encode a protein with two PHD-type zinc finger motifs (Bjorses, P. et al. (1998) Hum. Mol. Genet. 7:1547-1553).

Furthermore, the generation of multicellular organisms is based upon the induction and coordination of cell differentiation at the appropriate stages of development. Central to this process is differential gene expression, which confers the distinct identities of cells and tissues throughout the body. Failure to regulate gene expression during development could result in developmental disorders. Human developmental disorders caused by mutations in zinc finger-type transcriptional regulators include: urogenital developmental abnormalities associated with WT1; Greig cephalopolysyndactyly, Pallister-Hall syndrome, and postaxial polydactyly type A (GLI3), and Townes-Brocks syndrome, characterized by anal, renal, limb, and ear abnormalities (SALL1) (Engelkamp, D. and V. van Heyningen (1996) Curr. Opin. Genet. Dev. 6:334-342; Kohlhase, J. et al. (1999) Am. J. Hum. Genet. 64:435-445).

Human acute leukemias involve reciprocal chromosome translocations that fuse the ALL-1 gene located at chromosome region 11q23 to a series of partner genes positioned on a variety of human chromosomes. The fused genes encode chimeric proteins. The AF17 gene encodes a protein of 1093 amino acids, containing a leucine-zipper dimerization motif located 3' of the fusion point and a cysteine-rich domain at the N terminus that shows homology to a domain within the protein Br140 (peregrin) (Prasad R. et al. (1994) Proc. Natl. Acad. Sci. U S A 91:8107-8111).

SYNTHESIS OF NUCLEIC ACIDS

Polymerases

DNA and RNA replication are critical processes for cell replication and function. DNA and RNA replication are mediated by the enzymes DNA and RNA polymerase, respectively, by a "templating" process in which the nucleotide sequence of a DNA or RNA strand is copied by complementary base-pairing into a complementary nucleic acid sequence of either DNA or RNA. However, there are fundamental differences between the two processes.

DNA polymerase catalyzes the stepwise addition of a deoxyribonucleotide to the 3'-OH end of a polynucleotide strand (the primer strand) that is paired to a second (template) strand. The new DNA strand therefore grows in the 5' to 3' direction (Alberts, B. et al. (1994) The Molecular Biology of the Cell, Garland Publishing Inc., New York, NY, pp 251-254). The substrates for the polymerization reaction are the corresponding deoxynucleotide triphosphates which must base-pair with the correct nucleotide on the template strand in order to be recognized by the polymerase. Because DNA exists as a double-stranded helix, each of the two strands may serve as a template for the formation of a new complementary strand. Each of the two daughter cells of a dividing cell therefore inherits a new DNA double helix containing one old and one new strand. Thus, DNA is said to be replicated "semiconservatively" by DNA polymerase. In addition to the synthesis of new DNA, DNA polymerase is also involved in the repair of damaged DNA as discussed below under

"Ligases."

In contrast to DNA polymerase, RNA polymerase uses a DNA template strand to "transcribe" DNA into RNA using ribonucleotide triphosphates as substrates. Like DNA polymerization, RNA polymerization proceeds in a 5' to 3' direction by addition of a ribonucleoside monophosphate to the 3'-OH end of a growing RNA chain. DNA transcription generates messenger RNAs (mRNA) that carry information for protein synthesis, as well as the transfer, ribosomal, and other RNAs that have structural or catalytic functions. In eukaryotes, three discrete RNA polymerases synthesize the three different types of RNA (Alberts, *supra*, pp. 367-368). RNA polymerase I makes the large ribosomal RNAs, RNA polymerase II makes the mRNAs that will be translated into proteins, and RNA polymerase III makes a variety of small, stable RNAs, including 5S ribosomal RNA and the transfer RNAs (tRNA). In all cases, RNA synthesis is initiated by binding of the RNA polymerase to a promoter region on the DNA and synthesis begins at a start site within the promoter. Synthesis is completed at a stop (termination) signal in the DNA whereupon both the polymerase and the completed RNA chain are released.

<u>Ligases</u>

DNA repair is the process by which accidental base changes, such as those produced by oxidative damage, hydrolytic attack, or uncontrolled methylation of DNA, are corrected before replication or transcription of the DNA can occur. Because of the efficiency of the DNA repair process, fewer than one in a thousand accidental base changes causes a mutation (Alberts, *supra*, pp. 245-249). The three steps common to most types of DNA repair are (1) excision of the damaged or altered base or nucleotide by DNA nucleases, (2) insertion of the correct nucleotide in the gap left by the excised nucleotide by DNA polymerase using the complementary strand as the template and, (3) sealing the break left between the inserted nucleotide(s) and the existing DNA strand by DNA ligase. In the last reaction, DNA ligase uses the energy from ATP hydrolysis to activate the 5' end of the broken phosphodiester bond before forming the new bond with the 3'-OH of the DNA strand. In Bloom's syndrome, an inherited human disease, individuals are partially deficient in DNA ligation and consequently have an increased incidence of cancer (Alberts, *supra* p. 247).

<u>Nucleases</u>

Nucleases comprise enzymes that hydrolyze both DNA (DNase) and RNA (Rnase). They serve different purposes in nucleic acid metabolism. Nucleases hydrolyze the phosphodiester bonds between adjacent nucleotides either at internal positions (endonucleases) or at the terminal 3' or 5' nucleotide positions (exonucleases). A DNA exonuclease activity in DNA polymerase, for example, serves to remove improperly paired nucleotides attached to the 3'-OH end of the growing DNA strand by the polymerase and thereby serves a "proofreading" function. As mentioned above, DNA endonuclease activity is involved in the excision step of the DNA repair process.

RNases also serve a variety of functions. For example, RNase P is a ribonucleoprotein enzyme which cleaves the 5' end of pre-tRNAs as part of their maturation process. RNase H digests the RNA strand of an RNA/DNA hybrid. Such hybrids occur in cells invaded by retroviruses, and RNase H is an important enzyme in the retroviral replication cycle. Pancreatic RNase secreted by the pancreas into the intestine hydrolyzes RNA present in ingested foods. RNase activity in serum and cell extracts is elevated in a variety of cancers and infectious diseases (Schein, C.H. (1997) Nat. Biotechnol. 15:529-536). Regulation of RNase activity is being investigated as a means to control tumor angiogenesis, allergic reactions, viral infection and replication, and fungal infections.

MODIFICATION OF NUCLEIC ACIDS

Methylases

Methylation of specific nucleotides occurs in both DNA and RNA, and serves different functions in the two macromolecules. Methylation of cytosine residues to form 5-methyl cytosine in DNA occurs specifically in CG sequences which are base-paired with one another in the DNA double-helix. The pattern of methylation is passed from generation to generation during DNA replication by an enzyme called "maintenance methylase" that acts preferentially on those CG sequences that are base-paired with a CG sequence that is already methylated. Such methylation appears to distinguish active from inactive genes by preventing the binding of regulatory proteins that "turn on" the gene, but permiting the binding of proteins that inactivate the gene (Alberts, *supra* pp. 448-451). In RNA metabolism, "tRNA methylase" produces one of several nucleotide modifications in tRNA that affect the conformation and base-pairing of the molecule and facilitate the recognition of the appropriate mRNA codons by specific tRNAs. The primary methylation pattern is the dimethylation of guanine residues to form N,N-dimethyl guanine.

Helicases and Single-stranded Binding Proteins

Helicases are enzymes that destabilize and unwind double helix structures in both DNA and RNA. Since DNA replication occurs more or less simultaneously on both strands, the two strands must first separate to generate a replication "fork" for DNA polymerase to act on. Two types of replication proteins contribute to this process, DNA helicases and single-stranded binding proteins. DNA helicases hydrolyze ATP and use the energy of hydrolysis to separate the DNA strands. Single-stranded binding proteins (SSBs) then bind to the exposed DNA strands, without covering the bases, thereby temporarily stabilizing them for templating by the DNA polymerase (Alberts, *supra*, pp. 255-256).

RNA helicases also alter and regulate RNA conformation and secondary structure. Like the DNA helicases, RNA helicases utilize energy derived from ATP hydrolysis to destabilize and unwind RNA duplexes. The most well-characterized and ubiquitous family of RNA helicases is the DEAD-box family, so named for the conserved B-type ATP-binding motif which is diagnostic of proteins in this family. Over 40 DEAD-box helicases have been identified in organisms as diverse as bacteria, insects, yeast, amphibians, mammals, and plants. DEAD-box helicases function in diverse processes such as translation initiation, splicing, ribosome assembly, and RNA editing, transport, and stability. Examples of these RNA helicases include yeast Drs1 protein, which is involved in ribosomal RNA processing; yeast TIF1 and TIF2 and mammalian eIF-4A, which are essential to the initiation of RNA translation; and human p68 antigen, which regulates cell growth and division (Ripmaster, T.L. et al. (1992) Proc. Natl. Acad. Sci. USA 89:11131-11135; Chang, T.-H. et al. (1990) Proc. Natl. Acad. Sci. USA 87:1571-1575). These RNA helicases demonstrate strong sequence homology over a stretch of some 420 amino acids. Included among these conserved sequences are the consensus sequence for the A motif of an ATP binding protein; the "DEAD box" sequence, associated with ATPase activity; the sequence SAT, associated with the actual helicase unwinding region; and an octapeptide consensus sequence, required for RNA binding and ATP hydrolysis (Pause, A. et al. (1993) Mol. Cell Biol. 13:6789-6798). Differences outside of these conserved regions are believed to reflect differences in the functional roles of individual proteins (Chang, T.H. et al. (1990) Proc. Natl. Acad. Sci. USA 87:1571-1575).

Some DEAD-box helicases play tissue- and stage-specific roles in spermatogenesis and embryogenesis. Overexpression of the DEAD-box 1 protein (DDX1) may play a role in the progression of neuroblastoma (Nb) and retinoblastoma (Rb) tumors (Godbout, R. et al. (1998) J. Biol. Chem. 273:21161-21168). These observations suggest that DDX1 may promote or enhance tumor progression by altering the normal secondary structure and expression levels of RNA in cancer cells. Other DEAD-box helicases have been implicated either directly or indirectly in tumorigenesis. (Discussed in Godbout, *supra.*) For example, murine p68 is mutated in ultraviolet light-induced tumors, and human DDX6 is located at a chromosomal breakpoint associated with B-cell lymphoma. Similarly, a chimeric protein comprised of DDX10 and NUP98, a nucleoporin protein, may be involved in the pathogenesis of certain myeloid malignancies.

Topoisomerases

Besides the need to separate DNA strands prior to replication, the two strands must be "unwound" from one another prior to their separation by DNA helicases. This function is performed by proteins known as DNA topoisomerases. DNA topoisomerase effectively acts as a reversible nuclease that hydrolyzes a phosphodiesterase bond in a DNA strand, permits the two strands to rotate freely about one another to remove the strain of the helix, and then rejoins the original phosphodiester bond between the two strands. Topoisomerases are essential enzymes responsible for the topological rearrangement of DNA brought about by transcription, replication, chromatin formation, recombination, and chromosome segregation. Superhelical coils are introduced into DNA by the passage of processive enzymes such as RNA polymerase, or by the separation of DNA strands by a helicase prior to replication. Knotting and concatenation can occur in the process of DNA synthesis, storage, and repair. All topoisomerases work by breaking a phosphodiester bond in the ribose-phosphate backbone of DNA. A catalytic tyrosine residue on the enzyme makes a nucleophilic attack on the scissile phosphodiester bond, resulting in a reaction intermediate in which a covalent bond is formed between the enzyme and one end of the broken strand. A tyrosine-DNA phosphodiesterase functions in DNA repair by hydrolyzing this bond in occasional dead-end topoisomerase I-DNA intermediates (Pouliot, J.J. et al. (1999) Science 286:552-555).

Two types of DNA topoisomerase exist, types I and II. Type I topoisomerases work as monomers, making a break in a single strand of DNA while type II topoisomerases, working as homodimers, cleave both strands. DNA Topoisomerase I causes a single-strand break in a DNA helix to allow the rotation of the two strands of the helix about the remaining phosphodiester bond in the opposite strand. DNA topoisomerase II causes a transient break in both strands of a DNA helix where two double helices cross over one another. This type of topoisomerase can efficiently separate two interlocked DNA circles (Alberts, *supra*, pp.260-262). Type II topoisomerases are largely confined to proliferating cells in eukaryotes, such as cancer cells. For this reason they are targets for anticancer drugs. Topoisomerase II has been implicated in multi-drug resistance (MDR) as it appears to aid in the repair of DNA damage inflicted by DNA binding agents such as doxorubicin and vincristine.

The topoisomerase I family includes topoisomerases I and III (topo I and topo III). The crystal structure of human topoisomerase I suggests that rotation about the intact DNA strand is partially controlled by the enzyme. In this "controlled rotation" model, protein-DNA interactions limit the rotation, which is driven by torsional strain in the DNA (Stewart, L. et al. (1998) Science 379:1534-1541). Structurally, topo I can be recognized by its catalytic tyrosine residue and a number of other conserved residues in the active site region. Topo I is thought to function during transcription. Two topo IIIs are known in humans, and they are homologous to prokaryotic topoisomerase I, with a conserved tyrosine and active site signature specific to this family. Topo III has been suggested to play a role in meiotic recombination. A mouse topo III is highly expressed in testis tissue and its expression increases with the increase in the number of cells in pachytene (Seki, T. et al. (1998) J. Biol. Chem. 273:28553-28556).

The topoisomerase II family includes two isozymes (IIα and IIβ) encoded by different genes. Topo II cleaves double stranded DNA in a reproducible, nonrandom fashion, preferentially in an AT rich region, but the basis of cleavage site selectivity is not known. Structurally, topo II is made up of four domains, the first two of which are structurally similar and probably distantly homologous to similar domains in eukaryotic topo I. The second domain bears the catalytic tyrosine, as well as a highly conserved pentapeptide. The IIα isoform appears to be responsible for unlinking DNA during chromosome segregation. Cell lines expressing IIα but not IIβ suggest that IIβ is dispensable in cellular processes; however, IIβ knockout mice died perinatally due to a failure in neural development. That the major abnormalities occurred in predominantly late developmental events (neurogenesis) suggests that IIβ is needed not at mitosis, but rather during DNA repair (Yang, X. et al. (2000) Science 287:131-134).

Topoisomerases have been implicated in a number of disease states, and topoisomerase poisons have proven to be effective anti-tumor drugs for some human malignancies. Topo I is mislocalized in Fanconi's anemia, and may be involved in the chromosomal breakage seen in this disorder (Wunder, E. (1984) Hum. Genet. 68:276-281). Overexpression of a truncated topo III in ataxia-telangiectasia (A-T) cells partially suppresses the A-T phenotype, probably through a dominant negative mechanism. This suggests that topo III is deregulated in A-T (Fritz, E. et al. (1997) Proc. Natl. Acad. Sci. USA 94:4538-4542). Topo III also interacts with the Bloom's Syndrome gene product, and has been suggested to have a role as a tumor suppressor (Wu, L. et al. (2000) J. Biol. Chem. 275:9636-9644). Aberrant topo II activity is often associated with cancer or increased cancer risk. Greatly lowered topo II activity has been found in some, but not all A-T cell lines (Mohamed, R. et al. (1987) Biochem. Biophys. Res. Commun. 149:233-238). On the other hand, topo II can break DNA in the region of the A-T gene (ATM), which controls all DNA damage-responsive cell cycle checkpoints (Kaufmann, W.K. (1998) Proc. Soc. Exp. Biol. Med. 217:327-334). The ability of topoisomerases to break DNA has been used as the basis of antitumor drugs. Topoisomerase poisons act by increasing the number of dead-end covalent DNA-enzyme complexes in the cell, ultimately triggering cell death pathways (Fortune, J.M. and N. Osheroff (2000) Prog. Nucleic Acid Res. Mol. Biol. 64:221-253; Guichard, S.M. and M.K. Danks (1999) Curr. Opin. Oncol. 11:482-489). Antibodies against topo I are found in the serum of systemic sclerosis patients, and the levels of the antibody may be used as a marker of pulmonary involvement in the disease (Diot, E. et al. (1999) Chest 116:715-720). Finally, the DNA binding region of human topo I has been used as a DNA delivery vehicle for gene therapy (Chen, T.Y. et al. (2000) Appl. Microbiol. Biotechnol. 53:558-567).

Recombinases

Genetic recombination is the process of rearranging DNA sequences within an organism's genome to provide genetic variation for the organism in response to changes in the environment. DNA recombination allows variation in the particular combination of genes present in an individual's genome, as well as the timing and level of expression of these genes. (See Alberts, *supra* pp. 263-273.) Two broad classes of genetic recombination are commonly recognized, general recombination and site-specific recombination. General recombination involves genetic exchange between any homologous pair of DNA sequences usually located on two copies of the same chromosome. The process is aided by enzymes, recombinases, that "nick" one strand of a DNA duplex more or less randomly and permit exchange with a complementary strand on another duplex. The process does not normally change the arrangement of genes in a chromosome. In site-specific recombination, the recombinase recognizes specific nucleotide sequences present in one or both of the recombining molecules. Base-pairing is not involved in this form of recombination and therefore it does not require DNA homology between the recombining molecules. Unlike general recombination, this form of recombination can alter the relative positions of nucleotide sequences in chromosomes.

RNA METABOLISM

Ribonucleic acid (RNA) is a linear single-stranded polymer of four nucleotides, ATP, CTP, UTP, and GTP. In most organisms, RNA is transcribed as a copy of deoxyribonucleic acid (DNA), the genetic material of the organism. In retroviruses RNA rather than DNA serves as the genetic material. RNA copies of the genetic material encode proteins or serve various structural, catalytic, or regulatory roles in organisms. RNA is classified according to its cellular localization and function. Messenger RNAs (mRNAs) encode polypeptides. Ribosomal RNAs (rRNAs) are assembled, along with ribosomal proteins, into ribosomes, which are cytoplasmic particles that translate mRNA into polypeptides. Transfer RNAs (tRNAs) are cytosolic adaptor molecules that function in mRNA translation by recognizing both an mRNA codon and the amino acid that matches that codon. Heterogeneous nuclear RNAs (hnRNAs) include mRNA precursors and other nuclear RNAs of various sizes. Small nuclear RNAs (snRNAs) are a part of the nuclear spliceosome complex that removes intervening, non-coding sequences (introns) and rejoins exons in pre-mRNAs.

Proteins are associated with RNA during its transcription from DNA, RNA processing, and translation of mRNA into protein. Proteins are also associated with RNA as it is used for structural, catalytic, and regulatory purposes.

RNA Processing

Ribosomal RNAs (rRNAs) are assembled, along with ribosomal proteins, into ribosomes, which are cytoplasmic particles that translate messenger RNA (mRNA) into polypeptides. The eukaryotic ribosome is composed of a 60S (large) subunit and a 40S (small) subunit, which together form the 80S ribosome. In addition to the 18S, 28S, 5S, and 5.8S rRNAs, ribosomes contain from 50 to over 80 different ribosomal proteins, depending on the organism. Ribosomal proteins are classified according to which subunit they belong (i.e., L, if associated with the large 60S large subunit or S if associated with the small 40S subunit). *E. coli* ribosomes have been the most thoroughly studied and contain 50 proteins, many of which are conserved in all life forms. The structures of nine ribosomal proteins have been solved to less than 3.0D resolution (i.e., S5, S6, S17, L1, L6, L9, L12, L14, L30), revealing common motifs, such as b-a-b protein folds in addition to acidic and basic RNA-binding motifs positioned between b-strands. Most ribosomal proteins are believed to contact rRNA directly (reviewed in Liljas, A. and Garber, M. (1995) Curr. Opin. Struct. Biol. 5:721–727; see also Woodson, S.A. and Leontis, N.B. (1998) Curr. Opin. Struct. Biol. 8:294-300; Ramakrishnan, V. and White, S.W. (1998) Trends Biochem. Sci. 23:208-212).

Ribosomal proteins may undergo post-translational modifications or interact with other ribosome-associated proteins to regulate translation. For example, the highly homologous 40S ribosomal protein S6 kinases (S6K1 and S6K2) play a key role in the regulation of cell growth by controlling the biosynthesis of translational components which make up the protein synthetic apparatus (including the ribosomal proteins). In the case of S6K1, at least eight phosphorylation sites are believed to mediate kinase activation in a hierarchical fashion (Dufner and Thomas (1999) Exp. Cell. Res. 253:100-109). Some of the ribosomal proteins, including L1, also function as translational repressors by binding to polycistronic mRNAs encoding ribosomal proteins (reviewed in Liljas, *supra* and Garber, *supra*).

Recent evidence suggests that a number of ribosomal proteins have secondary functions independent of their involvement in protein biosynthesis. These proteins function as regulators of cell proliferation and, in some instances, as inducers of cell death. For example, the expression of human ribosomal protein L13a has been shown to induce apoptosis by arresting cell growth in the G2/M phase of the cell cycle. Inhibition of expression of L13a induces apoptosis in target cells, which suggests that this protein is necessary, in the appropriate amount, for cell survival. Similar results have been obtained in yeast where inactivation of yeast homologues of L13a, rp22 and rp23, results in severe growth retardation and death. A closely related ribosomal protein, L7, arrests cells in G1 and also induces apoptosis. Thus, it appears that a subset of ribosomal proteins may function as cell cycle checkpoints and compose a new family of cell proliferation regulators.

Mapping of individual ribosomal proteins on the surface of intact ribosomes is accomplished using 3D immunocryoelectronmicroscopy, whereby antibodies raised against specific ribosomal proteins are visualized. Progress has been made toward the mapping of L1, L7, and L12 while the structure of the intact ribosome has been solved to only 20-25D resolution and inconsistencies exist among different crude structures (Frank, J. (1997) Curr. Opin. Struct. Biol. 7:266-272).

Three distinct sites have been identified on the ribosome. The aminoacyl-tRNA acceptor site (A site) receives charged tRNAs (with the exception of the initiator-tRNA). The peptidyl-tRNA site (P site) binds the nascent polypeptide as the amino acid from the A site is added to the elongating chain. Deacylated tRNAs bind in the exit site (E site) prior to their release from the ribosome. The structure of the ribosome is reviewed in Stryer, L. (1995) Biochemistry, W.H. Freeman and Company, New York NY, pp. 888-9081; Lodish, H. et al. (1995) Molecular Cell Biology, Scientific American Books, New York NY, pp. 119-138; and Lewin, B (1997) Genes VI, Oxford University Press, Inc. New York, NY).

Various proteins are necessary for processing of transcribed RNAs in the nucleus. Pre-mRNA processing steps include capping at the 5' end with methylguanosine, polyadenylating the 3' end, and splicing to remove introns. The primary RNA transript from DNA is a faithful copy of the gene containing both exon and intron sequences, and the latter sequences must be cut out of the RNA transcript to produce a mRNA that codes for a protein. This "splicing" of the mRNA sequence takes place in the nucleus with the aid of a large, multicomponent ribonucleoprotein complex known as a spliceosome. The spliceosomal complex is comprised of five small nuclear ribonucleoprotein particles (snRNPs) designated U1, U2, U4, U5, and U6. Each snRNP contains a single species of snRNA and about ten proteins. The RNA components of some snRNPs recognize and base-pair with intron consensus sequences. The protein components mediate spliceosome assembly and the splicing reaction. Autoantibodies to snRNP proteins are found in the blood of patients with systemic lupus erythematosus (Stryer, L. (1995) Biochemistry, W.H. Freeman and Company, New York NY, p. 863).

Heterogeneous nuclear ribonucleoproteins (hnRNPs) have been identified that have roles in splicing, exporting of the mature RNAs to the cytoplasm, and mRNA translation (Biamonti, G. et al. (1998) Clin. Exp. Rheumatol. 16:317-326). Some examples of hnRNPs include the yeast proteins Hrp1p, involved in cleavage and polyadenylation at the 3' end of the RNA; Cbp80p, involved in capping the 5' end of the RNA; and Npl3p, a homolog of mammalian hnRNP A1, involved in export of mRNA from the nucleus (Shen, E.C. et al. (1998) Genes Dev. 12:679-691). HnRNPs have been shown to be important targets of the autoimmune response in rheumatic diseases (Biamonti, *supra*).

Many snRNP and hnRNP proteins are characterized by an RNA recognition motif (RRM). (Reviewed in Birney, E. et al. (1993) Nucleic Acids Res. 21:5803-5816.) The RRM is about 80 amino acids in length and forms four β-strands and two α-helices arranged in an α/β sandwich. The RRM contains a core RNP-1 octapeptide motif along with surrounding conserved sequences. In addition to snRNP proteins, examples of RNA-binding proteins which contain the above motifs include heteronuclear ribonucleoproteins which stabilize nascent RNA and factors which regulate alternative splicing. Alternative splicing factors include developmentally regulated proteins, specific examples of which have been identified in lower eukaryotes such as *Drosophila melanogaster* and *Caenorhabditis elegans*. These proteins play key roles in developmental processes such as pattern formation and sex determination, respectively. (See, for example, Hodgkin, J. et al. (1994) Development 120:3681-3689.)

The 3' ends of most eukaryote mRNAs are also posttranscriptionally modified by polyadenylation. Polyadenylation proceeds through two enzymatically distinct steps: (i) the endonucleolytic cleavage of nascent mRNAs at *cis*-acting polyadenylation signals in the 3'-untranslated (non-coding) region and (ii) the addition of a poly(A) tract to the 5' mRNA fragment. The presence of *cis*-acting RNA sequences is necessary for both steps. These sequences include 5'-AAUAAA-3' located 10-30 nucleotides upstream of the cleavage site and a less well-conserved GU-or U-rich sequence element located 10-30 nucleotides downstream of the cleavage site. Cleavage stimulation factor (CstF), cleavage factor I (CF I), and cleavage factor II (CF II) are involved in the cleavage reaction while cleavage and polyadenylation specificity factor (CPSF) and poly(A) polymerase (PAP) are necessary for both cleavage and polyadenylation. An additional enzyme, poly(A)-binding protein II (PAB II), promotes poly(A) tract elongation (Rüegsegger, U. et al. (1996) J. Biol. Chem. 271:6107-6113; and references within).

TRANSLATION

Correct translation of the genetic code depends upon each amino acid forming a linkage with the appropriate transfer RNA (tRNA). The aminoacyl-tRNA synthetases (aaRSs) are essential proteins found in all living organisms. The aaRSs are responsible for the activation and correct attachment of an amino acid with its cognate tRNA, as the first step in protein biosynthesis. Prokaryotic organisms have at least twenty different types of aaRSs, one for each different amino acid, while eukaryotes usually have two aaRSs, a cytosolic form and a mitochondrial form, for each different amino acid. The 20 aaRS enzymes can be divided into two structural classes. Class I enzymes add amino acids to the 2' hydroxyl at the 3' end of tRNAs while Class II enzymes add amino acids to the 3' hydroxyl at the 3' end of tRNAs. Each class is characterized by a distinctive topology of the catalytic domain. Class I enzymes contain a catalytic domain based on the nucleotide-binding Rossman 'fold'. In particular, a consensus tetrapeptide motif is highly conserved (Prosite Document PDOC00161, Aminoacyl-transfer RNA synthetases class-I signature). Class I enzymes are specific for arginine, cysteine, glutamic acid, glutamine, isoleucine, leucine, methionine, tyrosine, tryptophan, and valine. Class II enzymes contain a central catalytic domain, which consists of a seven-stranded antiparallel ß-sheet domain, as well as N- and C- terminal regulatory domains. Class II enzymes are separated into two groups based on the heterodimeric or homodimeric structure of the enzyme; the latter group is further subdivided by the structure of the N- and C-terminal regulatory domains (Hartlein, M. and Cusack, S. (1995) J. Mol. Evol. 40:519-530). Class II enzymes are specific for alanine, asparagine, aspartic acid, glycine, histidine, lysine, phenylalanine, proline, serine, and threonine.

Certain aaRSs also have editing functions. IleRS, for example, can misactivate valine to form Val-$tRNA^{Ile}$, but this product is cleared by a hydrolytic activity that destroys the mischarged product. This editing activity is located within a second catalytic site found in the connective polypeptide 1 region (CP1), a long insertion sequence within the Rossman fold domain of Class I enzymes (Schimmel, P. et al. (1998) FASEB J. 12:1599-1609). AaRSs also play a role in tRNA processing. It has been shown that mature tRNAs are charged with their respective amino acids in the nucleus before export to the cytoplasm, and charging may serve as a quality control mechanism to insure the tRNAs are functional (Martinis, S.A. et al. (1999) EMBO J. 18:4591-4596).

Under optimal conditions, polypeptide synthesis proceeds at a rate of approximately 40 amino acid residues per second. The rate of misincorporation during translation in on the order of 10^{-4} and is primarily the result of aminoacyl-t-RNAs being charged with the incorrect amino acid. Incorrectly charged tRNA are toxic to cells as they result in the incorporation of incorrect amino acid residues into an elongating polypeptide. The rate of translation is presumed to be a compromise between the optimal rate of elongation and the need for translational fidelity. Mathematical calculations predict that 10^{-4} is indeed the maximum acceptable error rate for protein synthesis in a biological system (reviewed in Stryer, *supra*; and Watson, J. et al. (1987) The Benjamin/Cummings Publishing Co., Inc. Menlo Park, CA). A particularly error prone aminoacyl-tRNA charging event is the charging of $tRNA^{Gln}$ with Gln. A mechanism exits for the correction of this mischarging event which likely has its origins in evolution. Gln was among the last of the 20 naturally occurring amino acids used in polypeptide synthesis to appear in nature. Gram positive eubacteria, cyanobacteria, Archeae, and eukaryotic organelles possess a noncanonical pathway for the synthesis of Gln-$tRNA^{Gln}$ based on the transformation of Glu-$tRNA^{Gln}$ (synthesized by Glu-tRNA synthetase, GluRS) using the enzyme Glu-$tRNA^{Gln}$ amidotransferase (Glu-AdT). The reactions involved in the transamidation pathway are as follows (Curnow, A.W. et al. (1997) Nucleic Acids Symposium 36:2-4):

$$\text{tRNA}^{\text{Gln}} + \text{Glu} + \text{ATP} \xrightarrow{\text{GluRS}} \text{Glu-tRNA}^{\text{Gln}} + \text{AMP} + \text{PP}_i$$

$$\text{Glu-tRNA}^{\text{Gln}} + \text{Gln} + \text{ATP} \xrightarrow{\text{Glu-AdT}} \text{Gln-tRNA}^{\text{Gln}} + \text{Glu} + \text{ADP} + \text{P}$$

A similar enzyme, Asp-$tRNA^{Asn}$ amidotransferase, exists in Archaea, which transforms Asp-$tRNA^{Asn}$ to Asn-$tRNA^{Asn}$. Formylase, the enzyme that transforms Met-$tRNA^{fMet}$ to fMet-$tRNA^{fMet}$ in eubacteria, is likely to be a related enzyme. A hydrolytic activity has also been identified that destroys mischarged Val-$tRNA^{Ile}$ (Schimmel, P. et al. (1998) FASEB J. 12:1599-1609). One likely scenario for the evolution of Glu-AdT in primitive life forms is the absence of a specific glutaminyl-tRNA synthetase (GlnRS), requiring an alternative pathway for the synthesis of Gln-$tRNA^{Gln}$. In fact, deletion of the Glu-AdT operon in Gram positive bacteria is lethal (Curnow, A.W. et al. (1997) Proc. Natl. Acad. Sci. USA 94:11819-11826). The existence of GluRS activity in other organisms has been inferred by the high degree of conservation in translation machinery in nature; however, GluRS has not been identified in all organisms, including *Homo sapiens*. Such an enzyme would be responsible for ensuring translational fidelity and reducing the synthesis of defective polypeptides.

In addition to their function in protein synthesis, specific aminoacyl tRNA synthetases also play roles in cellular fidelity, RNA splicing, RNA trafficking, apoptosis, and transcriptional and translational regulation. For example, human tyrosyl-tRNA synthetase can be proteolytically cleaved into two fragments with distinct cytokine activities. The carboxy-terminal domain exhibits monocyte and leukocyte chemotaxis activity as well as stimulating production of myeloperoxidase, tumor necrosis factor-α, and tissue factor. The N-terminal domain binds to the interleukin-8 type A receptor and functions as an interleukin-8-like cytokine. Human tyrosyl-tRNA synthetase is secreted from apoptotic tumor cells and may accelerate apoptosis (Wakasugi, K., and Schimmel, P. (1999) Science 284:147-151). Mitochondrial *Neurospora crassa* TyrRS and *S. cerevisiae* LeuRS are essential factors for certain group I intron splicing activities, and human mitochondrial LeuRS can substitute for the yeast LeuRS in a yeast null strain. Certain bacterial aaRSs are involved in regulating their own transcription or translation (Martinis, *supra*). Several aaRSs are able to synthesize diadenosine oligophosphates, a class of signalling molecules with roles in cell proliferation, differentiation, and apoptosis (Kisselev, L.L et al. (1998) FEBS Lett. 427:157-163; Vartanian, A. et al. (1999) FEBS Lett. 456:175-180).

Autoantibodies against aminoacyl-tRNAs are generated by patients with autoimmune diseases such as rheumatic arthritis, dermatomyositis and polymyositis, and correlate strongly with complicating interstitial lung disease (ILD) (Freist, W. et al. (1999) Biol. Chem. 380:623-646; Freist, W. et al. (1996) Biol. Chem. Hoppe Seyler 377:343-356). These antibodies appear to be generated in response to viral infection, and coxsackie virus has been used to induce experimental viral myositis in animals.

Comparison of aaRS structures between humans and pathogens has been useful in the design of novel antibiotics (Schimmel, *supra*). Genetically engineered aaRSs have been utilized to allow site-specific incorporation of unnatural amino acids into proteins *in vivo* (Liu, D.R. et al. (1997) Proc. Natl. Acad. Sci. USA 94:10092-10097).

tRNA Modifications

The modified ribonucleoside, pseudouridine (ψ), is present ubiquitously in the anticodon regions of transfer RNAs (tRNAs), large and small ribosomal RNAs (rRNAs), and small nuclear RNAs (snRNAs). y is the most common of the modified nucleosides (i.e., other than G, A, U, and C) present in tRNAs. Only a few yeast tRNAs that are not involved in protein synthesis do not contain ψ (Cortese, R. et al. (1974) J. Biol. Chem. 249:1103-1108). The enzyme responsible for the conversion of uridine to ψ, pseudouridine synthase (pseudouridylate synthase), was first isolated from *Salmonella typhimurium* (Arena, F. et al. (1978) Nucleic Acids Res. 5:4523-4536). The enzyme has since been isolated from a number of mammals, including steer and mice (Green, C.J. et al. (1982) J. Biol. Chem. 257:3045-52; and Chen, J. and Patton, J.R. (1999) RNA 5:409-419). tRNA pseudouridine synthases have been the most extensively studied members of the family. They require a thiol donor (e.g., cysteine) and a monovalent cation (e.g., ammonia or potassium) for optimal activity. Additional cofactors or high energy molecules (e.g., ATP or GTP) are not required (Green, *supra*). Other eukaryotic pseudouridine synthases have been identified that appear to be specific for rRNA (reviewed in Smith, C.M. and Steitz, J.A. (1997) Cell 89:669-672) and a dual-specificity enzyme has been identified that uses both tRNA and rRNA substrates (Wrzesinski, J. et al. (1995) RNA 1: 437-448). The absence of ψ in the anticodon loop of tRNAs results in reduced growth in both bacteria (Singer, C.E. et al. (1972) Nature New Biol. 238:72-74) and yeast (Lecointe, F. (1998) J. Biol. Chem. 273:1316-1323), although the genetic defect is not lethal.

Another ribonucleoside modification that occurs primarily in eukaryotic cells is the conversion of guanosine to N^2,N^2-dimethylguanosine ($m^2{}_2G$) at position 26 or 10 at the base of the D-stem of cytosolic and mitochondrial tRNAs. This posttranscriptional modification is believed to stabilize tRNA structure by preventing the formation of alternative tRNA secondary and tertiary structures. Yeast $tRNA^{Asp}$ is unusual in that it does not contain this modification. The modification does not occur in eubacteria, presumably because the structure of tRNAs in these cells and organelles is sequence constrained and does not require posttranscriptional modification to prevent the formation of alternative structures (Steinberg, S. and Cedergren, R. (1995) RNA 1:886-891, and references within). The enzyme responsible for the conversion of guanosine to $m^2{}_2G$ is a 63 kDa *S*-adenosylmethionine (SAM)-dependent tRNA N^2,N^2-dimethyl-guanosine methyltransferase (also referred to as the *TRM1* gene product and herein referred to as TRM) (Edqvist, J. (1995) Biochimie 77:54-61). The enzyme localizes to both the nucleus and the mitochondria (Li, J-M. et al. (1989) J. Cell Biol. 109:1411-1419). Based on studies with TRM from *Xenopus laevis*, there appears to be a requirement for base pairing at positions C11-G24 and G10-C25 immediately preceding the G26 to be modified, with other structural features of the tRNA also being required for the proper presentation of the G26 substrate (Edqvist. J. et al. (1992) Nucleic Acids Res. 20:6575-6581). Studies in yeast suggest that cells carrying a weak ochre tRNA suppressor (sup3-i) are unable to suppress translation termination in the absence of TRM activity, suggesting a role for TRM in modifying the frequency of suppression in eukaryotic cells (Niederberger, C. et al. (1999) FEBS Lett. 464:67-70), in addition to the more general function of ensuring the proper three-dimensional structures for tRNA.

Translation Initiation

Initiation of translation can be divided into three stages. The first stage brings an initiator transfer RNA (Met-$tRNA_f$) together with the 40S ribosomal subunit to form the 43S preinitiation complex. The second stage binds the 43S preinitiation complex to the mRNA, followed by migration of the complex to the correct AUG initiation codon. The third stage brings the 60S ribosomal subunit to the 40S subunit to generate an 80S ribosome at the inititation codon. Regulation of translation primarily involves the first and second stage in the initiation process (V.M. Pain (1996) Eur. J. Biochem. 236:747-771).

Several initiation factors, many of which contain multiple subunits, are involved in bringing an initiator tRNA and the 40S ribosomal subunit together. eIF2, a guanine nucleotide binding protein, recruits the initiator tRNA to the 40S ribosomal subunit. Only when eIF2 is bound to GTP does it associate with the initiator tRNA. eIF2B, a guanine nucleotide exchange protein, is responsible for converting eIF2 from the GDP-bound inactive form to the GTP-bound active form. Two other factors, eIF1A and eIF3 bind and stabilize the 40S subunit by interacting with the 18S ribosomal RNA and specific ribosomal structural proteins. eIF3 is also involved in association of the 40S ribosomal subunit with mRNA. The Met-$tRNA_f$, eIF1A, eIF3, and 40S ribosomal subunit together make up the 43S preinitiation complex (Pain, *supra*).

Additional factors are required for binding of the 43S preinitiation complex to an mRNA molecule, and the process is regulated at several levels. eIF4F is a complex consisting of three proteins: eIF4E, eIF4A, and eIF4G. eIF4E recognizes and binds to the mRNA 5'-terminal m^7GTP cap, eIF4A is a bidirectional RNA-dependent helicase, and eIF4G is a scaffolding polypeptide. eIF4G has three binding domains. The N-terminal third of eIF4G interacts with eIF4E, the central third interacts with eIF4A, and the C-terminal third interacts with eIF3 bound to the 43S preinitiation complex. Thus, eIF4G acts as a bridge between the 40S ribosomal subunit and the mRNA (M.W. Hentze (1997) Science 275:500-501).

The ability of eIF4F to initiate binding of the 43S preinitiation complex is regulated by structural features of the mRNA. The mRNA molecule has an untranslated region (UTR) between the 5' cap and the AUG start codon. In some mRNAs this region forms secondary structures that impede binding of the 43S preinitiation complex. The helicase activity of eIF4A is thought to function in removing this secondary structure to facilitate binding of the 43S preinitiation complex (Pain, *supra*).

Translation Elongation

Elongation is the process whereby additional amino acids are joined to the initiator methionine to form the complete polypeptide chain. The elongation factors EF1 α, EF1 $\beta\gamma$, and EF2 are involved in elongating the polypeptide chain following initiation. EF1 α is a GTP-binding protein. In EF1 α's GTP-bound form, it brings an aminoacyl-tRNA to the ribosome's A site. The amino acid attached to the newly arrived aminoacyl-tRNA forms a peptide bond with the initiatior methionine. The GTP on EF1 α is hydrolyzed to GDP, and EF1 α -GDP dissociates from the ribosome. EF1 $\beta\gamma$ binds EF1 α -GDP and induces the dissociation of GDP from EF1 α, allowing EF1 α to bind GTP and a new cycle to begin.

As subsequent aminoacyl-tRNAs are brought to the ribosome, EF-G, another GTP-binding protein, catalyzes the translocation of tRNAs from the A site to the P site and finally to the E site of the ribosome. This allows the ribosome and the mRNA to remain attached during translation.

Translation Termination

The release factor eRF carries out termination of translation. eRF recognizes stop codons in the mRNA, leading to the release of the polypeptide chain from the ribosome.

Expression profiling

Microarrays are analytical tools used in bioanalysis. A microarray has a plurality of molecules spatially distributed over, and stably associated with, the surface of a solid support. Microarrays of polypeptides, polynucleotides, and/or antibodies have been developed and find use in a variety of applications, such as gene sequencing, monitoring gene expression, gene mapping, bacterial identification, drug discovery, and combinatorial chemistry.

One area in particular in which microarrays find use is in gene expression analysis. Array technology can provide a simple way to explore the expression of a single polymorphic gene or the expression profile of a large number of related or unrelated genes. When the expression of a single gene is examined, arrays are employed to detect the expression of a specific gene or its variants. When an expression profile is examined, arrays provide a platform for identifying genes that are tissue specific, are affected by a substance being tested in a toxicology assay, are part of a signaling cascade, carry out housekeeping functions, or are specifically related to a particular genetic predisposition, condition, disease, or disorder.

Colon cancer

While soft tissue sarcomas are relatively rare, more than 50% of new patients diagnosed with the disease will die from it. The molecular pathways leading to the development of sarcomas are relatively unknown, due to the rarity of the disease and variation in pathology. Colon cancer evolves through a multi-step process whereby pre-malignant colonocytes undergo a relatively defined sequence of events leading to tumor formation. Several factors participate in the process of tumor progression and malignant transformation including genetic factors, mutations, and selection.

To understand the nature of gene alterations in colorectal cancer, a number of studies have focused on the inherited syndromes. The first, Familial Adenomatous Polyposis (FAP), is caused by mutations in the Adenomatous Polyposis Coli gene (APC), resulting in truncated or inactive forms of the protein. This tumor suppressor gene has been mapped to chromosome 5q. The second known inherited syndrome is hereditary nonpolyposis colorectal cancer (HNPCC), which is caused by mutations in mismatch repair genes.

Although hereditary colon cancer syndromes occur in a small percentage of the population, and most colorectal cancers are considered sporadic, knowledge from studies of the hereditary syndromes can be applied broadly. For instance, somatic mutations in APC occur in at least 80% of sporadic colon tumors. APC mutations are thought to be the initiating event in disease progression. Other mutations occur subsequently. Approximately 50% of colorectal cancers contain activating mutations in ras, while 85% contain inactivating mutations in p53. Changes in all of these genes lead to gene expression changes in colon cancer. Less is understood about downstream targets of these mutations and the role they may play in cancer development and progression.

Breast cancer

More than 180,000 new cases of breast cancer are diagnosed each year, and the mortality rate for breast cancer approaches 10% of all deaths in females between the ages of 45-54 (Gish, K. (1999) AWIS Magazine 28:7-10). However the survival rate based on early diagnosis of localized breast cancer is extremely high (97%), compared with the advanced stage of the disease in which the tumor has spread beyond the breast (22%). Current procedures for clinical breast examination are lacking in sensitivity and specificity, and efforts are underway to develop comprehensive gene expression profiles for breast cancer that may be used in conjunction with conventional screening methods to improve diagnosis and prognosis of this disease (Perou, C.M. et al. (2000) Nature 406:747-752).

Mutations in two genes, BRCA1 and BRCA2, are known to greatly predispose a woman to breast cancer and may be passed on from parents to children (Gish, *supra*). However, this type of hereditary breast cancer accounts for only about 5% to 9% of breast cancers, while the vast majority of breast cancer is due to non-inherited mutations that occur in breast epithelial cells.

The relationship between expression of epidermal growth factor (EGF) and its receptor, EGFR, to human mammary carcinoma has been particularly well studied. (See Khazaie, K. et al. (1993) Cancer and Metastasis Rev. 12:255-274, and references cited therein for a review of this area.) Overexpression of EGFR, particularly coupled with down-regulation of the estrogen receptor, is a marker of poor prognosis in breast cancer patients. In addition, EGFR expression in breast tumor metastases is frequently elevated relative to the primary tumor, suggesting that EGFR is involved in tumor progression and metastasis. This is supported by accumulating evidence that EGF has effects on cell functions related to metastatic potential, such as cell motility, chemotaxis, secretion and differentiation. Changes in expression of other members of the erbB receptor family, of which EGFR is one, have also been implicated in breast cancer. The abundance of erbB receptors, such as HER-2/neu, HER-3, and HER-4, and their ligands in breast cancer points to their functional importance in the pathogenesis of the disease, and may therefore provide targets for therapy of the disease (Bacus, S.S. et al. (1994) Am. J. Clin. Pathol. 102:S13-S24). Other known markers of breast cancer include a human secreted frizzled protein mRNA that is downregulated in breast tumors; the matrix Gla protein which is overexpressed in human breast carcinoma cells; Drg1 or RTP, a gene whose expression is diminished in colon, breast, and prostate tumors; maspin, a tumor suppressor gene downregulated in invasive breast carcinomas; and CaN19, a member of the S100 protein family, all of which are down-regulated in mammary carcinoma cells relative to normal mammary epithelial cells (Zhou, Z. et al. (1998) Int. J. Cancer 78:95-99; Chen, L. et al. (1990) Oncogene 5:1391-1395; Ulrix,

W. et al (1999) FEBS Lett 455:23-26; Sager, R. et al. (1996) Curr. Top. Microbiol. Immunol. 213:51-64; and Lee, S.W. et al. (1992) Proc. Natl. Acad. Sci. USA 89:2504-2508).

Cell lines derived from human mammary epithelial cells at various stages of breast cancer provide a useful model to study the process of malignant transformation and tumor progression as it has been shown that these cell lines retain many of the properties of their parental tumors for lengthy culture periods (Wistuba, I.I. et al. (1998) Clin. Cancer Res. 4:2931-2938). Such a model is particularly useful for comparing phenotypic and molecular characteristics of human mammary epithelial cells at various stages of malignant transformation.

Epidermal growth factor (EGF) was originally discovered in crude preparations of nerve growth factor prepared from mouse submaxillary glands as an activity that induced early eyelid opening, incisor eruption, hair growth inhibition, and stunting of growth when injected into newborn mice. Human EGF was isolated from urine based on its inhibitory effect on gastric secretion and named urogastrone, accordingly. EGF is prototypic of a family of growth factors that are derived from membrane-anchored precursors. All members of this family are characterized by the presence of at least one EGF structural unit (defined by the presence of a conserved 6-cysteine motif that forms three disulfide bonds) in their extracellular domain. A wide variety of *in vitro* and *in vivo* biological effects have been ascribed to EGF and other members of the EGF family. The best documented activity of EGF is its ability to promote proliferation and differentiation of mesenchymal and epithelial cells. *In vitro*, EGF is a mitogen for fibroblasts, epithelial and endothelial cells, and promotes colony formation of epidermal cells in culture. *In vivo*, EGF induces epithelial development, promotes angiogenesis, and inhibits gastric acid secretion.

EGF is a potent mitogen for normal as well as malignant mammary epithelial cells. In addition, EGF was found to induce tumor progression, and is highly expressed in breast carcinoma cells. EGF produces its effect by binding to its receptor, the protein tyrosine kinase EGF receptor (EGFR) also known as erbB2. Ligation of EGF to the EGFR activates the tyrosine kinase domain of EGFR and the phosphorylation of multiple substrates, thereby regulating gene expression. Since EGF plays an important role in growth as well as malignant progression, we investigated the effect of EGF on gene expression in multiple breast carcinoma cell lines. However, more recent studies have indicated that the EGFR may participate in signaling cascades unrelated to the EGF pathway.

Lung cancer

Lung cancer is the leading cause of cancer death in the United States, affecting more than 100,000 men and 50,000 women each year. Nearly 90% of the patients diagnosed with lung cancer are cigarette smokers. Tobacco smoke contains thousands of noxious substances that induce carcinogen metabolizing enzymes and covalent DNA adduct formation in the exposed bronchial epithelium. In nearly 80% of patients diagnosed with lung cancer, metastasis has already occurred.

Most commonly lung cancers metastasize to pleura, brain, bone, pericardium, and liver. The decision to treat with surgery, radiation therapy, or chemotherapy is made on the basis of tumor histology, response to growth factors or hormones, and sensitivity to inhibitors or drugs. With current treatments, most patients die within one year of diagnosis. Earlier diagnosis and a systematic approach to identification, staging, and treatment of lung cancer could positively affect patient outcome.

Lung cancers progress through a series of morphologically distinct stages from hyperplasia to invasive carcinoma. Malignant lung cancers are divided into two groups comprising four histopathological classes. The Non Small Cell Lung Carcinoma (NSCLC) group includes squamous cell carcinomas, adenocarcinomas, and large cell carcinomas and accounts for about 70% of all lung cancer cases. Adenocarcinomas typically arise in the peripheral airways and often form mucin secreting glands. Squamous cell carcinomas typically arise in proximal airways. The histogenesis of squamous cell carcinomas may be related to chronic inflammation and injury to the bronchial epithelium, leading to squamous metaplasia. The Small Cell Lung Carcinoma (SCLC) group accounts for about 20% of lung cancer cases. SCLCs typically arise in proximal airways and exhibit a number of paraneoplastic syndromes including inappropriate production of adrenocorticotropin and anti-diuretic hormone.

Lung cancer cells accumulate numerous genetic lesions, many of which are associated with cytologically visible chromosomal aberrations. The high frequency of chromosomal deletions associated with lung cancer may reflect the role of multiple tumor suppressor loci in the etiology of this disease. Deletion of the short arm of chromosome 3 is found in over 90% of cases and represents one of the earliest genetic lesions leading to lung cancer. Deletions at chromosome arms 9p and 17p are also common. Other frequently observed genetic lesions include overexpression of telomerase, activation of oncogenes such as K-ras and c-myc, and inactivation of tumor suppressor genes such as RB, p53 and CDKN2.

Genes differentially regulated in lung cancer have been identified by a variety of methods. Using mRNA differential display technology, Manda et al. (1999; Genomics 51:5-14) identified five genes differentially expressed in lung cancer cell lines compared to normal bronchial epithelial cells. Among the known genes, pulmonary surfactant apoprotein A and alpha 2 macroglobulin were down regulated whereas nm23H1 was upregulated. Petersen et al. (2000; Int J. Cancer, 86:512-517) used suppression subtractive hybridization to identify 552 clones differentially expressed in lung tumor derived cell lines, 205 of which represented known genes. Among the known genes, thrombospondin-1, fibronectin, intercellular adhesion molecule 1, and cytokeratins 6 and 18 were previously observed to be differentially expressed in lung cancers. Wang et al. (2000; Oncogene 19:1519-1528) used a combination of microarray analysis and subtractive hybridization to identify 17 genes differentially overexpresssed in squamous cell carcinoma compared with normal lung epithelium. Among the known genes they identified were keratin isoform 6, KOC, SPRC, IGFb2, connexin 26, plakofillin 1 and cytokeratin 13.

Ovarian cancer

Ovarian cancer is the leading cause of death from a gynecologic cancer. The majority of ovarian cancers are derived from epithelial cells, and 70% of patients with epithelial ovarian cancers present with late-stage disease. As a result, the long-term survival rate for this disease is very low. Identification of early-stage markers for ovarian cancer would significantly increase the survival rate. Genetic variations involved in ovarian cancer development include mutation of p53 and microsatellite instability. Gene expression patterns likely vary when normal ovary is compared to ovarian tumors.

Prostate cancer

Prostate cancer is a common malignancy in men over the age of 50, and the incidence increases with age. In the US, there are approximately 132,000 newly diagnosed cases of prostate cancer and more than 33,000 deaths from the disorder each year.

Once cancer cells arise in the prostate, they are stimulated by testosterone to a more rapid growth. Thus, removal of the testes can indirectly reduce both rapid growth and metastasis of the cancer. Over 95 percent of prostatic cancers are adenocarcinomas which originate in the prostatic acini. The remaining 5 percent are divided between squamous cell and transitional cell carcinomas, both of which arise in the prostatic ducts or other parts of the prostate gland.

As with most tumors, prostate cancer develops through a multistage progression ultimately resulting in an aggressive tumor phenotype. The initial step in tumor progression involves the hyperproliferation of normal luminal and/or basal epithelial cells. Androgen responsive cells become hyperplastic and evolve into early-stage tumors. Although early-stage tumors are often androgen sensitive and respond to androgen ablation, a population of androgen independent cells evolve from the hyperplastic population. These cells represent a more advanced form of prostate tumor that may become invasive and potentially become metastatic to the bone, brain, or lung. A variety of genes may be differentially expressed during tumor progression. For example, loss of heterozygosity (LOH) is frequently observed on chromosome 8p in prostate cancer. Fluorescence *in situ* hybridization (FISH) revealed a deletion for at least 1 locus on 8p in 29 (69%) tumors, with a significantly higher frequency of the deletion on 8p21.2-p21.1 in advanced prostate cancer than in localized prostate cancer, implying that deletions on 8p22-p21.3 play an important role in tumor differentiation, while 8p21.2-p21.1 deletion plays a role in progression of prostate cancer (Oba, K. et al. (2001) Cancer Genet. Cytogenet. 124: 20-26).

A primary diagnostic marker for prostate cancer is prostate specific antigen (PSA). PSA is a tissue-specific serine protease almost exclusively produced by prostatic epithelial cells. The quantity of PSA correlates with the number and volume of the prostatic epithelial cells, and consequently, the levels of PSA are an excellent indicator of abnormal prostate growth. Men with prostate cancer exhibit an early linear increase in PSA levels followed by an exponential increase prior to diagnosis. However, since PSA levels are also influenced by factors such as inflammation, androgen and other growth factors, some scientists maintain that changes in PSA levels are not useful in detecting individual cases of prostate cancer.

Current areas of cancer research provide additional prospects for markers as well as potential therapeutic targets for prostate cancer. Several growth factors have been shown to play a critical role in tumor development, growth, and progression. The growth factors Epidermal Growth Factor (EGF), Fibroblast Growth Factor (FGF), and Tumor Growth Factor alpha (TGFα) are important in the growth of normal as well as hyperproliferative prostate epithelial cells, particularly at early stages of tumor development and progression, and affect signaling pathways in these cells in various ways (Lin, J. et al. (1999) Cancer Res. 59:2891-2897; Putz, T. et al. (1999) Cancer Res. 59:227-233). The TGF-β family of growth factors are generally expressed at increased levels in human cancers and the high expression levels in many cases correlates with advanced stages of malignancy and poor survival (Gold, L.I. (1999) Crit. Rev. Oncog. 10:303-360). Finally, there are human cell lines representing both the androgen-dependent stage of prostate cancer (LNCap) as well as the androgen-independent, hormone refractory stage of the disease (PC3 and DU-145) that have proved useful in studying gene expression patterns associated with the progression of prostate cancer, and the effects of cell treatments on these expressed genes (Chung, T.D. (1999) Prostate 15:199-207).

Inflammation and Immune Response

Human umbilical vein endothelial cells (HUVEC) are used in the study of vascular and other physiology. Atherosclerosis is a pathological condition characterized by a chronic local inflammatory response within the vessel wall of major arteries. Disease progression results in the formation of atherosclerotic lesions, unstable plaques which occasionally rupture, precipitating a catastrophic thrombotic occlusion of the vessel lumen. Atherosclerosis and the associated coronary artery disease and cerebral stroke represent the most common causes of death in industrialized nations. Although certain key risk factors have been identified, a full molecular characterization that elucidates the causes and identifies all potential therapeutic targets for this complex disease has not been achieved. Molecular characterization of atherosclerosis requires identification of the genes that contribute to lesion growth, stability, dissolution, rupture and induction of occlusive vessel thrombi.

Blood vessel walls are composed of two tissue layers: an endothelial cell (EC) layer which comprises the lumenal surface of the vessel, and an underlying vascular smooth muscle cell (VSMC) layer. Through dynamic interactions with each other and with surrounding tissues, the vascular endothelium and smooth muscle tissues maintain vascular tone, control selective permeability of the vascular wall, direct vessel remodeling and angiogenesis, and modulate inflammatory and immune responses.

The inflammatory response is a complex vascular reaction mediated by numerous cytokines, chemokines, growth factors, and other signaling molecules expressed by activated ECs, VSMCs and leukocytes. Inflammation protects the organism during trauma and infection, but can also lead to pathological conditions such as atherosclerosis. The pro-inflammatory cytokines, interleukin (IL)-1 and tumor necrosis factor (TNF), are secreted by a small number of activated macrophages or other cells and can set off a cascade of vascular changes, largely through their ability to alter gene expression patterns in ECs and VSMCs. These vascular changes include vasodilation and increased permeability of microvasculature, edema, and leukocyte extravasation and transmigration across the vessel wall. Ultimately, leukocytes, particularly neutrophils and monocytes/macrophages, accumulate in the extravascular space, where they remove injurious agents by phagocytosis and oxidative killing, a process accompanied by release of toxic factors, such as proteases and reactive oxygen species.

IL-1 and TNF induce pro-inflammatory, thrombotic, and anti-apoptotic changes in gene expression by signaling through receptors on the surface of ECs and VSMCs; these receptors activate transcription factors such as NF*k*B as well as AP-1, IRF-1, and NF-GMa, leading to alterations in gene expression. Genes known to be differentially regulated in EC by IL-1 and TNF include E selectin, VCAM-1, ICAM-1, PAF, I*k*Bα, IAP-1, MCP-1, eotaxin, ENA-78, G-CSF, A20, ICE, and complement C3 component. A key event in inflammation, adhesion and transmigration of blood leukocytes across the vascular endothelium, for example, is mediated by increased expression of E selectin, P selectin, ICAM-1, and VCAM-1 on activated endothelium.

Several investigators have examined changes in vascular cell gene expression associated with various inflammatory diseases or model systems. Examining human umbilical vein endothelial cells (HUVEC) activated by recombinant TNFα or conditioned medium from activated human primary monocytes, Horrevoets et al. (1999; Blood 93:3418-3431) identified 106 differentially regulated genes. In a similar approach, deVries et al. (2000; J. Biol. Chem. 275:23939-23947) identified 40 differentially regulated genes in umbilical cord artery-derived smooth muscle cells activated by conditioned media from cultured macrophages after stimulation with oxidized LDL particles. In both studies, many of the identified genes were already known to be involved in inflammation. Comparing expression profiles from inflammatory diseased tissues, cultured macrophages, chondrocyte cell lines, primary chondrocytes, and synoviocytes, Heller et al. (1997; Proc. Natl. Acad. Sci. USA 94:2150-2155) identifed candidate genes involved in inflammatory responses, including TNF, IL-1 IL-6, IL-8 G-CSF, RANTES, and V-CAM. From this candidate gene set, tissue inhibitor of metalloproteinase 1, ferritin light chain, and manganese superoxide dismutase were found to be differentially expressed in rheumatoid arthritis (RA) relative to inflammatory bowel disease (IBD). Further, IL-3, chemokine Groα, and metalloproteinase matrix metallo-elastase were expressed in both RA and IBD. Most recently, in an analysis of cultured aortic smooth muscle cells treated with TNFα, Haley et al. (2000; Circulation 102:2185-2189) found a 20-fold increase in eotaxin, an eosinophil chemotactic factor. The overexpression of eotaxin and its receptor CCR3 in atherosclerotic lesions was confirmed by northern analysis.

Tumor necrosis factor-alpha (TNF-alpha) is a proinflammatory cytokine. It mediates immune regulation and inflammatory responses through various intermediates, including protein kinases, protein phosphatases, reactive oxygen intermediates, phospholipases, proteases, sphingomyelinases and transcription factors. TNF-α-related cytokines generate cellular responses including differentiation, proliferation, cell death, and activation of nuclear factor-κB (NF-κB) (Smith, C.A. et al. (1994) Cell 76:959-962), through their interaction with distinct cell surface receptors (TNFRs). NF-κB is a transcription factor that induces genes involved in physiological processes such as response to injury and infection. (For a review of TNF-α in the NF-κB activation pathway see Bowie and O'Neill (2000) Biochem. Pharmacol. 59:13-23.)

TNF-alpha is upregulated when the endothelium is physically disrupted or functionally perturbed by events such as postischemic reperfusion, acute and chronic inflammation, atherosclerosis, diabetes and chronic arterial hypertension. Inflammatory stimulation sets the stage for later tissue repair. Elevated TNF-alpha initially increases, and then inhibits, the activity of a number of key enzymes including protein-tyrosine kinase (PTKase) and protein-tyrosine phosphatase (Holden, R.J. et al. (1999) Med. Hypotheses 52:319-23).

Development of atherosclerosis involves inflammatory responses induced by circulating lipoprotein. Lipoproteins, such as low-density lipoprotein (LDL), accumulate in the extracellular space of the vascular intima and undergo modifications including oxidation of LDL to Ox-LDL, most avidly in the sub-endothelial space where circulating antioxidant defenses are less effective. Mononuclear phagocytes enter the intima, differentiate into macrophages, and ingest modified lipids including Ox-LDL. During Ox-LDL uptake, macrophages produce cytokines including TNF-α, as well as interleukin-1 and growth factors (e.g. M-CSF, VEGF, and PDGF-BB), that elicit further events in atherogenesis such as smooth muscle cell proliferation and production of extracellular matrix by vascular endothelium. These macrophages may also activate genes in endothelium and smooth muscle tissue involved in inflammation and tissue differentiation, including superoxide dismutatse (SOD), IL-8, and ICAM-1.

Non-atherosclerotic vascular endothelium not only mediates vascular dilatation but prevents platelet adhesion and activation, blocks thrombin formation, mitigates fibrin deposition, and attenuates adhesion and transmigration of inflammatory leukocytes. When the endothelium is physically disrupted, or perturbed by events such as postischemic reperfusion, acute and chronic inflammation, atherosclerosis, diabetes and chronic arterial hypertension, it acts in the opposite manner. The perturbed or proinflammatory state is characterised by vaso-constriction, platelet and leukocyte activation and adhesion (involving externalisation, expression and upregulation of, for example, von Willebrand factor, platelet activating factor, P-selectin, ICAM-1, IL-8, MCP-1, and TNF-α), promotion of thrombin formation, coagulation and deposition of fibrin at the vascular wall (expression of tissue factor, PAI-1, and phosphatidyl serine) and, in platelet-leukocyte coaggregates, additional inflammatory interactions via attachment of platelet CD40-ligand to endothelial, monocyte and B-cell CD40. Thrombin formation and inflammatory stimulation set the stage for later tissue repair, but limiting procoagulatory, prothrombotic actions of a dysfunctional vascular endothelium may be the goal of clinical interventions (for review, see Becker et al. (2000) Z Kardiol 89:160-167).

<u>Peroxisome Proliferator-activated Receptor Gamma (PPARγ) Agonist</u>

Thiazolidinediones (TZDs) act as agonists for the peroxisome-proliferator-activated receptor gamma (PPARγ), a member of the nuclear hormone receptor superfamily. TZDs reduce hyperglycemia, hyperinsulinemia, and hypertension, in part by promoting glucose metabolism and inhibiting gluconeogenesis. Roles for PPARγ and its agonists have been demonstrated in a wide range of pathological conditions including diabetes, obesity, hypertension, atherosclerosis, polycystic ovarian syndrome, and cancers such as breast, prostate, liposarcoma, and colon cancer.

The mechanism by which TZDs and other PPARγ agonists enhance insulin sensitivity is not fully understood, but may involve the ability of PPARγ to promote adipogenesis. When ectopically expressed in cultured preadipocytes, PPARγ is a potent inducer of adipocyte differentiation. TZDs, in combination with insulin and other factors, can also enhance differentiation of human preadipocytes in culture (Adams et al. (1997) J. Clin. Invest. 100:3149-3153). The relative potency of different TZDs in promoting adipogenesis *in vitro* is proportional to both their insulin sensitizing effects *in vivo*, and their ability to bind and activate PPARγ *in vitro*. Interestingly, adipocytes derived from omental adipose depots are refractory to the effects of TZDs. It has therefore been suggested that the insulin sensitizing effects of TZDs may result from their ability to promote adipogenesis in subcutaneous adipose depots (Adams et al., *supra*). Further, dominant negative mutations in the PPARγ gene have been identified in two non-obese subjects with severe insulin resistance, hypertension, and overt non-insulin dependent diabetes mellitus (NIDDM) (Barroso et al. (1998) Nature 402:880-883).

NIDDM is the most common form of diabetes mellitus, a chronic metabolic disease that affects 143 million people worldwide. NIDDM is characterized by abnormal glucose and lipid metabolism that results from a combination of peripheral insulin resistance and defective insulin secretion. NIDDM has a complex, progressive etiology and a high degree of heritability. Numerous complications of diabetes including heart disease, stroke, renal failure, retinopathy, and peripheral neuropathy contribute to the high rate of morbidity and mortality.

At the molecular level, PPARγ functions as a ligand activated transcription factor. In the presence of ligand, PPARγ forms a heterodimer with the retinoid X receptor (RXR) which then activates transcription of target genes containing one or more copies of a PPARγ response element (PPRE). Many genes important in lipid storage and metabolism contain PPREs and have been identified as PPARγ targets, including PEPCK, aP2, LPL, ACS, and FAT-P (Auwerx, J. (1999) Diabetologia 42:1033-1049). Multiple ligands for PPARγ have been identified. These include a variety of fatty acid metabolites; synthetic drugs belonging to the TZD class, such as Pioglitazone and Rosiglitazone (BRL49653); and certain non-glitazone tyrosine analogs such as GI262570 and GW1929. The prostaglandin derivative 15-dPGJ2 is a potent endogenous ligand for PPARγ.

Expression of PPARγ is very high in adipose but barely detectable in skeletal muscle, the primary site for insulin stimulated glucose disposal in the body. PPARγ is also moderately expressed in large intestine, kidney, liver, vascular smooth muscle, hematopoietic cells, and macrophages. The high expression of PPARγ in adipose tissue suggests that the insulin sensitizing effects of TZDs may result from alterations in the expression of one or more PPARγ regulated genes in adipose tissue. Identification of PPARγ target genes will contribute to better drug design and the development of novel therapeutic strategies for diabetes, obesity, and other conditions.

Systematic attempts to identify PPARγ target genes have been made in several rodent models of obesity and diabetes (Suzuki et al. (2000) Jpn. J. Pharmacol. 84:113-123; Way et al. (2001) Endocrinology 142:1269-1277). However, a serious drawback of the rodent gene expression studies is that significant differences exist between human and rodent models of adipogenesis, diabetes, and obesity (Taylor (1999) Cell 97:9-12; Gregoire et al. (1998) Physiol. Reviews 78:783-809). Therefore, an unbiased approach to identifying TZD regulated genes in primary cultures of human tissues is necessary to fully elucidate the molecular basis for diseases associated with PPARγ activity.

There is a need in the art for new compositions, including nucleic acids and proteins, for the diagnosis, prevention, and treatment of cell proliferative, neurological, developmental, autoimmune/inflammatory, and lipid metabolism disorders, and infections.

SUMMARY OF THE INVENTION

Various embodiments of the invention provide purified polypeptides, nucleic acid-associated proteins, referred to collectively as 'NAAP' and individually as 'NAAP-1,' 'NAAP-2,' 'NAAP-3,' 'NAAP-4,' 'NAAP-5,' 'NAAP-6,' 'NAAP-7,' 'NAAP-8,' 'NAAP-9,' 'NAAP-10,' 'NAAP-11,' 'NAAP-12,' 'NAAP-13,' 'NAAP-14,' 'NAAP-15,' 'NAAP-16,' 'NAAP-17,' 'NAAP-18,' 'NAAP-19,' 'NAAP-20,' 'NAAP-21,' 'NAAP-22,' 'NAAP-23,' 'NAAP-24,' 'NAAP-25,' 'NAAP-26,'

'NAAP-27,' 'NAAP-28,' 'NAAP-29,' 'NAAP-30,' 'NAAP-31,' 'NAAP-32,' 'NAAP-33,' 'NAAP-34,' and 'NAAP-35,' and methods for using these proteins and their encoding polynucleotides for the detection, diagnosis, and treatment of diseases and medical conditions. Embodiments also provide methods for utilizing the purified nucleic acid-associated proteins and/or their encoding polynucleotides for facilitating the drug discovery process, including determination of efficacy, dosage, toxicity, and pharmacology. Related embodiments provide methods for utilizing the purified nucleic acid-associated proteins and/or their encoding polynucleotides for investigating the pathogenesis of diseases and medical conditions.

An embodiment provides an isolated polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. Another embodiment provides an isolated polypeptide comprising an amino acid sequence of SEQ ID NO:1-35.

Still another embodiment provides an isolated polynucleotide encoding a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. In another embodiment, the polynucleotide encodes a polypeptide selected from the group consisting of SEQ ID NO:1-35. In an alternative embodiment, the polynucleotide is selected from the group consisting of SEQ ID NO:36-70.

Still another embodiment provides a recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide encoding a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ

ID NO:1-35. Another embodiment provides a cell transformed with the recombinant polynucleotide. Yet another embodiment provides a transgenic organism comprising the recombinant polynucleotide.

Another embodiment provides a method for producing a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. The method comprises a) culturing a cell under conditions suitable for expression of the polypeptide, wherein said cell is transformed with a recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide encoding the polypeptide, and b) recovering the polypeptide so expressed.

Yet another embodiment provides an isolated antibody which specifically binds to a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

Still yet another embodiment provides an isolated polynucleotide selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). In other embodiments, the polynucleotide can comprise at least about 20, 30, 40, 60, 80, or 100 contiguous nucleotides.

Yet another embodiment provides a method for detecting a target polynucleotide in a sample, said target polynucleotide being selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). The method comprises a) hybridizing the sample with a probe comprising at least 20 contiguous nucleotides comprising a sequence complementary to said target polynucleotide in the sample, and which probe specifically hybridizes to said target polynucleotide, under conditions whereby a hybridization complex is formed between said probe and said target polynucleotide or fragments thereof, and b) detecting the presence or absence of said hybridization complex. In a related embodiment, the method can include detecting the amount of the hybridization complex. In still other embodiments, the probe can comprise at least about 20, 30, 40, 60, 80, or 100 contiguous nucleotides.

Still yet another embodiment provides a method for detecting a target polynucleotide in a sample, said target polynucleotide being selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). The method comprises a) amplifying said target polynucleotide or fragment thereof using polymerase chain reaction amplification, and b) detecting the presence or absence of said amplified target polynucleotide or fragment thereof. In a related embodiment, the method can include detecting the amount of the amplified target polynucleotide or fragment thereof.

Another embodiment provides a composition comprising an effective amount of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and a pharmaceutically acceptable excipient. In one embodiment, the composition can comprise an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. Other embodiments provide a method of treating a disease or condition associated with decreased or abnormal expression of functional NAAP, comprising administering to a patient in need of such treatment the composition.

Yet another embodiment provides a method for screening a compound for effectiveness as an agonist of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. The method comprises a) contacting a sample comprising the polypeptide with a compound, and b) detecting agonist activity in the sample. Another embodiment provides a composition comprising an agonist compound identified by the method and a pharmaceutically acceptable excipient. Yet another embodiment provides a method of treating a disease or condition associated with decreased expression of functional NAAP, comprising administering to a patient in need of such treatment the composition.

Still yet another embodiment provides a method for screening a compound for effectiveness as an antagonist of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. The method comprises a) contacting a sample comprising the polypeptide with a compound, and b) detecting antagonist activity in the sample. Another embodiment provides a composition comprising an antagonist compound identified by the method and a pharmaceutically acceptable excipient. Yet another embodiment provides a method of treating a disease or condition associated with overexpression of functional NAAP, comprising administering to a patient in need of such treatment the composition.

Another embodiment provides a method of screening for a compound that specifically binds to a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. The method comprises a) combining the polypeptide with at least one test compound under suitable conditions, and b) detecting binding of the polypeptide to the test compound, thereby identifying a compound that specifically binds to the polypeptide.

Yet another embodiment provides a method of screening for a compound that modulates the activity of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35. The method comprises a) combining the polypeptide with at least one test compound under conditions permissive for the activity of the polypeptide, b) assessing the activity of the polypeptide in the presence of the test compound, and c) comparing the activity of the polypeptide in the presence of the test compound with the activity of the polypeptide in the absence of the test compound, wherein a change in the activity of the polypeptide in the presence of the test compound is indicative of a compound that modulates the activity of the polypeptide.

Still yet another embodiment provides a method for screening a compound for effectiveness in altering expression of a target polynucleotide, wherein said target polynucleotide comprises a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, the method comprising a) contacting a sample comprising the target polynucleotide with a compound, b) detecting altered expression of the target polynucleotide, and c) comparing the expression of the target polynucleotide in the presence of varying amounts of the compound and in the absence of the compound.

Another embodiment provides a method for assessing toxicity of a test compound, said method comprising a) treating a biological sample containing nucleic acids with the test compound; b) hybridizing the nucleic acids of the treated biological sample with a probe comprising at least 20 contiguous nucleotides of a polynucleotide selected from the group consisting of i) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, ii) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, iii) a polynucleotide having a sequence complementary to i), iv) a polynucleotide complementary to the polynucleotide of ii), and v) an RNA equivalent of i)-iv). Hybridization occurs under conditions whereby a specific hybridization complex is formed between said probe and a target polynucleotide in the biological sample, said target polynucleotide selected from the group consisting of i) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, ii) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70, iii) a polynucleotide complementary to the polynucleotide of i), iv) a polynucleotide complementary to the polynucleotide of ii), and v) an RNA equivalent of i)-

iv). Alternatively, the target polynucleotide can comprise a fragment of a polynucleotide selected from the group consisting of i)-v) above; c) quantifying the amount of hybridization complex; and d) comparing the amount of hybridization complex in the treated biological sample with the amount of hybridization complex in an untreated biological sample, wherein a difference in the amount of hybridization complex in the treated biological sample is indicative of toxicity of the test compound.

BRIEF DESCRIPTION OF THE TABLES

Table 1 summarizes the nomenclature for full length polynucleotide and polypeptide embodiments of the invention.

Table 2 shows the GenBank identification number and annotation of the nearest GenBank homolog, and the PROTEOME database identification numbers and annotations of PROTEOME database homologs, for polypeptide embodiments of the invention. The probability scores for the matches between each polypeptide and its homolog(s) are also shown.

Table 3 shows structural features of polypeptide embodiments, including predicted motifs and domains, along with the methods, algorithms, and searchable databases used for analysis of the polypeptides.

Table 4 lists the cDNA and/or genomic DNA fragments which were used to assemble polynucleotide embodiments, along with selected fragments of the polynucleotides.

Table 5 shows representative cDNA libraries for polynucleotide embodiments.

Table 6 provides an appendix which describes the tissues and vectors used for construction of the cDNA libraries shown in Table 5.

Table 7 shows the tools, programs, and algorithms used to analyze polynucleotides and polypeptides, along with applicable descriptions, references, and threshold parameters.

Table 8 shows single nucleotide polymorphisms found in polynucleotide sequences of the invention, along with allele frequencies in different human populations.

DESCRIPTION OF THE INVENTION

Before the present proteins, nucleic acids, and methods are described, it is understood that embodiments of the invention are not limited to the particular machines, instruments, materials, and methods described, as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the invention.

As used herein and in the appended claims, the singular forms "a," "an," and "the" include plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to "a host cell" includes a plurality of such host cells, and a reference to "an antibody" is a reference to one or more antibodies and equivalents thereof known to those skilled in the art, and so forth.

Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any machines, materials, and methods similar or equivalent to those described herein can be used to practice or test the present invention, the preferred machines, materials and methods are now described. All publications mentioned herein are cited for the purpose of describing and disclosing the cell lines, protocols, reagents and vectors which are reported in the publications and which might be used in connection with various embodiments of the invention. Nothing herein is to be construed as an admission that the invention is not entitled to antedate such disclosure by virtue of prior invention.

DEFINITIONS

"NAAP" refers to the amino acid sequences of substantially purified NAAP obtained from any species, particularly a mammalian species, including bovine, ovine, porcine, murine, equine, and human, and from any source, whether natural, synthetic, semi-synthetic, or recombinant.

The term "agonist" refers to a molecule which intensifies or mimics the biological activity of NAAP. Agonists may include proteins, nucleic acids, carbohydrates, small molecules, or any other compound or composition which modulates the activity of NAAP either by directly interacting with NAAP or by acting on components of the biological pathway in which NAAP participates.

An "allelic variant" is an alternative form of the gene encoding NAAP. Allelic variants may result from at least one mutation in the nucleic acid sequence and may result in altered mRNAs or in polypeptides whose structure or function may or may not be altered. A gene may have none, one, or many allelic variants of its naturally occurring form. Common mutational changes which give rise to allelic variants are generally ascribed to natural deletions, additions, or substitutions of nucleotides. Each of these types of changes may occur alone, or in combination with the others, one or more times in a given sequence.

"Altered" nucleic acid sequences encoding NAAP include those sequences with deletions, insertions, or substitutions of different nucleotides, resulting in a polypeptide the same as NAAP or a polypeptide with at least one functional characteristic of NAAP. Included within this definition are polymorphisms which may or may not be readily detectable using a particular oligonucleotide probe of the polynucleotide encoding NAAP, and improper or unexpected hybridization to allelic variants, with a locus other than the normal chromosomal locus for the polynucleotide encoding NAAP. The encoded protein may also be "altered," and may contain deletions, insertions, or substitutions of amino acid residues which produce a silent change and result in a functionally equivalent NAAP. Deliberate amino acid substitutions may be made on the basis of one or more similarities in polarity, charge, solubility, hydrophobicity, hydrophilicity, and/or the amphipathic nature of the residues, as long as the biological or immunological activity of NAAP is retained. For example, negatively charged amino acids may include aspartic acid and glutamic acid, and positively charged amino acids may include lysine and arginine. Amino acids with uncharged polar side chains having similar hydrophilicity values may include: asparagine and glutamine; and serine and threonine. Amino acids with uncharged side chains having similar hydrophilicity values may include: leucine, isoleucine, and valine; glycine and alanine; and phenylalanine and tyrosine.

The terms "amino acid" and "amino acid sequence" can refer to an oligopeptide, a peptide, a polypeptide, or a protein sequence, or a fragment of any of these, and to naturally occurring or synthetic molecules. Where "amino acid sequence" is recited to refer to a sequence of a naturally occurring protein molecule, "amino acid sequence" and like terms are not meant to limit the amino acid sequence to the complete native amino acid sequence associated with the recited protein molecule.

"Amplification" relates to the production of additional copies of a nucleic acid. Amplification may be carried out using polymerase chain reaction (PCR) technologies or other nucleic acid amplification technologies well known in the art.

The term "antagonist" refers to a molecule which inhibits or attenuates the biological activity of NAAP. Antagonists may include proteins such as antibodies, anticalins, nucleic acids, carbohydrates, small molecules, or any other compound or composition which modulates the activity of NAAP either by directly interacting with NAAP or by acting on components of the biological pathway in which NAAP participates.

The term "antibody" refers to intact immunoglobulin molecules as well as to fragments thereof, such as Fab, $F(ab')_2$, and Fv fragments, which are capable of binding an epitopic determinant. Antibodies that bind NAAP polypeptides can be prepared using intact polypeptides or using fragments containing small peptides of interest as the immunizing antigen. The polypeptide or oligopeptide used to immunize an animal (e.g., a mouse, a rat, or a rabbit) can be derived from the translation of RNA, or synthesized chemically, and can be conjugated to a carrier protein if desired. Commonly used carriers that are chemically coupled to peptides include bovine serum albumin, thyroglobulin, and keyhole limpet hemocyanin (KLH). The coupled peptide is then used to immunize the animal.

The term "antigenic determinant" refers to that region of a molecule (i.e., an epitope) that makes contact with a particular antibody. When a protein or a fragment of a protein is used to immunize a host animal, numerous regions of the protein may induce the production of antibodies which bind specifically to antigenic determinants (particular regions or three-dimensional structures on the protein). An antigenic determinant may compete with the intact antigen (i.e., the immunogen used to elicit the immune response) for binding to an antibody.

The term "aptamer" refers to a nucleic acid or oligonucleotide molecule that binds to a specific molecular target. Aptamers are derived from an *in vitro* evolutionary process (e.g., SELEX (Systematic Evolution of Ligands by EXponential Enrichment), described in U.S. Patent No. 5,270,163), which selects for target-specific aptamer sequences from large combinatorial libraries. Aptamer compositions may be double-stranded or single-stranded, and may include deoxyribonucleotides, ribonucleotides, nucleotide derivatives, or other nucleotide-like molecules. The nucleotide components of an aptamer may have modified sugar groups (e.g., the 2'-OH group of a ribonucleotide may be replaced by 2'-F or 2'-NH_2), which may improve a desired property, e.g., resistance to nucleases or longer lifetime in blood. Aptamers may be conjugated to other molecules, e.g., a high molecular weight carrier to slow clearance of the aptamer from the circulatory system. Aptamers may be specifically cross-linked to their cognate ligands, e.g., by photo-activation of a cross-linker (Brody, E.N. and L. Gold (2000) J. Biotechnol. 74:5-13).

The term "intramer" refers to an aptamer which is expressed *in vivo*. For example, a vaccinia virus-based RNA expression system has been used to express specific RNA aptamers at high levels in the cytoplasm of leukocytes (Blind, M. et al. (1999) Proc. Natl. Acad. Sci. USA 96:3606-3610).

The term "spiegelmer" refers to an aptamer which includes L-DNA, L-RNA, or other left-handed nucleotide derivatives or nucleotide-like molecules. Aptamers containing left-handed nucleotides are resistant to degradation by naturally occurring enzymes, which normally act on substrates containing right-handed nucleotides.

The term "antisense" refers to any composition capable of base-pairing with the "sense" (coding) strand of a polynucleotide having a specific nucleic acid sequence. Antisense compositions may include DNA; RNA; peptide nucleic acid (PNA); oligonucleotides having modified backbone linkages such as phosphorothioates, methylphosphonates, or benzylphosphonates; oligonucleotides having modified sugar groups such as 2'-methoxyethyl sugars or 2'-methoxyethoxy sugars; or oligonucleotides having modified bases such as 5-methyl cytosine, 2'-deoxyuracil, or 7-deaza-2'-deoxyguanosine. Antisense molecules may be produced by any method including chemical synthesis or transcription. Once introduced into a cell, the complementary antisense molecule base-pairs with a naturally occurring nucleic acid sequence produced by the cell to form duplexes which block either transcription or translation. The designation "negative" or "minus" can refer to the antisense strand, and the designation "positive" or "plus" can refer to the sense strand of a reference DNA molecule.

The term "biologically active" refers to a protein having structural, regulatory, or biochemical functions of a naturally occurring molecule. Likewise, "immunologically active" or "immunogenic" refers to the capability of the natural, recombinant, or synthetic NAAP, or of any oligopeptide thereof, to induce a specific immune response in appropriate animals or cells and to bind with specific antibodies.

"Complementary" describes the relationship between two single-stranded nucleic acid sequences that anneal by base-pairing. For example, 5'-AGT-3' pairs with its complement, 3'-TCA-5'.

A "composition comprising a given polynucleotide" and a "composition comprising a given polypeptide" can refer to any composition containing the given polynucleotide or polypeptide. The composition may comprise a dry formulation or an aqueous solution. Compositions comprising polynucleotides encoding NAAP or fragments of NAAP may be employed as hybridization probes. The probes may be stored in freeze-dried form and may be associated with a stabilizing agent such as a carbohydrate. In hybridizations, the probe may be deployed in an aqueous solution containing salts (e.g., NaCl), detergents (e.g., sodium dodecyl sulfate; SDS), and other components (e.g., Denhardt's solution, dry milk, salmon sperm DNA, etc.).

"Consensus sequence" refers to a nucleic acid sequence which has been subjected to repeated DNA sequence analysis to resolve uncalled bases, extended using the XL-PCR kit (Applied Biosystems, Foster City CA) in the 5' and/or the 3' direction, and resequenced, or which has been assembled from one or more overlapping cDNA, EST, or genomic DNA fragments using a computer program for fragment assembly, such as the GELVIEW fragment assembly system (Accelrys, Burlington MA) or Phrap (University of Washington, Seattle WA). Some sequences have been both extended and assembled to produce the consensus sequence.

"Conservative amino acid substitutions" are those substitutions that are predicted to least interfere with the properties of the original protein, i.e., the structure and especially the function of the protein is conserved and not significantly changed by such substitutions. The table below shows amino acids which may be substituted for an original amino acid in a protein and which are regarded as conservative amino acid substitutions.

Original Residue	Conservative Substitution
Ala	Gly, Ser
Arg	His, Lys
Asn	Asp, Gln, His
Asp	Asn, Glu
Cys	Ala, Ser
Gln	Asn, Glu, His
Glu	Asp, Gln, His
Gly	Ala
His	Asn, Arg, Gln, Glu
Ile	Leu, Val
Leu	Ile, Val
Lys	Arg, Gln, Glu
Met	Leu, Ile
Phe	His, Met, Leu, Trp, Tyr
Ser	Cys, Thr
Thr	Ser, Val
Trp	Phe, Tyr

Tyr	His, Phe, Trp
Val	Ile, Leu, Thr

Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

A "deletion" refers to a change in the amino acid or nucleotide sequence that results in the absence of one or more amino acid residues or nucleotides.

The term "derivative" refers to a chemically modified polynucleotide or polypeptide. Chemical modifications of a polynucleotide can include, for example, replacement of hydrogen by an alkyl, acyl, hydroxyl, or amino group. A derivative polynucleotide encodes a polypeptide which retains at least one biological or immunological function of the natural molecule. A derivative polypeptide is one modified by glycosylation, pegylation, or any similar process that retains at least one biological or immunological function of the polypeptide from which it was derived.

A "detectable label" refers to a reporter molecule or enzyme that is capable of generating a measurable signal and is covalently or noncovalently joined to a polynucleotide or polypeptide.

"Differential expression" refers to increased or upregulated; or decreased, downregulated, or absent gene or protein expression, determined by comparing at least two different samples. Such comparisons may be carried out between, for example, a treated and an untreated sample, or a diseased and a normal sample.

"Exon shuffling" refers to the recombination of different coding regions (exons). Since an exon may represent a structural or functional domain of the encoded protein, new proteins may be assembled through the novel reassortment of stable substructures, thus allowing acceleration of the evolution of new protein functions.

A "fragment" is a unique portion of NAAP or a polynucleotide encoding NAAP which can be identical in sequence to, but shorter in length than, the parent sequence. A fragment may comprise up to the entire length of the defined sequence, minus one nucleotide/amino acid residue. For example, a fragment may comprise from about 5 to about 1000 contiguous nucleotides or amino acid residues. A fragment used as a probe, primer, antigen, therapeutic molecule, or for other purposes, may be at least 5, 10, 15, 16, 20, 25, 30, 40, 50, 60, 75, 100, 150, 250 or at least 500 contiguous nucleotides or amino acid residues in length. Fragments may be preferentially selected from certain regions of a molecule. For example, a polypeptide fragment may comprise a certain length of contiguous amino acids selected from the first 250 or 500 amino acids (or first 25% or 50%) of a polypeptide as shown in a certain defined sequence. Clearly these lengths are exemplary, and any length that is supported by the specification, including the Sequence Listing, tables, and figures, may be encompassed by the present embodiments.

A fragment of SEQ ID NO:36-70 can comprise a region of unique polynucleotide sequence that specifically identifies SEQ ID NO:36-70, for example, as distinct from any other sequence in the genome from which the fragment was obtained. A fragment of SEQ ID NO:36-70 can be employed in one or more embodiments of methods of the invention, for example, in hybridization and amplification technologies and in analogous methods that distinguish SEQ ID NO:36-70 from related polynucleotides. The precise length of a fragment of SEQ ID NO:36-70 and the region of SEQ ID NO:36-70 to which the fragment corresponds are routinely determinable by one of ordinary skill in the art based on the intended purpose for the fragment.

A fragment of SEQ ID NO:1-35 is encoded by a fragment of SEQ ID NO:36-70. A fragment of SEQ ID NO:1-35 can comprise a region of unique amino acid sequence that specifically identifies SEQ ID NO:1-35. For example, a fragment of SEQ ID NO:1-35 can be used as an immunogenic peptide for the development of antibodies that specifically recognize SEQ ID NO:1-35. The precise length of a fragment of SEQ ID NO:1-35 and the region of SEQ ID NO:1-35 to which the fragment corresponds can be determined based on the intended purpose for the fragment using one or more analytical methods described herein or otherwise known in the art.

A "full length" polynucleotide is one containing at least a translation initiation codon (e.g., methionine) followed by an open reading frame and a translation termination codon. A "full length" polynucleotide sequence encodes a "full length" polypeptide sequence.

"Homology" refers to sequence similarity or, alternatively, sequence identity, between two or more polynucleotide sequences or two or more polypeptide sequences.

The terms "percent identity" and "% identity," as applied to polynucleotide sequences, refer to the percentage of identical nucleotide matches between at least two polynucleotide sequences aligned using a standardized algorithm. Such an algorithm may insert, in a standardized and reproducible way, gaps in the sequences being compared in order to optimize alignment between two sequences, and therefore achieve a more meaningful comparison of the two sequences.

Percent identity between polynucleotide sequences may be determined using one or more computer algorithms or programs known in the art or described herein. For example, percent identity can be determined using the default parameters of the CLUSTAL V algorithm as incorporated into the MEGALIGN version 3.12e sequence alignment program. This program is part of the LASERGENE software package, a suite of molecular biological analysis programs (DNASTAR, Madison WI). CLUSTAL V is described in Higgins, D.G. and P.M. Sharp (1989; CABIOS 5:151-153) and in Higgins, D.G. et al. (1992; CABIOS 8:189-191). For pairwise alignments of polynucleotide sequences, the default parameters are set as follows: Ktuple=2, gap penalty=5, window=4, and "diagonals saved"=4. The "weighted" residue weight table is selected as the default.

Alternatively, a suite of commonly used and freely available sequence comparison algorithms which can be used is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S.F. et al. (1990) J. Mol. Biol. 215:403-410), which is available from several sources, including the NCBI, Bethesda, MD, and on the Internet at http://www.ncbi.nlm.nih.gov/BLAST/. The BLAST software suite includes various sequence analysis programs including "blastn," that is used to align a known polynucleotide sequence with other polynucleotide sequences from a variety of databases. Also available is a tool called "BLAST 2 Sequences" that is used for direct pairwise comparison of two nucleotide sequences. "BLAST 2 Sequences" can be accessed and used interactively at http://www.ncbi.nlm.nih.gov/gorf/bl2.html. The "BLAST 2 Sequences" tool can be used for both blastn and blastp (discussed below). BLAST programs are commonly used with gap and other parameters set to default settings. For example, to compare two nucleotide sequences, one may use blastn with the "BLAST 2 Sequences" tool Version 2.0.12 (April-21-2000) set at default parameters. Such default parameters may be, for example:

Matrix: BLOSUM62
Reward for match: 1
Penalty for mismatch: -2
Open Gap: 5 and Extension Gap: 2 penalties
Gap x drop-off: 50
Expect: 10
Word Size: 11
Filter: on

Percent identity may be measured over the length of an entire defined sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined sequence, for instance, a fragment of at least 20, at least 30, at least 40, at least 50, at least 70, at least 100, or at least 200 contiguous nucleotides. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures, or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

Nucleic acid sequences that do not show a high degree of identity may nevertheless encode similar amino acid sequences due to the degeneracy of the genetic code. It is understood that changes in a nucleic acid sequence can be made using this degeneracy to produce multiple nucleic acid sequences that all encode substantially the same protein.

The phrases "percent identity" and "% identity," as applied to polypeptide sequences, refer to the percentage of identical residue matches between at least two polypeptide sequences aligned using a standardized algorithm. Methods of polypeptide sequence alignment are well-known. Some alignment methods take into account conservative amino acid substitutions. Such conservative substitutions, explained in more detail above, generally preserve the charge and hydrophobicity at the site of substitution, thus preserving the structure (and therefore function) of the polypeptide. The phrases "percent similarity" and "% similarity," as applied to polypeptide sequences, refer to the percentage of residue matches, including identical residue matches and conservative substitutions, between at least two polypeptide sequences aligned using a standardized algorithm. In contrast, conservative substitutions are not included in the calculation of percent identity between polypeptide sequences.

Percent identity between polypeptide sequences may be determined using the default parameters of the CLUSTAL V algorithm as incorporated into the MEGALIGN version 3.12e sequence alignment program (described and referenced above). For pairwise alignments of polypeptide sequences using CLUSTAL V, the default parameters are set as follows: Ktuple=1, gap penalty=3, window=5, and "diagonals saved"=5. The PAM250 matrix is selected as the default residue weight table.

Alternatively the NCBI BLAST software suite may be used. For example, for a pairwise comparison of two polypeptide sequences, one may use the "BLAST 2 Sequences" tool Version 2.0.12 (April-21-2000) with blastp set at default parameters. Such default parameters may be, for example:

Matrix: BLOSUM62

Open Gap: 11 and Extension Gap: 1 penalties

Gap x drop-off: 50

Expect: 10

Word Size: 3

Filter: on

Percent identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

"Human artificial chromosomes" (HACs) are linear microchromosomes which may contain DNA sequences of about 6 kb to 10 Mb in size and which contain all of the elements required for chromosome replication, segregation and maintenance.

The term "humanized antibody" refers to an antibody molecule in which the amino acid sequence in the non-antigen binding regions has been altered so that the antibody more closely resembles a human antibody, and still retains its original binding ability.

"Hybridization" refers to the process by which a polynucleotide strand anneals with a complementary strand through base pairing under defined hybridization conditions. Specific hybridization is an indication that two nucleic acid sequences share a high degree of complementarity. Specific hybridization complexes form under permissive annealing conditions and remain hybridized after the "washing" step(s). The washing step(s) is particularly important in determining the stringency of the hybridization process, with more stringent conditions allowing less non-specific binding, i.e., binding between pairs of nucleic acid strands that are not perfectly matched. Permissive conditions for annealing of nucleic acid sequences are routinely determinable by one of ordinary skill in the art and may be consistent among hybridization experiments, whereas wash conditions may be varied among experiments to achieve the desired stringency, and therefore hybridization specificity. Permissive annealing conditions occur, for example, at 68°C in the presence of about 6 x SSC, about 1% (w/v) SDS, and about 100 μg/ml sheared, denatured salmon sperm DNA.

Generally, stringency of hybridization is expressed, in part, with reference to the temperature under which the wash step is carried out. Such wash temperatures are typically selected to be about 5°C to 20°C lower than the thermal melting point (T_m) for the specific sequence at a defined ionic strength and pH. The T_m is the temperature (under defined ionic strength and pH) at which 50% of the target sequence hybridizes to a perfectly matched probe. An equation for calculating T_m and conditions for nucleic acid hybridization are well known and can be found in Sambrook, J. and D.W. Russell (2001; <u>Molecular Cloning: A Laboratory Manual</u>, 3rd ed., vol. 1-3, Cold Spring Harbor Press, Cold Spring Harbor NY, ch. 9).

High stringency conditions for hybridization between polynucleotides of the present invention include wash conditions of 68°C in the presence of about 0.2 x SSC and about 0.1% SDS, for 1 hour. Alternatively, temperatures of about 65°C, 60°C, 55°C, or 42°C may be used. SSC concentration may be varied from about 0.1 to 2 x SSC, with SDS being present at about 0.1%. Typically, blocking reagents are used to block non-specific hybridization. Such blocking reagents include, for instance, sheared and denatured salmon sperm DNA at about 100-200 μg/ml. Organic solvent, such as formamide at a concentration of about 35-50% v/v, may also be used under particular circumstances, such as for RNA:DNA hybridizations. Useful variations on these wash conditions will be readily apparent to those of ordinary skill in the art. Hybridization, particularly under high stringency conditions, may be suggestive of evolutionary similarity between the nucleotides. Such similarity is strongly indicative of a similar role for the nucleotides and their encoded polypeptides.

The term "hybridization complex" refers to a complex formed between two nucleic acids by virtue of the formation of hydrogen bonds between complementary bases. A hybridization complex may be formed in solution (e.g., C_0t or R_0t analysis) or formed between one nucleic acid present in solution and another nucleic acid immobilized on a solid support (e.g., paper, membranes, filters, chips, pins or glass slides, or any other appropriate substrate to which cells or their nucleic acids have been fixed).

The words "insertion" and "addition" refer to changes in an amino acid or polynucleotide sequence resulting in the addition of one or more amino acid residues or nucleotides, respectively.

"Immune response" can refer to conditions associated with inflammation, trauma, immune disorders, or infectious or genetic disease, etc. These conditions can be characterized by expression of various factors, e.g., cytokines, chemokines, and other signaling molecules, which may affect cellular and systemic defense systems.

An "immunogenic fragment" is a polypeptide or oligopeptide fragment of NAAP which is capable of eliciting an immune response when introduced into a living organism, for example, a mammal. The term "immunogenic fragment" also includes any polypeptide or oligopeptide fragment of NAAP which is useful in any of the antibody production methods disclosed herein or known in the art.

The term "microarray" refers to an arrangement of a plurality of polynucleotides, polypeptides, antibodies, or other chemical compounds on a substrate.

The terms "element" and "array element" refer to a polynucleotide, polypeptide, antibody, or other chemical compound having a unique and defined position on a microarray.

The term "modulate" refers to a change in the activity of NAAP. For example, modulation may cause an increase or a decrease in protein activity, binding characteristics, or any other biological, functional, or immunological properties of NAAP.

The phrases "nucleic acid" and "nucleic acid sequence" refer to a nucleotide, oligonucleotide, polynucleotide, or any fragment thereof. These phrases also refer to DNA or RNA of genomic or synthetic origin which may be single-stranded or double-stranded and may represent the sense or the antisense strand, to peptide nucleic acid (PNA), or to any DNA-like or RNA-like material.

"Operably linked" refers to the situation in which a first nucleic acid sequence is placed in a functional relationship with a second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be in close proximity or contiguous and, where necessary to join two protein coding regions, in the same reading frame.

"Peptide nucleic acid" (PNA) refers to an antisense molecule or anti-gene agent which comprises an oligonucleotide of at least about 5 nucleotides in length linked to a peptide backbone of amino acid residues ending in lysine. The terminal lysine confers solubility to the composition.

PNAs preferentially bind complementary single stranded DNA or RNA and stop transcript elongation, and may be pegylated to extend their lifespan in the cell.

"Post-translational modification" of an NAAP may involve lipidation, glycosylation, phosphorylation, acetylation, racemization, proteolytic cleavage, and other modifications known in the art. These processes may occur synthetically or biochemically. Biochemical modifications will vary by cell type depending on the enzymatic milieu of NAAP.

"Probe" refers to nucleic acids encoding NAAP, their complements, or fragments thereof, which are used to detect identical, allelic or related nucleic acids. Probes are isolated oligonucleotides or polynucleotides attached to a detectable label or reporter molecule. Typical labels include radioactive isotopes, ligands, chemiluminescent agents, and enzymes. "Primers" are short nucleic acids, usually DNA oligonucleotides, which may be annealed to a target polynucleotide by complementary base-pairing. The primer may then be extended along the target DNA strand by a DNA polymerase enzyme. Primer pairs can be used for amplification (and identification) of a nucleic acid, e.g., by the polymerase chain reaction (PCR).

Probes and primers as used in the present invention typically comprise at least 15 contiguous nucleotides of a known sequence. In order to enhance specificity, longer probes and primers may also be employed, such as probes and primers that comprise at least 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, or at least 150 consecutive nucleotides of the disclosed nucleic acid sequences. Probes and primers may be considerably longer than these examples, and it is understood that any length supported by the specification, including the tables, figures, and Sequence Listing, may be used.

Methods for preparing and using probes and primers are described in, for example, Sambrook, J. and D.W. Russell (2001; Molecular Cloning: A Laboratory Manual, 3rd ed., vol. 1-3, Cold Spring Harbor Press, Cold Spring Harbor NY), Ausubel, F.M. et al. (1999; Short Protocols in Molecular Biology, 4th ed., John Wiley & Sons, New York NY), and Innis, M. et al. (1990; PCR Protocols, A Guide to Methods and Applications, Academic Press, San Diego CA). PCR primer pairs can be derived from a known sequence, for example, by using computer programs intended for that purpose such as Primer (Version 0.5, 1991, Whitehead Institute for Biomedical Research, Cambridge MA).

Oligonucleotides for use as primers are selected using software known in the art for such purpose. For example, OLIGO 4.06 software is useful for the selection of PCR primer pairs of up to 100 nucleotides each, and for the analysis of oligonucleotides and larger polynucleotides of up to 5,000 nucleotides from an input polynucleotide sequence of up to 32 kilobases. Similar primer selection programs have incorporated additional features for expanded capabilities. For example, the PrimOU primer selection program (available to the public from the Genome Center at University of Texas South West Medical Center, Dallas TX) is capable of choosing specific primers from megabase sequences and is thus useful for designing primers on a genome-wide scope. The Primer3 primer selection program (available to the public from the Whitehead Institute/MIT Center for Genome Research, Cambridge MA) allows the user to input a "mispriming library," in which sequences to avoid as primer binding sites are user-specified. Primer3 is useful, in particular, for the selection of oligonucleotides for microarrays. (The source code for the latter two primer selection programs may also be obtained from their respective sources and modified to meet the user's specific needs.) The PrimeGen program (available to the public from the UK Human Genome Mapping Project Resource Centre, Cambridge UK) designs primers based on multiple sequence alignments, thereby allowing selection of primers that hybridize to either the most conserved or least conserved regions of aligned nucleic acid sequences. Hence, this program is useful for identification of both unique and conserved oligonucleotides and polynucleotide fragments. The oligonucleotides and polynucleotide fragments identified by any of the above selection methods are useful in hybridization technologies, for example, as PCR or sequencing primers, microarray elements, or specific probes to identify fully or partially complementary polynucleotides in a sample of nucleic acids. Methods of oligonucleotide selection are not limited to those described above.

A "recombinant nucleic acid" is a nucleic acid that is not naturally occurring or has a sequence that is made by an artificial combination of two or more otherwise separated segments of sequence. This artificial combination is often accomplished by chemical synthesis or, more commonly, by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques such as those described in Sambrook and Russell (*supra*). The term recombinant includes nucleic acids that have been altered solely by addition, substitution, or deletion of a portion of the nucleic acid. Frequently, a recombinant nucleic acid may include a nucleic acid sequence operably linked to a promoter sequence. Such a recombinant nucleic acid may be part of a vector that is used, for example, to transform a cell.

Alternatively, such recombinant nucleic acids may be part of a viral vector, e.g., based on a vaccinia virus, that could be use to vaccinate a mammal wherein the recombinant nucleic acid is expressed, inducing a protective immunological response in the mammal.

A "regulatory element" refers to a nucleic acid sequence usually derived from untranslated regions of a gene and includes enhancers, promoters, introns, and 5' and 3' untranslated regions (UTRs). Regulatory elements interact with host or viral proteins which control transcription, translation, or RNA stability.

"Reporter molecules" are chemical or biochemical moieties used for labeling a nucleic acid, amino acid, or antibody. Reporter molecules include radionuclides; enzymes; fluorescent, chemiluminescent, or chromogenic agents; substrates; cofactors; inhibitors; magnetic particles; and other moieties known in the art.

An "RNA equivalent," in reference to a DNA molecule, is composed of the same linear sequence of nucleotides as the reference DNA molecule with the exception that all occurrences of the nitrogenous base thymine are replaced with uracil, and the sugar backbone is composed of ribose instead of deoxyribose.

The term "sample" is used in its broadest sense. A sample suspected of containing NAAP, nucleic acids encoding NAAP, or fragments thereof may comprise a bodily fluid; an extract from a cell, chromosome, organelle, or membrane isolated from a cell; a cell; genomic DNA, RNA, or cDNA, in solution or bound to a substrate; a tissue; a tissue print; etc.

The terms "specific binding" and "specifically binding" refer to that interaction between a protein or peptide and an agonist, an antibody, an antagonist, a small molecule, or any natural or synthetic binding composition. The interaction is dependent upon the presence of a particular structure of the protein, e.g., the antigenic determinant or epitope, recognized by the binding molecule. For example, if an antibody is specific for epitope "A," the presence of a polypeptide comprising the epitope A, or the presence of free unlabeled A, in a reaction containing free labeled A and the antibody will reduce the amount of labeled A that binds to the antibody.

The term "substantially purified" refers to nucleic acid or amino acid sequences that are removed from their natural environment and are isolated or separated, and are at least about 60% free, preferably at least about 75% free, and most preferably at least about 90% free from other components with which they are naturally associated.

A "substitution" refers to the replacement of one or more amino acid residues or nucleotides by different amino acid residues or nucleotides, respectively.

"Substrate" refers to any suitable rigid or semi-rigid support including membranes, filters, chips, slides, wafers, fibers, magnetic or nonmagnetic beads, gels, tubing, plates, polymers, microparticles and capillaries. The substrate can have a variety of surface forms, such as wells, trenches, pins, channels and pores, to which polynucleotides or polypeptides are bound.

A "transcript image" or "expression profile" refers to the collective pattern of gene expression by a particular cell type or tissue under given conditions at a given time.

"Transformation" describes a process by which exogenous DNA is introduced into a recipient cell. Transformation may occur under natural or artificial conditions according to various methods well known in the art, and may rely on any known method for the insertion of foreign nucleic acid sequences into a prokaryotic or eukaryotic host cell. The method for transformation is selected based on the type of host cell being transformed and may include, but is not limited to, bacteriophage or viral infection, electroporation, heat shock, lipofection, and particle bombardment. The term "transformed cells" includes stably transformed cells in which the inserted DNA is capable of replication either as an autonomously replicating plasmid or as part of the host chromosome, as well as transiently transformed cells which express the inserted DNA or RNA for limited periods of time.

A "transgenic organism," as used herein, is any organism, including but not limited to animals and plants, in which one or more of the cells of the organism contains heterologous nucleic acid introduced by way of human intervention, such as by transgenic techniques well known in the art. The nucleic acid is introduced into the cell, directly or indirectly by introduction into a precursor of the cell, by way of deliberate genetic manipulation, such as by microinjection or by infection with a recombinant virus. In another embodiment, the nucleic acid can be introduced by infection with a recombinant viral vector, such as a lentiviral vector (Lois, C. et al. (2002) Science 295:868-872). The term genetic manipulation does not include classical cross-breeding, or *in vitro* fertilization, but rather is directed to the introduction of a recombinant DNA molecule. The transgenic organisms contemplated in accordance with the present invention include bacteria, cyanobacteria, fungi, plants and animals. The isolated DNA of the present invention can be introduced into the host by methods known in the art, for example infection, transfection, transformation or transconjugation. Techniques for transferring the DNA of the present invention into such organisms are widely known and provided in references such as Sambrook and Russell (*supra*).

A "variant" of a particular nucleic acid sequence is defined as a nucleic acid sequence having at least 40% sequence identity to the particular nucleic acid sequence over a certain length of one of the nucleic acid sequences using blastn with the "BLAST 2 Sequences" tool Version 2.0.9 (May-07-1999) set at default parameters. Such a pair of nucleic acids may show, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length. A variant may be described as, for example, an "allelic" (as defined above), "splice," "species," or "polymorphic" variant. A splice variant may have significant identity to a reference molecule, but will generally have a greater or lesser number of polynucleotides due to alternate splicing during mRNA processing. The corresponding polypeptide may possess additional functional domains or lack domains that are present in the reference molecule. Species variants are polynucleotides that vary from one species to another. The resulting polypeptides will generally have significant amino acid identity relative to each other. A polymorphic variant is a variation in the polynucleotide sequence of a particular gene between individuals of a given species. Polymorphic variants also may encompass "single nucleotide polymorphisms" (SNPs) in which the polynucleotide sequence varies by one nucleotide base. The presence of SNPs may be indicative of, for example, a certain population, a disease state, or a propensity for a disease state.

A "variant" of a particular polypeptide sequence is defined as a polypeptide sequence having at least 40% sequence identity or sequence similarity to the particular polypeptide sequence over a certain length of one of the polypeptide sequences using blastp with the "BLAST 2 Sequences" tool Version 2.0.9 (May-07-1999) set at default parameters. Such a pair of polypeptides may show, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity or sequence similarity over a certain defined length of one of the polypeptides.

THE INVENTION

Various embodiments of the invention include new human nucleic acid-associated proteins (NAAP), the polynucleotides encoding NAAP, and the use of these compositions for the diagnosis, treatment, or prevention of cell proliferative, neurological, developmental, autoimmune/inflammatory, and lipid metabolism disorders, and infections.

Table 1 summarizes the nomenclature for the full length polynucleotide and polypeptide embodiments of the invention. Each polynucleotide and its corresponding polypeptide are correlated to a single Incyte project identification number (Incyte Project ID). Each polypeptide sequence is denoted by both a polypeptide sequence identification number (Polypeptide SEQ ID NO:) and an Incyte polypeptide sequence number (Incyte Polypeptide ID) as shown. Each polynucleotide sequence is denoted by both a polynucleotide sequence identification number (Polynucleotide SEQ ID NO:) and an Incyte polynucleotide consensus sequence number (Incyte Polynucleotide ID) as shown. Column 6 shows the Incyte ID numbers of physical, full length clones corresponding to the polypeptide and polynucleotide sequences of the invention. The full length clones encode polypeptides which have at least 95% sequence identity to the polypeptide sequences shown in column 3.

Table 2 shows sequences with homology to polypeptide embodiments of the invention as identified by BLAST analysis against the GenBank protein (genpept) database and the PROTEOME database. Columns 1 and 2 show the polypeptide sequence identification number (Polypeptide SEQ ID NO:) and the corresponding Incyte polypeptide sequence number (Incyte Polypeptide ID) for polypeptides of the invention. Column 3 shows the GenBank identification number (GenBank ID NO:) of the nearest GenBank homolog and the PROTEOME database identification numbers (PROTEOME ID NO:) of the nearest PROTEOME database homologs. Column 4 shows the probability scores for the matches between each polypeptide and its homolog(s). Column 5 shows the annotation of the GenBank and PROTEOME database homolog(s) along with relevant citations where applicable, all of which are expressly incorporated by reference herein.

Table 3 shows various structural features of the polypeptides of the invention. Columns 1 and 2 show the polypeptide sequence identification number (SEQ ID NO:) and the corresponding

Incyte polypeptide sequence number (Incyte Polypeptide ID) for each polypeptide of the invention. Column 3 shows the number of amino acid residues in each polypeptide. Column 4 shows potential phosphorylation sites, and column 5 shows potential glycosylation sites, as determined by the MOTIFS program of the GCG sequence analysis software package (Accelrys, Burlington MA). Column 6 shows amino acid residues comprising signature sequences, domains, and motifs. Column 7 shows analytical methods for protein structure/function analysis and in some cases, searchable databases to which the analytical methods were applied.

Together, Tables 2 and 3 summarize the properties of polypeptides of the invention, and these properties establish that the claimed polypeptides are nucleic acid-associated proteins. For example, SEQ ID NO:7 is 95% identical, from residue M1 to residue K195, and 91% identical, from residue D151 to residue S445, to NF-kappa-B transcription factor (GenBank ID g189504) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 3.7E-242, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:7 also has homology to proteins that are localized to the nuclear and cytoplasmic region, are DNA-binding transcription factors, and are involved in cell differentiation, response to infection and apoptosis inhibition, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:7 also contains a Rel homology domain as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM database of conserved protein families/domains. (See Table 3.) Data from BLIMPS and MOTIFS analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:7 is a transcriptional activator, and a member of the NF-kappa-B/Rel/dorsal family. In another example, SEQ ID NO:23 is 99% identical, from residue M8 to residue R243, to *Homo sapiens* nuclear orphan receptor LXR-alpha (GenBank ID g726513) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 3.6E-127, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:23 also has homology to proteins that are localized to the nuclear region, are ligand-dependent nuclear receptor DNA-binding transcriptional activators, involved in RXR receptor and PPAR-γ and PPAR-α signaling, cholesterol and lipid metabolism, as well as TNF synthesis, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:23 also contains a C4 zinc finger nuclear hormone receptor domain as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS, MOTIFS, and PROFILESCAN analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:23 is a LXR-alpha nuclear orphan receptor. In another example, SEQ ID NO:25 is 95% identical, from residue M1 to residue S550, to *Mus* *musculus* nuclear transcription factor RelA (GenBank ID g9049520) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 2.2E-287, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:25 also has homology to DNA-binding transcriptional repressors localized to the nucleus, and more specifically, the RELA subunit of transcription factor NF-kappa-B, which is involved in development, immune responses, apoptosis inhibition, and is implicated in metastatic melanoma and Crohn's disease, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:25 also contains a Rel homology domain, as well as Ig-like, plexins, transcription factors, and IPT/TIG domains as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS and MOTIFS analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:25 is a subunit of NF-kappa-B DNA-binding transcriptional repressor. In another example, SEQ ID NO:27 is 49% identical, from residue C120 to residue D481, to human zinc finger protein (GenBank ID g186774) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 2.8e-106, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:27 also has homology to a DNA-binding protein which contains seven C2H2 type zinc finger domains, as well as to a protein containing a kruppel-associated box (KRAB) domain and twelve C2H2 type zinc finger domains, and which has moderate similarity to human ZNF136, which represses transcription when fused to the heterologous KRAB B subdomain of human ZNF10, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:27 also contains a zinc finger domain, a zinc finger, C2H2 type domain, and a KRAB box domain as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM/SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS and MOTIFS analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:27 is a zinc finger protein. In another example, SEQ ID NO:31 is 92% identical, from residue E176 to residue S580, to human Sp140 protein (GenBank ID g1669498) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 2.1e-242, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:31 also has homology to proteins that are localized to the nuclear region, and are Sp-type nuclear antigen proteins, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:31 also contains SAND, Sp100, PHD zinc finger, and bromo domains as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) The foregoing provides evidence that SEQ ID NO:31 is a nuclear antigen protein. In another example, SEQ ID NO:34 is 90% identical, from residue M1 to residue P220, and 95% identical, from residue D219 to residue Y410, to *Coturnix coturnix* FLI transcription factor (GenBank ID g3269303) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 1.1e-212, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:34 also has homology to proteins that are members of the erythroblast transformation specific (ETS) family of DNA-binding transcription factors, which is associated with Ewing sarcoma and related subtypes of primitive neuroectodermal tumors in humans and erythroleukemia in mice, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:34 also contains ETS, and Sterile alpha motif (SAM)/Pointed domains, as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS, MOTIFS, and PROFILESCAN analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:34 is an ETS transcription factor. SEQ ID NO:1-6, SEQ ID NO:8-22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28-30, SEQ ID NO:32-33, and SEQ ID NO:35 were analyzed and annotated in a similar manner. The algorithms and parameters for the analysis of SEQ ID NO:1-35 are described in Table 7.

As shown in Table 4, the full length polynucleotide embodiments were assembled using cDNA sequences or coding (exon) sequences derived from genomic DNA, or any combination of these two types of sequences. Column 1 lists the polynucleotide sequence identification number (Polynucleotide SEQ ID NO:), the corresponding Incyte polynucleotide consensus sequence number (Incyte ID) for each polynucleotide of the invention, and the length of each polynucleotide sequence in basepairs. Column 2 shows the nucleotide start (5′) and stop (3′) positions of the cDNA and/or genomic sequences used to assemble the full length polynucleotide embodiments, and of fragments of the polynucleotides which are useful, for example, in hybridization or amplification technologies that identify SEQ ID NO:36-70 or that distinguish between SEQ ID NO:36-70 and related polynucleotides.

The polynucleotide fragments described in Column 2 of Table 4 may refer specifically, for example, to Incyte cDNAs derived from tissue-specific cDNA libraries or from pooled cDNA libraries. Alternatively, the polynucleotide fragments described in column 2 may refer to GenBank cDNAs or ESTs which contributed to the assembly of the full length polynucleotides. In addition, the polynucleotide fragments described in column 2 may identify sequences derived from the ENSEMBL (The Sanger Centre, Cambridge, UK) database (*i.e.*, those sequences including the designation "ENST"). Alternatively, the polynucleotide fragments described in column 2 may be derived from the NCBI RefSeq Nucleotide Sequence Records Database (*i.e.*, those sequences including the designation "NM" or "NT") or the NCBI RefSeq Protein Sequence Records (*i.e.*, those sequences including the designation "NP"). Alternatively, the polynucleotide fragments described in column 2 may refer to assemblages of both cDNA and Genscan-predicted exons brought together by an "exon stitching" algorithm. For example, a polynucleotide sequence identified as FL_*XXXXXX*_N_1_N_2_*YYYYY*_N_3_N_4 represents a "stitched" sequence in which *XXXXXX* is the identification number of the cluster of sequences to which the algorithm was applied, and *YYYYY* is the number of the prediction generated by the algorithm, and $N_{1,2,3...}$, if present, represent specific exons that may have been manually edited during analysis (See Example V). Alternatively, the polynucleotide fragments in column 2 may refer to assemblages of exons brought together by an "exon-stretching" algorithm. For example, a polynucleotide sequence identified as FL*XXXXXX*_g*AAAAA*_g*BBBBB*_1_*N* is a "stretched" sequence, with *XXXXXX* being the Incyte project identification number, g*AAAAA* being the GenBank identification number of the human genomic sequence to which the "exon-stretching" algorithm was applied, g*BBBBB* being the GenBank identification number or NCBI RefSeq identification number of the nearest GenBank protein homolog, and *N* referring to specific exons (See Example V). In instances where a RefSeq sequence was used as a protein homolog for the "exon-stretching" algorithm, a RefSeq identifier (denoted by "NM," "NP," or "NT") may be used in place of the GenBank identifier (*i.e.*, g*BBBBB*).

Alternatively, a prefix identifies component sequences that were hand-edited, predicted from genomic DNA sequences, or derived from a combination of sequence analysis methods. The following Table lists examples of component sequence prefixes and corresponding sequence analysis methods associated with the prefixes (see Example IV and Example V).

Prefix	Type of analysis and/or examples of programs
GNN, GFG, ENST	Exon prediction from genomic sequences using, for example, GENSCAN (Stanford University, CA, USA) or FGENES (Computer Genomics Group, The Sanger Centre, Cambridge, UK).
GBI	Hand-edited analysis of genomic sequences.
FL	Stitched or stretched genomic sequences (see Example V).
INCY	Full length transcript and exon prediction from mapping of EST sequences to the genome. Genomic location and EST composition data are combined to predict the exons and resulting transcript.

In some cases, Incyte cDNA coverage redundant with the sequence coverage shown in Table 4 was obtained to confirm the final consensus polynucleotide sequence, but the relevant Incyte cDNA identification numbers are not shown.

Table 5 shows the representative cDNA libraries for those full length polynucleotides which were assembled using Incyte cDNA sequences. The representative cDNA library is the Incyte cDNA library which is most frequently represented by the Incyte cDNA sequences which were used to assemble and confirm the above polynucleotides. The tissues and vectors which were used to construct the cDNA libraries shown in Table 5 are described in Table 6.

Table 8 shows single nucleotide polymorphisms (SNPs) found in polynucleotide sequences of the invention, along with allele frequencies in different human populations. Columns 1 and 2 show the polynucleotide sequence identification number (SEQ ID NO:) and the corresponding Incyte project identification number (PID) for polynucleotides of the invention. Column 3 shows the Incyte identification number for the EST in which the SNP was detected (EST ID), and column 4 shows the identification number for the SNP (SNP ID). Column 5 shows the position within the EST sequence at which the SNP is located (EST SNP), and column 6 shows the position of the SNP within the full-length polynucleotide sequence (CB1 SNP). Column 7 shows the allele found in the EST sequence. Columns 8 and 9 show the two alleles found at the SNP site. Column 10 shows the amino acid encoded by the codon including the SNP site, based upon the allele found in the EST. Columns 11-14 show the frequency of allele 1 in four different human populations. An entry of n/d (not detected) indicates that the frequency of allele 1 in the population was too low to be detected, while n/a (not available) indicates that the allele frequency was not determined for the population.

The invention also encompasses NAAP variants. Various embodiments of NAAP variants can have at least about 80%, at least about 90%, or at least about 95% amino acid sequence identity to the NAAP amino acid sequence, and can contain at least one functional or structural characteristic of NAAP.

Various embodiments also encompass polynucleotides which encode NAAP. In a particular embodiment, the invention encompasses a polynucleotide sequence comprising a sequence selected from the group consisting of SEQ ID NO:36-70, which encodes NAAP. The polynucleotide sequences of SEQ ID NO:36-70, as presented in the Sequence Listing, embrace the equivalent RNA sequences, wherein occurrences of the nitrogenous base thymine are replaced with uracil, and the sugar backbone is composed of ribose instead of deoxyribose.

The invention also encompasses variants of a polynucleotide encoding NAAP. In particular, such a variant polynucleotide will have at least about 70%, or alternatively at least about 85%, or even at least about 95% polynucleotide sequence identity to a polynucleotide encoding NAAP. A particular aspect of the invention encompasses a variant of a polynucleotide comprising a sequence selected from the group consisting of SEQ ID NO:36-70 which has at least about 70%, or alternatively at least about 85%, or even at least about 95% polynucleotide sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO:36-70. Any one of the polynucleotide variants described above can encode a polypeptide which contains at least one functional or structural characteristic of NAAP.

In addition, or in the alternative, a polynucleotide variant of the invention is a splice variant of a polynucleotide encoding NAAP. A splice variant may have portions which have significant sequence identity to a polynucleotide encoding NAAP, but will generally have a greater or lesser number of nucleotides due to additions or deletions of blocks of sequence arising from alternate splicing during mRNA processing. A splice variant may have less than about 70%, or alternatively less than about 60%, or alternatively less than about 50% polynucleotide sequence identity to a polynucleotide encoding NAAP over its entire length; however, portions of the splice variant will have at least about 70%, or alternatively at least about 85%, or alternatively at least about 95%, or alternatively 100% polynucleotide sequence identity to portions of the polynucleotide encoding NAAP. For example, a polynucleotide comprising a sequence of SEQ ID NO:56 and a polynucleotide comprising a sequence of SEQ ID NO:57 are splice variants of each other. Any one of the splice variants described above can encode a polypeptide which contains at least one functional or structural characteristic of NAAP.

It will be appreciated by those skilled in the art that as a result of the degeneracy of the genetic code, a multitude of polynucleotide sequences encoding NAAP, some bearing minimal similarity to the polynucleotide sequences of any known and naturally occurring gene, may be produced. Thus, the invention contemplates each and every possible variation of polynucleotide sequence that could be made by selecting combinations based on possible codon choices. These combinations are made in accordance with the standard triplet genetic code as applied to the polynucleotide sequence of naturally occurring NAAP, and all such variations are to be considered as being specifically disclosed.

Although polynucleotides which encode NAAP and its variants are generally capable of hybridizing to polynucleotides encoding naturally occurring NAAP under appropriately selected conditions of stringency, it may be advantageous to produce polynucleotides encoding NAAP or its derivatives possessing a substantially different codon usage, e.g., inclusion of non-naturally occurring codons. Codons may be selected to increase the rate at which expression of the peptide occurs in a particular prokaryotic or eukaryotic host in accordance with the frequency with which particular codons are utilized by the host. Other reasons for substantially altering the nucleotide sequence encoding NAAP and its derivatives without altering the encoded amino acid sequences include the production of RNA transcripts having more desirable properties, such as a greater half-life, than transcripts produced from the naturally occurring sequence.

The invention also encompasses production of polynucleotides which encode NAAP and NAAP derivatives, or fragments thereof, entirely by synthetic chemistry. After production, the synthetic polynucleotide may be inserted into any of the many available expression vectors and cell systems using reagents well known in the art. Moreover, synthetic chemistry may be used to introduce mutations into a polynucleotide encoding NAAP or any fragment thereof.

Embodiments of the invention can also include polynucleotides that are capable of hybridizing to the claimed polynucleotides, and, in particular, to those having the sequences shown in SEQ ID NO:36-70 and fragments thereof, under various conditions of stringency (Wahl, G.M. and S.L. Berger (1987) Methods Enzymol. 152:399-407; Kimmel, A.R. (1987) Methods Enzymol. 152:507-511). Hybridization conditions, including annealing and wash conditions, are described in "Definitions."

Methods for DNA sequencing are well known in the art and may be used to practice any of the embodiments of the invention. The methods may employ such enzymes as the Klenow fragment of DNA polymerase I, SEQUENASE (US Biochemical, Cleveland OH), Taq polymerase (Applied Biosystems), thermostable T7 polymerase (Amersham Biosciences, Piscataway NJ), or combinations of polymerases and proofreading exonucleases such as those found in the ELONGASE amplification system (Invitrogen, Carlsbad CA). Preferably, sequence preparation is automated with machines such as the MICROLAB 2200 liquid transfer system (Hamilton, Reno NV), PTC200 thermal cycler (MJ Research, Watertown MA) and ABI CATALYST 800 thermal cycler (Applied Biosystems). Sequencing is then carried out using either the ABI 373 or 377 DNA sequencing system (Applied Biosystems), the MEGABACE 1000 DNA sequencing system (Amersham Biosciences), or other systems known in the art. The resulting sequences are analyzed using a variety of algorithms which are well known in the art (Ausubel et al., *supra*, ch. 7; Meyers, R.A. (1995) Molecular Biology and Biotechnology, Wiley VCH, New York NY, pp. 856-853).

The nucleic acids encoding NAAP may be extended utilizing a partial nucleotide sequence and employing various PCR-based methods known in the art to detect upstream sequences, such as promoters and regulatory elements. For example, one method which may be employed, restriction-site PCR, uses universal and nested primers to amplify unknown sequence from genomic DNA within a cloning vector (Sarkar, G. (1993) PCR Methods Applic. 2:318-322). Another method, inverse PCR, uses primers that extend in divergent directions to amplify unknown sequence from a circularized template. The template is derived from restriction fragments comprising a known genomic locus and surrounding sequences (Triglia, T. et al. (1988) Nucleic Acids Res. 16:8186). A third method, capture PCR, involves PCR amplification of DNA fragments adjacent to known sequences in human and yeast artificial chromosome DNA (Lagerstrom, M. et al. (1991) PCR Methods Applic. 1:111-119). In this method, multiple restriction enzyme digestions and ligations may be used to insert an engineered double-stranded sequence into a region of unknown sequence before performing PCR. Other methods which may be used to retrieve unknown sequences are known in the art (Parker, J.D. et al. (1991) Nucleic Acids Res. 19:3055-3060). Additionally, one may use PCR, nested primers, and PROMOTERFINDER libraries (BD Clontech, Palo Alto CA) to walk genomic DNA. This procedure avoids the need to screen libraries and is useful in finding intron/exon junctions. For all PCR-based methods, primers may be designed using commercially available software, such as OLIGO 4.06 primer analysis software (National Biosciences, Plymouth MN) or another appropriate program, to be about 22 to 30 nucleotides in length, to have a GC content of about 50% or more, and to anneal to the template at temperatures of about 68°C to 72°C.

When screening for full length cDNAs, it is preferable to use libraries that have been size-selected to include larger cDNAs. In addition, random-primed libraries, which often include sequences containing the 5' regions of genes, are preferable for situations in which an oligo d(T) library does not yield a full-length cDNA. Genomic libraries may be useful for extension of sequence into 5' non-transcribed regulatory regions.

Capillary electrophoresis systems which are commercially available may be used to analyze the size or confirm the nucleotide sequence of sequencing or PCR products. In particular, capillary sequencing may employ flowable polymers for electrophoretic separation, four different nucleotide-specific, laser-stimulated fluorescent dyes, and a charge coupled device camera for detection of the emitted wavelengths. Output/light intensity may be converted to electrical signal using appropriate software (e.g., GENOTYPER and SEQUENCE NAVIGATOR, Applied Biosystems), and the entire process from loading of samples to computer analysis and electronic data display may be computer controlled. Capillary electrophoresis is especially preferable for sequencing small DNA fragments which may be present in limited amounts in a particular sample.

In another embodiment of the invention, polynucleotides or fragments thereof which encode NAAP may be cloned in recombinant DNA molecules that direct expression of NAAP, or fragments or functional equivalents thereof, in appropriate host cells. Due to the inherent degeneracy of the genetic code, other polynucleotides which encode substantially the same or a functionally equivalent polypeptides may be produced and used to express NAAP.

The polynucleotides of the invention can be engineered using methods generally known in the art in order to alter NAAP-encoding sequences for a variety of purposes including, but not limited to, modification of the cloning, processing, and/or expression of the gene product. DNA shuffling by random fragmentation and PCR reassembly of gene fragments and synthetic oligonucleotides may be used to engineer the nucleotide sequences. For example, oligonucleotide-mediated site-directed mutagenesis may be used to introduce mutations that create new restriction sites, alter glycosylation patterns, change codon preference, produce splice variants, and so forth.

The nucleotides of the present invention may be subjected to DNA shuffling techniques such as MOLECULARBREEDING (Maxygen Inc., Santa Clara CA; described in U.S. Patent No. 5,837,458; Chang, C.-C. et al. (1999) Nat. Biotechnol. 17:793-797; Christians, F.C. et al. (1999) Nat.

Biotechnol. 17:259-264; and Crameri, A. et al. (1996) Nat. Biotechnol. 14:315-319) to alter or improve the biological properties of NAAP, such as its biological or enzymatic activity or its ability to bind to other molecules or compounds. DNA shuffling is a process by which a library of gene variants is produced using PCR-mediated recombination of gene fragments. The library is then subjected to selection or screening procedures that identify those gene variants with the desired properties. These preferred variants may then be pooled and further subjected to recursive rounds of DNA shuffling and selection/screening. Thus, genetic diversity is created through "artificial" breeding and rapid molecular evolution. For example, fragments of a single gene containing random point mutations may be recombined, screened, and then reshuffled until the desired properties are optimized. Alternatively, fragments of a given gene may be recombined with fragments of homologous genes in the same gene family, either from the same or different species, thereby maximizing the genetic diversity of multiple naturally occurring genes in a directed and controllable manner.

In another embodiment, polynucleotides encoding NAAP may be synthesized, in whole or in part, using one or more chemical methods well known in the art (Caruthers, M.H. et al. (1980) Nucleic Acids Symp. Ser. 7:215-223; Horn, T. et al. (1980) Nucleic Acids Symp. Ser. 7:225-232). Alternatively, NAAP itself or a fragment thereof may be synthesized using chemical methods known in the art. For example, peptide synthesis can be performed using various solution-phase or solid-phase techniques (Creighton, T. (1984) Proteins, Structures and Molecular Properties, WH Freeman, New York NY, pp. 55-60; Roberge, J.Y. et al. (1995) Science 269:202-204). Automated synthesis may be achieved using the ABI 431A peptide synthesizer (Applied Biosystems). Additionally, the amino acid sequence of NAAP, or any part thereof, may be altered during direct synthesis and/or combined with sequences from other proteins, or any part thereof, to produce a variant polypeptide or a polypeptide having a sequence of a naturally occurring polypeptide.

The peptide may be substantially purified by preparative high performance liquid chromatography (Chiez, R.M. and F.Z. Regnier (1990) Methods Enzymol. 182:392-421). The composition of the synthetic peptides may be confirmed by amino acid analysis or by sequencing (Creighton, *supra*, pp. 28-53).

In order to express a biologically active NAAP, the polynucleotides encoding NAAP or derivatives thereof may be inserted into an appropriate expression vector, i.e., a vector which contains the necessary elements for transcriptional and translational control of the inserted coding sequence in a suitable host. These elements include regulatory sequences, such as enhancers, constitutive and inducible promoters, and 5' and 3' untranslated regions in the vector and in polynucleotides encoding NAAP. Such elements may vary in their strength and specificity. Specific initiation signals may also be used to achieve more efficient translation of polynucleotides encoding NAAP. Such signals include the ATG initiation codon and adjacent sequences, e.g. the Kozak sequence. In cases where a polynucleotide sequence encoding NAAP and its initiation codon and upstream regulatory sequences are inserted into the appropriate expression vector, no additional transcriptional or translational control signals may be needed. However, in cases where only coding sequence, or a fragment thereof, is inserted, exogenous translational control signals including an in-frame ATG initiation codon should be provided by the vector. Exogenous translational elements and initiation codons may be of various origins, both natural and synthetic. The efficiency of expression may be enhanced by the inclusion of enhancers appropriate for the particular host cell system used (Scharf, D. et al. (1994) Results Probl. Cell Differ. 20:125-162).

Methods which are well known to those skilled in the art may be used to construct expression vectors containing polynucleotides encoding NAAP and appropriate transcriptional and translational control elements. These methods include *in vitro* recombinant DNA techniques, synthetic techniques, and *in vivo* genetic recombination (Sambrook and Russell, *supra*, ch. 1-4, and 8; Ausubel et al., *supra*, ch. 1, 3, and 15).

A variety of expression vector/host systems may be utilized to contain and express polynucleotides encoding NAAP. These include, but are not limited to, microorganisms such as bacteria transformed with recombinant bacteriophage, plasmid, or cosmid DNA expression vectors; yeast transformed with yeast expression vectors; insect cell systems infected with viral expression vectors (e.g., baculovirus); plant cell systems transformed with viral expression vectors (e.g., cauliflower mosaic virus, CaMV, or tobacco mosaic virus, TMV) or with bacterial expression vectors (e.g., Ti or pBR322 plasmids); or animal cell systems (Sambrook and Russell, *supra*; Ausubel et al., *supra*; Van Heeke, G. and S.M. Schuster (1989) J. Biol. Chem. 264:5503-5509; Engelhard, E.K. et al. (1994) Proc. Natl. Acad. Sci. USA 91:3224-3227; Sandig, V. et al. (1996) Hum. Gene Ther. 7:1937-1945; Takamatsu, N. (1987) EMBO J. 6:307-311; The McGraw Hill Yearbook of Science and Technology (1992) McGraw Hill, New York NY, pp. 191-196; Logan, J. and T. Shenk (1984) Proc. Natl. Acad. Sci. USA 81:3655-3659; Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355). Expression vectors derived from retroviruses, adenoviruses, or herpes or vaccinia viruses, or from various bacterial plasmids, may be used for delivery of polynucleotides to the targeted organ, tissue, or cell population (Di Nicola, M. et al. (1998) Cancer Gen. Ther. 5:350-356; Yu, M. et al. (1993) Proc. Natl. Acad. Sci. USA 90:6340-6344; Buller, R.M. et al. (1985) Nature 317:813-815; McGregor, D.P. et al. (1994) Mol. Immunol. 31:219-226; Verma, I.M. and N. Somia (1997) Nature 389:239-242). The invention is not limited by the host cell employed.

In bacterial systems, a number of cloning and expression vectors may be selected depending upon the use intended for polynucleotides encoding NAAP. For example, routine cloning, subcloning, and propagation of polynucleotides encoding NAAP can be achieved using a multifunctional *E. coli* vector such as PBLUESCRIPT (Stratagene, La Jolla CA) or PSPORT1 plasmid (Invitrogen). Ligation of polynucleotides encoding NAAP into the vector's multiple cloning site disrupts the *lacZ* gene, allowing a colorimetric screening procedure for identification of transformed bacteria containing recombinant molecules. In addition, these vectors may be useful for *in vitro* transcription, dideoxy sequencing, single strand rescue with helper phage, and creation of nested deletions in the cloned sequence (Van Heeke, G. and S.M. Schuster (1989) J. Biol. Chem. 264:5503-5509). When large quantities of NAAP are needed, e.g. for the production of antibodies, vectors which direct high level expression of NAAP may be used. For example, vectors containing the strong, inducible SP6 or T7 bacteriophage promoter may be used.

Yeast expression systems may be used for production of NAAP. A number of vectors containing constitutive or inducible promoters, such as alpha factor, alcohol oxidase, and PGH promoters, may be used in the yeast *Saccharomyces cerevisiae* or *Pichia pastoris*. In addition, such vectors direct either the secretion or intracellular retention of expressed proteins and enable integration of foreign polynucleotide sequences into the host genome for stable propagation (Ausubel et al., *supra*; Bitter, G.A. et al. (1987) Methods Enzymol. 153:516-544; Scorer, C.A. et al. (1994) Bio/Technology 12:181-184).

Plant systems may also be used for expression of NAAP. Transcription of polynucleotides encoding NAAP may be driven by viral promoters, e.g., the 35S and 19S promoters of CaMV used alone or in combination with the omega leader sequence from TMV (Takamatsu, N. (1987) EMBO J. 6:307-311). Alternatively, plant promoters such as the small subunit of RUBISCO or heat shock promoters may be used (Coruzzi, G. et al. (1984) EMBO J. 3:1671-1680; Broglie, R. et al. (1984) Science 224:838-843; Winter, J. et al. (1991) Results Probl. Cell Differ. 17:85-105). These constructs can be introduced into plant cells by direct DNA transformation or pathogen-mediated transfection (<u>The McGraw Hill Yearbook of Science and Technology</u> (1992) McGraw Hill, New York NY, pp. 191-196).

In mammalian cells, a number of viral-based expression systems may be utilized. In cases where an adenovirus is used as an expression vector, polynucleotides encoding NAAP may be ligated into an adenovirus transcription/translation complex consisting of the late promoter and tripartite leader sequence. Insertion in a non-essential E1 or E3 region of the viral genome may be used to obtain infective virus which expresses NAAP in host cells (Logan, J. and T. Shenk (1984) Proc. Natl. Acad. Sci. USA 81:3655-3659). In addition, transcription enhancers, such as the Rous sarcoma virus (RSV) enhancer, may be used to increase expression in mammalian host cells. SV40 or EBV-based vectors may also be used for high-level protein expression.

Human artificial chromosomes (HACs) may also be employed to deliver larger fragments of DNA than can be contained in and expressed from a plasmid. HACs of about 6 kb to 10 Mb are constructed and delivered via conventional delivery methods (liposomes, polycationic amino polymers, or vesicles) for therapeutic purposes (Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355).

For long term production of recombinant proteins in mammalian systems, stable expression of NAAP in cell lines is preferred. For example, polynucleotides encoding NAAP can be transformed into cell lines using expression vectors which may contain viral origins of replication and/or endogenous expression elements and a selectable marker gene on the same or on a separate vector. Following the introduction of the vector, cells may be allowed to grow for about 1 to 2 days in enriched media before being switched to selective media. The purpose of the selectable marker is to confer resistance to a selective agent, and its presence allows growth and recovery of cells which successfully express the introduced sequences. Resistant clones of stably transformed cells may be propagated using tissue culture techniques appropriate to the cell type.

Any number of selection systems may be used to recover transformed cell lines. These include, but are not limited to, the herpes simplex virus thymidine kinase and adenine phosphoribosyltransferase genes, for use in *tk*$^-$ and *apr*$^-$ cells, respectively (Wigler, M. et al. (1977) Cell 11:223-232; Lowy, I. et al. (1980) Cell 22:817-823). Also, antimetabolite, antibiotic, or herbicide resistance can be used as the basis for selection. For example, *dhfr* confers resistance to methotrexate; *neo* confers resistance to the aminoglycosides neomycin and G-418; and *als* and *pat* confer resistance to chlorsulfuron and phosphinotricin acetyltransferase, respectively (Wigler, M. et al. (1980) Proc. Natl. Acad. Sci. USA 77:3567-3570; Colbere-Garapin, F. et al. (1981) J. Mol. Biol. 150:1-14). Additional selectable genes have been described, e.g., *trpB* and *hisD*, which alter cellular requirements for metabolites (Hartman, S.C. and R.C. Mulligan (1988) Proc. Natl. Acad. Sci. USA 85:8047-8051). Visible markers, e.g., anthocyanins, green fluorescent proteins (GFP; BD Clontech), β-glucuronidase and its substrate β-glucuronide, or luciferase and its substrate luciferin may be used. These markers can be used not only to identify transformants, but also to quantify the amount of transient or stable protein expression attributable to a specific vector system (Rhodes, C.A. (1995) Methods Mol. Biol. 55:121-131).

Although the presence/absence of marker gene expression suggests that the gene of interest is also present, the presence and expression of the gene may need to be confirmed. For example, if the sequence encoding NAAP is inserted within a marker gene sequence, transformed cells containing polynucleotides encoding NAAP can be identified by the absence of marker gene function. Alternatively, a marker gene can be placed in tandem with a sequence encoding NAAP under the control of a single promoter. Expression of the marker gene in response to induction or selection usually indicates expression of the tandem gene as well.

In general, host cells that contain the polynucleotide encoding NAAP and that express NAAP may be identified by a variety of procedures known to those of skill in the art. These procedures include, but are not limited to, DNA-DNA or DNA-RNA hybridizations, PCR amplification, and protein bioassay or immunoassay techniques which include membrane, solution, or chip based technologies for the detection and/or quantification of nucleic acid or protein sequences.

Immunological methods for detecting and measuring the expression of NAAP using either specific polyclonal or monoclonal antibodies are known in the art. Examples of such techniques include enzyme-linked immunosorbent assays (ELISAs), radioimmunoassays (RIAs), and fluorescence activated cell sorting (FACS). A two-site, monoclonal-based immunoassay utilizing monoclonal antibodies reactive to two non-interfering epitopes on NAAP is preferred, but a competitive binding assay may be employed. These and other assays are well known in the art (Hampton, R. et al. (1990) Serological Methods, a Laboratory Manual, APS Press, St. Paul MN, Sect. IV; Coligan, J.E. et al. (1997) Current Protocols in Immunology, Greene Pub. Associates and Wiley-Interscience, New York NY; Pound, J.D. (1998) Immunochemical Protocols, Humana Press, Totowa NJ).

A wide variety of labels and conjugation techniques are known by those skilled in the art and may be used in various nucleic acid and amino acid assays. Means for producing labeled hybridization or PCR probes for detecting sequences related to polynucleotides encoding NAAP include oligolabeling, nick translation, end-labeling, or PCR amplification using a labeled nucleotide. Alternatively, polynucleotides encoding NAAP, or any fragments thereof, may be cloned into a vector for the production of an mRNA probe. Such vectors are known in the art, are commercially available, and may be used to synthesize RNA probes *in vitro* by addition of an appropriate RNA polymerase such as T7, T3, or SP6 and labeled nucleotides. These procedures may be conducted using a variety of commercially available kits, such as those provided by Amersham Biosciences, Promega (Madison WI), and US Biochemical. Suitable reporter molecules or labels which may be used for ease of detection include radionuclides, enzymes, fluorescent, chemiluminescent, or chromogenic agents, as well as substrates, cofactors, inhibitors, magnetic particles, and the like.

Host cells transformed with polynucleotides encoding NAAP may be cultured under conditions suitable for the expression and recovery of the protein from cell culture. The protein produced by a transformed cell may be secreted or retained intracellularly depending on the sequence and/or the vector used. As will be understood by those of skill in the art, expression vectors containing polynucleotides which encode NAAP may be designed to contain signal sequences which direct secretion of NAAP through a prokaryotic or eukaryotic cell membrane.

In addition, a host cell strain may be chosen for its ability to modulate expression of the inserted polynucleotides or to process the expressed protein in the desired fashion. Such modifications of the polypeptide include, but are not limited to, acetylation, carboxylation, glycosylation, phosphorylation, lipidation, and acylation. Post-translational processing which cleaves a "prepro" or "pro" form of the protein may also be used to specify protein targeting, folding, and/or activity. Different host cells which have specific cellular machinery and characteristic mechanisms for post-translational activities (e.g., CHO, HeLa, MDCK, HEK293, and WI38) are available from the American Type Culture Collection (ATCC, Manassas VA) and may be chosen to ensure the correct modification and processing of the foreign protein.

In another embodiment of the invention, natural, modified, or recombinant polynucleotides encoding NAAP may be ligated to a heterologous sequence resulting in translation of a fusion protein in any of the aforementioned host systems. For example, a chimeric NAAP protein containing a heterologous moiety that can be recognized by a commercially available antibody may facilitate the screening of peptide libraries for inhibitors of NAAP activity. Heterologous protein and peptide moieties may also facilitate purification of fusion proteins using commercially available affinity matrices. Such moieties include, but are not limited to, glutathione S-transferase (GST), maltose binding protein (MBP), thioredoxin (Trx), calmodulin binding peptide (CBP), 6-His, FLAG, *c-myc*, and hemagglutinin (HA). GST, MBP, Trx, CBP, and 6-His enable purification of their cognate fusion proteins on immobilized glutathione, maltose, phenylarsine oxide, calmodulin, and metal-chelate resins, respectively. FLAG, *c-myc*, and hemagglutinin (HA) enable immunoaffinity purification of fusion proteins using commercially available monoclonal and polyclonal antibodies that specifically recognize these epitope tags. A fusion protein may also be engineered to contain a proteolytic cleavage site located between the NAAP encoding sequence and the heterologous protein sequence, so that NAAP may be cleaved away from the heterologous moiety following purification. Methods for fusion protein expression and purification are discussed in Ausubel et al. (*supra*, ch. 10 and 16). A variety of commercially available kits may also be used to facilitate expression and purification of fusion proteins.

In another embodiment, synthesis of radiolabeled NAAP may be achieved *in vitro* using the TNT rabbit reticulocyte lysate or wheat germ extract system (Promega). These systems couple transcription and translation of protein-coding sequences operably associated with the T7, T3, or SP6 promoters. Translation takes place in the presence of a radiolabeled amino acid precursor, for example, ^{35}S-methionine.

NAAP, fragments of NAAP, or variants of NAAP may be used to screen for compounds that specifically bind to NAAP. One or more test compounds may be screened for specific binding to NAAP. In various embodiments, 1, 2, 3, 4, 5, 10, 20, 50, 100, or 200 test compounds can be screened for specific binding to NAAP. Examples of test compounds can include antibodies, anticalins, oligonucleotides, proteins (e.g., ligands or receptors), or small molecules.

In related embodiments, variants of NAAP can be used to screen for binding of test compounds, such as antibodies, to NAAP, a variant of NAAP, or a combination of NAAP and/or one or more variants NAAP. In an embodiment, a variant of NAAP can be used to screen for compounds that bind to a variant of NAAP, but not to NAAP having the exact sequence of a sequence of SEQ ID NO:1-35. NAAP variants used to perform such screening can have a range of about 50% to about 99% sequence identity to NAAP, with various embodiments having 60%, 70%, 75%, 80%, 85%, 90%, and 95% sequence identity.

In an embodiment, a compound identified in a screen for specific binding to NAAP can be closely related to the natural ligand of NAAP, e.g., a ligand or fragment thereof, a natural substrate, a structural or functional mimetic, or a natural binding partner (Coligan, J.E. et al. (1991) Current Protocols in Immunology 1(2):Chapter 5). In another embodiment, the compound thus identified can be a natural ligand of a receptor NAAP (Howard, A.D. et al. (2001) Trends Pharmacol. Sci.22:132-140; Wise, A. et al. (2002) Drug Discovery Today 7:235-246).

In other embodiments, a compound identified in a screen for specific binding to NAAP can be closely related to the natural receptor to which NAAP binds, at least a fragment of the receptor, or a fragment of the receptor including all or a portion of the ligand binding site or binding pocket. For example, the compound may be a receptor for NAAP which is capable of propagating a signal, or a decoy receptor for NAAP which is not capable of propagating a signal (Ashkenazi, A. and V.M. Divit (1999) Curr. Opin. Cell Biol. 11:255-260; Mantovani, A. et al. (2001) Trends Immunol. 22:328-336). The compound can be rationally designed using known techniques. Examples of such techniques include those used to construct the compound etanercept (ENBREL; Amgen Inc., Thousand Oaks CA), which is efficacious for treating rheumatoid arthritis in humans. Etanercept is an engineered p75 tumor necrosis factor (TNF) receptor dimer linked to the Fc portion of human IgG_1 (Taylor, P.C. et al. (2001) Curr. Opin. Immunol. 13:611-616).

In one embodiment, two or more antibodies having similar or, alternatively, different specificities can be screened for specific binding to NAAP, fragments of NAAP, or variants of NAAP. The binding specificity of the antibodies thus screened can thereby be selected to identify particular fragments or variants of NAAP. In one embodiment, an antibody can be selected such that its binding specificity allows for preferential identification of specific fragments or variants of NAAP. In another embodiment, an antibody can be selected such that its binding specificity allows for preferential diagnosis of a specific disease or condition having increased, decreased, or otherwise abnormal production of NAAP.

In an embodiment, anticalins can be screened for specific binding to NAAP, fragments of NAAP, or variants of NAAP. Anticalins are ligand-binding proteins that have been constructed based on a lipocalin scaffold (Weiss, G.A. and H.B. Lowman (2000) Chem. Biol. 7:R177-R184; Skerra, A. (2001) J. Biotechnol. 74:257-275). The protein architecture of lipocalins can include a beta-barrel having eight antiparallel beta-strands, which supports four loops at its open end. These loops form the natural ligand-binding site of the lipocalins, a site which can be re-engineered *in vitro* by amino acid substitutions to impart novel binding specificities. The amino acid substitutions can be made using methods known in the art or described herein, and can include conservative substitutions (e.g., substitutions that do not alter binding specificity) or substitutions that modestly, moderately, or significantly alter binding specificity.

In one embodiment, screening for compounds which specifically bind to, stimulate, or inhibit NAAP involves producing appropriate cells which express NAAP, either as a secreted protein or on the cell membrane. Preferred cells can include cells from mammals, yeast, *Drosophila*, or *E. coli*. Cells expressing NAAP or cell membrane fractions which contain NAAP are then contacted with a test compound and binding, stimulation, or inhibition of activity of either NAAP or the compound is analyzed.

An assay may simply test binding of a test compound to the polypeptide, wherein binding is detected by a fluorophore, radioisotope, enzyme conjugate, or other detectable label. For example, the assay may comprise the steps of combining at least one test compound with NAAP, either in solution or affixed to a solid support, and detecting the binding of NAAP to the compound. Alternatively, the assay may detect or measure binding of a test compound in the presence of a labeled competitor. Additionally, the assay may be carried out using cell-free preparations, chemical libraries, or natural product mixtures, and the test compound(s) may be free in solution or affixed to a solid support.

An assay can be used to assess the ability of a compound to bind to its natural ligand and/or to inhibit the binding of its natural ligand to its natural receptors. Examples of such assays include radio-labeling assays such as those described in U.S. Patent No. 5,914,236 and U.S. Patent No. 6,372,724. In a related embodiment, one or more amino acid substitutions can be introduced into a polypeptide compound (such as a receptor) to improve or alter its ability to bind to its natural ligands (Matthews, D.J. and J.A. Wells. (1994) Chem. Biol. 1:25-30). In another related embodiment, one or more amino acid substitutions can be introduced into a polypeptide compound (such as a ligand) to improve or alter its ability to bind to its natural receptors (Cunningham, B.C. and J.A. Wells (1991) Proc. Natl. Acad. Sci. USA 88:3407-3411; Lowman, H.B. et al. (1991) J. Biol. Chem. 266:10982-10988).

NAAP, fragments of NAAP, or variants of NAAP may be used to screen for compounds that modulate the activity of NAAP. Such compounds may include agonists, antagonists, or partial or inverse agonists. In one embodiment, an assay is performed under conditions permissive for NAAP activity, wherein NAAP is combined with at least one test compound, and the activity of NAAP in the presence of a test compound is compared with the activity of NAAP in the absence of the test compound. A change in the activity of NAAP in the presence of the test compound is indicative of a compound that modulates the activity of NAAP. Alternatively, a test compound is combined with an *in vitro* or cell-free system comprising NAAP under conditions suitable for NAAP activity, and the assay is performed. In either of these assays, a test compound which modulates the activity of NAAP may do so indirectly and need not come in direct contact with the test compound. At least one and up to a plurality of test compounds may be screened.

In another embodiment, polynucleotides encoding NAAP or their mammalian homologs may be "knocked out" in an animal model system using homologous recombination in embryonic stem (ES) cells. Such techniques are well known in the art and are useful for the generation of animal models of human disease (see, e.g., U.S. Patent No. 5,175,383 and U.S. Patent No. 5,767,337). For example, mouse ES cells, such as the mouse 129/SvJ cell line, are derived from the early mouse embryo and grown in culture. The ES cells are transformed with a vector containing the gene of interest disrupted by a marker gene, e.g., the neomycin phosphotransferase gene (*neo*; Capecchi, M.R. (1989) Science 244:1288-1292). The vector integrates into the corresponding region of the host genome by homologous recombination. Alternatively, homologous recombination takes place using the Cre-loxP system to knockout a gene of interest in a tissue- or developmental stage-specific manner (Marth, J.D. (1996) Clin. Invest. 97:1999-2002; Wagner, K.U. et al. (1997) Nucleic Acids Res. 25:4323-4330). Transformed ES cells are identified and microinjected into mouse cell blastocysts such as those from the C57BL/6 mouse strain. The blastocysts are surgically transferred to pseudopregnant dams, and the resulting chimeric progeny are genotyped and bred to produce heterozygous or homozygous strains. Transgenic animals thus generated may be tested with potential therapeutic or toxic agents.

Polynucleotides encoding NAAP may also be manipulated *in vitro* in ES cells derived from human blastocysts. Human ES cells have the potential to differentiate into at least eight separate cell lineages including endoderm, mesoderm, and ectodermal cell types. These cell lineages differentiate into, for example, neural cells, hematopoietic lineages, and cardiomyocytes (Thomson, J.A. et al. (1998) Science 282:1145-1147).

Polynucleotides encoding NAAP can also be used to create "knockin" humanized animals (pigs) or transgenic animals (mice or rats) to model human disease. With knockin technology, a region of a polynucleotide encoding NAAP is injected into animal ES cells, and the injected sequence integrates into the animal cell genome. Transformed cells are injected into blastulae, and the blastulae are implanted as described above. Transgenic progeny or inbred lines are studied and treated with potential pharmaceutical agents to obtain information on treatment of a human disease. Alternatively, a mammal inbred to overexpress NAAP, e.g., by secreting NAAP in its milk, may also serve as a convenient source of that protein (Janne, J. et al. (1998) Biotechnol. Annu. Rev. 4:55-74).

THERAPEUTICS

Chemical and structural similarity, e.g., in the context of sequences and motifs, exists between regions of NAAP and nucleic acid-associated proteins. In addition, examples of tissues expressing NAAP can be found in Table 6 and can also be found in Example XI. Therefore, NAAP appears to play a role in cell proliferative, neurological, developmental, autoimmune/inflammatory, and lipid metabolism disorders, and infections. In the treatment of disorders associated with increased NAAP expression or activity, it is desirable to decrease the expression or activity of NAAP. In the treatment of disorders associated with decreased NAAP expression or activity, it is desirable to increase the expression or activity of NAAP.

Therefore, in one embodiment, NAAP or a fragment or derivative thereof may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of NAAP. Examples of such disorders include, but are not limited to, a cell proliferative disorder such as actinic keratosis, arteriosclerosis, atherosclerosis, bursitis, cirrhosis, hepatitis, mixed connective tissue disease (MCTD), myelofibrosis, paroxysmal nocturnal hemoglobinuria, polycythemia vera, psoriasis, primary thrombocythemia, and cancers including adenocarcinoma, leukemia, lymphoma, melanoma, myeloma, sarcoma, teratocarcinoma, and, in particular, a cancer of the adrenal gland, bladder, bone, bone marrow, brain, breast, cervix, colon, gall bladder, ganglia, gastrointestinal tract, heart, kidney, liver, lung, muscle, ovary, pancreas, parathyroid, penis, prostate, salivary glands, skin, spleen, testis, thymus, thyroid, and uterus; a neurological disorder such as epilepsy, ischemic cerebrovascular disease, stroke, cerebral neoplasms, Alzheimer's disease, Pick's disease, Huntington's disease, dementia, Parkinson's disease and other extrapyramidal disorders, amyotrophic lateral sclerosis and other motor neuron disorders, progressive neural muscular atrophy, retinitis pigmentosa, hereditary ataxias, multiple sclerosis and other demyelinating diseases, bacterial and viral meningitis, brain abscess, subdural empyema, epidural abscess, suppurative intracranial thrombophlebitis, myelitis and radiculitis, viral central nervous system disease, prion diseases including kuru, Creutzfeldt-Jakob disease, and Gerstmann-Straussler-Scheinker syndrome, fatal familial insomnia, nutritional and metabolic diseases of the nervous system, neurofibromatosis, tuberous sclerosis, cerebelloretinal hemangioblastomatosis, encephalotrigeminal syndrome, mental retardation and other developmental disorder of the central nervous system, cerebral palsy, a neuroskeletal disorder, an autonomic nervous system disorder, a cranial nerve disorder, a spinal cord disease, muscular dystrophy and other neuromuscular disorder, a peripheral nervous system disorder, dermatomyositis and polymyositis, inherited, metabolic, endocrine, and toxic myopathy, myasthenia gravis, periodic paralysis, a mental disorder including mood, anxiety, and schizophrenic disorder, seasonal affective disorder (SAD), akathesia, amnesia, catatonia, diabetic neuropathy, tardive dyskinesia, dystonias, paranoid psychoses, postherpetic neuralgia, and Tourette's disorder; a developmental disorder such as renal tubular acidosis, anemia, Cushing's syndrome, achondroplastic dwarfism, Duchenne and Becker muscular dystrophy, epilepsy, gonadal dysgenesis, WAGR syndrome (Wilms' tumor, aniridia, genitourinary abnormalities, and mental retardation), Smith-Magenis syndrome, myelodysplastic syndrome, hereditary mucoepithelial dysplasia, hereditary keratodermas, hereditary neuropathies such as Charcot-Marie-Tooth disease and neurofibromatosis, hypothyroidism, hydrocephalus, seizure disorders such as Syndenham's chorea and cerebral palsy, spina bifida, anencephaly, craniorachischisis, congenital glaucoma, cataract, and sensorineural hearing loss; an autoimmune/inflammatory disorder such as acquired immunodeficiency syndrome (AIDS), Addison's disease, adult respiratory distress syndrome, allergies, ankylosing spondylitis, amyloidosis, anemia, asthma, atherosclerosis, autoimmune hemolytic anemia, autoimmune thyroiditis, autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED), bronchitis, cholecystitis, contact dermatitis, Crohn's disease, atopic dermatitis, dermatomyositis, diabetes mellitus, emphysema, episodic lymphopenia with lymphocytotoxins, erythroblastosis fetalis, erythema nodosum, atrophic gastritis, glomerulonephritis, Goodpasture's syndrome, gout, Graves' disease, Hashimoto's thyroiditis, hypereosinophilia, irritable bowel syndrome, multiple sclerosis, myasthenia gravis, myocardial or pericardial inflammation, osteoarthritis, osteoporosis, pancreatitis, polymyositis, psoriasis, Reiter's syndrome, rheumatoid arthritis, scleroderma, Sjögren's syndrome, systemic anaphylaxis, systemic lupus erythematosus, systemic sclerosis, thrombocytopenic purpura, ulcerative colitis, uveitis, Werner syndrome, complications of cancer, hemodialysis, and extracorporeal circulation, viral, bacterial, fungal, parasitic, protozoal, and helminthic infections, and trauma; a disorder of lipid metabolism such as fatty liver, cholestasis, primary biliary cirrhosis, carnitine deficiency, carnitine palmitoyltransferase deficiency, myoadenylate deaminase deficiency, hypertriglyceridemia, lipid storage disorders such Fabry's disease, Gaucher's disease, Niemann-Pick's disease, metachromatic leukodystrophy, adrenoleukodystrophy, GM_2 gangliosidosis, and ceroid lipofuscinosis, abetalipoproteinemia, Tangier disease, hyperlipoproteinemia, diabetes mellitus, lipodystrophy, lipomatoses, acute panniculitis, disseminated fat necrosis, adiposis dolorosa, lipoid adrenal hyperplasia, minimal change disease, lipomas, atherosclerosis, hypercholesterolemia, hypercholesterolemia with hypertriglyceridemia, primary hypoalphalipoproteinemia, hypothyroidism, renal disease, liver disease, lecithin:cholesterol acyltransferase deficiency, cerebrotendinous xanthomatosis, sitosterolemia, hypocholesterolemia, Tay-Sachs disease, Sandhoff's disease, hyperlipidemia, hyperlipemia, lipid myopathies, and obesity; and an infection, such as those caused by a viral agent classified as adenovirus, arenavirus, bunyavirus, calicivirus, coronavirus, filovirus, hepadnavirus, herpesvirus, flavivirus, orthomyxovirus, parvovirus, papovavirus, paramyxovirus, picornavirus, poxvirus, reovirus, retrovirus, rhabdovirus, or togavirus; and an infection, such as those caused by a viral agent classified as adenovirus, arenavirus, bunyavirus, calicivirus, coronavirus, filovirus, hepadnavirus, herpesvirus, flavivirus, orthomyxovirus, parvovirus, papovavirus, paramyxovirus, picornavirus, poxvirus, reovirus, retrovirus, rhabdovirus, or togavirus; an infection caused by a bacterial agent classified as pneumococcus, staphylococcus, streptococcus, bacillus, corynebacterium, clostridium, meningococcus, gonococcus, listeria, moraxella, kingella, haemophilus, legionella, bordetella, gram-negative enterobacterium including shigella, salmonella, or campylobacter, pseudomonas, vibrio, brucella, francisella, yersinia, bartonella, norcardium, actinomyces, mycobacterium, spirochaetale, rickettsia, chlamydia, or mycoplasma; an infection caused by a fungal agent classified as aspergillus, blastomyces, dermatophytes, cryptococcus, coccidioides, malasezzia, histoplasma, or other mycosis-causing fungal agent; and an infection caused by a parasite classified as plasmodium or malaria-causing, parasitic entamoeba, leishmania, trypanosoma, toxoplasma, pneumocystis carinii, intestinal protozoa such as giardia, trichomonas, tissue nematode such as trichinella, intestinal nematode such as ascaris, lymphatic filarial nematode, trematode such as schistosoma, and cestode such as tapeworm.

In another embodiment, a vector capable of expressing NAAP or a fragment or derivative thereof may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of NAAP including, but not limited to, those described above.

In a further embodiment, a composition comprising a substantially purified NAAP in conjunction with a suitable pharmaceutical carrier may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of NAAP including, but not limited to, those provided above.

In still another embodiment, an agonist which modulates the activity of NAAP may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of NAAP including, but not limited to, those listed above.

In a further embodiment, an antagonist of NAAP may be administered to a subject to treat or prevent a disorder associated with increased expression or activity of NAAP. Examples of such disorders include, but are not limited to, those cell proliferative, neurological, developmental, autoimmune/inflammatory, and lipid metabolism disorders, and infections described above. In one aspect, an antibody which specifically binds NAAP may be used directly as an antagonist or indirectly as a targeting or delivery mechanism for bringing a pharmaceutical agent to cells or tissues which express NAAP.

In an additional embodiment, a vector expressing the complement of the polynucleotide encoding NAAP may be administered to a subject to treat or prevent a disorder associated with increased expression or activity of NAAP including, but not limited to, those described above.

In other embodiments, any protein, agonist, antagonist, antibody, complementary sequence, or vector embodiments may be administered in combination with other appropriate therapeutic agents. Selection of the appropriate agents for use in combination therapy may be made by one of ordinary skill in the art, according to conventional pharmaceutical principles. The combination of therapeutic agents may act synergistically to effect the treatment or prevention of the various disorders described above. Using this approach, one may be able to achieve therapeutic efficacy with lower dosages of each agent, thus reducing the potential for adverse side effects.

An antagonist of NAAP may be produced using methods which are generally known in the art. In particular, purified NAAP may be used to produce antibodies or to screen libraries of pharmaceutical agents to identify those which specifically bind NAAP. Antibodies to NAAP may also be generated using methods that are well known in the art. Such antibodies may include, but are not limited to, polyclonal, monoclonal, chimeric, and single chain antibodies, Fab fragments, and fragments produced by a Fab expression library. In an embodiment, neutralizing antibodies (i.e., those which inhibit dimer formation) can be used therapeutically. Single chain antibodies (e.g., from camels or llamas) may be potent enzyme inhibitors and may have application in the design of peptide mimetics, and in the development of immuno-adsorbents and biosensors (Muyldermans, S. (2001) J. Biotechnol. 74:277-302).

For the production of antibodies, various hosts including goats, rabbits, rats, mice, camels, dromedaries, llamas, humans, and others may be immunized by injection with NAAP or with any fragment or oligopeptide thereof which has immunogenic properties. Depending on the host species, various adjuvants may be used to increase immunological response. Such adjuvants include, but are not limited to, Freund's, mineral gels such as aluminum hydroxide, and surface active substances such as lysolecithin, pluronic polyols, polyanions, peptides, oil emulsions, KLH, and dinitrophenol. Among adjuvants used in humans, BCG (bacilli Calmette-Guerin) and *Corynebacterium parvum* are especially preferable.

It is preferred that the oligopeptides, peptides, or fragments used to induce antibodies to NAAP have an amino acid sequence consisting of at least about 5 amino acids, and generally will consist of at least about 10 amino acids. It is also preferable that these oligopeptides, peptides, or fragments are substantially identical to a portion of the amino acid sequence of the natural protein. Short stretches of NAAP amino acids may be fused with those of another protein, such as KLH, and antibodies to the chimeric molecule may be produced.

Monoclonal antibodies to NAAP may be prepared using any technique which provides for the production of antibody molecules by continuous cell lines in culture. These include, but are not limited to, the hybridoma technique, the human B-cell hybridoma technique, and the EBV-hybridoma technique (Kohler, G. et al. (1975) Nature 256:495-497; Kozbor, D. et al. (1985) J. Immunol. Methods 81:31-42; Cote, R.J. et al. (1983) Proc. Natl. Acad. Sci. USA 80:2026-2030; Cole, S.P. et al. (1984) Mol. Cell Biol. 62:109-120).

In addition, techniques developed for the production of "chimeric antibodies," such as the splicing of mouse antibody genes to human antibody genes to obtain a molecule with appropriate antigen specificity and biological activity, can be used (Morrison, S.L. et al. (1984) Proc. Natl. Acad. Sci. USA 81:6851-6855; Neuberger, M.S. et al. (1984) Nature 312:604-608; Takeda, S. et al. (1985) Nature 314:452-454). Alternatively, techniques described for the production of single chain antibodies may be adapted, using methods known in the art, to produce NAAP-specific single chain antibodies. Antibodies with related specificity, but of distinct idiotypic composition, may be generated by chain shuffling from random combinatorial immunoglobulin libraries (Burton, D.R. (1991) Proc. Natl. Acad. Sci. USA 88:10134-10137).

Antibodies may also be produced by inducing *in vivo* production in the lymphocyte population or by screening immunoglobulin libraries or panels of highly specific binding reagents as disclosed in the literature (Orlandi, R. et al. (1989) Proc. Natl. Acad. Sci. USA 86:3833-3837; Winter, G. et al. (1991) Nature 349:293-299).

Antibody fragments which contain specific binding sites for NAAP may also be generated. For example, such fragments include, but are not limited to, $F(ab')_2$ fragments produced by pepsin digestion of the antibody molecule and Fab fragments generated by reducing the disulfide bridges of the F(ab')2 fragments. Alternatively, Fab expression libraries may be constructed to allow rapid and easy identification of monoclonal Fab fragments with the desired specificity (Huse, W.D. et al. (1989) Science 246:1275-1281).

Various immunoassays may be used for screening to identify antibodies having the desired specificity. Numerous protocols for competitive binding or immunoradiometric assays using either polyclonal or monoclonal antibodies with established specificities are well known in the art. Such immunoassays typically involve the measurement of complex formation between NAAP and its specific antibody. A two-site, monoclonal-based immunoassay utilizing monoclonal antibodies reactive to two non-interfering NAAP epitopes is generally used, but a competitive binding assay may also be employed (Pound, *supra*).

Various methods such as Scatchard analysis in conjunction with radioimmunoassay techniques may be used to assess the affinity of antibodies for NAAP. Affinity is expressed as an association constant, K_a, which is defined as the molar concentration of NAAP-antibody complex divided by the molar concentrations of free antigen and free antibody under equilibrium conditions. The K_a determined for a preparation of polyclonal antibodies, which are heterogeneous in their affinities for multiple NAAP epitopes, represents the average affinity, or avidity, of the antibodies for NAAP. The K_a determined for a preparation of monoclonal antibodies, which are monospecific for a particular NAAP epitope, represents a true measure of affinity. High-affinity antibody preparations with K_a ranging from about 10^9 to 10^{12} L/mole are preferred for use in immunoassays in which the NAAP-antibody complex must withstand rigorous manipulations. Low-affinity antibody preparations with K_a ranging from about 10^6 to 10^7 L/mole are preferred for use in immunopurification and similar procedures which ultimately require dissociation of NAAP, preferably in active form, from the antibody (Catty, D. (1988) Antibodies, Volume I: A Practical Approach, IRL Press, Washington DC; Liddell, J.E. and A. Cryer (1991) A Practical Guide to Monoclonal Antibodies, John Wiley & Sons, New York NY).

The titer and avidity of polyclonal antibody preparations may be further evaluated to determine the quality and suitability of such preparations for certain downstream applications. For example, a polyclonal antibody preparation containing at least 1-2 mg specific antibody/ml, preferably 5-10 mg specific antibody/ml, is generally employed in procedures requiring precipitation of NAAP-antibody complexes. Procedures for evaluating antibody specificity, titer, and avidity, and guidelines for antibody quality and usage in various applications, are generally available (Catty, *supra*; Coligan et al., *supra*).

In another embodiment of the invention, polynucleotides encoding NAAP, or any fragment or complement thereof, may be used for therapeutic purposes. In one aspect, modifications of gene expression can be achieved by designing complementary sequences or antisense molecules (DNA, RNA, PNA, or modified oligonucleotides) to the coding or regulatory regions of the gene encoding NAAP. Such technology is well known in the art, and antisense oligonucleotides or larger fragments can be designed from various locations along the coding or control regions of sequences encoding NAAP (Agrawal, S., ed. (1996) Antisense Therapeutics, Humana Press, Totawa NJ).

In therapeutic use, any gene delivery system suitable for introduction of the antisense sequences into appropriate target cells can be used. Antisense sequences can be delivered intracellularly in the form of an expression plasmid which, upon transcription, produces a sequence complementary to at least a portion of the cellular sequence encoding the target protein (Slater, J.E. et al. (1998) J. Allergy Clin. Immunol. 102:469-475; Scanlon, K.J. et al. (1995) FASEB J. 9:1288-1296). Antisense sequences can also be introduced intracellularly through the use of viral vectors, such as retrovirus and adeno-associated virus vectors (Miller, A.D. (1990) Blood 76:271-278; Ausubel et al., *supra*; Uckert, W. and W. Walther (1994) Pharmacol. Ther. 63:323-347). Other gene delivery mechanisms include liposome-derived systems, artificial viral envelopes, and other systems known in the art (Rossi, J.J. (1995) Br. Med. Bull. 51:217-225; Boado, R.J. et al. (1998) J. Pharm. Sci. 87:1308-1315; Morris, M.C. et al. (1997) Nucleic Acids Res. 25:2730-2736).

In another embodiment of the invention, polynucleotides encoding NAAP may be used for somatic or germline gene therapy. Gene therapy may be performed to (i) correct a genetic deficiency (e.g., in the cases of severe combined immunodeficiency (SCID)-X1 disease characterized by X-linked inheritance (Cavazzana-Calvo, M. et al. (2000) Science 288:669-672), severe combined immunodeficiency syndrome associated with an inherited adenosine deaminase (ADA) deficiency

(Blaese, R.M. et al. (1995) Science 270:475-480; Bordignon, C. et al. (1995) Science 270:470-475), cystic fibrosis (Zabner, J. et al. (1993) Cell 75:207-216; Crystal, R.G. et al. (1995) Hum. Gene Therapy 6:643-666; Crystal, R.G. et al. (1995) Hum. Gene Therapy 6:667-703), thalassamias, familial hypercholesterolemia, and hemophilia resulting from Factor VIII or Factor IX deficiencies (Crystal, R.G. (1995) Science 270:404-410; Verma, I.M. and N. Somia (1997) Nature 389:239-242)), (ii) express a conditionally lethal gene product (e.g., in the case of cancers which result from unregulated cell proliferation), or (iii) express a protein which affords protection against intracellular parasites (e.g., against human retroviruses, such as human immunodeficiency virus (HIV) (Baltimore, D. (1988) Nature 335:395-396; Poeschla, E. et al. (1996) Proc. Natl. Acad. Sci. USA 93:11395-11399), hepatitis B or C virus (HBV, HCV); fungal parasites, such as *Candida albicans* and *Paracoccidioides brasiliensis*; and protozoan parasites such as *Plasmodium falciparum* and *Trypanosoma cruzi*). In the case where a genetic deficiency in NAAP expression or regulation causes disease, the expression of NAAP from an appropriate population of transduced cells may alleviate the clinical manifestations caused by the genetic deficiency.

In a further embodiment of the invention, diseases or disorders caused by deficiencies in NAAP are treated by constructing mammalian expression vectors encoding NAAP and introducing these vectors by mechanical means into NAAP-deficient cells. Mechanical transfer technologies for use with cells *in vivo* or *ex vitro* include (i) direct DNA microinjection into individual cells, (ii) ballistic gold particle delivery, (iii) liposome-mediated transfection, (iv) receptor-mediated gene transfer, and (v) the use of DNA transposons (Morgan, R.A. and W.F. Anderson (1993) Annu. Rev. Biochem. 62:191-217; Ivics, Z. (1997) Cell 91:501-510; Boulay, J.-L. and H. Récipon (1998) Curr. Opin. Biotechnol. 9:445-450).

Expression vectors that may be effective for the expression of NAAP include, but are not limited to, the PCDNA 3.1, EPITAG, PRCCMV2, PREP, PVAX, PCR2-TOPOTA vectors (Invitrogen, Carlsbad CA), PCMV-SCRIPT, PCMV-TAG, PEGSH/PERV (Stratagene, La Jolla CA), and PTET-OFF, PTET-ON, PTRE2, PTRE2-LUC, PTK-HYG (BD Clontech, Palo Alto CA). NAAP may be expressed using (i) a constitutively active promoter, (e.g., from cytomegalovirus (CMV), Rous sarcoma virus (RSV), SV40 virus, thymidine kinase (TK), or β-actin genes), (ii) an inducible promoter (e.g., the tetracycline-regulated promoter (Gossen, M. and H. Bujard (1992) Proc. Natl. Acad. Sci. USA 89:5547-5551; Gossen, M. et al. (1995) Science 268:1766-1769; Rossi, F.M.V. and H.M. Blau (1998) Curr. Opin. Biotechnol. 9:451-456), commercially available in the T-REX plasmid (Invitrogen)); the ecdysone-inducible promoter (available in the plasmids PVGRXR and PIND; Invitrogen); the FK506/rapamycin inducible promoter; or the RU486/mifepristone inducible promoter (Rossi, F.M.V. and H.M. Blau, *supra*)), or (iii) a tissue-specific promoter or the native promoter of the endogenous gene encoding NAAP from a normal individual.

Commercially available liposome transformation kits (e.g., the PERFECT LIPID TRANSFECTION KIT, available from Invitrogen) allow one with ordinary skill in the art to deliver polynucleotides to target cells in culture and require minimal effort to optimize experimental parameters. In the alternative, transformation is performed using the calcium phosphate method (Graham, F.L. and A.J. Eb (1973) Virology 52:456-467), or by electroporation (Neumann, E. et al. (1982) EMBO J. 1:841-845). The introduction of DNA to primary cells requires modification of these standardized mammalian transfection protocols.

In another embodiment of the invention, diseases or disorders caused by genetic defects with respect to NAAP expression are treated by constructing a retrovirus vector consisting of (i) the polynucleotide encoding NAAP under the control of an independent promoter or the retrovirus long terminal repeat (LTR) promoter, (ii) appropriate RNA packaging signals, and (iii) a Rev-responsive element (RRE) along with additional retrovirus *cis*-acting RNA sequences and coding sequences required for efficient vector propagation. Retrovirus vectors (e.g., PFB and PFBNEO) are commercially available (Stratagene) and are based on published data (Riviere, I. et al. (1995) Proc. Natl. Acad. Sci. USA 92:6733-6737), incorporated by reference herein. The vector is propagated in an appropriate vector producing cell line (VPCL) that expresses an envelope gene with a tropism for receptors on the target cells or a promiscuous envelope protein such as VSVg (Armentano, D. et al. (1987) J. Virol. 61:1647-1650; Bender, M.A. et al. (1987) J. Virol. 61:1639-1646; Adam, M.A. and A.D. Miller (1988) J. Virol. 62:3802-3806; Dull, T. et al. (1998) J. Virol. 72:8463-8471; Zufferey, R. et al. (1998) J. Virol. 72:9873-9880). U.S. Patent No. 5,910,434 to Rigg ("Method for obtaining retrovirus packaging cell lines producing high transducing efficiency retroviral supernatant") discloses a method for obtaining retrovirus packaging cell lines and is hereby incorporated by reference. Propagation of retrovirus vectors, transduction of a population of cells (e.g., $CD4^+$ T-cells), and the return of transduced cells to a patient are procedures well known to persons skilled in the art of gene therapy and have been well documented (Ranga, U. et al. (1997) J. Virol. 71:7020-7029; Bauer, G. et al. (1997) Blood 89:2259-2267; Bonyhadi, M.L. (1997) J. Virol. 71:4707-4716; Ranga, U. et al. (1998) Proc. Natl. Acad. Sci. USA 95:1201-1206; Su, L. (1997) Blood 89:2283-2290).

In an embodiment, an adenovirus-based gene therapy delivery system is used to deliver polynucleotides encoding NAAP to cells which have one or more genetic abnormalities with respect to the expression of NAAP. The construction and packaging of adenovirus-based vectors are well known to those with ordinary skill in the art. Replication defective adenovirus vectors have proven to be versatile for importing genes encoding immunoregulatory proteins into intact islets in the pancreas (Csete, M.E. et al. (1995) Transplantation 27:263-268). Potentially useful adenoviral vectors are described in U.S. Patent No. 5,707,618 to Armentano ("Adenovirus vectors for gene therapy"), hereby incorporated by reference. For adenoviral vectors, see also Antinozzi, P.A. et al. (1999; Annu. Rev. Nutr. 19:511-544) and Verma, I.M. and N. Somia (1997; Nature 18:389:239-242).

In another embodiment, a herpes-based, gene therapy delivery system is used to deliver polynucleotides encoding NAAP to target cells which have one or more genetic abnormalities with respect to the expression of NAAP. The use of herpes simplex virus (HSV)-based vectors may be especially valuable for introducing NAAP to cells of the central nervous system, for which HSV has a tropism. The construction and packaging of herpes-based vectors are well known to those with ordinary skill in the art. A replication-competent herpes simplex virus (HSV) type 1-based vector has been used to deliver a reporter gene to the eyes of primates (Liu, X. et al. (1999) Exp. Eye Res. 169:385-395). The construction of a HSV-1 virus vector has also been disclosed in detail in U.S. Patent No. 5,804,413 to DeLuca ("Herpes simplex virus strains for gene transfer"), which is hereby incorporated by reference. U.S. Patent No. 5,804,413 teaches the use of recombinant HSV d92 which consists of a genome containing at least one exogenous gene to be transferred to a cell under the control of the appropriate promoter for purposes including human gene therapy. Also taught by this patent are the construction and use of recombinant HSV strains deleted for ICP4, ICP27 and ICP22. For HSV vectors, see also Goins, W.F. et al. (1999; J. Virol. 73:519-532) and Xu, H. et al. (1994; Dev. Biol. 163:152-161). The manipulation of cloned herpesvirus sequences, the generation of recombinant virus following the transfection of multiple plasmids containing different segments of the large herpesvirus genomes, the growth and propagation of herpesvirus, and the infection of cells with herpesvirus are techniques well known to those of ordinary skill in the art.

In another embodiment, an alphavirus (positive, single-stranded RNA virus) vector is used to deliver polynucleotides encoding NAAP to target cells. The biology of the prototypic alphavirus, Semliki Forest Virus (SFV), has been studied extensively and gene transfer vectors have been based on the SFV genome (Garoff, H. and K.-J. Li (1998) Curr. Opin. Biotechnol. 9:464-469). During alphavirus RNA replication, a subgenomic RNA is generated that normally encodes the viral capsid proteins. This subgenomic RNA replicates to higher levels than the full length genomic RNA, resulting in the overproduction of capsid proteins relative to the viral proteins with enzymatic activity (e.g., protease and polymerase). Similarly, inserting the coding sequence for NAAP into the alphavirus genome in place of the capsid-coding region results in the production of a large number of NAAP-coding RNAs and the synthesis of high levels of NAAP in vector transduced cells. While alphavirus infection is typically associated with cell lysis within a few days, the ability to establish a persistent infection in hamster normal kidney cells (BHK-21) with a variant of Sindbis virus (SIN) indicates that the lytic replication of alphaviruses can be altered to suit the needs of the gene therapy application (Dryga, S.A. et al. (1997) Virology 228:74-83). The wide host range of alphaviruses will allow the introduction of NAAP into a variety of cell types. The specific transduction of a subset of cells in a population may require the sorting of cells prior to transduction. The methods of manipulating infectious cDNA clones of alphaviruses, performing alphavirus cDNA and RNA transfections, and performing alphavirus infections, are well known to those with ordinary skill in the art.

Oligonucleotides derived from the transcription initiation site, e.g., between about positions -10 and +10 from the start site, may also be employed to inhibit gene expression. Similarly, inhibition can be achieved using triple helix base-pairing methodology. Triple helix pairing is useful because it causes inhibition of the ability of the double helix to open sufficiently for the binding of polymerases, transcription factors, or regulatory molecules. Recent therapeutic advances using triplex DNA have been described in the literature (Gee, J.E. et al. (1994) in Huber, B.E. and B.I. Carr, Molecular and Immunologic Approaches, Futura Publishing, Mt. Kisco NY, pp. 163-177). A complementary sequence or antisense molecule may also be designed to block translation of mRNA by preventing the transcript from binding to ribosomes.

Ribozymes, enzymatic RNA molecules, may also be used to catalyze the specific cleavage of RNA. The mechanism of ribozyme action involves sequence-specific hybridization of the ribozyme molecule to complementary target RNA, followed by endonucleolytic cleavage. For example, engineered hammerhead motif ribozyme molecules may specifically and efficiently catalyze endonucleolytic cleavage of RNA molecules encoding NAAP.

Specific ribozyme cleavage sites within any potential RNA target are initially identified by scanning the target molecule for ribozyme cleavage sites, including the following sequences: GUA, GUU, and GUC. Once identified, short RNA sequences of between 15 and 20 ribonucleotides, corresponding to the region of the target gene containing the cleavage site, may be evaluated for secondary structural features which may render the oligonucleotide inoperable. The suitability of candidate targets may also be evaluated by testing accessibility to hybridization with complementary oligonucleotides using ribonuclease protection assays.

Complementary ribonucleic acid molecules and ribozymes may be prepared by any method known in the art for the synthesis of nucleic acid molecules. These include techniques for chemically synthesizing oligonucleotides such as solid phase phosphoramidite chemical synthesis. Alternatively, RNA molecules may be generated by *in vitro* and *in vivo* transcription of DNA molecules encoding NAAP. Such DNA sequences may be incorporated into a wide variety of vectors with suitable RNA polymerase promoters such as T7 or SP6. Alternatively, these cDNA constructs that synthesize complementary RNA, constitutively or inducibly, can be introduced into cell lines, cells, or tissues.

RNA molecules may be modified to increase intracellular stability and half-life. Possible modifications include, but are not limited to, the addition of flanking sequences at the 5' and/or 3' ends of the molecule, or the use of phosphorothioate or 2' O-methyl rather than phosphodiesterase linkages within the backbone of the molecule. This concept is inherent in the production of PNAs and can be extended in all of these molecules by the inclusion of nontraditional bases such as inosine, queosine, and wybutosine, as well as acetyl-, methyl-, thio-, and similarly modified forms of adenine, cytidine, guanine, thymine, and uridine which are not as easily recognized by endogenous endonucleases.

In other embodiments of the invention, the expression of one or more selected polynucleotides of the present invention can be altered, inhibited, decreased, or silenced using RNA interference (RNAi) or post-transcriptional gene silencing (PTGS) methods known in the art. RNAi is a post-transcriptional mode of gene silencing in which double-stranded RNA (dsRNA) introduced into a targeted cell specifically suppresses the expression of the homologous gene (i.e., the gene bearing the sequence complementary to the dsRNA). This effectively knocks out or substantially reduces the expression of the targeted gene. PTGS can also be accomplished by use of DNA or DNA fragments as well. RNAi methods are described by Fire, A. et al. (1998; Nature 391:806-811) and Gura, T. (2000; Nature 404:804-808). PTGS can also be initiated by introduction of a complementary segment of DNA into the selected tissue using gene delivery and/or viral vector delivery methods described herein or known in the art.

RNAi can be induced in mammalian cells by the use of small interfering RNA also known as siRNA. siRNA are shorter segments of dsRNA (typically about 21 to 23 nucleotides in length) that result *in vivo* from cleavage of introduced dsRNA by the action of an endogenous ribonuclease. siRNA appear to be the mediators of the RNAi effect in mammals. The most effective siRNAs appear to be 21 nucleotide dsRNAs with 2 nucleotide 3' overhangs. The use of siRNA for inducing RNAi in mammalian cells is described by Elbashir, S.M. et al. (2001; Nature 411:494-498).

siRNA can be generated indirectly by introduction of dsRNA into the targeted cell. Alternatively, siRNA can be synthesized directly and introduced into a cell by transfection methods and agents described herein or known in the art (such as liposome-mediated transfection, viral vector methods, or other polynucleotide delivery/introductory methods). Suitable siRNAs can be selected by examining a transcript of the target polynucleotide (e.g., mRNA) for nucleotide sequences downstream from the AUG start codon and recording the occurrence of each nucleotide and the 3' adjacent 19 to 23 nucleotides as potential siRNA target sites, with sequences having a 21 nucleotide length being preferred. Regions to be avoided for target siRNA sites include the 5' and 3' untranslated regions (UTRs) and regions near the start codon (within 75 bases), as these may be richer in regulatory protein binding sites. UTR-binding proteins and/or translation initiation complexes may interfere with binding of the siRNP endonuclease complex. The selected target sites for siRNA can then be compared to the appropriate genome database (e.g., human, etc.) using BLAST or other sequence comparison algorithms known in the art. Target sequences with significant homology to other coding sequences can be eliminated from consideration. The selected siRNAs can be produced by chemical synthesis methods known in the art or by *in vitro* transcription using commercially available methods and kits such as the SILENCER siRNA construction kit (Ambion, Austin TX).

In alternative embodiments, long-term gene silencing and/or RNAi effects can be induced in selected tissue using expression vectors that continuously express siRNA. This can be accomplished using expression vectors that are engineered to express hairpin RNAs (shRNAs) using methods known in the art (see, e.g., Brummelkamp, T.R. et al. (2002) Science 296:550-553; and Paddison, P.J. et al. (2002) Genes Dev. 16:948-958). In these and related embodiments, shRNAs can be delivered to target cells using expression vectors known in the art. An example of a suitable expression vector for delivery of siRNA is the PSILENCER1.0-U6 (circular) plasmid (Ambion). Once delivered to the target tissue, shRNAs are processed *in vivo* into siRNA-like molecules capable of carrying out gene-specific silencing.

In various embodiments, the expression levels of genes targeted by RNAi or PTGS methods can be determined by assays for mRNA and/or protein analysis. Expression levels of the mRNA of a targeted gene can be determined, for example, by northern analysis methods using the NORTHERNMAX-GLY kit (Ambion); by microarray methods; by PCR methods; by real time PCR methods; and by other RNA/polynucleotide assays known in the art or described herein. Expression levels of the protein encoded by the targeted gene can be determined, for example, by microarray methods; by polyacrylamide gel electrophoresis; and by Western analysis using standard techniques known in the art.

An additional embodiment of the invention encompasses a method for screening for a compound which is effective in altering expression of a polynucleotide encoding NAAP. Compounds which may be effective in altering expression of a specific polynucleotide may include, but are not limited to, oligonucleotides, antisense oligonucleotides, triple helix-forming oligonucleotides, transcription factors and other polypeptide transcriptional regulators, and non-macromolecular chemical entities which are capable of interacting with specific polynucleotide sequences. Effective compounds may alter polynucleotide expression by acting as either inhibitors or promoters of polynucleotide expression. Thus, in the treatment of disorders associated with increased NAAP expression or activity, a compound which specifically inhibits expression of the polynucleotide encoding NAAP may be therapeutically useful, and in the treatment of disorders associated with decreased NAAP expression or activity, a compound which specifically promotes expression of the polynucleotide encoding NAAP may be therapeutically useful.

In various embodiments, one or more test compounds may be screened for effectiveness in altering expression of a specific polynucleotide. A test compound may be obtained by any method commonly known in the art, including chemical modification of a compound known to be effective in altering polynucleotide expression; selection from an existing, commercially-available or proprietary library of naturally-occurring or non-natural chemical compounds; rational design of a compound based on chemical and/or structural properties of the target polynucleotide; and selection from a library of chemical compounds created combinatorially or randomly. A sample comprising a polynucleotide encoding NAAP is exposed to at least one test compound thus obtained. The sample may comprise, for example, an intact or permeabilized cell, or an *in vitro* cell-free or reconstituted biochemical system. Alterations in the expression of a polynucleotide encoding NAAP are assayed by any method commonly known in the art. Typically, the expression of a specific nucleotide is detected by hybridization with a probe having a nucleotide sequence complementary to the sequence of the polynucleotide encoding NAAP. The amount of hybridization may be quantified, thus forming the basis for a comparison of the expression of the polynucleotide both with and without exposure to one or more test compounds. Detection of a change in the expression of a polynucleotide exposed to a test compound indicates that the test compound is effective in altering the expression of the polynucleotide. A screen for a compound effective in altering expression of a specific polynucleotide can be carried out, for example, using a *Schizosaccharomyces pombe* gene expression system (Atkins, D. et al. (1999) U.S. Patent No. 5,932,435; Arndt, G.M. et al. (2000) Nucleic Acids Res. 28:E15) or a human cell line such as HeLa cell (Clarke, M.L. et al. (2000) Biochem. Biophys. Res. Commun. 268:8-13). A particular embodiment of the present invention involves screening a combinatorial library of oligonucleotides (such as deoxyribonucleotides, ribonucleotides, peptide nucleic acids, and modified oligonucleotides) for antisense activity against a specific polynucleotide sequence (Bruice, T.W. et al. (1997) U.S. Patent No. 5,686,242; Bruice, T.W. et al. (2000) U.S. Patent No. 6,022,691).

Many methods for introducing vectors into cells or tissues are available and equally suitable for use *in vivo*, *in vitro*, and *ex vivo*. For *ex vivo* therapy, vectors may be introduced into stem cells taken from the patient and clonally propagated for autologous transplant back into that same patient. Delivery by transfection, by liposome injections, or by polycationic amino polymers may be achieved using methods which are well known in the art (Goldman, C.K. et al. (1997) Nat. Biotechnol. 15:462-466).

Any of the therapeutic methods described above may be applied to any subject in need of such therapy, including, for example, mammals such as humans, dogs, cats, cows, horses, rabbits, and monkeys.

An additional embodiment of the invention relates to the administration of a composition which generally comprises an active ingredient formulated with a pharmaceutically acceptable excipient. Excipients may include, for example, sugars, starches, celluloses, gums, and proteins. Various formulations are commonly known and are thoroughly discussed in the latest edition of

Remington's Pharmaceutical Sciences (Maack Publishing, Easton PA). Such compositions may consist of NAAP, antibodies to NAAP, and mimetics, agonists, antagonists, or inhibitors of NAAP.

In various embodiments, the compositions described herein, such as pharmaceutical compositions, may be administered by any number of routes including, but not limited to, oral, intravenous, intramuscular, intra-arterial, intramedullary, intrathecal, intraventricular, pulmonary, transdermal, subcutaneous, intraperitoneal, intranasal, enteral, topical, sublingual, or rectal means.

Compositions for pulmonary administration may be prepared in liquid or dry powder form. These compositions are generally aerosolized immediately prior to inhalation by the patient. In the case of small molecules (e.g. traditional low molecular weight organic drugs), aerosol delivery of fast-acting formulations is well-known in the art. In the case of macromolecules (e.g. larger peptides and proteins), recent developments in the field of pulmonary delivery via the alveolar region of the lung have enabled the practical delivery of drugs such as insulin to blood circulation (see, e.g., Patton, J.S. et al., U.S. Patent No. 5,997,848). Pulmonary delivery allows administration without needle injection, and obviates the need for potentially toxic penetration enhancers.

Compositions suitable for use in the invention include compositions wherein the active ingredients are contained in an effective amount to achieve the intended purpose. The determination of an effective dose is well within the capability of those skilled in the art.

Specialized forms of compositions may be prepared for direct intracellular delivery of macromolecules comprising NAAP or fragments thereof. For example, liposome preparations containing a cell-impermeable macromolecule may promote cell fusion and intracellular delivery of the macromolecule. Alternatively, NAAP or a fragment thereof may be joined to a short cationic N-terminal portion from the HIV Tat-1 protein. Fusion proteins thus generated have been found to transduce into the cells of all tissues, including the brain, in a mouse model system (Schwarze, S.R. et al. (1999) Science 285:1569-1572).

For any compound, the therapeutically effective dose can be estimated initially either in cell culture assays, e.g., of neoplastic cells, or in animal models such as mice, rats, rabbits, dogs, monkeys, or pigs. An animal model may also be used to determine the appropriate concentration range and route of administration. Such information can then be used to determine useful doses and routes for administration in humans.

A therapeutically effective dose refers to that amount of active ingredient, for example NAAP or fragments thereof, antibodies of NAAP, and agonists, antagonists or inhibitors of NAAP, which ameliorates the symptoms or condition. Therapeutic efficacy and toxicity may be determined by standard pharmaceutical procedures in cell cultures or with experimental animals, such as by calculating the ED_{50} (the dose therapeutically effective in 50% of the population) or LD_{50} (the dose lethal to 50% of the population) statistics. The dose ratio of toxic to therapeutic effects is the therapeutic index, which can be expressed as the LD_{50}/ED_{50} ratio. Compositions which exhibit large therapeutic indices are preferred. The data obtained from cell culture assays and animal studies are used to formulate a range of dosage for human use. The dosage contained in such compositions is preferably within a range of circulating concentrations that includes the ED_{50} with little or no toxicity. The dosage varies within this range depending upon the dosage form employed, the sensitivity of the patient, and the route of administration.

The exact dosage will be determined by the practitioner, in light of factors related to the subject requiring treatment. Dosage and administration are adjusted to provide sufficient levels of the active moiety or to maintain the desired effect. Factors which may be taken into account include the severity of the disease state, the general health of the subject, the age, weight, and gender of the subject, time and frequency of administration, drug combination(s), reaction sensitivities, and response to therapy. Long-acting compositions may be administered every 3 to 4 days, every week, or biweekly depending on the half-life and clearance rate of the particular formulation.

Normal dosage amounts may vary from about 0.1 μg to 100,000 μg, up to a total dose of about 1 gram, depending upon the route of administration. Guidance as to particular dosages and methods of delivery is provided in the literature and generally available to practitioners in the art. Those skilled in the art will employ different formulations for nucleotides than for proteins or their inhibitors. Similarly, delivery of polynucleotides or polypeptides will be specific to particular cells, conditions, locations, etc.

DIAGNOSTICS

In another embodiment, antibodies which specifically bind NAAP may be used for the diagnosis of disorders characterized by expression of NAAP, or in assays to monitor patients being treated with NAAP or agonists, antagonists, or inhibitors of NAAP. Antibodies useful for diagnostic purposes may be prepared in the same manner as described above for therapeutics. Diagnostic assays for NAAP include methods which utilize the antibody and a label to detect NAAP in human body fluids or in extracts of cells or tissues. The antibodies may be used with or without modification, and may be labeled by covalent or non-covalent attachment of a reporter molecule. A wide variety of reporter molecules, several of which are described above, are known in the art and may be used.

A variety of protocols for measuring NAAP, including ELISAs, RIAs, and FACS, are known in the art and provide a basis for diagnosing altered or abnormal levels of NAAP expression. Normal or standard values for NAAP expression are established by combining body fluids or cell extracts taken from normal mammalian subjects, for example, human subjects, with antibodies to NAAP under conditions suitable for complex formation. The amount of standard complex formation may be quantitated by various methods, such as photometric means. Quantities of NAAP expressed in subject, control, and disease samples from biopsied tissues are compared with the standard values.

Deviation between standard and subject values establishes the parameters for diagnosing disease.

In another embodiment of the invention, polynucleotides encoding NAAP may be used for diagnostic purposes. The polynucleotides which may be used include oligonucleotides, complementary RNA and DNA molecules, and PNAs. The polynucleotides may be used to detect and quantify gene expression in biopsied tissues in which expression of NAAP may be correlated with disease. The diagnostic assay may be used to determine absence, presence, and excess expression of NAAP, and to monitor regulation of NAAP levels during therapeutic intervention.

In one aspect, hybridization with PCR probes which are capable of detecting polynucleotides, including genomic sequences, encoding NAAP or closely related molecules may be used to identify nucleic acid sequences which encode NAAP. The specificity of the probe, whether it is made from a highly specific region, e.g., the 5' regulatory region, or from a less specific region, e.g., a conserved motif, and the stringency of the hybridization or amplification will determine whether the probe identifies only naturally occurring sequences encoding NAAP, allelic variants, or related sequences.

Probes may also be used for the detection of related sequences, and may have at least 50% sequence identity to any of the NAAP encoding sequences. The hybridization probes of the subject invention may be DNA or RNA and may be derived from the sequence of SEQ ID NO:36-70 or from genomic sequences including promoters, enhancers, and introns of the NAAP gene.

Means for producing specific hybridization probes for polynucleotides encoding NAAP include the cloning of polynucleotides encoding NAAP or NAAP derivatives into vectors for the production of mRNA probes. Such vectors are known in the art, are commercially available, and may be used to synthesize RNA probes *in vitro* by means of the addition of the appropriate RNA polymerases and the appropriate labeled nucleotides. Hybridization probes may be labeled by a variety of reporter groups, for example, by radionuclides such as ^{32}P or ^{35}S, or by enzymatic labels, such as alkaline phosphatase coupled to the probe via avidin/biotin coupling systems, and the like.

Polynucleotides encoding NAAP may be used for the diagnosis of disorders associated with expression of NAAP. Examples of such disorders include, but are not limited to, a cell proliferative disorder such as actinic keratosis, arteriosclerosis, atherosclerosis, bursitis, cirrhosis, hepatitis, mixed connective tissue disease (MCTD), myelofibrosis, paroxysmal nocturnal hemoglobinuria, polycythemia vera, psoriasis, primary thrombocythemia, and cancers including adenocarcinoma, leukemia, lymphoma, melanoma, myeloma, sarcoma, teratocarcinoma, and, in particular, a cancer of the adrenal gland, bladder, bone, bone marrow, brain, breast, cervix, colon, gall bladder, ganglia, gastrointestinal tract, heart, kidney, liver, lung, muscle, ovary, pancreas, parathyroid, penis, prostate, salivary glands, skin, spleen, testis, thymus, thyroid, and uterus; a neurological disorder such as epilepsy, ischemic cerebrovascular disease, stroke, cerebral neoplasms, Alzheimer's disease, Pick's disease, Huntington's disease, dementia, Parkinson's disease and other extrapyramidal disorders, amyotrophic lateral sclerosis and other motor neuron disorders, progressive neural muscular atrophy, retinitis pigmentosa, hereditary ataxias, multiple sclerosis and other demyelinating diseases, bacterial and viral meningitis, brain abscess, subdural empyema, epidural abscess, suppurative intracranial thrombophlebitis, myelitis and radiculitis, viral central nervous system disease, prion diseases including kuru, Creutzfeldt-Jakob disease, and Gerstmann-Straussler-Scheinker syndrome, fatal familial insomnia, nutritional and metabolic diseases of the nervous system, neurofibromatosis, tuberous sclerosis, cerebelloretinal hemangioblastomatosis, encephalotrigeminal syndrome, mental retardation and other developmental disorder of the central nervous system, cerebral palsy, a neuroskeletal disorder, an autonomic nervous system disorder, a cranial nerve disorder, a spinal cord disease, muscular dystrophy and other neuromuscular disorder, a peripheral nervous system disorder, dermatomyositis and polymyositis, inherited, metabolic, endocrine, and toxic myopathy, myasthenia gravis, periodic paralysis, a mental disorder including mood, anxiety, and schizophrenic disorder, seasonal affective disorder (SAD), akathesia, amnesia, catatonia, diabetic neuropathy, tardive dyskinesia, dystonias, paranoid psychoses, postherpetic neuralgia, and Tourette's disorder; a developmental disorder such as renal tubular acidosis, anemia, Cushing's syndrome, achondroplastic dwarfism, Duchenne and Becker muscular dystrophy, epilepsy, gonadal dysgenesis, WAGR syndrome (Wilms' tumor, aniridia, genitourinary abnormalities, and mental retardation), Smith-Magenis syndrome, myelodysplastic syndrome, hereditary mucoepithelial dysplasia, hereditary keratodermas, hereditary neuropathies such as Charcot-Marie-Tooth disease and neurofibromatosis, hypothyroidism, hydrocephalus, seizure disorders such as Syndenham's chorea and cerebral palsy, spina bifida, anencephaly, craniorachischisis, congenital glaucoma, cataract, and sensorineural hearing loss; an autoimmune/inflammatory disorder such as acquired immunodeficiency syndrome (AIDS), Addison's disease, adult respiratory distress syndrome, allergies, ankylosing spondylitis, amyloidosis, anemia, asthma, atherosclerosis, autoimmune hemolytic anemia, autoimmune thyroiditis, autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED), bronchitis, cholecystitis, contact dermatitis, Crohn's disease, atopic dermatitis, dermatomyositis, diabetes mellitus, emphysema, episodic lymphopenia with lymphocytotoxins, erythroblastosis fetalis, erythema nodosum, atrophic gastritis, glomerulonephritis, Goodpasture's syndrome, gout, Graves' disease, Hashimoto's thyroiditis, hypereosinophilia, irritable bowel syndrome, multiple sclerosis, myasthenia gravis, myocardial or pericardial inflammation, osteoarthritis, osteoporosis, pancreatitis, polymyositis, psoriasis, Reiter's syndrome, rheumatoid arthritis, scleroderma, Sjögren's syndrome, systemic anaphylaxis, systemic lupus erythematosus, systemic sclerosis, thrombocytopenic purpura, ulcerative colitis, uveitis, Werner syndrome, complications of cancer, hemodialysis, and extracorporeal circulation, viral, bacterial, fungal, parasitic, protozoal, and helminthic infections, and trauma; a disorder of lipid metabolism such as fatty liver, cholestasis, primary biliary cirrhosis, carnitine deficiency, carnitine palmitoyltransferase deficiency, myoadenylate deaminase deficiency, hypertriglyceridemia, lipid storage disorders such Fabry's disease, Gaucher's disease, Niemann-Pick's disease, metachromatic leukodystrophy, adrenoleukodystrophy, GM_2 gangliosidosis, and ceroid lipofuscinosis, abetalipoproteinemia, Tangier disease, hyperlipoproteinemia, diabetes mellitus, lipodystrophy, lipomatoses, acute panniculitis, disseminated fat necrosis, adiposis dolorosa, lipoid adrenal hyperplasia, minimal change disease, lipomas, atherosclerosis, hypercholesterolemia, hypercholesterolemia with hypertriglyceridemia, primary hypoalphalipoproteinemia, hypothyroidism, renal disease, liver disease, lecithin:cholesterol acyltransferase deficiency, cerebrotendinous xanthomatosis, sitosterolemia, hypocholesterolemia, Tay-Sachs disease, Sandhoff's disease, hyperlipidemia, hyperlipemia, lipid myopathies, and obesity; and an infection, such as those caused by a viral agent classified as adenovirus, arenavirus, bunyavirus, calicivirus, coronavirus, filovirus, hepadnavirus, herpesvirus, flavivirus, orthomyxovirus, parvovirus, papovavirus, paramyxovirus, picornavirus, poxvirus, reovirus, retrovirus, rhabdovirus, or togavirus; and an infection, such as those caused by a viral agent classified as adenovirus, arenavirus, bunyavirus, calicivirus, coronavirus, filovirus, hepadnavirus, herpesvirus, flavivirus, orthomyxovirus, parvovirus, papovavirus, paramyxovirus, picornavirus, poxvirus, reovirus, retrovirus, rhabdovirus, or togavirus; an infection caused by a bacterial agent classified as pneumococcus, staphylococcus, streptococcus, bacillus, corynebacterium, clostridium, meningococcus, gonococcus, listeria, moraxella, kingella, haemophilus, legionella, bordetella, gram-negative enterobacterium including shigella, salmonella, or campylobacter, pseudomonas, vibrio, brucella, francisella, yersinia, bartonella, norcardium, actinomyces, mycobacterium, spirochaetale, rickettsia, chlamydia, or mycoplasma; an infection caused by a fungal agent classified as aspergillus, blastomyces, dermatophytes, cryptococcus, coccidioides, malasezzia, histoplasma, or other mycosis-causing fungal agent; and an infection caused by a parasite classified as plasmodium or malaria-causing, parasitic entamoeba, leishmania, trypanosoma, toxoplasma, pneumocystis carinii, intestinal protozoa such as giardia, trichomonas, tissue nematode such as trichinella, intestinal nematode such as ascaris, lymphatic filarial nematode, trematode such as schistosoma, and cestode such as tapeworm. Polynucleotides encoding NAAP may be used in Southern or northern analysis, dot blot, or other membrane-based technologies; in PCR technologies; in dipstick, pin, and multiformat ELISA-like assays; and in microarrays utilizing fluids or tissues from patients to detect altered NAAP expression. Such qualitative or quantitative methods are well known in the art.

In a particular embodiment, polynucleotides encoding NAAP may be used in assays that detect the presence of associated disorders, particularly those mentioned above. Polynucleotides complementary to sequences encoding NAAP may be labeled by standard methods and added to a fluid or tissue sample from a patient under conditions suitable for the formation of hybridization complexes. After a suitable incubation period, the sample is washed and the signal is quantified and compared with a standard value. If the amount of signal in the patient sample is significantly altered in comparison to a control sample then the presence of altered levels of polynucleotides encoding NAAP in the sample indicates the presence of the associated disorder. Such assays may also be used to evaluate the efficacy of a particular therapeutic treatment regimen in animal studies, in clinical trials, or to monitor the treatment of an individual patient.

In order to provide a basis for the diagnosis of a disorder associated with expression of NAAP, a normal or standard profile for expression is established. This may be accomplished by combining body fluids or cell extracts taken from normal subjects, either animal or human, with a sequence, or a fragment thereof, encoding NAAP, under conditions suitable for hybridization or amplification. Standard hybridization may be quantified by comparing the values obtained from normal subjects with values from an experiment in which a known amount of a substantially purified polynucleotide is used. Standard values obtained in this manner may be compared with values obtained from samples from patients who are symptomatic for a disorder. Deviation from standard values is used to establish the presence of a disorder.

Once the presence of a disorder is established and a treatment protocol is initiated, hybridization assays may be repeated on a regular basis to determine if the level of expression in the patient begins to approximate that which is observed in the normal subject. The results obtained from successive assays may be used to show the efficacy of treatment over a period ranging from several days to months.

With respect to cancer, the presence of an abnormal amount of transcript (either under- or overexpressed) in biopsied tissue from an individual may indicate a predisposition for the development of the disease, or may provide a means for detecting the disease prior to the appearance of actual clinical symptoms. A more definitive diagnosis of this type may allow health professionals to employ preventative measures or aggressive treatment earlier, thereby preventing the development or further progression of the cancer.

Additional diagnostic uses for oligonucleotides designed from the sequences encoding NAAP may involve the use of PCR. These oligomers may be chemically synthesized, generated enzymatically, or produced *in vitro*. Oligomers will preferably contain a fragment of a polynucleotide encoding NAAP, or a fragment of a polynucleotide complementary to the polynucleotide encoding NAAP, and will be employed under optimized conditions for identification of a specific gene or condition. Oligomers may also be employed under less stringent conditions for detection or quantification of closely related DNA or RNA sequences.

In a particular aspect, oligonucleotide primers derived from polynucleotides encoding NAAP may be used to detect single nucleotide polymorphisms (SNPs). SNPs are substitutions, insertions and deletions that are a frequent cause of inherited or acquired genetic disease in humans. Methods of SNP detection include, but are not limited to, single-stranded conformation polymorphism (SSCP) and fluorescent SSCP (fSSCP) methods. In SSCP, oligonucleotide primers derived from polynucleotides encoding NAAP are used to amplify DNA using the polymerase chain reaction (PCR). The DNA may be derived, for example, from diseased or normal tissue, biopsy samples, bodily fluids, and the like. SNPs in the DNA cause differences in the secondary and tertiary structures of PCR products in single-stranded form, and these differences are detectable using gel electrophoresis in non-denaturing gels. In fSCCP, the oligonucleotide primers are fluorescently labeled, which allows detection of the amplimers in high-throughput equipment such as DNA sequencing machines. Additionally, sequence database analysis methods, termed in silico SNP (isSNP), are capable of identifying polymorphisms by comparing the sequence of individual overlapping DNA fragments which assemble into a common consensus sequence. These computer-based methods filter out sequence variations due to laboratory preparation of DNA and sequencing errors using statistical models and automated analyses of DNA sequence chromatograms. In the alternative, SNPs may be detected and characterized by mass spectrometry using, for example, the high throughput MASSARRAY system (Sequenom, Inc., San Diego CA).

SNPs may be used to study the genetic basis of human disease. For example, at least 16 common SNPs have been associated with non-insulin-dependent diabetes mellitus. SNPs are also useful for examining differences in disease outcomes in monogenic disorders, such as cystic fibrosis, sickle cell anemia, or chronic granulomatous disease. For example, variants in the mannose-binding lectin, MBL2, have been shown to be correlated with deleterious pulmonary outcomes in cystic fibrosis. SNPs also have utility in pharmacogenomics, the identification of genetic variants that influence a patient's response to a drug, such as life-threatening toxicity. For example, a variation in N-acetyl transferase is associated with a high incidence of peripheral neuropathy in response to the anti-tuberculosis drug isoniazid, while a variation in the core promoter of the ALOX5 gene results in diminished clinical response to treatment with an anti-asthma drug that targets the 5-lipoxygenase pathway. Analysis of the distribution of SNPs in different populations is useful for investigating genetic drift, mutation, recombination, and selection, as well as for tracing the origins of populations and their migrations (Taylor, J.G. et al. (2001) Trends Mol. Med. 7:507-512; Kwok, P.-Y. and Z. Gu (1999) Mol. Med. Today 5:538-543; Nowotny, P. et al. (2001) Curr. Opin. Neurobiol. 11:637-641).

Methods which may also be used to quantify the expression of NAAP include radiolabeling or biotinylating nucleotides, coamplification of a control nucleic acid, and interpolating results from standard curves (Melby, P.C. et al. (1993) J. Immunol. Methods 159:235-244; Duplaa, C. et al. (1993) Anal. Biochem. 212:229-236). The speed of quantitation of multiple samples may be accelerated by running the assay in a high-throughput format where the oligomer or polynucleotide of interest is presented in various dilutions and a spectrophotometric or colorimetric response gives rapid quantitation.

In further embodiments, oligonucleotides or longer fragments derived from any of the polynucleotides described herein may be used as elements on a microarray. The microarray can be used in transcript imaging techniques which monitor the relative expression levels of large numbers of genes simultaneously as described below. The microarray may also be used to identify genetic variants, mutations, and polymorphisms. This information may be used to determine gene function, to understand the genetic basis of a disorder, to diagnose a disorder, to monitor progression/regression of disease as a function of gene expression, and to develop and monitor the activities of therapeutic agents in the treatment of disease. In particular, this information may be used to develop a pharmacogenomic profile of a patient in order to select the most appropriate and effective treatment regimen for that patient. For example, therapeutic agents which are highly effective and display the fewest side effects may be selected for a patient based on his/her pharmacogenomic profile.

In another embodiment, NAAP, fragments of NAAP, or antibodies specific for NAAP may be used as elements on a microarray. The microarray may be used to monitor or measure protein-protein interactions, drug-target interactions, and gene expression profiles, as described above.

A particular embodiment relates to the use of the polynucleotides of the present invention to generate a transcript image of a tissue or cell type. A transcript image represents the global pattern of gene expression by a particular tissue or cell type. Global gene expression patterns are analyzed by quantifying the number of expressed genes and their relative abundance under given conditions and at a given time (Seilhamer et al., "Comparative Gene Transcript Analysis," U.S. Patent No. 5,840,484; hereby expressly incorporated by reference herein). Thus a transcript image may be generated by hybridizing the polynucleotides of the present invention or their complements to the totality of transcripts or reverse transcripts of a particular tissue or cell type. In one embodiment, the hybridization takes place in high-throughput format, wherein the polynucleotides of the present invention or their complements comprise a subset of a plurality of elements on a microarray. The resultant transcript image would provide a profile of gene activity.

Transcript images may be generated using transcripts isolated from tissues, cell lines, biopsies, or other biological samples. The transcript image may thus reflect gene expression *in vivo*, as in the case of a tissue or biopsy sample, or *in vitro*, as in the case of a cell line.

Transcript images which profile the expression of the polynucleotides of the present invention may also be used in conjunction with *in vitro* model systems and preclinical evaluation of pharmaceuticals, as well as toxicological testing of industrial and naturally-occurring environmental compounds. All compounds induce characteristic gene expression patterns, frequently termed molecular fingerprints or toxicant signatures, which are indicative of mechanisms of action and toxicity (Nuwaysir, E.F. et al. (1999) Mol. Carcinog. 24:153-159; Steiner, S. and N.L. Anderson (2000) Toxicol. Lett. 112-113:467-471). If a test compound has a signature similar to that of a compound with known toxicity, it is likely to share those toxic properties. These fingerprints or signatures are most useful and refined when they contain expression information from a large number of genes and gene families. Ideally, a genome-wide measurement of expression provides the highest quality signature. Even genes whose expression is not altered by any tested compounds are important as well, as the levels of expression of these genes are used to normalize the rest of the expression data. The normalization procedure is useful for comparison of expression data after treatment with different compounds. While the assignment of gene function to elements of a toxicant signature aids in interpretation of toxicity mechanisms, knowledge of gene function is not necessary for the statistical matching of signatures which leads to prediction of toxicity (see, for example, Press Release 00-02 from the National Institute of Environmental Health Sciences, released February 29, 2000, available at http://www.niehs.nih.gov/oc/news/toxchip.htm). Therefore, it is important and desirable in toxicological screening using toxicant signatures to include all expressed gene sequences.

In an embodiment, the toxicity of a test compound can be assessed by treating a biological sample containing nucleic acids with the test compound. Nucleic acids that are expressed in the treated biological sample are hybridized with one or more probes specific to the polynucleotides of the present invention, so that transcript levels corresponding to the polynucleotides of the present invention may be quantified. The transcript levels in the treated biological sample are compared with levels in an untreated biological sample. Differences in the transcript levels between the two samples are indicative of a toxic response caused by the test compound in the treated sample.

Another embodiment relates to the use of the polypeptides disclosed herein to analyze the proteome of a tissue or cell type. The term proteome refers to the global pattern of protein expression in a particular tissue or cell type. Each protein component of a proteome can be subjected individually to further analysis. Proteome expression patterns, or profiles, are analyzed by quantifying the number of expressed proteins and their relative abundance under given conditions and at a given time. A profile of a cell's proteome may thus be generated by separating and analyzing the polypeptides of a particular tissue or cell type. In one embodiment, the separation is achieved using two-dimensional gel electrophoresis, in which proteins from a sample are separated by isoelectric focusing in the first dimension, and then according to molecular weight by sodium dodecyl sulfate slab gel electrophoresis in the second dimension (Steiner and Anderson, *supra*). The proteins are visualized in the gel as discrete and uniquely positioned spots, typically by staining the gel with an agent such as Coomassie Blue or silver or fluorescent stains. The optical density of each protein spot is generally proportional to the level of the protein in the sample. The optical densities of equivalently positioned protein spots from different samples, for example, from biological samples either treated or untreated with a test compound or therapeutic agent, are compared to identify any changes in protein spot density related to the treatment. The proteins in the spots are partially sequenced using, for example, standard methods employing chemical or enzymatic cleavage followed by mass spectrometry. The identity of the protein in a spot may be determined by comparing its partial sequence, preferably of at least 5 contiguous amino acid residues, to the polypeptide sequences of interest. In some cases, further sequence data may be obtained for definitive protein identification.

A proteomic profile may also be generated using antibodies specific for NAAP to quantify the levels of NAAP expression. In one embodiment, the antibodies are used as elements on a microarray, and protein expression levels are quantified by contacting the microarray with the sample and detecting the levels of protein bound to each array element (Lueking, A. et al. (1999) Anal. Biochem. 270:103-111; Mendoze, L.G. et al. (1999) Biotechniques 27:778-788). Detection may be performed by a variety of methods known in the art, for example, by reacting the proteins in the sample with a thiol- or amino-reactive fluorescent compound and detecting the amount of fluorescence bound at each array element.

Toxicant signatures at the proteome level are also useful for toxicological screening, and should be analyzed in parallel with toxicant signatures at the transcript level. There is a poor correlation between transcript and protein abundances for some proteins in some tissues (Anderson, N.L. and J. Seilhamer (1997) Electrophoresis 18:533-537), so proteome toxicant signatures may be useful in the analysis of compounds which do not significantly affect the transcript image, but which alter the proteomic profile. In addition, the analysis of transcripts in body fluids is difficult, due to rapid degradation of mRNA, so proteomic profiling may be more reliable and informative in such cases.

In another embodiment, the toxicity of a test compound is assessed by treating a biological sample containing proteins with the test compound. Proteins that are expressed in the treated biological sample are separated so that the amount of each protein can be quantified. The amount of each protein is compared to the amount of the corresponding protein in an untreated biological sample. A difference in the amount of protein between the two samples is indicative of a toxic response to the test compound in the treated sample. Individual proteins are identified by sequencing the amino acid residues of the individual proteins and comparing these partial sequences to the polypeptides of the present invention.

In another embodiment, the toxicity of a test compound is assessed by treating a biological sample containing proteins with the test compound. Proteins from the biological sample are incubated with antibodies specific to the polypeptides of the present invention. The amount of protein recognized by the antibodies is quantified. The amount of protein in the treated biological sample is compared with the amount in an untreated biological sample. A difference in the amount of protein between the two samples is indicative of a toxic response to the test compound in the treated sample.

Microarrays may be prepared, used, and analyzed using methods known in the art (Brennan, T.M. et al. (1995) U.S. Patent No. 5,474,796; Schena, M. et al. (1996) Proc. Natl. Acad. Sci. USA 93:10614-10619; Baldeschweiler et al. (1995) PCT application WO95/25116; Shalon, D. et al. (1995) PCT application WO95/35505; Heller, R.A. et al. (1997) Proc. Natl. Acad. Sci. USA 94:2150-2155; Heller, M.J. et al. (1997) U.S. Patent No. 5,605,662). Various types of microarrays are well known and thoroughly described in Schena, M., ed. (1999; DNA Microarrays: A Practical Approach, Oxford University Press, London).

In another embodiment of the invention, nucleic acid sequences encoding NAAP may be used to generate hybridization probes useful in mapping the naturally occurring genomic sequence. Either coding or noncoding sequences may be used, and in some instances, noncoding sequences may be preferable over coding sequences. For example, conservation of a coding sequence among members of a multi-gene family may potentially cause undesired cross hybridization during chromosomal mapping. The sequences may be mapped to a particular chromosome, to a specific region of a chromosome, or to artificial chromosome constructions, e.g., human artificial chromosomes (HACs), yeast artificial chromosomes (YACs), bacterial artificial chromosomes (BACs), bacterial P1 constructions, or single chromosome cDNA libraries (Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355; Price, C.M. (1993) Blood Rev. 7:127-134; Trask, B.J. (1991) Trends Genet. 7:149-154). Once mapped, the nucleic acid sequences may be used to develop genetic linkage maps, for example, which correlate the inheritance of a disease state with the inheritance of a particular chromosome region or restriction fragment length polymorphism (RFLP) (Lander, E.S. and D. Botstein (1986) Proc. Natl. Acad. Sci. USA 83:7353-7357).

Fluorescent *in situ* hybridization (FISH) may be correlated with other physical and genetic map data (Heinz-Ulrich, et al. (1995) in Meyers, *supra*, pp. 965-968). Examples of genetic map data can be found in various scientific journals or at the Online Mendelian Inheritance in Man (OMIM) World Wide Web site. Correlation between the location of the gene encoding NAAP on a physical map and a specific disorder, or a predisposition to a specific disorder, may help define the region of DNA associated with that disorder and thus may further positional cloning efforts.

In situ hybridization of chromosomal preparations and physical mapping techniques, such as linkage analysis using established chromosomal markers, may be used for extending genetic maps. Often the placement of a gene on the chromosome of another mammalian species, such as mouse, may reveal associated markers even if the exact chromosomal locus is not known. This information is valuable to investigators searching for disease genes using positional cloning or other gene discovery techniques. Once the gene or genes responsible for a disease or syndrome have been crudely localized by genetic linkage to a particular genomic region, e.g., ataxia-telangiectasia to 11q22-23, any sequences mapping to that area may represent associated or regulatory genes for further investigation (Gatti, R.A. et al. (1988) Nature 336:577-580). The nucleotide sequence of the instant invention may also be used to detect differences in the chromosomal location due to translocation, inversion, etc., among normal, carrier, or affected individuals.

In another embodiment of the invention, NAAP, its catalytic or immunogenic fragments, or oligopeptides thereof can be used for screening libraries of compounds in any of a variety of drug screening techniques. The fragment employed in such screening may be free in solution, affixed to a solid support, borne on a cell surface, or located intracellularly. The formation of binding complexes between NAAP and the agent being tested may be measured.

Another technique for drug screening provides for high throughput screening of compounds having suitable binding affinity to the protein of interest (Geysen, et al. (1984) PCT application WO84/03564). In this method, large numbers of different small test compounds are synthesized on a solid substrate. The test compounds are reacted with NAAP, or fragments thereof, and washed. Bound NAAP is then detected by methods well known in the art. Purified NAAP can also be coated directly onto plates for use in the aforementioned drug screening techniques. Alternatively, non-neutralizing antibodies can be used to capture the peptide and immobilize it on a solid support.

In another embodiment, one may use competitive drug screening assays in which neutralizing antibodies capable of binding NAAP specifically compete with a test compound for binding NAAP. In this manner, antibodies can be used to detect the presence of any peptide which shares one or more antigenic determinants with NAAP.

In additional embodiments, the nucleotide sequences which encode NAAP may be used in any molecular biology techniques that have yet to be developed, provided the new techniques rely on properties of nucleotide sequences that are currently known, including, but not limited to, such properties as the triplet genetic code and specific base pair interactions.

Without further elaboration, it is believed that one skilled in the art can, using the preceding description, utilize the present invention to its fullest extent. The following embodiments are, therefore, to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever.

The disclosures of all patents, applications, and publications mentioned above and below, including U.S. Ser. No. 60/398,907, U.S. Ser. No. 60/407,068, U.S. Ser. No. 60/414,139, U.S. Ser. No. 60/424,094, U.S. Ser. No. 60/440,912, and U.S. Ser. No. 60/442,419, are hereby expressly incorporated by reference.

EXAMPLES

I. Construction of cDNA Libraries

Incyte cDNAs are derived from cDNA libraries described in the LIFESEQ database (Incyte, Palo Alto CA). Some tissues are homogenized and lysed in guanidinium isothiocyanate, while others are homogenized and lysed in phenol or in a suitable mixture of denaturants, such as TRIZOL (Invitrogen), a monophasic solution of phenol and guanidine isothiocyanate. The resulting lysates are centrifuged over CsCl cushions or extracted with chloroform. RNA is precipitated from the lysates with either isopropanol or sodium acetate and ethanol, or by other routine methods.

Phenol extraction and precipitation of RNA are repeated as necessary to increase RNA purity. In some cases, RNA is treated with DNase. For most libraries, poly(A)+ RNA is isolated using oligo d(T)-coupled paramagnetic particles (Promega), OLIGOTEX latex particles (QIAGEN, Chatsworth CA), or an OLIGOTEX mRNA purification kit (QIAGEN). Alternatively, RNA is isolated directly from tissue lysates using other RNA isolation kits, e.g., the POLY(A)PURE mRNA purification kit (Ambion, Austin TX).

In some cases, Stratagene is provided with RNA and constructs the corresponding cDNA libraries. Otherwise, cDNA is synthesized and cDNA libraries are constructed with the UNIZAP vector system (Stratagene) or SUPERSCRIPT plasmid system (Invitrogen), using the recommended procedures or similar methods known in the art (Ausubel et al., *supra*, ch. 5). Reverse transcription is initiated using oligo d(T) or random primers. Synthetic oligonucleotide adapters are ligated to double stranded cDNA, and the cDNA is digested with the appropriate restriction enzyme or enzymes. For most libraries, the cDNA is size-selected (300-1000 bp) using SEPHACRYL S1000, SEPHAROSE CL2B, or SEPHAROSE CL4B column chromatography (Amersham Biosciences) or preparative agarose gel electrophoresis. cDNAs are ligated into compatible restriction enzyme sites of the polylinker of a suitable plasmid, e.g., PBLUESCRIPT plasmid (Stratagene), PSPORT1 plasmid (Invitrogen, Carlsbad CA), PCDNA2.1 plasmid (Invitrogen), PBK-CMV plasmid (Stratagene), PCR2-TOPOTA plasmid (Invitrogen), PCMV-ICIS plasmid (Stratagene), pIGEN (Incyte, Palo Alto CA), pRARE (Incyte), or pINCY (Incyte), or derivatives thereof. Recombinant plasmids are transformed into competent *E. coli* cells including XL1-Blue, XL1-BlueMRF, or SOLR from Stratagene or DH5α, DH10B, or ElectroMAX DH10B from Invitrogen.

II. Isolation of cDNA Clones

Plasmids obtained as described in Example I are recovered from host cells by *in vivo* excision using the UNIZAP vector system (Stratagene) or by cell lysis. Plasmids are purified using at least one of the following: a Magic or WIZARD Minipreps DNA purification system (Promega); an AGTC Miniprep purification kit (Edge Biosystems, Gaithersburg MD); and QIAWELL 8 Plasmid, QIAWELL 8 Plus Plasmid, QIAWELL 8 Ultra Plasmid purification systems or the R.E.A.L. PREP 96 plasmid purification kit from QIAGEN. Following precipitation, plasmids are resuspended in 0.1 ml of distilled water and stored, with or without lyophilization, at 4°C.

Alternatively, plasmid DNA is amplified from host cell lysates using direct link PCR in a high-throughput format (Rao, V.B. (1994) Anal. Biochem. 216:1-14). Host cell lysis and thermal cycling steps are carried out in a single reaction mixture. Samples are processed and stored in 384-well plates, and the concentration of amplified plasmid DNA is quantified fluorometrically using PICOGREEN dye (Molecular Probes, Eugene OR) and a FLUOROSKAN II fluorescence scanner (Labsystems Oy, Helsinki, Finland).

III. Sequencing and Analysis

Incyte cDNA recovered in plasmids as described in Example II are sequenced as follows. Sequencing reactions are processed using standard methods or high-throughput instrumentation such as the ABI CATALYST 800 (Applied Biosystems) thermal cycler or the PTC-200 thermal cycler (MJ Research) in conjunction with the HYDRA microdispenser (Robbins Scientific) or the MICROLAB 2200 (Hamilton) liquid transfer system. cDNA sequencing reactions are prepared using reagents provided by Amersham Biosciences or supplied in ABI sequencing kits such as the ABI PRISM BIGDYE Terminator cycle sequencing ready reaction kit (Applied Biosystems). Electrophoretic separation of cDNA sequencing reactions and detection of labeled polynucleotides are carried out using the MEGABACE 1000 DNA sequencing system (Amersham Biosciences); the ABI PRISM 373 or 377 sequencing system (Applied Biosystems) in conjunction with standard ABI protocols and base calling software; or other sequence analysis systems known in the art. Reading frames within the cDNA sequences are identified using standard methods (Ausubel et al., *supra*, ch. 7). Some of the cDNA sequences are selected for extension using the techniques disclosed in Example VIII.

Polynucleotide sequences derived from Incyte cDNAs are validated by removing vector, linker, and poly(A) sequences and by masking ambiguous bases, using algorithms and programs based on BLAST, dynamic programming, and dinucleotide nearest neighbor analysis. The Incyte cDNA sequences or translations thereof are then queried against a selection of public databases such as the GenBank primate, rodent, mammalian, vertebrate, and eukaryote databases, and BLOCKS, PRINTS, DOMO, PRODOM; PROTEOME databases with sequences from *Homo sapiens*, *Rattus norvegicus*, *Mus musculus*, *Caenorhabditis elegans*, *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, and *Candida albicans* (Incyte, Palo Alto CA); hidden Markov model (HMM)-based protein family databases such as PFAM, INCY, and TIGRFAM (Haft, D.H. et al. (2001) Nucleic Acids Res. 29:41-43); and HMM-based protein domain databases such as SMART (Schultz, J. et al. (1998) Proc. Natl. Acad. Sci. USA 95:5857-5864; Letunic, I. et al. (2002) Nucleic Acids Res. 30:242-244). (HMM is a probabilistic approach which analyzes consensus primary structures of gene families; see, for example, Eddy, S.R. (1996) Curr. Opin. Struct. Biol. 6:361-365.) The queries are performed using programs based on BLAST, FASTA, BLIMPS, and HMMER. The Incyte cDNA sequences are assembled to produce full length polynucleotide sequences. Alternatively, GenBank cDNAs, GenBank ESTs, stitched sequences, stretched sequences, or Genscan-predicted coding sequences (see Examples IV and V) are used to extend Incyte cDNA assemblages to full length. Assembly is performed using programs based on Phred, Phrap, and Consed, and cDNA assemblages are screened for open reading frames using programs based on GeneMark, BLAST, and FASTA. The full length polynucleotide sequences are translated to derive the corresponding full length polypeptide sequences. Alternatively, a polypeptide may begin at any of the methionine residues of the full length translated polypeptide. Full length polypeptide sequences are subsequently analyzed by querying against databases such as the GenBank protein databases (genpept), SwissProt, the PROTEOME databases, BLOCKS, PRINTS, DOMO, PRODOM, Prosite, hidden Markov model (HMM)-based protein family databases such as PFAM, INCY, and TIGRFAM; and HMM-based protein domain databases such as SMART. Full length polynucleotide sequences are also analyzed using MACDNASIS PRO software (MiraiBio, Alameda CA) and LASERGENE software (DNASTAR). Polynucleotide and polypeptide sequence alignments are generated using default parameters specified by the CLUSTAL algorithm as incorporated into the MEGALIGN multisequence alignment program (DNASTAR), which also calculates the percent identity between aligned sequences.

Table 7 summarizes tools, programs, and algorithms used for the analysis and assembly of Incyte cDNA and full length sequences and provides applicable descriptions, references, and threshold parameters. The first column of Table 7 shows the tools, programs, and algorithms used, the second column provides brief descriptions thereof, the third column presents appropriate references, all of which are incorporated by reference herein in their entirety, and the fourth column presents, where applicable, the scores, probability values, and other parameters used to evaluate the strength of a match between two sequences (the higher the score or the lower the probability value, the greater the identity between two sequences).

The programs described above for the assembly and analysis of full length polynucleotide and polypeptide sequences are also used to identify polynucleotide sequence fragments from SEQ ID NO:36-70. Fragments from about 20 to about 4000 nucleotides which are useful in hybridization and amplification technologies are described in Table 4, column 2.

IV. Identification and Editing of Coding Sequences from Genomic DNA

Putative nucleic acid-associated proteins are initially identified by running the Genscan gene identification program against public genomic sequence databases (e.g., gbpri and gbhtg). Genscan is a general-purpose gene identification program which analyzes genomic DNA sequences from a variety of organisms (Burge, C. and S. Karlin (1997) J. Mol. Biol. 268:78-94; Burge, C. and S. Karlin (1998) Curr. Opin. Struct. Biol. 8:346-354). The program concatenates predicted exons to form an assembled cDNA sequence extending from a methionine to a stop codon. The output of Genscan is a FASTA database of polynucleotide and polypeptide sequences. The maximum range of sequence for Genscan to analyze at once is set to 30 kb. To determine which of these Genscan predicted cDNA sequences encode nucleic acid-associated proteins, the encoded polypeptides are analyzed by querying against PFAM models for nucleic acid-associated proteins. Potential nucleic acid-associated proteins are also identified by homology to Incyte cDNA sequences that have been annotated as nucleic acid-associated proteins. These selected Genscan-predicted sequences are then compared by BLAST analysis to the genpept and gbpri public databases. Where necessary, the Genscan-predicted sequences are then edited by comparison to the top BLAST hit from genpept to correct errors in the sequence predicted by Genscan, such as extra or omitted exons. BLAST analysis is also used to find any Incyte cDNA or public cDNA coverage of the Genscan-predicted sequences, thus providing evidence for transcription. When Incyte cDNA coverage is available, this information is used to correct or confirm the Genscan predicted sequence. Full length polynucleotide sequences are obtained by assembling Genscan-predicted coding sequences with Incyte cDNA sequences and/or public cDNA sequences using the assembly process described in Example III. Alternatively, full length polynucleotide sequences are derived entirely from edited or unedited Genscan-predicted coding sequences.

V. Assembly of Genomic Sequence Data with cDNA Sequence Data

"Stitched" Sequences

Partial cDNA sequences are extended with exons predicted by the Genscan gene identification program described in Example IV. Partial cDNAs assembled as described in Example III are mapped to genomic DNA and parsed into clusters containing related cDNAs and Genscan exon predictions from one or more genomic sequences. Each cluster is analyzed using an algorithm based on graph theory and dynamic programming to integrate cDNA and genomic information, generating possible splice variants that are subsequently confirmed, edited, or extended to create a full length sequence. Sequence intervals in which the entire length of the interval is present on more than one sequence in the cluster are identified, and intervals thus identified are considered to be equivalent by transitivity. For example, if an interval is present on a cDNA and two genomic sequences, then all three intervals are considered to be equivalent. This process allows unrelated but consecutive genomic sequences to be brought together, bridged by cDNA sequence. Intervals thus identified are then "stitched" together by the stitching algorithm in the order that they appear along their parent sequences to generate the longest possible sequence, as well as sequence variants. Linkages between intervals which proceed along one type of parent sequence (cDNA to cDNA or genomic sequence to genomic sequence) are given preference over linkages which change parent type (cDNA to genomic sequence). The resultant stitched sequences are translated and compared by BLAST analysis to the genpept and gbpri public databases. Incorrect exons predicted by Genscan are corrected by comparison to the top BLAST hit from genpept. Sequences are further extended with additional cDNA sequences, or by inspection of genomic DNA, when necessary.

"Stretched" Sequences

Partial DNA sequences are extended to full length with an algorithm based on BLAST analysis. First, partial cDNAs assembled as described in Example III are queried against public databases such as the GenBank primate, rodent, mammalian, vertebrate, and eukaryote databases using the BLAST program. The nearest GenBank protein homolog is then compared by BLAST analysis to either Incyte cDNA sequences or GenScan exon predicted sequences described in Example IV. A chimeric protein is generated by using the resultant high-scoring segment pairs (HSPs) to map the translated sequences onto the GenBank protein homolog. Insertions or deletions may occur in the chimeric protein with respect to the original GenBank protein homolog. The GenBank protein homolog, the chimeric protein, or both are used as probes to search for homologous genomic sequences from the public human genome databases. Partial DNA sequences are therefore "stretched" or extended by the addition of homologous genomic sequences. The resultant stretched sequences are examined to determine whether they contain a complete gene.

VI. Chromosomal Mapping of NAAP Encoding Polynucleotides

The sequences used to assemble SEQ ID NO:36-70 are compared with sequences from the Incyte LIFESEQ database and public domain databases using BLAST and other implementations of the Smith-Waterman algorithm. Sequences from these databases that matched SEQ ID NO:36-70 are assembled into clusters of contiguous and overlapping sequences using assembly algorithms such as Phrap (Table 7). Radiation hybrid and genetic mapping data available from public resources such as the Stanford Human Genome Center (SHGC), Whitehead Institute for Genome Research (WIGR), and Généthon are used to determine if any of the clustered sequences have been previously mapped. Inclusion of a mapped sequence in a cluster results in the assignment of all sequences of that cluster, including its particular SEQ ID NO:, to that map location.

Map locations are represented by ranges, or intervals, of human chromosomes. The map position of an interval, in centiMorgans, is measured relative to the terminus of the chromosome's p-arm. (The centiMorgan (cM) is a unit of measurement based on recombination frequencies between chromosomal markers. On average, 1 cM is roughly equivalent to 1 megabase (Mb) of DNA in humans, although this can vary widely due to hot and cold spots of recombination.) The cM distances are based on genetic markers mapped by Généthon which provide boundaries for radiation hybrid markers whose sequences were included in each of the clusters. Human genome maps and other resources available to the public, such as the NCBI "GeneMap'99" World Wide Web site (http://www.ncbi.nlm.nih.gov/genemap/), can be employed to determine if previously identified disease genes map within or in proximity to the intervals indicated above.

VII. Analysis of Polynucleotide Expression

Northern analysis is a laboratory technique used to detect the presence of a transcript of a gene and involves the hybridization of a labeled nucleotide sequence to a membrane on which RNAs from a particular cell type or tissue have been bound (Sambrook and Russell, *supra*, ch. 7; Ausubel et al., *supra*, ch. 4).

Analogous computer techniques applying BLAST are used to search for identical or related molecules in databases such as GenBank or LIFESEQ (Incyte). This analysis is much faster than multiple membrane-based hybridizations. In addition, the sensitivity of the computer search can be modified to determine whether any particular match is categorized as exact or similar. The basis of the search is the product score, which is defined as:

$$\frac{\text{BLAST Score x Percent Identity}}{5 \text{ x minimum } \{\text{length(Seq. 1), length(Seq. 2)}\}}$$

The product score takes into account both the degree of similarity between two sequences and the length of the sequence match. The product score is a normalized value between 0 and 100, and is calculated as follows: the BLAST score is multiplied by the percent nucleotide identity and the product is divided by (5 times the length of the shorter of the two sequences). The BLAST score is calculated by assigning a score of +5 for every base that matches in a high-scoring segment pair (HSP), and -4 for every mismatch. Two sequences may share more than one HSP (separated by gaps). If there is more than one HSP, then the pair with the highest BLAST score is used to calculate the product score. The product score represents a balance between fractional overlap and quality in a BLAST alignment. For example, a product score of 100 is produced only for 100% identity over the entire length of the shorter of the two sequences being compared. A product score of 70 is produced either by 100% identity and 70% overlap at one end, or by 88% identity and 100% overlap at the other. A product score of 50 is produced either by 100% identity and 50% overlap at one end, or 79% identity and 100% overlap.

Alternatively, polynucleotides encoding NAAP are analyzed with respect to the tissue sources from which they are derived. For example, some full length sequences are assembled, at least in part, with overlapping Incyte cDNA sequences (see Example III). Each cDNA sequence is derived from a cDNA library constructed from a human tissue. Each human tissue is classified into one of the following organ/tissue categories: cardiovascular system; connective tissue; digestive system; embryonic structures; endocrine system; exocrine glands; genitalia, female; genitalia, male; germ cells; hemic and immune system; liver; musculoskeletal system; nervous system; pancreas; respiratory system; sense organs; skin; stomatognathic system; unclassified/mixed; or urinary tract.

The number of libraries in each category is counted and divided by the total number of libraries across all categories. Similarly, each human tissue is classified into one of the following disease/condition categories: cancer, cell line, developmental, inflammation, neurological, trauma, cardiovascular, pooled, and other, and the number of libraries in each category is counted and divided by the total number of libraries across all categories. The resulting percentages reflect the tissue- and disease-specific expression of cDNA encoding NAAP. cDNA sequences and cDNA library/tissue information are found in the LIFESEQ database (Incyte, Palo Alto CA).

VIII. Extension of NAAP Encoding Polynucleotides

Full length polynucleotides are produced by extension of an appropriate fragment of the full length molecule using oligonucleotide primers designed from this fragment. One primer is synthesized to initiate 5' extension of the known fragment, and the other primer is synthesized to initiate 3' extension of the known fragment. The initial primers are designed using OLIGO 4.06 software (National Biosciences), or another appropriate program, to be about 22 to 30 nucleotides in length, to have a GC content of about 50% or more, and to anneal to the target sequence at temperatures of about 68 °C to about 72°C. Any stretch of nucleotides which would result in hairpin structures and primer-primer dimerizations is avoided.

Selected human cDNA libraries are used to extend the sequence. If more than one extension is necessary or desired, additional or nested sets of primers are designed.

High fidelity amplification is obtained by PCR using methods well known in the art. PCR is performed in 96-well plates using the PTC-200 thermal cycler (MJ Research, Inc.). The reaction mix contains DNA template, 200 nmol of each primer, reaction buffer containing Mg^{2+}, $(NH_4)_2SO_4$, and 2-mercaptoethanol, Taq DNA polymerase (Amersham Biosciences), ELONGASE enzyme (Invitrogen), and Pfu DNA polymerase (Stratagene), with the following parameters for primer pair PCI A and PCI B: Step 1: 94°C, 3 min; Step 2: 94°C, 15 sec; Step 3: 60°C, 1 min; Step 4: 68°C, 2 min; Step 5: Steps 2, 3, and 4 repeated 20 times; Step 6: 68°C, 5 min; Step 7: storage at 4°C. In the alternative, the parameters for primer pair T7 and SK+ are as follows: Step 1: 94°C, 3 min; Step 2: 94°C, 15 sec; Step 3: 57°C, 1 min; Step 4: 68°C, 2 min; Step 5: Steps 2, 3, and 4 repeated 20 times; Step 6: 68°C, 5 min; Step 7: storage at 4°C.

The concentration of DNA in each well is determined by dispensing 100 μl PICOGREEN quantitation reagent (0.25% (v/v) PICOGREEN; Molecular Probes, Eugene OR) dissolved in 1X TE and 0.5 μl of undiluted PCR product into each well of an opaque fluorimeter plate (Corning Costar, Acton MA), allowing the DNA to bind to the reagent. The plate is scanned in a Fluoroskan II (Labsystems Oy, Helsinki, Finland) to measure the fluorescence of the sample and to quantify the concentration of DNA. A 5 μl to 10 μl aliquot of the reaction mixture is analyzed by electrophoresis on a 1 % agarose gel to determine which reactions are successful in extending the sequence.

The extended nucleotides are desalted and concentrated, transferred to 384-well plates, digested with CviJI cholera virus endonuclease (Molecular Biology Research, Madison WI), and sonicated or sheared prior to religation into pUC 18 vector (Amersham Biosciences). For shotgun sequencing, the digested nucleotides are separated on low concentration (0.6 to 0.8%) agarose gels, fragments are excised, and agar digested with Agar ACE (Promega). Extended clones were religated using T4 ligase (New England Biolabs, Beverly MA) into pUC 18 vector (Amersham Biosciences), treated with Pfu DNA polymerase (Stratagene) to fill-in restriction site overhangs, and transfected into competent *E. coli* cells. Transformed cells are selected on antibiotic-containing media, and individual colonies are picked and cultured overnight at 37°C in 384-well plates in LB/2x carb liquid media.

The cells are lysed, and DNA is amplified by PCR using Taq DNA polymerase (Amersham Biosciences) and Pfu DNA polymerase (Stratagene) with the following parameters: Step 1: 94°C, 3 min; Step 2: 94°C, 15 sec; Step 3: 60°C, 1 min; Step 4: 72°C, 2 min; Step 5: steps 2, 3, and 4 repeated 29 times; Step 6: 72°C, 5 min; Step 7: storage at 4°C. DNA is quantified by PICOGREEN reagent (Molecular Probes) as described above. Samples with low DNA recoveries are reamplified using the same conditions as described above. Samples are diluted with 20% dimethysulfoxide (1:2, v/v), and sequenced using DYENAMIC energy transfer sequencing primers and the DYENAMIC DIRECT kit (Amersham Biosciences) or the ABI PRISM BIGDYE Terminator cycle sequencing ready reaction kit (Applied Biosystems).

In like manner, full length polynucleotides are verified using the above procedure or are used to obtain 5' regulatory sequences using the above procedure along with oligonucleotides designed for such extension, and an appropriate genomic library.

IX. Identification of Single Nucleotide Polymorphisms in NAAP Encoding Polynucleotides

Common DNA sequence variants known as single nucleotide polymorphisms (SNPs) are identified in SEQ ID NO:36-70 using the LIFESEQ database (Incyte). Sequences from the same gene are clustered together and assembled as described in Example III, allowing the identification of all sequence variants in the gene. An algorithm consisting of a series of filters is used to distinguish SNPs from other sequence variants. Preliminary filters remove the majority of basecall errors by requiring a minimum Phred quality score of 15, and remove sequence alignment errors and errors resulting from improper trimming of vector sequences, chimeras, and splice variants. An automated procedure of advanced chromosome analysis is applied to the original chromatogram files in the vicinity of the putative SNP. Clone error filters use statistically generated algorithms to identify errors introduced during laboratory processing, such as those caused by reverse transcriptase, polymerase, or somatic mutation. Clustering error filters use statistically generated algorithms to identify errors resulting from clustering of close homologs or pseudogenes, or due to contamination by non-human sequences. A final set of filters removes duplicates and SNPs found in immunoglobulins or T-cell receptors.

Certain SNPs are selected for further characterization by mass spectrometry using the high throughput MASSARRAY system (Sequenom, Inc.) to analyze allele frequencies at the SNP sites in four different human populations. The Caucasian population comprises 92 individuals (46 male, 46 female), including 83 from Utah, four French, three Venezualan, and two Amish individuals. The African population comprises 194 individuals (97 male, 97 female), all African Americans. The Hispanic population comprises 324 individuals (162 male, 162 female), all Mexican Hispanic. The Asian population comprises 126 individuals (64 male, 62 female) with a reported parental breakdown of 43% Chinese, 31% Japanese, 13% Korean, 5% Vietnamese, and 8% other Asian. Allele frequencies are first analyzed in the Caucasian population; in some cases those SNPs which show no allelic variance in this population are not further tested in the other three populations.

X. Labeling and Use of Individual Hybridization Probes

Hybridization probes derived from SEQ ID NO:36-70 are employed to screen cDNAs, genomic DNAs, or mRNAs. Although the labeling of oligonucleotides, consisting of about 20 base pairs, is specifically described, essentially the same procedure is used with larger nucleotide fragments. Oligonucleotides are designed using state-of-the-art software such as OLIGO 4.06 software (National Biosciences) and labeled by combining 50 pmol of each oligomer, 250 μCi of [γ-^{32}P] adenosine triphosphate (Amersham Biosciences), and T4 polynucleotide kinase (DuPont NEN, Boston MA). The labeled oligonucleotides are substantially purified using a SEPHADEX G-25 superfine size exclusion dextran bead column (Amersham Biosciences). An aliquot containing 10^7 counts per minute of the labeled probe is used in a typical membrane-based hybridization analysis of human genomic DNA digested with one of the following endonucleases: Ase I, Bgl II, Eco RI, Pst I, Xba I, or Pvu II (DuPont NEN).

The DNA from each digest is fractionated on a 0.7% agarose gel and transferred to nylon membranes (Nytran Plus, Schleicher & Schuell, Durham NH). Hybridization is carried out for 16 hours at 40°C. To remove nonspecific signals, blots are sequentially washed at room temperature under conditions of up to, for example, 0.1 x saline sodium citrate and 0.5% sodium dodecyl sulfate. Hybridization patterns are visualized using autoradiography or an alternative imaging means and compared.

XI. Microarrays

The linkage or synthesis of array elements upon a microarray can be achieved utilizing photolithography, piezoelectric printing (ink-jet printing; see, e.g., Baldeschweiler et al., *supra*), mechanical microspotting technologies, and derivatives thereof. The substrate in each of the aforementioned technologies should be uniform and solid with a non-porous surface (Schena, M., ed.

(1999) DNA Microarrays: A Practical Approach, Oxford University Press, London). Suggested substrates include silicon, silica, glass slides, glass chips, and silicon wafers. Alternatively, a procedure analogous to a dot or slot blot may also be used to arrange and link elements to the surface of a substrate using thermal, UV, chemical, or mechanical bonding procedures. A typical array may be produced using available methods and machines well known to those of ordinary skill in the art and may contain any appropriate number of elements (Schena, M. et al. (1995) Science 270:467-470; Shalon, D. et al. (1996) Genome Res. 6:639-645; Marshall, A. and J. Hodgson (1998) Nat. Biotechnol. 16:27-31).

Full length cDNAs, Expressed Sequence Tags (ESTs), or fragments or oligomers thereof may comprise the elements of the microarray. Fragments or oligomers suitable for hybridization can be selected using software well known in the art such as LASERGENE software (DNASTAR). The array elements are hybridized with polynucleotides in a biological sample. The polynucleotides in the biological sample are conjugated to a fluorescent label or other molecular tag for ease of detection. After hybridization, nonhybridized nucleotides from the biological sample are removed, and a fluorescence scanner is used to detect hybridization at each array element. Alternatively, laser desorbtion and mass spectrometry may be used for detection of hybridization. The degree of complementarity and the relative abundance of each polynucleotide which hybridizes to an element on the microarray may be assessed. In one embodiment, microarray preparation and usage is described in detail below.

Tissue or Cell Sample Preparation

Total RNA is isolated from tissue samples using the guanidinium thiocyanate method and poly(A)$^+$ RNA is purified using the oligo-(dT) cellulose method. Each poly(A)$^+$ RNA sample is reverse transcribed using MMLV reverse-transcriptase, 0.05 pg/μl oligo-(dT) primer (21mer), 1X first strand buffer, 0.03 units/μl RNase inhibitor, 500 μM dATP, 500 μM dGTP, 500 μM dTTP, 40 μM dCTP, 40 μM dCTP-Cy3 (BDS) or dCTP-Cy5 (Amersham Biosciences). The reverse transcription reaction is performed in a 25 ml volume containing 200 ng poly(A)$^+$ RNA with GEMBRIGHT kits (Incyte). Specific control poly(A)$^+$ RNAs are synthesized by *in vitro* transcription from non-coding yeast genomic DNA. After incubation at 37°C for 2 hr, each reaction sample (one with Cy3 and another with Cy5 labeling) is treated with 2.5 ml of 0.5M sodium hydroxide and incubated for 20 minutes at 85°C to the stop the reaction and degrade the RNA. Samples are purified using two successive CHROMA SPIN 30 gel filtration spin columns (BD Clontech, Palo Alto CA) and after combining, both reaction samples are ethanol precipitated using 1 ml of glycogen (1 mg/ml), 60 ml sodium acetate, and 300 ml of 100% ethanol. The sample is then dried to completion using a SpeedVAC (Savant Instruments Inc., Holbrook NY) and resuspended in 14 μl 5X SSC/0.2% SDS.

Microarray Preparation

Sequences of the present invention are used to generate array elements. Each array element is amplified from bacterial cells containing vectors with cloned cDNA inserts. PCR amplification uses primers complementary to the vector sequences flanking the cDNA insert. Array elements are amplified in thirty cycles of PCR from an initial quantity of 1-2 ng to a final quantity greater than 5 μg. Amplified array elements are then purified using SEPHACRYL-400 (Amersham Biosciences).

Purified array elements are immobilized on polymer-coated glass slides. Glass microscope slides (Corning) are cleaned by ultrasound in 0.1% SDS and acetone, with extensive distilled water washes between and after treatments. Glass slides are etched in 4% hydrofluoric acid (VWR Scientific Products Corporation (VWR), West Chester PA), washed extensively in distilled water, and coated with 0.05% aminopropyl silane (Sigma-Aldrich, St. Louis MO) in 95% ethanol. Coated slides are cured in a 110°C oven.

Array elements are applied to the coated glass substrate using a procedure described in U.S. Patent No. 5,807,522, incorporated herein by reference. 1 μl of the array element DNA, at an average concentration of 100 ng/μl, is loaded into the open capillary printing element by a high-speed robotic apparatus. The apparatus then deposits about 5 nl of array element sample per slide.

Microarrays are UV-crosslinked using a STRATALINKER UV-crosslinker (Stratagene). Microarrays are washed at room temperature once in 0.2% SDS and three times in distilled water. Non-specific binding sites are blocked by incubation of microarrays in 0.2% casein in phosphate buffered saline (PBS) (Tropix, Inc., Bedford MA) for 30 minutes at 60° C followed by washes in 0.2% SDS and distilled water as before.

Hybridization

Hybridization reactions contain 9 μl of sample mixture consisting of 0.2 μg each of Cy3 and Cy5 labeled cDNA synthesis products in 5X SSC, 0.2% SDS hybridization buffer. The sample mixture is heated to 65° C for 5 minutes and is aliquoted onto the microarray surface and covered with an 1.8 cm^2 coverslip. The arrays are transferred to a waterproof chamber having a cavity just slightly larger than a microscope slide. The chamber is kept at 100% humidity internally by the addition of 140 μl of 5X SSC in a corner of the chamber. The chamber containing the arrays is incubated for about 6.5 hours at 60° C. The arrays are washed for 10 min at 45° C in a first wash buffer (1X SSC, 0.1% SDS), three times for 10 minutes each at 45° C in a second wash buffer (0.1X SSC), and dried.

Detection

Reporter-labeled hybridization complexes are detected with a microscope equipped with an Innova 70 mixed gas 10 W laser (Coherent, Inc., Santa Clara CA) capable of generating spectral lines at 488 nm for excitation of Cy3 and at 632 nm for excitation of Cy5. The excitation laser light is focused on the array using a 20X microscope objective (Nikon, Inc., Melville NY). The slide containing the array is placed on a computer-controlled X-Y stage on the microscope and raster-scanned past the objective. The 1.8 cm x 1.8 cm array used in the present example is scanned with a resolution of 20 micrometers.

In two separate scans, a mixed gas multiline laser excites the two fluorophores sequentially. Emitted light is split, based on wavelength, into two photomultiplier tube detectors (PMT R1477, Hamamatsu Photonics Systems, Bridgewater NJ) corresponding to the two fluorophores. Appropriate filters positioned between the array and the photomultiplier tubes are used to filter the signals. The emission maxima of the fluorophores used are 565 nm for Cy3 and 650 nm for Cy5. Each array is typically scanned twice, one scan per fluorophore using the appropriate filters at the laser source, although the apparatus is capable of recording the spectra from both fluorophores simultaneously.

The sensitivity of the scans is typically calibrated using the signal intensity generated by a cDNA control species added to the sample mixture at a known concentration. A specific location on the array contains a complementary DNA sequence, allowing the intensity of the signal at that location to be correlated with a weight ratio of hybridizing species of 1:100,000. When two samples from different sources (e.g., representing test and control cells), each labeled with a different fluorophore, are hybridized to a single array for the purpose of identifying genes that are differentially expressed, the calibration is done by labeling samples of the calibrating cDNA with the two fluorophores and adding identical amounts of each to the hybridization mixture.

The output of the photomultiplier tube is digitized using a 12-bit RTI-835H analog-to-digital (A/D) conversion board (Analog Devices, Inc., Norwood MA) installed in an IBM-compatible PC computer. The digitized data are displayed as an image where the signal intensity is mapped using a linear 20-color transformation to a pseudocolor scale ranging from blue (low signal) to red (high signal). The data is also analyzed quantitatively. Where two different fluorophores are excited and measured simultaneously, the data are first corrected for optical crosstalk (due to overlapping emission spectra) between the fluorophores using each fluorophore's emission spectrum.

A grid is superimposed over the fluorescence signal image such that the signal from each spot is centered in each element of the grid. The fluorescence signal within each element is then integrated to obtain a numerical value corresponding to the average intensity of the signal. The software used for signal analysis is the GEMTOOLS gene expression analysis program (Incyte). Array elements that exhibit at least about a two-fold change in expression, a signal-to-background ratio of at least about 2.5, and an element spot size of at least about 40%, are considered to be differentially expressed.

Expression

SEQ ID NO:36, SEQ ID NO:52, and SEQ ID NO:55 showed differential expression in association with colon cancer, as determined by microarray analysis. Gene expression profiles were obtained by comparing the results of competitive hybridization experiments between normal colon tissue and tumorous colon tissue from the same donor (Huntsman Cancer Institute, Salt Lake City, UT). In a matched tissue experiment, the expression of SEQ ID NO:36 was increased by at least two-fold in tumorous colon tissue as compared to grossly uninvolved colon tissue originating from the same donor. In an additional separate matched tissue experiment, the expression of SEQ ID NO:52 was decreased by at least two-fold in tumorous colon tissue as compared to grossly uninvolved colon tissue originating from the same donor. In a third separate matched tissue experiment, the expression of SEQ ID NO:55 was increased by at least two-fold in the tumorous colon tissue as compared to grossly uninvolved colon tissue originating the from the same donor. Thus, in various embodiments, SEQ ID NO:36, SEQ ID NO:52, and SEQ ID NO:55 can each be used for one or more of the following: i) monitoring treatment of colon cancer, ii) diagnostic assays for colon cancer, and iii) developing therapeutics and/or other treatments for colon cancer.

In another example, SEQ ID NO:38, SEQ ID NO:43, and SEQ ID NO:52 showed differential expression in association with lung cancer, as determined by microarray analysis. Gene expression profiles were obtained by comparing the results of competitive hybridization experiments. Messenger RNA isolated from grossly uninvolved lung tissue with no visible abnormalities was compared to lung squamous cell adenocarcinoma tissue from matched donors (Roy Castle International Centre for Lung Cancer Research, Liverpool, UK). In a matched tissue experiment, the expression of SEQ ID NO:38 was increased by at least two-fold in lung squamous cell adenocarcinoma tissue as compared to normal lung tissue from the same donor. In a separate matched tissue experiment, the expression of SEQ ID NO:43 was increased by at least two-fold in tumorous lung tissue as compared to normal lung tissue from the same donor. In three additional, separate matched tissue experiments, the expression of SEQ ID NO:52 was increased by at least two-fold in lung squamous cell adenocarcinoma tissue as compared to normal lung tissue from matched donors. Thus, in various embodiments, SEQ ID NO:38, SEQ ID NO:43, and SEQ ID NO:52 can each be used for one or more of the following: i) monitoring treatment of lung cancer, ii) diagnostic assays for lung cancer, and iii) developing therapeutics and/or other treatments for lung cancer.

In another example, SEQ ID NO:49 showed differential expression in association with ovarian cancer, as determined by microarray analysis. A normal ovary from a 79-year-old female donor was compared to an ovarian tumor from the same donor (Huntsman Cancer Institute, Salt Lake City, UT). The expression of SEQ ID NO:49 was increased by two-fold in tumorous ovarian tissue as compared to normal ovarian tissue from the same donor. Therefore, in various embodiments, SEQ ID NO:49 can be used for one or more of the following: i) monitoring treatment of ovarian cancer, ii) diagnostic assays for ovarian cancer, and iii) developing therapeutics and/or other treatments for ovarian cancer.

In another example, SEQ ID NO:49 and SEQ ID NO:52 showed differential expression in association with breast cancer, as determined by microarray analysis. The gene expression profiles of a nonmalignant mammary epithelial (HMEC) cell line, and a nonmalignant mammary gland cell line isolated from a 36-year-old woman with fibrocystic breast disease (MCF-10A), were compared to the gene expression profiles of breast carcinoma lines at different stages of tumor progression. Cell lines compared included: a) MCF7, a nonmalignant breast adenocarcinoma cell line isolated from the pleural effusion of a 69-year-old female; b)T-47D, a breast carcinoma cell line isolated from a pleural effusion obtained from a 54-year-old female with an infiltrating ductal carcinoma of the breast; c)Sk-BR-3, a breast adenocarcinoma cell line isolated from a malignant pleural effusion of a 43-year-old female; d)BT-20, a breast carcinoma cell line derived *in vitro* from tumor mass isolated from a 74-year-old female; e)MDA-mb-231, a breast tumor cell line isolated from the pleural effusion of a 51-year old female; and f)MDA-mb-435S, a spindle shaped strain that evolved from the parent line (435) isolated from the pleural effusion of a 31-year-old female with metastatic, ductal adenocarcinoma of the breast. The expression of SEQ ID NO:49 was decreased by at least two-fold in T-47D, BT-20, MCF7, SKBr3, and MDA-mb-435S cells as compared to HMEC cells. The expression of SEQ ID NO:52 was decreased by at least two-fold in T-47D, MCF7, and SKBr3 cells as compared to HMEC and MCF-10A cells. Therefore, in various embodiments, SEQ ID NO:49 and SEQ ID NO:52 can each be used for one or more of the following: i) monitoring treatment of breast cancer, ii) diagnostic assays for breast cancer, and iii) developing therapeutics and/or other treatments for breast cancer.

In yet another example, SEQ ID NO:49 and SEQ ID NO:52 showed differential expression in association with prostate cancer, as determined by microarray analysis. Primary prostate epithelial cells were compared with prostate carcinomas representative of the different stages of tumor progression. Cell lines compared included: a) PrEC, a primary prostate epithelial cell line isolated from a normal donor, b) DU 145, a prostate carcinoma cell line isolated from a metastatic site in the brain of 69-year old male with widespread metastatic prostate carcinoma, c) LNCaP, a prostate carcinoma cell line isolated from a lymph node biopsy of a 50-year-old male with metastatic prostate carcinoma, and d) PC-3, a prostate adenocarcinoma cell line isolated from a metastatic site in the bone of a 62-year-old male with grade IV prostate adenocarcinoma. Cells grown under restrictive conditions were compared to normal PrECs grown under restrictive conditions. Cells were grown in basal media in the absence of growth factors and hormones, and separately, cells were grown under optimal growth conditions, in the presence of growth factors and nutrients. Under both conditions, the expression of SEQ ID NO:49 and SEQ ID NO:52 was increased by at least two-fold in cells from the DU 145 line as compared to cells from the PrEC line. Therefore, in various embodiments, SEQ ID

NO:49 and SEQ ID NO:52 can each be used for one or more of the following: i) monitoring treatment of prostate cancer, ii) diagnostic assays for prostate cancer, and iii) developing therapeutics and/or other treatments for prostate cancer.

In still another example, SEQ ID NO:36 showed differential expression in association with vascular inflammation and immune responses, as determined by microarray analysis. Human aortic endothelial cells (HMVECdNeos) were grown to 85% confluency and then treated with 10 ng/ml TNF-α for 1, 2, 4, 8, and 24 hours. TNF-α -treated cells were compared to untreated HMVECdNeos collected at 85% confluency (0 hour). The expression of SEQ ID NO:36 was decreased by at least two-fold in HMVECdNeos treated with TNF-α as compared to untreated HMVECdNeos grown to the same confluency. Thus, in various embodiments, SEQ ID NO:36 can be used for one or more of the following: i) monitoring treatment of vascular inflammation and immune responses, ii) diagnostic assays for vascular inflammation and immune responses, and iii) developing therapeutics and/or other treatments for vascular inflammation and immune responses.

In addition, SEQ ID NO:40 and SEQ ID NO:42 showed tissue-specific expression, as determined by microarray analysis. RNA samples isolated from a variety of normal human tissues were compared to a common reference sample. Tissues contributing to the reference sample were selected for their ability to provide a complete distribution of RNA in the human body and include brain (4%), heart (7%), kidney (3%), lung (8%), placenta (46%), small intestine (9%), spleen (3%), stomach (6%), testis (9%), and uterus (5%). The normal tissues assayed were obtained from at least three different donors. RNA from each donor was separately isolated and individually hybridized to the microarray. Since these hybridization experiments were conducted using a common reference sample, differential expression values are directly comparable from one tissue to another. The expression of SEQ ID NO:40 and SEQ ID NO:42 was increased by at least two-fold in blood leukocytes as compared to the reference sample. Therefore, in an embodiment, SEQ ID NO:40 and SEQ ID NO:42 can each be used as tissue markers for blood leukocytes.

SEQ ID NO:58 showed differential expression in association with disorders of metabolism and PPAR-γ signaling, as determined by microarray analysis. SEQ ID NO:58 also shares a high percent identity with NR1H3, a human DNA-binding transcriptional activator involved in PPAR-α signaling, cholesterol and lipid metabolism, and TNF synthesis. In addition, mutations in mouse Nr1h3 result in cholesterol accumulation. In this experiment, primary subcutaneous preadipocytes were isolated from the adipose tissue of a 40-year-old healthy female with a body mass index (BMI) of 32.47 (Donor 3801, obese), a 28-year-old healthy female with a BMI of 23.59 (Donor 3802, healthy), and a 36-year-old female with a BMI of 27.7 (Donor 3624, overweight, but otherwise healthy). The preadipocytes were separately cultured and induced to differentiate into adipocytes by exposing them to differentiation medium containing the active components PPAR-γ agonist and human insulin (Zen-Bio). Thiazolidinediones, or PPAR-γ agonists, can bind and activate the orphan nuclear receptor, PPAR-γ, and some of them have been proven to be able to induce human adipocyte differentiation. The preadipocytes were treated with human insulin and PPAR-γ agonist for 3 days and subsequently switched to medium containing insulin alone for a variety of time periods ranging from one to 20 days before the cells were collected for analysis. Differentiated adipocytes were compared to untreated preadipocytes maintained in culture in the absence of inducing agents. The expression of SEQ ID NO:58 was increased by a factor of at least two-fold in the cells isolated from the obese donor 48 hours after treatment of the preadipocytes with differentiation inducing medium. In contrast, the expression of SEQ ID NO:58 showed no differential expression at the 48 hour time point in the cells isolated from the healthy donor, and it was not until 1.1 weeks after treatment of the preadipocytes with differentiation inducing medium that the expression of SEQ ID NO:58 increased by a factor of at least two-fold. Therefore, in various embodiments, SEQ ID NO:58 can be used for one or more of the following: i) monitoring treatment of obesity, and/or lipid metabolism and associated disorders, ii) diagnostic assays for obesity, and/or lipid metabolism and associated disorders, and iii) developing therapeutics and/or other treatments for obesity, and/or lipid metabolism and associated disorders.

For example, SEQ ID NO:68-69 showed differential expression in association with lung cancer, as determined by microarray analysis. Normal lung tissue was compared to lung tumor tissue from the same donor (Roy Castle International Centre for Lung Cancer Research, Liverpool, UK). In two out of five patients sampled, SEQ ID NO:68 showed a two-fold decrease in expression in lung tissue from patients with lung squamous cell carcinoma as compared to matched microscopically normal tissue from the same donors. In one out of three patients sampled, SEQ ID NO:68 showed a two-fold decrease in expression in lung tissue from patients with lung cell adenocarcinoma as compared to matched microscopically normal tissue from the same donor. In one out of one patient sampled, SEQ ID NO:69 showed a two-fold decrease in expression in lung tissue from patients with lung cell adenocarcinoma as compared to matched microscopically normal tissue from the same donor. Therefore, in various embodiments, SEQ ID NO:68-69 can be used for one or more of the following: i) monitoring treatment of lung cancer, ii) diagnostic assays for lung cancer, and iii) developing therapeutics and/or other treatments for lung cancer.

In addition, SEQ ID NO:68 showed tissue-specific expression as determined by microarray analysis. RNA samples isolated from a variety of normal human tissues were compared to a common reference sample. Tissues contributing to the reference sample were selected for their ability to provide a complete distribution of RNA in the human body and include brain (4%), heart (7%), kidney (3%), lung (8%), placenta (46%), small intestine (9%), spleen (3%), stomach (6%), testis (9%), and uterus (5%). The normal tissues assayed were obtained from at least three different donors. RNA from each donor was separately isolated and individually hybridized to the microarray. Since these hybridization experiments were conducted using a common reference sample, differential expression values are directly comparable from one tissue to another. The expression of SEQ ID NO:68 was increased by at least two-fold in temporal cortex and hippocampus tissue as compared to the reference sample. Therefore, SEQ ID NO:68 can be used as a tissue marker for temporal cortex and hippocampus.

XII. Complementary Polynucleotides

Sequences complementary to the NAAP-encoding sequences, or any parts thereof, are used to detect, decrease, or inhibit expression of naturally occurring NAAP. Although use of oligonucleotides comprising from about 15 to 30 base pairs is described, essentially the same procedure is used with smaller or with larger sequence fragments. Appropriate oligonucleotides are designed using OLIGO 4.06 software (National Biosciences) and the coding sequence of NAAP. To inhibit transcription, a complementary oligonucleotide is designed from the most unique 5' sequence and used to prevent promoter binding to the coding sequence. To inhibit translation, a complementary oligonucleotide is designed to prevent ribosomal binding to the NAAP-encoding transcript.

XIII. Expression of NAAP

Expression and purification of NAAP is achieved using bacterial or virus-based expression systems. For expression of NAAP in bacteria, cDNA is subcloned into an appropriate vector containing an antibiotic resistance gene and an inducible promoter that directs high levels of cDNA transcription. Examples of such promoters include, but are not limited to, the *trp-lac* (*tac*) hybrid promoter and the T5 or T7 bacteriophage promoter in conjunction with the *lac* operator regulatory element. Recombinant vectors are transformed into suitable bacterial hosts, e.g., BL21(DE3). Antibiotic resistant bacteria express NAAP upon induction with isopropyl beta-D-thiogalactopyranoside (IPTG). Expression of NAAP in eukaryotic cells is achieved by infecting insect or mammalian cell lines with recombinant *Autographica californica* nuclear polyhedrosis virus (AcMNPV), commonly known as baculovirus. The nonessential polyhedrin gene of baculovirus is replaced with cDNA encoding NAAP by either homologous recombination or bacterial-mediated transposition involving transfer plasmid intermediates. Viral infectivity is maintained and the strong polyhedrin promoter drives high levels of cDNA transcription. Recombinant baculovirus is used to infect *Spodoptera frugiperda* (Sf9) insect cells in most cases, or human hepatocytes, in some cases. Infection of the latter requires additional genetic modifications to baculovirus (Engelhard, E.K. et al. (1994) Proc. Natl. Acad. Sci. USA 91:3224-3227; Sandig, V. et al. (1996) Hum. Gene Ther. 7:1937-1945).

In most expression systems, NAAP is synthesized as a fusion protein with, e.g., glutathione S-transferase (GST) or a peptide epitope tag, such as FLAG or 6-His, permitting rapid, single-step, affinity-based purification of recombinant fusion protein from crude cell lysates. GST, a 26-kilodalton enzyme from *Schistosoma japonicum*, enables the purification of fusion proteins on immobilized glutathione under conditions that maintain protein activity and antigenicity (Amersham Biosciences). Following purification, the GST moiety can be proteolytically cleaved from NAAP at specifically engineered sites. FLAG, an 8-amino acid peptide, enables immunoaffinity purification using commercially available monoclonal and polyclonal anti-FLAG antibodies (Eastman Kodak). 6-His, a stretch of six consecutive histidine residues, enables purification on metal-chelate resins (QIAGEN). Methods for protein expression and purification are discussed in Ausubel et al. (*supra*, ch. 10 and 16). Purified NAAP obtained by these methods can be used directly in the assays shown in Examples XVII, XVIII, and XIX, where applicable.

XIV. Functional Assays

NAAP function is assessed by expressing the sequences encoding NAAP at physiologically elevated levels in mammalian cell culture systems. cDNA is subcloned into a mammalian expression vector containing a strong promoter that drives high levels of cDNA expression. Vectors of choice include PCMV SPORT plasmid (Invitrogen, Carlsbad CA) and PCR3.1 plasmid (Invitrogen), both of which contain the cytomegalovirus promoter. 5-10 μg of recombinant vector are transiently transfected into a human cell line, for example, an endothelial or hematopoietic cell line, using either liposome formulations or electroporation. 1-2 μg of an additional plasmid containing sequences encoding a marker protein are co-transfected. Expression of a marker protein provides a means to distinguish transfected cells from nontransfected cells and is a reliable predictor of cDNA expression from the recombinant vector. Marker proteins of choice include, e.g., Green Fluorescent Protein (GFP; BD Clontech), CD64, or a CD64-GFP fusion protein. Flow cytometry (FCM), an automated, laser optics-based technique, is used to identify transfected cells expressing GFP or CD64-GFP and to evaluate the apoptotic state of the cells and other cellular properties. FCM detects and quantifies the uptake of fluorescent molecules that diagnose events preceding or coincident with cell death. These events include changes in nuclear DNA content as measured by staining of DNA with propidium iodide; changes in cell size and granularity as measured by forward light scatter and 90 degree side light scatter; down-regulation of DNA synthesis as measured by decrease in bromodeoxyuridine uptake; alterations in expression of cell surface and intracellular proteins as measured by reactivity with specific antibodies; and alterations in plasma membrane composition as measured by the binding of fluorescein-conjugated Annexin V protein to the cell surface. Methods in flow cytometry are discussed in Ormerod, M.G. (1994; Flow Cytometry, Oxford, New York NY).

The influence of NAAP on gene expression can be assessed using highly purified populations of cells transfected with sequences encoding NAAP and either CD64 or CD64-GFP. CD64 and CD64-GFP are expressed on the surface of transfected cells and bind to conserved regions of human immunoglobulin G (IgG). Transfected cells are efficiently separated from nontransfected cells using magnetic beads coated with either human IgG or antibody against CD64 (DYNAL, Lake Success NY). mRNA can be purified from the cells using methods well known by those of skill in the art. Expression of mRNA encoding NAAP and other genes of interest can be analyzed by northern analysis or microarray techniques.

XV. Production of NAAP Specific Antibodies

NAAP substantially purified using polyacrylamide gel electrophoresis (PAGE; see, e.g., Harrington, M.G. (1990) Methods Enzymol. 182:488-495), or other purification techniques, is used to immunize animals (e.g., rabbits, mice, etc.) and to produce antibodies using standard protocols.

Alternatively, the NAAP amino acid sequence is analyzed using LASERGENE software (DNASTAR) to determine regions of high immunogenicity, and a corresponding oligopeptide is synthesized and used to raise antibodies by means known to those of skill in the art. Methods for selection of appropriate epitopes, such as those near the C-terminus or in hydrophilic regions are well described in the art (Ausubel et al., *supra*, ch. 11).

Typically, oligopeptides of about 15 residues in length are synthesized using an ABI 431A peptide synthesizer (Applied Biosystems) using FMOC chemistry and coupled to KLH (Sigma-Aldrich, St. Louis MO) by reaction with N-maleimidobenzoyl-N-hydroxysuccinimide ester (MBS) to increase immunogenicity (Ausubel et al., *supra*). Rabbits are immunized with the oligopeptide-KLH complex in complete Freund's adjuvant. Resulting antisera are tested for antipeptide and anti-NAAP activity by, for example, binding the peptide or NAAP to a substrate, blocking with 1% BSA, reacting with rabbit antisera, washing, and reacting with radio-iodinated goat anti-rabbit IgG.

XVI. Purification of Naturally Occurring NAAP Using Specific Antibodies

Naturally occurring or recombinant NAAP is substantially purified by immunoaffinity chromatography using antibodies specific for NAAP. An immunoaffinity column is constructed by covalently coupling anti-NAAP antibody to an activated chromatographic resin, such as CNBr-activated SEPHAROSE (Amersham Biosciences). After the coupling, the resin is blocked and washed according to the manufacturer's instructions.

Media containing NAAP are passed over the immunoaffinity column, and the column is washed under conditions that allow the preferential absorbance of NAAP (e.g., high ionic strength buffers in the presence of detergent). The column is eluted under conditions that disrupt antibody/NAAP binding (e.g., a buffer of pH 2 to pH 3, or a high concentration of a chaotrope, such as urea or thiocyanate ion), and NAAP is collected.

XVII. Identification of Molecules Which Interact with NAAP

NAAP, or biologically active fragments thereof, are labeled with ^{125}I Bolton-Hunter reagent (Bolton, A.E. and W.M. Hunter (1973) Biochem. J. 133:529-539). Candidate molecules previously arrayed in the wells of a multi-well plate are incubated with the labeled NAAP, washed, and any wells with labeled NAAP complex are assayed. Data obtained using different concentrations of NAAP are used to calculate values for the number, affinity, and association of NAAP with the candidate molecules.

Alternatively, molecules interacting with NAAP are analyzed using the yeast two-hybrid system as described in Fields, S. and O. Song (1989; Nature 340:245-246), or using commercially available kits based on the two-hybrid system, such as the MATCHMAKER system (BD Clontech).

NAAP may also be used in the PATHCALLING process (CuraGen Corp., New Haven CT) which employs the yeast two-hybrid system in a high-throughput manner to determine all interactions between the proteins encoded by two large libraries of genes (Nandabalan, K. et al. (2000) U.S. Patent No. 6,057,101).

XVIII. Demonstration of NAAP Activity

NAAP activity is measured by its ability to stimulate transcription of a reporter gene (Liu, H.Y. et al. (1997) EMBO J. 16:5289-5298). The assay entails the use of a well characterized reporter gene construct, $LexA_{op}$-LacZ, that consists of LexA DNA transcriptional control elements ($LexA_{op}$) fused to sequences encoding the E. coli LacZ enzyme. The methods for constructing and expressing fusion genes, introducing them into cells, and measuring LacZ enzyme activity, are well known to those skilled in the art. Sequences encoding NAAP are cloned into a plasmid that directs the synthesis of a fusion protein, LexA-NAAP, consisting of NAAP and a DNA binding domain derived from the LexA transcription factor. The resulting plasmid, encoding a LexA-NAAP fusion protein, is introduced into yeast cells along with a plasmid containing the $LexA_{op}$-LacZ reporter gene. The amount of LacZ enzyme activity associated with LexA-NAAP transfected cells, relative to control cells, is proportional to the amount of transcription stimulated by the NAAP.

Alternatively, NAAP activity is measured by its ability to bind zinc. A 5-10 μM sample solution in 2.5 mM ammonium acetate solution at pH 7.4 is combined with 0.05 M zinc sulfate solution (Aldrich, Milwaukee WI) in the presence of 100 μM dithiothreitol with 10% methanol added. The sample and zinc sulfate solutions are allowed to incubate for 20 minutes. The reaction solution is passed through a VYDAC column (Grace Vydac, Hesperia, CA) with approximately 300 Angstrom bore size and 5 μM particle size to isolate zinc-sample complex from the solution, and into a mass spectrometer (PE Sciex, Ontario, Canada). Zinc bound to sample is quantified using the functional atomic mass of 63.5 Da observed by Whittal, R. M. et al. ((2000) Biochemistry 39:8406-8417).

In the alternative, a method to determine nucleic acid binding activity of NAAP involves a polyacrylamide gel mobility-shift assay. In preparation for this assay, NAAP is expressed by transforming a mammalian cell line such as COS7, HeLa or CHO with a eukaryotic expression vector containing NAAP cDNA. The cells are incubated for 48-72 hours after transformation under conditions appropriate for the cell line to allow expression and accumulation of NAAP. Extracts containing solubilized proteins can be prepared from cells expressing NAAP by methods well known in the art. Portions of the extract containing NAAP are added to [^{32}P]-labeled RNA or DNA. Radioactive nucleic acid can be synthesized *in vitro* by techniques well known in the art. The mixtures are incubated at 25°C in the presence of RNase- and DNase-inhibitors under buffered conditions for 5-10 minutes. After incubation, the samples are analyzed by polyacrylamide gel electrophoresis followed by autoradiography. The presence of a band on the autoradiogram indicates the formation of a complex between NAAP and the radioactive transcript. A band of similar mobility will not be present in samples prepared using control extracts prepared from untransformed cells.

In the alternative, a method to determine methylase activity of NAAP measures transfer of radiolabeled methyl groups between a donor substrate and an acceptor substrate. Reaction mixtures (50 μl final volume) contain 15 mM HEPES, pH 7.9, 1.5 mM $MgCl_2$, 10 mM dithiothreitol, 3% polyvinylalcohol, 1.5 μCi [*methyl*-^{3}H]AdoMet (0.375 μM AdoMet) (DuPont-NEN), 0.6 μg NAAP, and acceptor substrate (e.g., 0.4 μg [^{35}S]RNA, or 6-mercaptopurine (6-MP) to 1 mM final concentration). Reaction mixtures are incubated at 30°C for 30 minutes, then 65°C for 5 minutes.

Analysis of [*methyl*-^{3}H]RNA is as follows: (1) 50 μl of 2 x loading buffer (20 mM Tris-HCl, pH 7.6, 1 M LiCl, 1 mM EDTA, 1% sodium dodecyl sulphate (SDS)) and 50 μl oligo d(T)-cellulose (10 mg/ml in 1 x loading buffer) are added to the reaction mixture, and incubated at ambient temperature with shaking for 30 minutes. (2) Reaction mixtures are transferred to a 96-well filtration plate attached to a vacuum apparatus. (3) Each sample is washed sequentially with three 2.4 ml aliquots of 1 x oligo d(T) loading buffer containing 0.5% SDS, 0.1% SDS, or no SDS. (4) RNA is eluted with 300 μl of water into a 96-well collection plate, transferred to scintillation vials containing liquid scintillant, and radioactivity determined.

Analysis of [*methyl*-^{3}H]6-MP is as follows: (1) 500 μl 0.5 M borate buffer, pH 10.0, and then 2.5 ml of 20% (v/v) isoamyl alcohol in toluene are added to the reaction mixtures. (2) The samples are mixed by vigorous vortexing for ten seconds. (3) After centrifugation at 700g for 10 minutes, 1.5 ml of the organic phase is transferred to scintillation vials containing 0.5 ml absolute ethanol and liquid scintillant, and radioactivity determined. (4) Results are corrected for the extraction of 6-MP into the organic phase (approximately 41%).

In the alternative, type I topoisomerase activity of NAAP can be assayed based on the relaxation of a supercoiled DNA substrate. NAAP is incubated with its substrate in a buffer lacking

Mg^{2+} and ATP, the reaction is terminated, and the products are loaded on an agarose gel. Altered topoisomers can be distinguished from supercoiled substrate electrophoretically. This assay is specific for type I topoisomerase activity because Mg^{2+} and ATP are necessary cofactors for type II topoisomerases.

Type II topoisomerase activity of NAAP can be assayed based on the decatenation of a kinetoplast DNA (KDNA) substrate. NAAP is incubated with KDNA, the reaction is terminated, and the products are loaded on an agarose gel. Monomeric circular KDNA can be distinguished from catenated KDNA electrophoretically. Kits for measuring type I and type II topoisomerase activities are available commercially from Topogen (Columbus OH).

ATP-dependent RNA helicase unwinding activity of NAAP can be measured by the method described by Zhang and Grosse (1994; Biochemistry 33:3906-3912). The substrate for RNA unwinding consists of ^{32}P-labeled RNA composed of two RNA strands of 194 and 130 nucleotides in length containing a duplex region of 17 base-pairs. The RNA substrate is incubated together with ATP, Mg^{2+}, and varying amounts of NAAP in a Tris-HCl buffer, pH 7.5, at 37°C for 30 minutes. The single-stranded RNA product is then separated from the double-stranded RNA substrate by electrophoresis through a 10% SDS-polyacrylamide gel, and quantitated by autoradiography. The amount of single-stranded RNA recovered is proportional to the amount of NAAP in the preparation.

In the alternative, NAAP function is assessed by expressing the sequences encoding NAAP at physiologically elevated levels in mammalian cell culture systems. cDNA is subcloned into a mammalian expression vector containing a strong promoter that drives high levels of cDNA expression. Vectors of choice include pCMV SPORT (Life Technologies) and pCR3.1 (Invitrogen Corporation, Carlsbad CA), both of which contain the cytomegalovirus promoter. 5-10 μg of recombinant vector are transiently transfected into a human cell line, preferably of endothelial or hematopoietic origin, using either liposome formulations or electroporation. 1-2 μg of an additional plasmid containing sequences encoding a marker protein are co-transfected.

Expression of a marker protein provides a means to distinguish transfected cells from nontransfected cells and is a reliable predictor of cDNA expression from the recombinant vector. Marker proteins of choice include, e.g., Green Fluorescent Protein (GFP; CLONTECH), CD64, or a CD64-GFP fusion protein. Flow cytometry (FCM), an automated laser optics-based technique, is used to identify transfected cells expressing GFP or CD64-GFP and to evaluate the apoptotic state of the cells and other cellular properties.

FCM detects and quantifies the uptake of fluorescent molecules that diagnose events preceding or coincident with cell death. These events include changes in nuclear DNA content as measured by staining of DNA with propidium iodide; changes in cell size and granularity as measured by forward light scatter and 90 degree side light scatter; down-regulation of DNA synthesis as measured by decrease in bromodeoxyuridine uptake; alterations in expression of cell surface and intracellular proteins as measured by reactivity with specific antibodies; and alterations in plasma membrane composition as measured by the binding of fluorescein-conjugated Annexin V protein to the cell surface. Methods in flow cytometry are discussed in Ormerod, M. G. (1994) Flow Cytometry, Oxford, New York NY.

The influence of NAAP on gene expression can be assessed using highly purified populations of cells transfected with sequences encoding NAAP and either CD64 or CD64-GFP. CD64 and CD64-GFP are expressed on the surface of transfected cells and bind to conserved regions of human immunoglobulin G (IgG). Transfected cells are efficiently separated from nontransfected cells using magnetic beads coated with either human IgG or antibody against CD64 (DYNAL, Inc., Lake Success NY). mRNA can be purified from the cells using methods well known by those of skill in the art. Expression of mRNA encoding NAAP and other genes of interest can be analyzed by northern analysis or microarray techniques.

Pseudouridine synthase activity of NAAP is assayed using a tritium (^{3}H) release assay modified from Nurse et al. ((1995) RNA 1:102-112), which measures the release of ^{3}H from the C_5 position of the pyrimidine component of uridylate (U) when ^{3}H-radiolabeled U in RNA is isomerized to pseudouridine (ψ). A typical 500 μl assay mixture contains 50 mM HEPES buffer (pH 7.5), 100 mM ammonium acetate, 5 mM dithiothreitol, 1 mM EDTA, 30 units RNase inhibitor, and 0.1-4.2 μM [5-^{3}H]tRNA (approximately 1 μCi/nmol tRNA). The reaction is initiated by the addition of <5 μl of a concentrated solution of NAAP (or sample containing NAAP) and incubated for 5 min at 37 °C. Portions of the reaction mixture are removed at various times (up to 30 min) following the addition of NAAP and quenched by dilution into 1 ml 0.1 M HCl containing Norit-SA3 (12% w/v). The quenched reaction mixtures are centrifuged for 5 min at maximum speed in a microcentrifuge, and the supernatants are filtered through a plug of glass wool. The pellet is washed twice by resuspension in 1 ml 0.1 M HCl, followed by centrifugation. The supernatants from the washes are separately passed through the glass wool plug and combined with the original filtrate. A portion of the combined filtrate is mixed with scintillation fluid (up to 10 ml) and counted using a scintillation counter. The amount of ^{3}H released from the RNA and present in the soluble filtrate is proportional to the amount of peudouridine synthase activity in the sample (Ramamurthy, V. (1999) J. Biol. Chem. 274:22225-22230).

In the alternative, pseudouridine synthase activity of NAAP is assayed at 30 °C to 37 °C in a mixture containing 100 mM Tris-HCl (pH 8.0), 100 mM ammonium acetate, 5 mM $MgCl_2$, 2 mM dithiothreitol, 0.1 mM EDTA, and 1-2 fmol of [^{32}P]-radiolabeled runoff transcripts (generated *in vitro* by an appropriate RNA polymerase, i.e., T7 or SP6) as substrates. NAAP is added to initiate the reaction or omitted from the reaction in control samples. Following incubation, the RNA is extracted with phenol-chloroform, precipitated in ethanol, and hydrolyzed completely to 3-nucleotide monophosphates using RNase T_2. The hydrolysates are analyzed by two-dimensional thin layer chromatography, and the amount of ^{32}P radiolabel present in the ψMP and UMP spots are evaluated after exposing the thin layer chromatography plates to film or a PhosphorImager screen. Taking into account the relative number of uridylate residues in the substrate RNA, the relative amount ψMP and UMP are determined and used to calculate the relative amount of ψ per tRNA molecule (expressed in mol ψ /mol of tRNA or mol ψ /mol of tRNA/minute), which corresponds to the amount of pseudouridine synthase activity in the NAAP sample (Lecointe, F. et al. (1998) J. Biol. Chem. 273:1316-1323).

N^2,N^2-dimethylguanosine transferase (($m^2{}_2G$)methyltransferase) activity of NAAP is measured in a 160 μl reaction mixture containing 100 mM Tris-HCl (pH 7.5), 0.1 mM EDTA, 10 mM $MgCl_2$, 20 mM NH_4Cl, 1mM dithiothreitol, 6.2 μM *S*-adenosyl-L-[*methyl*-^{3}H]methionine (30-70 Ci/mM), 8 μg $m^2{}_2G$-deficient tRNA or wild type tRNA from yeast, and approximately 100 μg of purified NAAP or a sample comprising NAAP. The reactions are incubated at 30 °C for 90 min and chilled on ice. A portion of each reaction is diluted to 1 ml in water containing 100 μg BSA. 1 ml of 2 M HCl is added to each sample and the acid insoluble products are allowed to precipitate on ice for 20 min before being collected by filtration through glass fiber filters. The collected material is washed several times with HCl and quantitated using a liquid scintillation counter. The amount of ^{3}H incorporated into the $m^2{}_2G$-deficient, acid-insoluble tRNAs is proportional to the amount of N^2,N^2-dimethylguanosine transferase activity in the NAAP sample. Reactions comprising no substrate tRNAs, or wild-type tRNAs that have already been modified, serve as control reactions which should not yield acid-insoluble ^{3}H-labeled products.

Polyadenylation activity of NAAP is measured using an *in vitro* polyadenylation reaction. The reaction mixture is assembled on ice and comprises 10 μl of 5 mM dithiothreitol, 0.025% (v/v) NONIDET P-40, 50 mM creatine phosphate, 6.5% (w/v) polyvinyl alcohol, 0.5 unit/μl RNAGUARD (Pharmacia), 0.025 μg/μl creatine kinase, 1.25 mM cordycepin 5'-triphosphate, and 3.75 mM $MgCl_2$, in a total volume of 25 μl. 60 fmol of CstF, 50 fmol of CPSF, 240 fmol of PAP, 4 μl of crude or partially purified CF II and various amounts of amounts CF I are then added to the reaction mix. The volume is adjusted to 23.5 μl with a buffer containing 50 mM TrisHCl, pH 7.9, 10% (v/v) glycerol, and 0.1 mM Na-EDTA. The final ammonium sulfate concentration should be below 20 mM. The reaction is initiated (on ice) by the addition of 15 fmol of ^{32}P-labeled pre-mRNA template, along with 2.5 μg of unlabeled tRNA, in 1.5 μl of water. Reactions are then incubated at 30 °C for 75-90 min and stopped by the addition of 75 μl (approximately two-volumes) of proteinase K mix (0.2 M Tris-HCl, pH 7.9, 300 mM NaCl, 25 mM Na-EDTA, 2% (w/v) SDS), 1 μl of 10 mg/ml proteinase K, 0.25 μl of 20 mg/ml glycogen, and 23.75 μl of water). Following incubation, the RNA is precipitated with ethanol and analyzed on a 6% (w/v) polyacrylamide, 8.3 M urea sequencing gel. The dried gel is developed by autoradiography or using a phosphoimager. Cleavage activity is determined by comparing the amount of cleavage product to the amount of pre-mRNA template. The omission of any of the polypeptide components of the reaction and substitution of NAAP is useful for identifying the specific biological function of NAAP in pre-mRNA polyadenylation (Rüegsegger, U. et al. (1996) J. Biol. Chem. 271:6107-6113; and references within).

tRNA synthetase activity is measured as the aminoacylation of a substrate tRNA in the presence of [^{14}C]-labeled amino acid. NAAP is incubated with [^{14}C]-labeled amino acid and the appropriate cognate tRNA (for example, [^{14}C]alanine and tRNA^{ala}) in a buffered solution. ^{14}C-labeled product is separated from free [^{14}C]amino acid by chromatography, and the incorporated ^{14}C is quantified by scintillation counter. The amount of ^{14}C-labeled product detected is proportional to the activity of NAAP in this assay.

In the alternative, NAAP activity is measured by incubating a sample containing NAAP in a solution containing 1 mM ATP, 5 mM Hepes-KOH (pH 7.0), 2.5 mM KCl, 1.5 mM magnesium chloride, and 0.5 mM DTT along with misacylated [^{14}C]-Glu-tRNAGln (e.g., 1 μM) and a similar concentration of unlabeled L-glutamine. Following the quenching of the reaction with 3 M sodium acetate (pH 5.0), the mixture is extracted with an equal volume of water-saturated phenol, and the aqueous and organic phases are separated by centrifugation at 15,000 × g at room temperature for 1 min. The aqueous phase is removed and precipitated with 3 volumes of ethanol at -70°C for 15 min. The precipitated aminoacyl-tRNAs are recovered by centrifugation at 15,000 × g at 4°C for15 min. The pellet is resuspended in of 25 mM KOH, deacylated at 65°C for 10 min., neutralized with 0.1 M HCl (to final pH 6-7), and dried under vacuum. The dried pellet is resuspended in water and spotted onto a cellulose TLC plate. The plate is developed in either isopropanol/formic acid/water or ammonia/water/chloroform/ methanol. The image is subjected to densitometric analysis and the relative amounts of Glu and Gln are calculated based on the Rf values and relative intensities of the spots. NAAP activity is calculated based on the amount of Gln resulting from the transformation of Glu while acylated as Glu-tRNA^{Gln} (adapted from Curnow, A.W. et al. (1997) Proc. Natl. Acad. Sci. USA 94:11819-26).

XIX. Identification of NAAP Agonists and Antagonists

Agonists or antagonists of NAAP activation or inhibition may be tested using the assays described in section XVIII. Agonists cause an increase in NAAP activity and antagonists cause a decrease in NAAP activity.

Various modifications and variations of the described compositions, methods, and systems of the invention will be apparent to those skilled in the art without departing from the scope and spirit of the invention. It will be appreciated that the invention provides novel and useful proteins, and their encoding polynucleotides, which can be used in the drug discovery process, as well as methods for using these compositions for the detection, diagnosis, and treatment of diseases and conditions. Although the invention has been described in connection with certain embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Nor should the description of such embodiments be considered exhaustive or limit the invention to the precise forms disclosed. Furthermore, elements from one embodiment can be readily recombined with elements from one or more other embodiments. Such combinations can form a number of embodiments within the scope of the invention. It is intended that the scope of the invention be defined by the following claims and their equivalents.

Table 1

Incyte Project ID	Polypeptide SEQ ID NO:	Incyte Polypeptide ID	Polynucleotide SEQ ID NO:	Incyte Polynucleotide ID	Incyte Full Length Clones
7511189	1	7511189CD1	36	7511189CB1	3297796CA2, 90147103CA2, 90147466CA2
7511811	2	7511811CD1	37	7511811CB1	
7511870	3	7511870CD1	38	7511870CB1	
7510245	4	7510245CD1	39	7510245CB1	1918575CA2
7511072	5	7511072CD1	40	7511072CB1	1810565CA2
7512247	6	7512247CD1	41	7512247CB1	95223867CA2
7512244	7	7512244CD1	42	7512244CB1	
7512177	8	7512177CD1	43	7512177CB1	
7511867	9	7511867CD1	44	7511867CB1	7189330CA2
7512866	10	7512866CD1	45	7512866CB1	
7512572	11	7512572CD1	46	7512572CB1	90063160CA2
7512518	12	7512518CD1	47	7512518CB1	
7512669	13	7512669CD1	48	7512669CB1	5894844CA2
7512706	14	7512706CD1	49	7512706CB1	
7512872	15	7512872CD1	50	7512872CB1	90057121CA2, 90057237CA2
7512925	16	7512925CD1	51	7512925CB1	
7512927	17	7512927CD1	52	7512927CB1	6541623CA2
7512943	18	7512943CD1	53	7512943CB1	
7512955	19	7512955CD1	54	7512955CB1	
7512950	20	7512950CD1	55	7512950CB1	
7517524	21	7517524CD1	56	7517524CB1	6921369CA2
7519735	22	7519735CD1	57	7519735CB1	
7517880	23	7517880CD1	58	7517880CB1	90197651CA2
4524267	24	4524267CD1	59	4524267CB1	4524267CA2
7481640	25	7481640CD1	60	7481640CB1	
7515465	26	7515465CD1	61	7515465CB1	
4912891	27	4912891CD1	62	4912891CB1	
70552759	28	70552759CD1	63	70552759CB1	1665821CA2
7524194	29	7524194CD1	64	7524194CB1	

Table 1

Incyte Project ID	Polypeptide SEQ ID NO:	Incyte Polypeptide ID	Polynucleotide SEQ ID NO:	Incyte Polynucleotide ID	Incyte Full Length Clones
7523973	30	7523973CD1	65	7523973CB1	
7524553	31	7524553CD1	66	7524553CB1	
2666675	32	2666675CD1	67	2666675CB1	2666675CA2, 2872055CA2, 4904968CA2, 6314648CA2
4032169	33	4032169CD1	68	4032169CB1	
7524926	34	7524926CD1	69	7524926CB1	95200854CA2
7509493	35	7509493CD1	70	7509493CB1	

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
1	7511189CD1	g18643896	2.3E-20	[Homo sapiens] zinc finger protein
				Rosati, M. et al., Members of the zinc finger protein gene family sharing a conserved N-terminal module. Nucleic Acids Res. 19 (20), 5661-5667 (1991).
		622017| ZBRK1	4.0E-21	[Homo sapiens][Inhibitor or repressor; DNA-binding protein; Transcription factor][Nuclear] Zinc finger and BRCA1-interacting protein with a KRAB domain 1, binds BRCA1 and represses GADD45A transcription through intron 3, inhibits quiescent cells stimulated with growth factors from entering cell cycle and maintains the nondividing state of cells.
				Zheng, L. et al., Sequence-specific transcriptional corepressor function for BRCA1 through a novel zinc finger protein, ZBRK1. Mol Cell 6, 757-68 (2000).
				Ran, Q. et al., Characterization of a novel zinc finger gene with increased expression in nondividing normal human cells. Exp Cell Res 263, 156-162. (2001).
		790083| LOC90346	1.8E-21	[Homo sapiens] Protein containing a kruppel-associated box (KRAB) domain, has high similarity to a region of zinc finger and BRCA1-interacting protein with a KRAB domain 1 (human ZBRK1), which inhibits quiescent cells from entering cell cycle.
2	7511811CD1	g187143	3.0E-284	[Homo sapiens] DNA ligase I
				Barnes, D.E. et al., Human DNA ligase I cDNA: cloning and functional expression in Saccharomyces cerevisiae Proc. Natl. Acad. Sci. U.S.A. 87 (17), 6679-6683 (1990).
		339476| LIG1	2.4E-285	[Homo sapiens][Ligase; DNA-binding protein][Nuclear] DNA ligase I, a DNA replication and repair enzyme; mutations in the corresponding gene cause a Bloom syndrome-like phenotype with immmunodeficiency, growth retardation and predisposition to cancer.
				Bentley, D. et al., DNA ligase I is required for fetal liver erythropoiesis but is not essential for mammalian cell viability., Nat Genet 13, 489-91 (1996).

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Barnes, D. E. et al., Mutations in the DNA ligase I gene of an individual with immunodeficiencies and cellular hypersensitivity to DNA-damaging agents., Cell 69, 495-503 (1992).
		585205| Lig1	5.0E-214	[Mus musculus][Ligase; DNA-binding protein][Nuclear] DNA ligase I, an essential enzyme for cell viability; mutation of the human LIG1 gene is associated with a Bloom syndrome-like disease.
				Bentley, D. J. et al., DNA ligase I null mouse cells show normal DNA repair activity but altered DNA replication and reduced genome stability., J Cell Sci 115, 1551-61. (2002).
3	7511870CD1	g17132791	1.2E-60	[Nostoc sp. PCC 7120] asparaginyl-tRNA synthetase
		438905| SPBC1198.10c	4.8E-51	[Schizosaccharomyces pombe] Asparaginyl-tRNA synthetase, mitochondrial
		691150| FLJ23441	4.6E-46	[Homo sapiens] Protein containing a class II asparagine, aspartic acid and lysine tRNA synthetase (D, K and N) domain, has moderate similarity to a region of S. cerevisiae Ycr024p, which is a mitochondrial asparaginyl-tRNA synthetase.
4	7510245CD1	g10716076	2.7E-32	[Homo sapiens] testis-abundant finger protein
				Orimo, A. et al., Molecular cloning of testis-abundant finger Protein/Ring finger protein 23 (RNF23), a novel RING-B box-coiled coil-B30.2 protein on the class I region of the human MHC, Biochem. Biophys. Res. Commun. 276, 45-51 (2000).
		731619| MGC16175	2.9E-154	[Homo sapiens] Member of the B-box zinc finger family, contains a C3HC4 type (RING) zinc finger, has a region of high similarity to a region of ret finger protein (human RFP), which functions as a transcriptional repressor and may act in male germ cell development.
				Ishizaka, Y. et al., Activation of the ret-II oncogene without a sequence encoding a transmembrane domain and transforming activity of two ret-II oncogene products differing in carboxy-termini due to alternative splicing [published erratum appears in Oncogene 1989 Nov;4(11):1415], Oncogene 4, 789-94 (1989).

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		581167| Trim27	6.1E-31	[Mus musculus][Nuclear] Tripartite motif protein 27 (Ret finger protein), a member of B box zinc finger gene family, contains a RING finger, B box finger, and coiled-coil domains, may act in male germ cell development; human RFP fusion with human RET tyrosine kinase is oncogenic.
				Cao, T. et al., Ret finger protein is a normal component of PML nuclear bodies and interacts directly with PML., J Cell Sci , 1319-29. (1998).
5	7511072CD1	g881954	2.0E-33	[Homo sapiens] RNPL
				Derry, J. M. et al., RBM3, a novel human gene in Xp11.23 with a putative RNA-binding domain, Hum. Mol. Genet. 4, 2307-2311 (1995).
		567994| RBM3	1.6E-34	[Homo sapiens][RNA-binding protein] RNA binding motif protein 3, a member of the glycine-rich RNA binding protein family that may be involved in response to cold, and may activate transcription of late vaccinia virus genes.
				Danno, S. et al., Increased transcript level of RBM3, a member of the glycine-rich RNA-binding protein family, in human cells in response to cold stress., Biochem Biophys Res Commun 236, 804-7. (1997).
		587617| Rbm3	9.0E-34	[Mus musculus][RNA-binding protein] RNA binding motif protein 3, a member of the glycine-rich RNA binding protein family, a putative RNA binding protein that may play a role in spermatogenesis and in response to cold.
				Danno, S. et al., Decreased expression of mouse Rbm3, a cold-shock protein, in Sertoli cells of cryptorchid testis., Am J Pathol 156, 1685-92 (2000).
6	7512247CD1	g899109	7.0E-253	[Homo sapiens] histidyl-tRNA synthetase homologue
				O'Hanlon, T. P. et al., A novel gene oriented in a head-to-head configuration with the human histidyl-tRNA synthetase (HRS) gene encodes an mRNA that predicts a polypeptide homologous to HRS, Biochem. Biophys. Res. Commun. 210, 556-566 (1995).
		743656| HARSL	5.7E-254	[Homo sapiens] Histidyl-tRNA synthetase-like, putative histidyl-tRNA synthetase, may regulate translation, potentially binds ATP, histidine, and histidyl-tRNA.

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				O'Hanlon, T. P. et al., (supra)
		583169|Hars	2.9E-186	[Mus musculus][Ligase; tRNA synthetase; RNA-binding protein][Cytoplasmic] Histidyl-tRNA synthetase, a putative class II aminoacyl tRNA synthetase; autoantibodies are often found in mice with a myositis-like disease, and autoantibodies to human HARS are found in patients with the autoimmune muscle disease polymyositis.
				Blechynden, L. M. et al., Sequence and polymorphism analysis of the murine gene encoding histidyl- tRNA synthetase., Gene 178, 151-6 (1996).
7	7512244CD1	g189504	3.7E-242	[Homo sapiens] NF-kappa-B transcription factor
				Ruben, S. M. et al., Isolation of a rel-related human cDNA that potentially encodes the 65-kD subunit of NF-kappa B, Science 251, 1490-1493 (1991).
		625957|RELA	1.0E-231	[Homo sapiens][Transcription factor; DNA-binding protein; Small molecule-binding protein] [Nuclear; Cytoplasmic] V-rel reticuloendotheliosis viral oncogene homolog A, a subunit of transcription factor nuclear factor kappa B (NFKB1) involved in cell differentiation, response to infection and apoptosis inhibition, implicated in metastatic melanoma and Crohn's disease.
				Lieb, K. et al., The neuropeptide substance P activates transcription factor NF-kappa B and kappa B-dependent gene expression in human astrocytoma cells., J Immunol 159, 4952-8 (1997).
		581153|Rela	1.1E-205	[Mus musculus][Inhibitor or repressor; DNA-binding protein; Transcription factor] [Nuclear] Avian reticuloendotheliosis viral (v-rel) oncogene homolog A, subunit of transcription factor nuclear factor kappa B involved in development, immune responses, and apoptosis inhibition; human RELA is implicated in metastatic melanoma and Crohn's disease.
				Lavon, I. et al., High susceptibility to bacterial infection, but no liver dysfunction, in mice compromised for hepatocyte NF-kappaB activation., Nat Med 6, 573-7 (2000).
8	7512177CD1	g9650982	1.9E-93	[Homo sapiens] testis-specific RING Finger protein

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Yoshikawa, T. et al., Isolation of a cDNA for a novel human RING finger protein gene, RNF18, by the virtual transcribed sequence (VTS) approach(1), Biochim. Biophys. Acta 1493, 349-355 (2000).
		685431| MGC4827	7.6E-110	[Homo sapiens] Protein containing a C3HC4 type (RING) zinc finger, has a region of moderate similarity to a region of 52 kDa Sjogren syndrome antigen A1 (human SSA1), which is a part of Ro-SSA ribonucleoprotein complex recognized by autoantibodies in Sjogren's Syndrome.
		475757| TRIM17	2.3e-42	[Homo sapiens] RING finger protein 16, expressed only in the testis; predicted to function in carcinogenesis or cell transformation.
				Ogawa, S. et al., Chromosome mapping of RNF16 and rnf16, human, mouse and rat genes coding for testis RING finger protein (terf), a member of the RING finger family. Cytogenet Cell Genet 89, 56-8 (2000).
		753971|Trim11	1.2E-44	[Mus musculus] Member of the B-box zinc finger family, contains a SPRY (SPla and RYanodine Receptor) domain and a C3HC4 type (RING) zinc finger, has moderate similarity to sjogren syndrome antigen A1 (52 kD) (human SSA1).
				Ogawa, S. et al., Molecular cloning of a novel RING finger-B box-coiled coil (RBCC) protein, terf, expressed in the testis. Biochem Biophys Res Commun 251, 515-9 (1998).
9	7511867CD1	g2366752	2.9E-32	[Homo sapiens] Lysyl tRNA Synthetase
				Shiba, K. et al., Human lysyl-tRNA synthetase accepts N73 variants and rescues *E. coli* double-defective mutant, J. Biol. Chem., 272:22809-16 (1997).
		342552| KARS	2.2E-33	[Homo sapiens][Ligase; tRNA synthetase; RNA-binding protein][Cytoplasmic] Lysyl-tRNA synthetase, a class II aminoacyl tRNA synthetase, aminoacylates its cognate tRNAs with lysine for protein biosynthesis, packaged into HIV virions, exists as cytoplasmic and mitochondrial alternative forms.
				Cen, S. et al., Incorporation of lysyl-tRNA synthetase into human immunodeficiency virus type 1., J Virol 75, 5043-8. (2001).

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		753897|Kars	2.9E-24	[Mus musculus] Lysyl-tRNA synthetase, a putative class II aminoacyl tRNA synthetase, predicted to aminoacylate its cognate tRNAs with lysine for protein biosynthesis, suspected autoantigen in chronic GVHD, may be involved in amyotrophic lateral sclerosis.
				Gelpi, C. et al., Autoantibodies to a transfer RNA-associated protein in a murine model of chronic graft versus host disease., J Immunol 152, 1989-99. (1994).
10	7512866CD1	g14348588	6.1E-105	[Homo sapiens] KRAB zinc finger protein
				Mark, C. et al., Molecular cloning and preliminary functional analysis of two novel human KRAB zinc finger proteins, HKr18 and HKr19, DNA Cell Biol. 20, 275-286 (2001).
		730137|KR18	1.2E-109	[Homo sapiens] Protein with high similarity to zinc finger protein 85 (human ZNF85), which functions as a transcriptional co-repressor, contains twenty C2H2 type zinc finger domains and a kruppel-associated box (KRAB) domain.
		705227|ZNF268	4.1E-109	[Homo sapiens] Zinc finger protein 268, contains an N-terminal Kruppel associated box domain and 24 C-terminal zinc fingers, may play a role in embryogenesis.
				Gou, D. M. et al., Cloning and characterization of a novel Kruppel-like zinc finger gene, ZNF268, expressed in early human embryo., Biochim Biophys Acta 1518, 306-10. (2001).
11	7512572CD1	g13620881	2.2E-141	[Homo sapiens] mitochondrial ribosomal protein S5
				Suzuki, T. et al., Proteomic analysis of the mammalian mitochondrial ribosome. Identification of protein components in the 28 S small subunit. J. Biol. Chem. 276 (35), 33181-33195 (2001).
		709641|MRPS5	1.7E-142	[Homo sapiens] Member of the ribosomal S5 family, which are part of the small ribosomal subunit.
				Cavdar Koc, E. et al., The small subunit of the mammalian mitochondrial ribosome. Identification of the full complement of ribosomal proteins present., J Biol Chem 276, 19363-74 (2001).

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		241991| E02A10.1	1.2E-24	[Caenorhabditis elegans][Ribosomal subunit][Mitochondrial] Member of the ribosomal S5 protein family.
				Cavdar Koc, E. et al., (supra)
12	7512518CD1	g18027283	1.2E-35	[Homo sapiens] TAR RNA-binding protein TRBP2
				Bannwarth, S. et al., Organization of the human tarbp2 gene reveals two promoters that are repressed in an astrocytic cell line, J. Biol. Chem. 276, 48803-48813 (2001).
		568860| TARBP2	9.7E-37	[Homo sapiens][RNA-binding protein][Nuclear] TAR RNA-binding protein 2, an RNA binding protein that functions as a translational repressor during spermatogenesis, binds to the HIV-1 TAR RNA element and is required for HIV-1 gene expression, may also have oncogenic potential.
				Gatignol, A. et al., Characterization of a human TAR RNA-binding protein that activates the HIV-1 LTR., Science 251, 1597-600 (1991).
				Benkirane, M. et al., Oncogenic potential of TAR RNA binding protein TRBP and its regulatory interaction with RNA-dependent protein kinase PKR., Embo Journal 16, 611-24. (1997).
		581573| Tarbp2	6.8E-36	[Mus musculus][Inhibitor or repressor; RNA-binding protein; Translation factor][Cytoplasmic] TAR RNA-binding protein 2, an RNA binding protein that functions as a translational repressor during spermatogenesis; human TARBP2 binds to the HIV-1 TAR RNA element and is required for HIV-1 gene expression.
				Kozak, C. A. et al., Genetic mapping in human and mouse of the locus encoding TRBP, a protein that binds the TAR region of the human immunodeficiency virus (HIV-1)., Genomics 25, 66-72 (1995).
				Zhong, J. et al., A double-stranded RNA binding protein required for activation of repressed messages in mammalian germ cells., Nat Genet 22, 171-4. (1999).
13	7512669CD1	g7453577	3.4E-147	[Homo sapiens] protein arginine N-methyltransferase 1-variant 1

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Scorilas, A. et al., Genomic organization, physical mapping, and expression analysis of the human protein arginine methyltransferase 1 gene, Biochem. Biophys. Res. Commun. 278, 349-359 (2000).
		688988| Hrmt112	6.0E-153	[Rattus norvegicus][Transferase] Protein arginine N-methyl transferase-1, a member of the S-adenosyl-L-methionine-dependent protein arginine N-methyl transferase family, methylates arginine residues in proteins and is a transcriptional co-activator.
				Lin, W. J. et al., The mammalian immediate-early TIS21 protein and the leukemia-associated BTG1 protein interact with a protein-arginine N-methyltransferase., J Biol Chem 271, 15034-44 (1996).
				Koh, S. S. et al., Synergistic Enhancement of Nuclear Receptor Function by p160 Coactivators and Two Coactivators with Protein Methyltransferase Activities., J Biol Chem 276, 1089-1098. (2001).
		608844| Hrmt112	7.4E-152	[Mus musculus][Transferase] Protein arginine N-methyltransferase 1, methylates arginine residues within proteins and is required for early postimplantation development
				Pawlak, M. R. et al., Arginine N-methyltransferase 1 is required for early postimplantation mouse development, but cells deficient in the enzyme are viable., Mol Cell Biol 20, 4859-69 (2000).
14	7512706CD1	g7804468	8.2E-46	[Homo sapiens] methionyl tRNA synthetase
		742474| MARS	6.4E-47	[Homo sapiens][Ligase; tRNA synthetase; RNA-binding protein][Nuclear; Nuclear nucleolus; Cytoplasmic] Methionine-tRNA synthetase, putative tRNA synthetase that is predicted to aminoacylate its cognate tRNAs with methionine for protein biosynthesis, required for rRNA synthesis in the nucleolus.
				Ko, Y. G. et al., Nucleolar localization of human methionyl-tRNA synthetase and its role in ribosomal RNA synthesis., J Cell Biol 149, 567-74. (2000).
15	7512872CD1	g2815614	6.9E-151	[Homo sapiens] heterogeneous nuclear ribonucleoprotein D

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Dempsey, L. A. et al., The human HNRPD locus maps to 4q21 and encodes a highly conserved protein, Genomics 49, 378-384 (1998).
		580925| Hnrpd	5.8E-110	[Mus musculus][RNA-binding protein][Cytoplasmic] Heterogeneous nuclear ribonucleoprotein D, an RNA and DNA-binding protein which may play a role in RNA catabolism and telomere maintenance, component of the protein LR1, possibly involved in Ig class switch recombination.
				Blaxall, B. C. et al., Differential expression and localization of the mRNA binding proteins, AU-rich element mRNA binding protein (AUF1) and Hu antigen R (HuR), in neoplastic lung tissue., Mol Carcinog 28, 76-83 (2000).
		742520| HNRPD	1.9E-117	[Homo sapiens][RNA-binding protein][Nuclear; Cytoplasmic] Heterogeneous nuclear ribonucleoprotein D (AU-rich element RNA-binding protein 1 37kD), an RNA and DNA-binding protein which binds to AU-rich elements in the 3' untranslated regions of mRNAs and mediates their degradation.
				Eversole, A. et al., In vitro properties of the conserved mammalian protein hnRNP D suggest a role in telomere maintenance. Mol Cell Biol 20, 5425-32 (2000).
16	7512925CD1	g10179425	0.0	[Homo sapiens] Rb-associated protein
				Wen, H. et al., RBP95, a novel leucine zipper protein, binds to the retinoblastoma protein, Biochem. Biophys. Res. Commun. 275, 141-148 (2000).
		759988| RNF20	2.6E-153	[Homo sapiens] Protein containing a C3HC4 type (RING) zinc finger, which may mediate protein-protein interactions.
17	7512927CD1	g1464742	2.4E-12	[Homo sapiens] threonyl-tRNA synthetase
				Cruzen, M. E. et al., Nucleotide and deduced amino acid sequence of human threonyl-tRNA synthetase reveals extensive homology to the Escherichia coli and yeast enzymes, J. Biol. Chem. 266, 9919-9923 (1991).
		338406| TARS	1.9E-13	[Homo sapiens][Ligase; tRNA synthetase; RNA-binding protein][Cytoplasmic] Threonyl-tRNA synthetase, a class II aminoacyl tRNA synthetase, aminoacylates its cognate tRNAs with threonine for protein synthesis.

Table 2

<table>
<tr><th>Polypeptide SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>GenBank ID NO: or PROTEOME ID NO:</th><th>Probability Score</th><th>Annotation</th></tr>
<tr><td></td><td></td><td></td><td></td><td>Marguerie, C. et al., Polymyositis, pulmonary fibrosis and autoantibodies to aminoacyl-tRNA synthetase enzymes., Q J Med 77, 1019-38. (1990).</td></tr>
<tr><td></td><td></td><td>749850|
D15Wsu59e</td><td>3.2E-13</td><td>[Mus musculus] Protein with strong similarity to threonyl-tRNA synthetase (human TARS), which may be associated with several autoimmune disorders, contains tRNA synthetase class II core (G, H, P, S and T), TGS (ThrRS, GTPase, and SpoT) and anticodon binding domains.</td></tr>
<tr><td>18</td><td>7512943CD1</td><td>g4092859</td><td>3.7E-98</td><td>[Homo sapiens] p53 regulated PA26-T2 nuclear protein</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Buckbinder, L. et al., Gene regulation by temperature-sensitive p53 mutants: identification of p53 response genes, Proc. Natl. Acad. Sci. U.S.A. 91, 10640-10644 (1994).</td></tr>
<tr><td></td><td></td><td>569356|
PA26</td><td>1.0E-80</td><td>[Homo sapiens][Nuclear] p53 regulated PA26 nuclear protein, induced in a p53 (TP53)-dependent manner by genotoxic stress, induced by growth arrest, widely expressed and primarily localized to the nucleus, may function in regulating cell growth.</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Velasco-Miguel, S. et al., PA26, a novel target of the p53 tumor suppressor and member of the GADD family of DNA damage and growth arrest inducible genes., Oncogene 18, 127-37 (1999).</td></tr>
<tr><td>19</td><td>7512955CD1</td><td>g16876833</td><td>7.2E-47</td><td>[Homo sapiens] Similar to mitochondrial ribosomal protein L24</td></tr>
<tr><td></td><td></td><td>742444|
MRPL24</td><td>5.8E-48</td><td>[Homo sapiens] Mitochondrial ribosomal protein L24, a putative component of the mitochondrial large ribosomal subunit, may function in protein biosynthesis.</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Koc, E. C. et al., The large subunit of the mammalian mitochondrial ribosome. Analysis of the complement of ribosomal proteins present. J Biol Chem 276, 43958-69 (2001).</td></tr>
<tr><td>20</td><td>7512950CD1</td><td>g1045574</td><td>0.0</td><td>[Homo sapiens] cleavage and polyadenylation specificity factor</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Murthy, K. G. et al., The 160-kD subunit of human cleavage-polyadenylation specificity factor coordinates pre-mRNA 3'-end formation, Genes Dev. 9, 2672-2683 (1995).</td></tr>
</table>

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Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		606152| CPSF1	0.0	[Homo sapiens] Protein with low similarity to S. pombe SPBC1709.08, which is a putative cleavage and polyadenylation specificity factor subunit.
				Kleiman, F. E. et al., The BARD1-CstF-50 interaction links mRNA 3' end formation to DNA damage and tumor suppression., Cell 104, 743-53. (2001).
		754125| Cpsf1	0.0	[Mus musculus] Protein with weak similarity to S. cerevisiae Cft1p, which is a component of pre-mRNA cleavage factor II.
21	7517524CD1	g4336734	1.2E-49	[Mus musculus] Pax transcription activation domain interacting protein PTIP.
				Lechner, M. S. et al., PTIP, a novel BRCT domain-containing protein interacts with Pax2 and is associated with active chromatin, Nucleic Acids Res. 28, 2741-2751 (2000).
		608310| Paxip1	9.0E-51	[Mus musculus][Regulatory subunit][Nuclear; Nuclear matrix] Pax transactivation domain interacting protein, contains BRCA1C terminus domains, mediates Pax2-dependent transcription activation, associated with active chromatin and nuclear matrix, may be involved in cell cycle control in response to DNA damage.
				Lechner, M. S. et al. (supra)
22	7519735CD1	g4336734	0.0	[Mus musculus] Pax transcription activation domain interacting protein PTIP.
				Lechner, M. S. et al. (supra)
		608310| Paxip1	0.0	[Mus musculus][Regulatory subunit][Nuclear; Nuclear matrix] Pax transactivation domain interacting protein, contains BRCA1C terminus domains, mediates Pax2-dependent transcription activation, associated with active chromatin and nuclear matrix, may be involved in cell cycle control in response to DNA damage.
				Lechner, M. S. et al. (supra)
		426794| PAXIP1L	3.2E-269	[Homo sapiens] Pax transactivation domain interacting protein, contains two BRCA1C terminus domains and may act in cell cycle control, also mediates Pax2-dependent transactivation.
				Lechner, M. S. et al. (supra)
23	7517880CD1	g726513	3.6E-127	[Homo sapiens] nuclear orphan receptor LXR-alpha

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Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Willy, P. J. et al., LXR, a nuclear receptor that defines a distinct retinoid response pathway, Genes Dev. 9, 1033-1045 (1995).
		342610| NR1H3	2.8E-128	[Homo sapiens][Activator; Transcription factor; DNA-binding protein; Receptor (signalling)][Nuclear] Nuclear receptor subfamily 1 group H member 3, a ligand-dependent nuclear receptor transcription factor, acts in RXR receptor and PPAR-alpha signaling, cholesterol/lipid metabolism, and TNF synthesis; mutations in mouse Nr1h3 result in cholesterol accumulation.
				Tamura, K. et al., LXRalpha functions as a cAMP-responsive transcriptional regulator of gene expression, Proc Natl Acad Sci U S A 97, 8513-8 (2000).
				Venkateswaran, A. et al., Control of cellular cholesterol efflux by the nuclear oxysterol receptor LXR-alpha, Proc Natl Acad Sci U S A 97, 12097-102 (2000).
				Chinetti, G. et al., PPAR-alpha and PPAR-gamma activators induce cholesterol removal from human macrophage foam cells through stimulation of the ABCA1 pathway., Nat Med 7, 53-58. (2001).
		587195| Nr1h3	3.3E-109	[Mus musculus][Activator; DNA-binding protein; Transcription factor; Receptor (signalling)][Nuclear] Nuclear receptor subfamily 1 group H member 3, a ligand-dependent nuclear receptor transcription factor, acts in RXR receptor and PPAR-gamma signaling as well as cholesterol, fatty acid, and bile metabolism; mutations result in cholesterol accumulation.
				Peet, D. J. et al., Cholesterol and bile acid metabolism are impaired in mice lacking the nuclear oxysterol receptor LXR alpha., Cell 93, 693-704 (1998).
				Wan, Y. J. et al., Hepatocyte-specific mutation establishes retinoid X receptor alpha as a heterodimeric integrator of multiple physiological processes in the liver., Mol Cell Biol 20, 4436-44 (2000).
				Repa, J. J. et al., Regulation of mouse sterol regulatory element-binding protein-1c gene (SREBP-1c) by oxysterol receptors, LXRalpha and LXRbeta, Genes And Development 14, 2819-30 (2000).
24	4524267CD1	g9650982	4.8E-31	[Homo sapiens] testis-specific RING Finger protein

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Yoshikawa, T. et al., Isolation of a cDNA for a novel human RING finger protein gene, RNF18, by the virtual transcribed sequence (VTS) approach(1), Biochim. Biophys. Acta 1493, 349-355 (2000).
		685431| MGC4827	9.0E-36	[Homo sapiens] Protein containing a C3HC4 type (RING) zinc finger, has a region of moderate similarity to a region of 52 kDa Sjogren syndrome antigen A1 (human SSA1), which is a part of Ro-SSA ribonucleoprotein complex recognized by autoantibodies in Sjogren's Syndrome
		753971| Trim11	3.6E-15	[Mus musculus] Member of the B-box zinc finger family, contains a SPRY (SPla and RYanodine Receptor) domain and a C3HC4 type (RING) zinc finger, has moderate similarity to sjogren syndrome antigen A1 (52 kD) (human SSA1)
		475757| TRIM17	4.9E-15	[Homo sapiens] RING finger protein 16, expressed only in the testis; predicted to function in carcinogenesis or cell transformation
				Ogawa, S. et al., Molecular cloning of a novel RING finger-B box-coiled coil (RBCC) protein, terf, expressed in the testis., Biochem Biophys Res Commun 251, 515-9 (1998).
				Ogawa, S. et al., Chromosome mapping of RNF16 and rnf16, human, mouse and rat genes coding for testis RING finger protein (terf), a member of the RING finger family., Cytogenet Cell Genet 89, 56-8 (2000).
25	7481640CD1	g9049520	2.2E-287	[Mus musculus] nuclear transcription factor RelA
				Lemmer, E. R. et al., Genomic structure and chromosome location of the mouse RelA p65 gene (Rela), Cytogenet. Cell Genet. 89, 129-132 (2000).
		581153| Rela	1.7E-288	[Mus musculus][Inhibitor or repressor; DNA-binding protein; Transcription factor][Nuclear] Avian reticuloendotheliosis viral (v-rel) oncogene homolog A, subunit of transcription factor nuclear factor kappa B involved in development, immune responses, and apoptosis inhibition; human RELA is implicated in metastatic melanoma and Crohn's disease

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Alcamo, E. et al., Targeted mutation of TNF receptor I rescues the RelA-deficient mouse and reveals a critical role for NF-kappaB in leukocyte recruitment., J Immunol 167, 1592-600. (2001).
				Beg, A. A. et al., Embryonic lethality and liver degeneration in mice lacking the RelA component of NF-kappa B., Nature 376, 167-70 (1995).
		625957|RELA	3.7E-259	[Homo sapiens][Transcription factor; DNA-binding protein; Small molecule-binding protein][Nuclear; Cytoplasmic] V-rel reticuloendotheliosis viral oncogene homolog A, a subunit of transcription factor nuclear factor kappa B (NFKB1) involved in cell differentiation, response to infection and apoptosis inhibition, implicated in metastatic melanoma and Crohn's disease
				Lieb, K. et al., The neuropeptide substance P activates transcription factor NF-kappa B and kappa B-dependent gene expression in human astrocytoma cells., J Immunol 159, 4952-8 (1997).
26	7515465CD1	g21666261	0.0	[Homo sapiens] (AF380349) F-box DNA helicase 1
				Kim, J. et al., The novel human DNA helicase hFBH1 is an F-box protein, J. Biol. Chem. 277, 24530-24537 (2002).
		476927|Fbxo18	0.0	[Mus musculus][Protein conjugation factor] Protein containing an F-box domain, which serve as a link between a target protein and a ubiquitin-conjugating enzyme, has strong similarity to uncharacterized human FLJ14590
		731923|FLJ14590	0.0	[Homo sapiens] Has strong similarity to uncharacterized mouse Fbxo18
27	4912891CD1	g186774	2.8E-106	[Homo sapiens] zinc finger protein
				Bellefroid, E. J. et al., The evolutionarily conserved Kruppel-associated box domain defines a subfamily of eukaryotic multifingered proteins, Proc. Natl. Acad. Sci. U.S.A. 88, 3608-3612 (1991)
		599086|FLJ20079	1.2E-166	[Homo sapiens][DNA-binding protein][Nuclear] Protein containing seven C2H2 type zinc finger domains, which bind nucleic acids

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		685415|MGC2663	3.0E-108	[Homo sapiens] Protein containing a kruppel-associated box (KRAB) domain and twelve C2H2 type zinc finger domains, has moderate similarity to human ZNF136, which represses transcription when fused to the heterologous KRAB B subdomain of human ZNF10
28	70552759CD1	g5059039	4.2E-183	[Homo sapiens] KRAB-zinc finger protein SZF1-2
				Liu, C. et al., SZF1: a novel KRAB-zinc finger gene expressed in CD34+ stem/progenitor cells, Exp. Hematol. 27, 313-325 (1999)
		476533|SZF1	1.3E-176	[Homo sapiens][Inhibitor or repressor;DNA-binding protein][Nuclear] Stem cell zinc finger 1, a transcriptional repressor with a KRAB domain and four C2H2 type zinc fingers, expressed in CD34+ stem/progenitor cells, and may function in hematopoietic stem/progenitor cell differentiation
				Liu, C. et al., SZF1: a novel KRAB-zinc finger gene expressed in CD34+ stem/progenitor cells., Exp Hematol 27, 313-25 (1999).
		606504|PRDM9	3.6E-64	[Homo sapiens][DNA-binding protein] Protein containing fourteen C2H2 type zinc finger domains, which bind nucleic acids, and a SET domain, which are involved in chromatin organization
29	7524194CD1	g27451600	1.0E-118	KOX31-like zinc finger protein [Homo sapiens]
		g18376583	1.0E-91	[Homo sapiens] dJ874C20.2 (novel C2H2 type zinc finger protein)
		690702|MGC4179	1.5E-136	[Homo sapiens] Protein containing a SCAN domain, which mediate homo- or hetero-oligomerization, has moderate similarity to a region of zinc finger protein 202 (human ZNF202), which is a transcriptional repressor for genes involved in lipid metabolism
		599370|FLJ20417	1.1E-122	[Homo sapiens][DNA-binding protein][Nuclear] Protein containing six C2H2 type zinc finger domains, which bind nucleic acids
30	7523973CD1	g23491772	0.0	HLH-PAS transcription factor NXF [Homo sapiens]

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		586633|Sim2	1.3E-21	[Mus musculus][Inhibitor or repressor; DNA-binding protein;Transcription factor] Single-minded (Drosophila) homolog 2, may act as a transcriptional corepressor via competition with Sim1 to form heterodimers with Arnt, regulates expression of sonic hedgehog (Shh); overexpression causes a mild impairment of learning and memory
				Ema, M. et al. Two new members of the murine Sim gene family are transcriptional repressors and show different expression patterns during mouse embryogenesis. Mol. Cell. Biol. 16, 5865-5875 (1996)
		568742|SIM2	8.6E-22	[Homo sapiens][Inhibitor or repressor; DNA-binding protein;Transcription factor] Single-minded (Drosophila) homolog 2, may regulate transcription, may be involved in central nervous system development and regionalization, overexpression may contribute to some aspects of the Down Syndrome phenotype, such as impaired learning and memory
				Fan, C. M. et al. Expression patterns of two murine homologs of Drosophila single-minded suggest possible roles in embryonic patterning and in the pathogenesis of Down syndrome [published erratum appears in Mol. Cell. Neurosci. 1996 Jun; 7(6):519], Mol. Cell. Neurosci. 7, 1-16 (1996)
31	7524553CD1	g1669498	2.1E-242	[Homo sapiens] Sp140 protein
				Bloch, D. B. et al. Identification and characterization of a leukocyte-specific component of the nuclear body. J. Biol. Chem. 271, 29198-29204 (1996)
		428464|SP140	1.5E-243	[Homo sapiens][Transcription factor][Nuclear] Protein of unknown function, has strong similarity to a region of nuclear body protein Sp140 (human SP140), which is a lymphoid-specific protein that associates with the PML transcription factor and nuclear antigen SP100 in nuclear bodies
				Kaul, S. et al. Properties of the glucocorticoid modulatory element binding proteins GMEB-1 and -2: potential new modifiers of glucocorticoid receptor transactivation and members of the family of KDWK proteins. Mol. Endocrinol. 14, 1010-1027 (2000)

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		789625|SP100	2.3E-110	[Homo sapiens][Nuclear; Nuclear nucleolus] Nuclear antigen SP100, an interferon-regulated transcriptional regulator that localizes to nuclear PML bodies and other nuclear structures, may bind DNA, sequestered in HSV-1 infection, autoantigen recognized by serum of primary biliary cirrhosis patients
				Szostecki, C. et al.. Isolation and characterization of cDNA encoding a human nuclear antigen predominantly recognized by autoantibodies from patients with primary biliary cirrhosis. J. Immunol. 145, 4338-4347 (1990)
32	2666675CD1	g4455608	8.0E-72	[Homo sapiens] dJ742C19.2 (APOBEC1 (Apolipoprotein B mRNA editing protein) and Phorbolin 1 LIKE)
		613317|AICDA	4.5E-25	[Homo sapiens][Hydrolase] Activation-induced cytidine deaminase, a putative RNA editing enzyme that functions in class switch recombination and somatic hypermutation during B cell terminal differentiation; deficiency causes hyper-IgM syndrome (HIGM2)
				Revy, P. et al. Activation-induced cytidine deaminase (AID) deficiency causes the autosomal recessive form of the Hyper-IgM syndrome (HIGM2). Cell 102, 565-575 (2000)
		429630|Aicda	1.2E-22	[Mus musculus][Hydrolase] Activation induced cytidine deaminase, a putative RNA editing enzyme that functions in class switch recombination and somatic hypermutation during B cell terminal differentiation; deficiency in human AICDA leads to hyper-IgM syndrome (HIGM2)
				Muramatsu, M. et al. Specific expression of activation-induced cytidine deaminase (AID), a novel member of the RNA-editing deaminase family in germinal center B cells. J. Biol. Chem. 274, 18470-18476 (1999)
33	4032169CD1	g21665855	4.1E-265	[Homo sapiens] zinc finger protein 25
				Hearn, T., Organisation, expression and evolution of Kruppel-type zinc finger genes in human chromosomal region 10p11.2-q11.2, Thesis (2000) Department of Human Genetics, University of Newcastle, Newcastle upon Tyne, United Kingdom

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
				Guy, J. et al., Genomic sequence and transcriptional profile of the boundary between pericentromeric satellites and genes on human chromosome arm 10p, Genome Res.
		568956| AF020591	8.3E-134	[Homo sapiens][Inhibitor or repressor;Transcription factor] Protein containing 26 C2H2 type zinc finger domains and two kruppel-associated box (KRAB) domains, has moderate similarity to zinc finger protein 234 (human ZNF234), which is a transcriptional corepressor that interacts with TRIM28
		435075| ZNF41	3.3E-141	[Homo sapiens][Inhibitor or repressor;Transcription factor] Zinc finger protein 41, a putative transcription factor that may function as a transcriptional repressor
				Rosati, M. et al., Coding region intron/exon organization, alternative splicing, and X-chromosome inactivation of the KRAB/FPB-domain-containing human zinc finger gene ZNF41., Cytogenet Cell Genet 85, 291-6 (1999).
34	7524926CD1	g3269303	1.1E-212	[Coturnix coturnix] FLI transcription factor
				Mager, A. M. et al., The avian fli gene is specifically expressed during embryogenesis in a subset of neural crest cells giving rise to mesenchyme, Int. J. Dev. Biol. 42, 561-572 (1998)
		568748| FLI1	1.4E-119	[Homo sapiens][Transcription factor; DNA-binding protein] Friend leukemia virus integration 1, a member of the Ets family of transcription factors; fusion of the corresponding gene is associated with Ewing sarcoma and related subtypes of primitive neuroectodermal tumors
				Spyropoulos, D. D. et al., Hemorrhage, impaired hematopoiesis, and lethality in mouse embryos carrying a targeted disruption of the Fli1 transcription factor., Mol Cell Biol 20, 5643-52 (2000).
				Hart, A. et al., Fli-1 is required for murine vascular and megakaryocytic development and is hemizygously deleted in patients with thrombocytopenia., Immunity 13, 167-77 (2000).

Table 2

Polypeptide SEQ ID NO:	Incyte Polypeptide ID	GenBank ID NO: or PROTEOME ID NO:	Probability Score	Annotation
		582889| Fli1	8.6E-118	[Mus musculus][Adhesin/agglutinin; Transcription factor; DNA-binding protein] Friend leukemia virus integration 1, a member of the Ets family of DNA-binding proteins; rearrangement of the corresponding gene in cell lines is associated with erythroleukemia induced by Friend murine leukemia virus
				Ben-David, Y. et al., Erythroleukemia induction by Friend murine leukemia virus: insertional activation of a new member of the ets gene family, Fli-1, closely linked to c-ets-1., Genes And Development 5, 908-18 (1991).
35	7509493CD1	g4056411	0.0	[Homo sapiens] Human homolog of Mus musculus wizS protein [AA 171-934]
		586149| Wiz	0.0	[Mus musculus][DNA-binding protein][Nuclear] Widely-interspaced zinc finger motifs, contains widely spaced Kruppel-type zinc finger motifs, may play distinct transcriptional roles in cerebellar granule cells and extracerebellar regions
				Matsumoto, K. et al., Molecular cloning and distinct developmental expression pattern of spliced forms of a novel zinc finger gene wiz in the mouse cerebellum., Brain Res Mol Brain Res 61, 179-89 (1998).

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
1	7511189CD1	93	T9 T18 T77	N40	krueppel associated box: L8-M68	HMMER_SMART
					KRAB box: L8-G48	HMMER_PFAM
					ZINC FINGER METAL-BINDING DNA-BINDING PROTEIN FINGER ZINC NUCLEAR REPEAT TRANSCRIPTION REGULATION PD001562: L8-D70	BLAST_PRODOM
					KRAB BOX DOMAIN DM00605 |P51523|5-79: Q5-P74 |I48689|11-85: Q5-D70 |P51786|24-86: S7-W67 |P52738|3-77: Q5-M68	BLAST_DOMO
2	7511811CD1	569	S27 S47 S66 S104 S109 S141 S163 S191 S199 S201 S303 S323 S368 S447 S552 T97 T123 T132 T159 T165 T171 T190 T195 T207 T233 T362 T389 T420 T494 T506	N26 N102	DNA LIGASE I POLYDEOXYRIBONUCLEOTIDE SYNTHASE ATP REPAIR REPLICATION RECOMBINATION CELL PD017676: M1-P288	BLAST_PRODOM
					LIGASE DNA ATP POLY-DEOXY-RIBONUCLEOTIDE SYNTHASE REPAIR REPLICATION RECOMBINATION CELL DIVISION PD004574: V379-T470	BLAST_PRODOM
					LIGASE DNA ATP POLY-DEOXY-RIBONUCLEOTIDE SYNTHASE REPAIR REPLICATION RECOMBINATION CELL DIVISION PD004907: Y289-D378	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					DNA; ATP; LIGASE; POLY-DEOXY-RIBONUCLEOTIDE; DM04655|P18858|19-283: K19-G284	BLAST_DOMO
					ATP-DEPENDENT DNA LIGASE DM01008 |P18858|285-901: Q285-P536 |I48921|283-899: Q285-P536 |P51892|438-1054: Q285-P536	BLAST_DOMO
3	7511870CD1	346	S10 S25 S29 S52 S55 S68 S79 S90 S103 S200 S249 S264 S268 S334 S339 T184 T241	N38 N186 N342	tRNA synthetases class II (D, K and N): P135-K281	HMMER_PFAM
					OB-fold nucleic acid binding domain: I44-I118	HMMER_PFAM
					Asparaginyl-tRNA synthetase: K27-S346	HMMER_TIGRFAM
					Aminoacyl-transfer RNA synthetases class-II signatures: Q222-M285	PROFILESCAN
					Aspartyl-tRNA synthetase signature PR01042: L217-S229, T233-N246	BLIMPS_PRINTS
					tRNA synthetases class II. PF00152: I44-D66, R159-I183, S220-F256	BLIMPS_PFAM
					SYNTHETASE AMINOACYL-TRNA LIGASE PROTEIN BIOSYNTHESIS ATP-BINDING ASPARTATE-TRNA ASPARTYL-TRNA ASPRS LYSYL-TRNA PD000871: R148-K281	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					AMINOACYL-TRANSFER RNA SYNTHETASES CLASS II DM00328 |P52276|70-512: I44-K341 |P17242|19-464: I44-F318 |P47359|21-455: I44-K300 |P25345|34-490: I44-N317	BLAST_DOMO
					Aminoacyl-transfer RNA synthetases class-II signature 1: F242-E260	MOTIFS
4	7510245CD1	287	S49 S81 S194 S254 S277 T13 T112 T191	N219	B-Box-type zinc finger, protein interaction: N222-L263	HMMER_SMART
					Ring finger: C20-C186	HMMER_SMART
					B-box zinc finger: N222-L263	HMMER_PFAM
					Zinc finger, C3HC4 type (RING finger): C20-S49, T182-C186	HMMER_PFAM
					Zinc finger, C3HC4 type (RING finger), signature: L14-E60 E161-S194	PROFILESCAN
					Zinc finger, C3HC4 type (RING finger) IPB001841: C35-V44	BLIMPS_BLOCKS
					B-box zinc finger signature PR01406: K235-S252, S254-Q268	BLIMPS_PRINTS
					RFP TRANSFORMING PROTEIN DM02346 |P14373|61-366: R192-K272 |P19474|59-337: R193-K272	BLAST_DOMO
					ZINC FINGER, C3HC4 TYPE, DM00063 |A43906|137-187: E16-W48 F181-S189 |I49642|6-56: L14-W48 F181-F190	BLAST_DOMO
					Zinc finger, C3HC4 type (RING finger), signature: C35-V44	MOTIFS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
5	7511072CD1	90	S2 S75 T43 T55		signal_cleavage: M1-A60	SPSCAN
					RNA recognition motif: K7-A82	HMMER_SMART
					RNA recognition motif, RRM, RBD: L8-G74	HMMER_PFAM
					RNA-binding region RNP-1 (RNA recognition motif) IPB000504: L8-V35	BLIMPS_BLOCKS
					Eukaryotic putative RNA-binding region RNP-1 signature: F26-G76	PROFILESCAN
					RIBONUCLEOPROTEIN REPEAT DM00012 |P98179|1-83: M1-G74 |P38159|3-84: G6-G74 |S46286|34-117: S2-G74 |B49418|3-84: G6-G74	BLAST_DOMO
					Eukaryotic putative RNA-binding region RNP-1 signature: R47-F54	MOTIFS
6	7512247CD1	470	S127 S161 S237 S279 T59 T103 T105 T400		signal_cleavage: M1-L20	SPSCAN
					tRNA synthetase class II core domain: G62-D207	HMMER_PFAM
					Histidyl-tRNA synthetase: N54-H470	HMMER_TIGRFAM
					hisS_second: histidyl-tRNA synthetase: F55-V465	HMMER_TIGRFAM
					WHEP-TRS domain IPB000738: T63-L78, D266-H275	BLIMPS_BLOCKS
					Aminoacyl-transfer RNA synthetases class-II signatures: M142-S201	PROFILESCAN
					SYNTHETASE AMINOACYL-TRNA HISTIDYL-TRNA PROTEIN LIGASE ATP-BINDING BIOSYNTHESIS HISTIDINE-TRNA HISRS KINASE PD025567: L114-M221	BLAST_PRODOM

Table 3

<table>
<tr><th>SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>Amino Acid Residues</th><th>Potential Phosphorylation Sites</th><th>Potential Glycosylation Sites</th><th>Signature Sequences, Domains and Motifs</th><th>Analytical Methods and Databases</th></tr>
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|P12081|64-257: D65-G259</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>HISTIDINE-TRNA LIGASE DM03615
|P49590|260-505: L260-G468
|P12081|259-504: L260-G468</td><td>BLAST_DOMO</td></tr>
<tr><td>7</td><td>7512244CD1</td><td>445</td><td>S131 S169 S423 S430 T54 T60 T71 T164 T199 T202 T329 Y100</td><td>N350</td><td>signal_cleavage: M1-A16</td><td>SPSCAN</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>Rel homology domain (RHD): G18-N186</td><td>HMMER_PFAM</td></tr>
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</table>

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					P65 SUBUNIT TRANSCRIPTION FACTOR NUCLEAR NF-KAPPA-B DNA-BINDING REGULATION ACTIVATOR PROTEIN PD022771: P240-S445	BLAST_PRODOM
					PROTEIN NUCLEAR PHOSPHORYLATION FACTOR TRANSCRIPTION REGULATION ACTIVATOR DNA-BINDING NF-KAPPA-B SUBUNIT PD002738: G18-E147, V72-N186, Y152-R198	BLAST_PRODOM
					FACTOR NUCLEAR PROTEIN TRANSCRIPTION REGULATION ACTIVATOR DNA-BINDING PHOSPHORYLATION HOMOLOG P65 PD002729: P19-D94	BLAST_PRODOM
					TRANSCRIPTION FACTOR P65 NUCLEAR NF-KAPPA-B SUBUNIT DNA-BINDING REGULATION ACTIVATOR PROTEIN PD036449: T199-A239	BLAST_PRODOM
					TRANSFORMING; REL; DM06957|Q04206|375-550: Q269-S445	BLAST_DOMO
					REL DM00785 |Q04865|11-343: P19-K195 |Q04206|12-345: E12-K195, D151-P240, P287-P309 |P98152|18-352: E12-N186, V163-P244, P275-P289, P285-P309, A302-P322	BLAST_DOMO
					NF-kappa-B/Rel/dorsal domain signature: F34-G40	MOTIFS
8	7512177CD1	375	S81 S87 S117 S146 S261 S344 T95 T99 T162	N349 N359	B-Box-type zinc finger, protein interaction domain: S88-A129	HMMER_SMART
					Ring finger: C15-C55	HMMER_SMART
					Domain in SPLA and the Ryanodine Receptor: S261-H374	HMMER_SMART

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SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					SPRY domain: S261-C372	HMMER_PFAM
					B-box zinc finger: S88-A129	HMMER_PFAM
					Zinc finger, C3HC4 type (RING finger): C15-C55	HMMER_PFAM
					Zinc finger, C3HC4 type (RING finger) IPB001841: C30-F39	BLIMPS_BLOCKS
					B-box zinc finger signature PR01406: K101-S118, E120-E134	BLIMPS_PRINTS
					Butyrophylin C-terminal DUF signature PR01407: F208-R225, R225-P242, A246-V270, Q323-D347, I354-C372	BLIMPS_PRINTS
					RFP TRANSFORMING PROTEIN DM01944 |P18892|355-477: S261-C371 |P19474|339-465: S261-F370 |P14373|368-492: S261-F370	BLAST_DOMO
9	7511867CD1	96	S19 S44	N57	SYNTHETASE LYSYL-TRNA LYSINE-TRNA LIGASE LYSRS AMINOACYL-TRNA PROTEIN BIOSYNTHESIS ATP-BINDING CYTOPLASMIC PD013348: E15-N74	BLAST_PRODOM
10	7512866CD1	430	S19 S93 S162 S190 S218 S333 S390 T44 T53 T112 T126 T150 T328 T349 T399 Y408	N247 N334	signal_cleavage: M1-R21	SPSCAN
					Signal Peptide: M1-Q20	HMMER
					Krueppel associated box: V43-T103	HMMER_SMART
					Zinc finger: Y320-H342, Y208-H230, Y348-H370, Y376-H398, Y404-H426, Y292-H314, Y236-H258, Y180-H202, N152-H174, Y264-H286	HMMER_SMART
					KRAB box: V43-G83	HMMER_PFAM

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SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Zinc finger, C2H2 type: Y320-H342, Y208-H230, Y376-H398, Y404-H426, Y348-H370, Y236-H258, Y292-H314, Y180-H202, N152-H174, Y264-H286	HMMER_PFAM
					Zinc finger, C2H2 type IPB000822: C322-C350	BLIMPS_BLOCKS
					C2H2-type zinc finger signature PR00048: P347-S360, V391-G400	BLIMPS_PRINTS
					PROTEIN ZINC-FINGER METAL-BINDING DNA-BINDING ZINC FINGER PATERNALLY EXPRESSED ZN-FINGER PW1 PD017719: R144-F385, G176-H426, G288-R428	BLAST_PRODOM
					ZINC-FINGER METAL-BINDING DNA-BINDING PROTEIN FINGER ZINC NUCLEAR REPEAT TRANSCRIPTION REGULATION PD001562: V43-E104	BLAST_PRODOM
					ZINC-FINGER DNA-BINDING PROTEIN METAL-BINDING NUCLEAR ZINC FINGER TRANSCRIPTION REGULATION REPEAT PD000072: E153-C213, R178-C241, K206-C269, K234-C297, Y292-C353, P319-C381, K346-C409, K374-Q429	BLAST_PRODOM
					PROTEIN ZINC-FINGER METAL-BINDI. PD00066: H338-C350	BLIMPS_PRODOM
					PROTEIN ZINC FINGER ZINC-FINGER METAL-BINDING NUCLEAR PD01066: F45-G83	BLIMPS_PRODOM
					KRAB BOX DOMAIN DM00605 |P52736|1-72: V43-P114 |I48689|11-85: V43-P114 |Q05481|10-83: G41-Q97 |P51786|24-86: K40-V101	BLAST_DOMO
					ATP/GTP-binding site motif A (P-loop): G181-S188	MOTIFS

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SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Zinc finger, C2H2 type, domain: C154-H174, C182-H202, C210-H230, C238-H258, C266-H286, C294-H314, C322-H342, C350-H370, C378-H398, C406-H426	MOTIFS
11	7512572CD1	268	S133 S192 S224 T100		signal_cleavage: M1-G20	SPSCAN
					Signal Peptide: M1-A19, M1-G20, M1-G24, M1-L29	HMMER
					Ribosomal protein S5, C-terminal domain: I132-G205	HMMER_PFAM
					Ribosomal protein S5 IPB000851: T127-K173	BLIMPS_BLOCKS
					RIBOSOMAL PROTEIN S5 DM00432 |Q10234|215-375: F94-G205 |P33759|134-296: G97-G205	BLAST_DOMO
12	7512518CD1	79	S2 T63 T67	N61	RNA-BINDING PROTEIN RNA-BINDING REPEAT NUCLEAR TAR TRANSACTIVATION RESPONSIVE PROTAMINE1 PD153298: M1-T30	BLAST_PRODOM
					DOUBLE-STRANDED RNA BINDING DOMAIN DM01051|S27945|1-92: S18-D70	BLAST_DOMO
13	7512669CD1	285	S21 S103 T30 T57 T173 T217 T238 T241 Y172		ubiE/COQ5 methyltransferase family IPB000339: L75-M86	BLIMPS_BLOCKS
					ARGININE N-METHYL-TRANSFERASE TRANSFERASE METHYL-TRANSFERASE PROTEIN INTERFERON RECEPTOR 1-BOUND ALTERNATIVE SPLICING PD011237: E168-Q272	BLAST_PRODOM
					ARGININE NMETHYL-TRANSFERASE TRANSFERASE METHYL-TRANSFERASE PROTEIN INTERFERON RECEPTOR 1-BOUND ALTERNATIVE SPLICING PD009274: E27-N65	BLAST_PRODOM
14	7512706CD1	110	S73		signal_cleavage: M1-G24	SPSCAN
					Signal Peptide: M1-G24	HMMER

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					METHIONYL-TRNA SYNTHETASE EC 6.1.1.10 METHIONINE-TRNA LIGASE METRS AMINOACYL-TRNA PROTEIN BIOSYNTHESIS ATP-BINDING PD124857: M1-Q93	BLAST_PRODOM
15	7512872CD1	287	S22 S71 S231 S275 S283 T89 T90 T108 T115 T174 T200 T210 Y225	N198	signal_cleavage: M1-G24	SPSCAN
					RNA recognition motif: K79-K151, K164-K236	HMMER_SMART
					RNA recognition motif, RRM, RBD: M80-P150, I165-V237	HMMER_PFAM
					RNA-binding region RNP-1 (RNA recognition motif) IPB000504: M80-C107	BLIMPS_BLOCKS
					Eukaryotic putative RNA-binding region RNP-1 signature: K91-R152	PROFILESCAN
					PROTEIN NUCLEAR RIBONUCLEOPROTEIN PD02784: K72-T108, T115-K157, E160-G205	BLIMPS_PRODOM
					RIBONUCLEOPROTEIN D HETEROGENEOUS NUCLEAR PROTEIN NUCLEOPROTEIN RNA-BINDING ELEMENT HNRNP-BINDING PD010372: K224-Y287	BLAST_PRODOM
					P37 AUF1 RNA-BINDING PROTEIN PD057907: M1-M80	BLAST_PRODOM
					RIBONUCLEOPROTEIN REPEAT DM00012 |I49070|145-227: K72-L143, K157-S240 |A44192|155-237: K72-L143, K157-S240 |I49070|61-143: E74-M156, K164-V228 |A44192|71-153: E74-M156, K164-V228	BLAST_DOMO

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Eukaryotic putative RNA-binding region RNP-1 signature: R119-F126, R204-F211	MOTIFS
16	7512925CD1	948	S23 S42 S43 S114 S120 S172 S173 S199 S215 S376 S556 S576 S585 S600 S632 S649 S665 S718 S756 S791 S819 T31 T266 T277 T278 T407 T644 T747 T850	N400	PROTEIN ZINC-FINGER CHROMOSOME III DNA-BINDING NUCLEAR PD039832: M790-R945	BLAST_PRODOM
					PROTEIN ZINC FINGER KIAA0661 R05D3.4 CHROMOSOME III DNA-BINDING NUCLEAR PD039832: M790-R945	BLAST_PRODOM
					PROTEIN ZINC-FINGER PD134879: V627-K943	BLAST_PRODOM
17	7512927CD1	94	S10		THREONYL-TRNA SYNTHETASE, CYTOPLASMIC EC 6.1.1.3 THREONINE-TRNA LIGASE THRRS AMINOACYL-TRNA SYNTHETASE PROTEIN BIOSYNTHESIS ATP-BINDING PD093205: M13-E46	BLAST_PRODOM
					THREONINE-TRNA LIGASE DM04411|P26639|1-67: M13-E46	BLAST_DOMO
18	7512943CD1	189		N83	signal_cleavage: M1-C26	SPSCAN
					Signal Peptide: M1-G28	HMMER
19	7512955CD1	96	S28 S50		signal_cleavage: M1-L15	SPSCAN
					Signal Peptide: M1-P16	HMMER
					KOW motif, found in ribosomal proteins: L56-G89	HMMER_PFAM
					Ribosomal protein L24 IPB000302: L56-N92	BLIMPS_BLOCKS
					Ribosomal protein L24 signature: G59-V76	MOTIFS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
20	7512950CD1	1395	S26 S97 S104 S168 S338 S356 S419 S518 S578 S725 S758 S766 S1059 S1146 S1169 S1224 S1276 S1363 T12 T51 T138 T248 T397 T525 T555 T561 T588 T739 T828 T908 T1003 T1086 T1307 T1376 T1388	N24 N224 N246 N290 N791	CPSF A subunit region: Q1022-N1329	HMMER_PFAM
					FACTOR CLEAVAGE POLYADENYLATION SPECIFICITY SUBUNIT C CPSF NUCLEAR PD025689: M1-V895 F892-K1109 Y941-E947	BLAST_PRODOM
					PROTEIN FACTOR DNA CLEAVAGE POLYADENYLATION SPECIFICITY SUBUNIT NUCLEAR REPAIR XERODERMA PD005035: K1111-E1387	BLAST_PRODOM
					ATP/GTP-binding site motif A (P-loop): G429-S436	MOTIFS
21	7517524CD1	147	S116 S120 S134 Y42		BRCA1 C Terminus (BRCT) domain: E10-P93	HMMER_PFAM
22	7519735CD1	1069	S110 S157 S171 S208 S213 S222 S235 S238 S243	N340 N375 N1018	breast cancer carboxy-terminal domain: E603-K684, G703-N779, C96-K173, T858-Q937, S970-Q1059	HMMER_SMART
			S255 S322 S621 S719 S795 S827 S845 S846 S959		BRCA1 C Terminus (BRCT) domain: C96-P183, G703-R790, E603-H694, L861-L947, E10-P93, S970-F1068	HMMER_PFAM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
			S963 S1008 S1020 S1065 T137 T174		Octamer-binding transcription factor signature PR00029: A496-Q511, Q520-Q535	BLIMPS_PRINTS
			T240 T267 T337 T628 T645 T858 T862 T872 T911 T990 Y42 Y181 Y945		PROTEIN REPEAT SIGNAL PRECURSOR PRION GLYCOPROTEIN NUCLEAR GPI ANCHOR BRAIN MAJOR PD001091: S390-Q619, P396-P596, Q406-Q590	BLAST_PRODOM
					SIMILARITY TO DROSOPHILA HOMEOTIC GENE REGULATOR BRM PD106589: Q405-F605	BLAST_PRODOM
					SEX; DETERMINING; DM08509 SEX-DETERMINING REGION Y PROTEIN: |Q05738|69-394: N383-D595, Q361-Q580, I359-D595, N375-D595, W331-Q577 SEX-DETERMINING PROTEIN: |S35565|69-395: N383-D595, Q361-Q580, I359-D595, N375-D595, W331-Q577	BLAST_DOMO
					SERUM RESPONSE FACTOR DNA-BINDING DOMAIN DM00242|P11746|16-285: E189-E225, E192-Q233, E371-Q508, M400-Q578, Q437-Q590, L419-Q587, Q451-Q590, Q480-P596	BLAST_DOMO
					ETS DNA-BINDING DOMAIN DM04000|B53225|1-770: C169-D254, N343-Q379, T395-H495, L401-Q438, Q432-P570, P439-H556, P490-A597, Q519-D595, H358-Q413, Q402-A515, Q500-V568, H476-P59, Q493-H5566, E198-P484, L419-A458	BLAST_DOMO
23	7517880CD1	261	S23 S235 S237 T75 T243 T247		C4 zinc finger in nuclear hormone receptor: N102-E173	HMMER_SMART
					Zinc finger, C4 type (two domains): E103-E178	HMMER_PFAM
					C4-type steroid receptor zinc finger IPB001628: C105-V133, Y139-V175	BLIMPS_BLOCKS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Nuclear hormones receptors DNA-binding region signature: S36-T151	PROFILESCAN
					C4-type steroid receptor zinc finger signature PR00047: C105-S121, S121-G136, R154-L162, L162-M170	BLIMPS_PRINTS
					Vitamin D receptor signature PR00350: C105-S121, C122-C141	BLIMPS_PRINTS
					NUCLEAR RECEPTOR TRANSCRIPTION REGULATION DNA-BINDING PROTEIN ZINC-FINGER ORPHAN LXR-ALPHA RLD1 PD029416: M8-E103	BLAST_PRODOM
					PROTEIN RECEPTOR NUCLEAR TRANSCRIPTION REGULATION DNA-BINDING ZINC-FINGER HORMONE FAMILY MULTIGENE PD000035: L104-M170	BLAST_PRODOM
					NUCLEAR RECEPTOR TRANSCRIPTION REGULATION DNA-BINDING PROTEIN ZINC-FINGER ORPHAN LXR-ALPHA RLD1 PD032282: R171-P211	BLAST_PRODOM
					RECEPTOR NUCLEAR PROTEIN TRANSCRIPTION REGULATION DNA-BINDING ZINC-FINGER ORPHAN UBIQUITOUSLY EXPRESSED PD010811: Q212-T243	BLAST_PRODOM
					NUCLEAR HORMONES RECEPTORS DNA-BINDING REGION DM00047 |I38975|68-375: T75-T243 |P55055|77-389: P95-P215, Q212-T243 |I49021|68-374: P95-Q229 |P34021|253-580: K93-P200, E191-D256	BLAST_DOMO

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Type-1 copper (blue) proteins signature: G136-M149	MOTIFS
					Nuclear hormones receptors DNA-binding region signature: C105-R131	MOTIFS
					Zinc finger, C2H2 type, domain: C122-H142	MOTIFS
24	4524267CD1	125	S58 S62 S78 S84		Zinc finger, C3HC4 type (RING finger): C15-C42	HMMER_PFAM
					Zinc finger, C3HC4 type (RING finger) IPB001841: C30-L39	BLIMPS_BLOCKS
					Zinc finger, C3HC4 type (RING finger), signature: F9-N69	PROFILESCAN
					Zinc finger, C3HC4 type (RING finger), signature: C30-L39	MOTIFS
25	7481640CD1	550	S131 S169 S240 S276 S487 S507 S528 S535 T54 T60 T71 T164 T305 T308 T434 Y100	N456	Ig-like, plexins, transcription factors: E193-L289	HMMER_SMART
					Rel homology domain (RHD): G18-N186	HMMER_PFAM
					IPT/TIG domain: L194-L289	HMMER_PFAM
					NF-kappa-B/Rel/dorsal family IPB000451: P19-G48, H58-G98, D185-F228, W233-Y257	BLIMPS_BLOCKS
					Transcription factor NF-KB signature PR00057: E25-S42, L175-P189, G208-F228, F239-Y257, L280-D294	BLIMPS_PRINTS
					PROTEIN NUCLEAR PHOSPHORYLATION FACTOR TRANSCRIPTION REGULATION ACTIVATOR DNA-BINDING NF-KAPPA-B SUBUNIT PD002738: G18-E147, Q119-R304, I312-P344	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					P65 SUBUNIT TRANSCRIPTION FACTOR NUCLEAR NF-KAPPA-B DNA-BINDING REGULATION ACTIVATOR PROTEIN PD022771: P346-S550	BLAST_PRODOM
					FACTOR NUCLEAR PROTEIN TRANSCRIPTION REGULATION ACTIVATOR DNA-BINDING PHOSPHORYLATION HOMOLOG P65 PD002729: P19-D94	BLAST_PRODOM
					TRANSCRIPTION FACTOR P65 NUCLEAR NF-KAPPA-B SUBUNIT DNA-BINDING REGULATION ACTIVATOR PROTEIN PD036449: T305-A345	BLAST_PRODOM
					REL DM00785 |Q04206|12-345: E12-P346 |Q04865|11-343: P19-T339 |P98152|18-352: E12-Q347 |P01126|9-340: S17-P346	BLAST_DOMO
					NF-kappa-B/Rel/dorsal domain signature: F34-G40	MOTIFS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
26	7515465CD1	1043	S51 S56 S85 S89 S120 S124 S126 S127 S138 S144 S271 S314 S357 S372 S402 S525 S566 S593 S701 S909 S911 S988 S1001 T10 T33 T47 T137 T170 T180 T299 T376 T467 T679 T714 T726 T762 T826 T979 Y710	N235 N400 N491	F-box domain: S210-L258	HMMER_PFAM
					ATP/GTP-binding site motif A (P-loop): A463-T470	MOTIFS
27	4912891CD1	486	S92 S252 S289 S360 S405 S406 S433 S461 S465 T23 T42 T51 T61 T88 T164 T248 T255 T274 Y230	N144 N264 N292	krueppel associated box: V41-N100	HMMER_SMART
					zinc finger: Y365-H387, H309-H331, F281-H303, Y197-H219, Y337-H359, F421-H443, F449-H471, Y393-H415, C118-R138	HMMER_SMART
					KRAB box: V41-E81	HMMER_PFAM
					Zinc finger, C2H2 type: Y365-H387, H309-H331, F449-H471, Y337-H359, F281-H303, F421-H443, Y393-H415, Y197-H219, C118-Q140	HMMER_PFAM
					Zinc finger, C2H2 type IPB000822: C311-C339	BLIMPS_BLOCKS
					C2H2-type zinc finger signature PR00048: P364-S377	BLIMPS_PRINTS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					PROTEIN ZINC-FINGER METAL-BINDING PD00066: H467-C479	BLIMPS_PRODOM
					PROTEIN ZINC FINGER ZINC-FINGER METAL-BINDING PD01066: F43-E81	BLIMPS_PRODOM
					PROTEIN ZINC FINGER METAL BINDING DNA BINDING ZINC FINGER PATERNALLY EXPRESSED ZN-FINGER PW1 PD017719: G277-E478 G305-C479 G361-K480 L224-F458 P196-L431 Y197-H387	BLAST_PRODOM
					ZINC FINGER DNA BINDING PROTEIN METAL BINDING NUCLEAR ZINC FINGER TRANSCRIPTION REGULATION REPEAT PD000072: K279-C342 K307-C370 K363-C426 K335-C398 K391-C454	BLAST_PRODOM
					KRAB BOX DOMAIN DM00605|P51523|5-79: Q38-G96	BLAST_DOMO
					ZINC FINGER, C2H2 TYPE, DOMAIN DM00002|P08042|314-358: C314-H359 C426-H471	BLAST_DOMO
					ZINC FINGER, C2H2 TYPE, DOMAIN DM00002|Q05481|789-829: Q301-E341 R357-E397 R441-D481	BLAST_DOMO
					ZINC FINGER, C2H2 TYPE, DOMAIN DM00002|Q05481|831-885: C426-D481	BLAST_DOMO
					Zinc_Finger_C2h2: C199-H219, C283-H303, C311-H331, C339-H359, C367-H387, C395-H415, C423-H443, C451-H471	MOTIFS
28	70552759CD1	364	S53 S73 S77 S98 S133 S157 S196 S204 S242 S247 S258 S314 T36 T45 T216	N194	krueppel associated box: V35-P88	HMMER_SMART

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					zinc finger: F332-H354, Y304-H326, Q248-H270, Y276-H298	HMMER_SMART
					KRAB box: V35-A75	HMMER_PFAM
					Zinc finger, C2H2 type: F332-H354, Q248-H270, Y276-H298, Y304-H326	HMMER_PFAM
					Zinc finger, C2H2 type IPB000822: C250-C278	BLIMPS_BLOCKS
					C2H2-type zinc finger signature PR00048: P303-K316, L347-G356	BLIMPS_PRINTS
					Vertebrate metallothionein signature PR00860: C306-C315	BLIMPS_PRINTS
					PROTEIN ZINC-FINGER METAL-BINDING PD00066: H266-C278	BLIMPS_PRODOM
					PROTEIN ZINC FINGER ZINC-FINGER METAL-BINDING PD01066: F37-A75	BLIMPS_PRODOM
					ZINC FINGER PROTEIN 133 TRANSCRIPTION REGULATION DNA BINDING REPRESSOR ZINC FINGER METAL BINDING NUCLEAR PD101192: A75-V215	BLAST_PRODOM
					ZINC FINGER DNA BINDING PROTEIN METAL BINDING NUCLEAR ZINC FINGER TRANSCRIPTION REGULATION REPEAT PD000072: K274-C337	BLAST_PRODOM
					ZINC FINGER, C2H2 TYPE, DOMAIN DM00002|P08042|314-358: C253-H298	BLAST_DOMO
					KRAB BOX DOMAIN DM00605|Q03923|1-75: L32-C84	BLAST_DOMO
					Zinc_Finger_C2h2: C250-H270, C278-H298, C306-H326, C334-H354	MOTIFS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
29	7524194CD1	697	S14 S72 S126 S146 S246 S316 S359 S461 S468 S559 S664 S689 T161 T241 T374 T460	N502 N563 N587 N615 N616 N643 N644	leucine rich region: S33-A142	HMMER_SMART
					zinc finger: Y577-H599, H549-H571, Y605-H627, S521-H543, Y633-H655, Y661-H683	HMMER_SMART
					SCAN domain: P31-V123	HMMER_PFAM
					Zinc finger, C2H2 type: Y577-H599, S521-H543, H549-H571, Y633-H655, Y661-H683, Y605-H627	HMMER_PFAM
					Zinc finger, C2H2 type IPB000822: C523-C551	BLIMPS_BLOCKS
					C2H2-type zinc finger signature PR00048: P576-S589	BLIMPS_PRINTS
					PROTEIN ZINC-FINGER METAL-BINDING PD00066: H595-C607	BLIMPS_PRODOM
					PROTEIN ZINC FINGER ZINC-FINGER METAL-BINDING PD01066: R214-T252	BLIMPS_PRODOM
					PROTEIN ZINC FINGER METAL BINDING DNA BINDING NUCLEAR ZINC FINGER PATERNALLY EXPRESSED ZN FINGER PD035222: E154-E198 E518-G601	BLAST_PRODOM
					PROTEIN ZINC FINGER METAL BINDING DNA BINDING ZINC FINGER PATERNALLY EXPRESSED ZN FINGER PW1 PD017719: C523-Q680 G545-H683	BLAST_PRODOM
					ZINC FINGER DNA BINDING PROTEIN METAL BINDING NUCLEAR ZINC FINGER TRANSCRIPTION REGULATION REPEAT PD000072: K519-C582 K547-C610	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					ZINC FINGER METAL BINDING ZINC FINGER PROTEIN DNA BINDING NUCLEAR TRANSCRIPTION REGULATION REPEAT PD004640: K16-S146	BLAST_PRODOM
					P18; FINGER; ZINC; DM03735|P49910|45-90: E34-L79	BLAST_DOMO
					P18; DM03974|S37648|57-189: L80-N187	BLAST_DOMO
					P18; DM03974|P49910|92-271: L80-P160 K416-K481	BLAST_DOMO
					ZINC FINGER, C2H2 TYPE, DOMAIN DM00002|P08042|314-358: C582-H627 C526-H571 C554-H599	BLAST_DOMO
					Zinc_Finger_C2h2: C523-H543, C551-H571, C579-H599, C607-H627, C635-H655, C663-H683	MOTIFS
30	7523973CD1	802	S4, S70, S100, S108, S118, S139, S156, S161, S315, S321, S345, S352, S377, S385, S406, S439, S493, S496, S625, S718, S747, S752, S790, S796, T50, T130, T143, T153, T454, T465, T476, T507, T518, T548, T708, T762, Y401	N152, N671	PAS domain: Q72-A140, S211-A271	HMMER_SMART

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					AH; MINDED; SINGLE; HYPOXIA; DM03763 |P30561|27-503:S9-L135 A114-T339 G433-S474 |I38972|18-524:T5-P484 |P35869|28-512:S9-L135 R148-T339 |P05709|1-564:M1-L137 M216-L322	BLAST_DOMO
31	7524553CD1	580	S9, S95, S111, S120, S143, S153, S157, S171, S187, S237, S246, S275, S288, S356, S394, S436, S546, T85, T227, T273, T337, Y526	N118, N201, N202, N392, N393	SAND domain: D293-P374	HMMER_PFAM
					Sp100 domain: G47-D150	HMMER_PFAM
					Bromodomain: C469-R558	HMMER_PFAM
					PHD-finger: E405-K449	HMMER_PFAM
					bromo domain: C469-A572	HMMER_SMART
					SAND domain: P301-P374	HMMER_SMART
					PHD-zinc finger: E405-R447	HMMER_SMART
					Bromodomain IPB001487: L508-Y547	BLIMPS_BLOCKS
					PROTEIN NUCLEAR SP100 SUPPRESSIN DEAF1 RELATED REGULATOR AUTOANTIGEN TRANSCRIPTIONAL BROMODOMAIN PD004075: E294-R365	BLAST_PRODOM
					SP100 PROTEIN NUCLEAR AUTOANTIGEN BROMODOMAIN DNA BINDING ALTERNATIVE SPLICING SPECKLED ANTIGEN PD005359: S31-K145	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					NUCLEAR PROTEIN BROMODOMAIN DNA BINDING ALTERNATIVE SPLICING AUTOANTIGEN LYSP100 LYMPOID RESTRICTED HOMOLOG PD017631: V177-D293	BLAST_PRODOM
					NUCLEAR PROTEIN BROMODOMAIN DNA BINDING LYSP100 LYMPHOID RESTRICTED HOMOLOG OF SP100 ALTERNATIVE PD021223: R401-L474	BLAST_PRODOM
					SP10_HUMAN 1 NUCLEAR AUTOANTIGEN SP-100: DM06712 |P23497|28-237:P28-G175 |Q99388|1-207:S31-S157 C421-P452	BLAST_DOMO
32	2666675CD1	183	T8	N38	Cytidine and deoxycytidylate deaminases zinc-binding region signature IPB002125: H54-I60, Y79-C88	BLIMPS_BLOCKS
					DEAMINASE ZINC HYDROLASE BIOSYNT PD01010: H54-I63, Q75-C85	BLIMPS_PRODOM
					EDITING; MRNA; APOLIPOPROTEIN; APOB; DM04741 |A53853|1-236:R26-H151 |P51908|1-228:R26-H151 |P41238|1-235:F13-H151	BLAST_DOMO
					Cytidine and deoxycytidylate deaminases zinc-binding region signature: H54-L92	MOTIFS
33	4032169CD1	456	S84, S156, S268, S408, T9, T18, T114, T365, T421, T449, Y370, Y426	N40	KRAB box: V8-G48	HMMER_PFAM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					Zinc finger, C2H2 type: C118-H140, Y146-H168, Y174-H196, Y202-H224, F230-H252, Y258-H280, Y286-H308, Y314-H336, F342-H364, Y370-H392, Y398-H420, Y426-R448	HMMER_PFAM
					kruеppel associated box: V8-I68	HMMER_SMART
					zinc finger: C118-H140, Y146-H168, Y174-H196, Y202-H224, F230-H252, Y258-H280, Y286-H308, Y314-H336, F342-H364, Y370-H392, Y398-H420, Y426-K446	HMMER_SMART
					Zinc finger, C2H2 type IPB000822: C288-C316	BLIMPS_BLOCKS
					C2H2-type zinc finger signature PR00048: P257-K270, L329-G338	BLIMPS_PRINTS
					PROTEIN ZINC FINGER ZINC-FINGER METAL-BINDING NU PD01066: L10-G48	BLIMPS_PRODOM
					PROTEIN ZINC-FINGER METAL-BINDING PD00066: L10-G48, H388-C400	BLIMPS_PRODOM
					ZINC-FINGER DNA-BINDING METAL-BINDING NUCLEAR TRANSCRIPTION REPEAT REGULATION FACTOR PD000072: K256-C319, K368-C431, K284-C347, K312-C375, K228-C291, K340-C403, R170-C235, K200-C263, K142-C207	BLAST_PRODOM
					ZINC FINGER ZINC-FINGER DNA-BINDING METAL-BINDING NUCLEAR TRANSCRIPTION REGULATION REPEAT FIS PD001562: V8-E70	BLAST_PRODOM
					PROTEIN ZINC FINGER METAL-BINDING DNA-BINDING ZINC FINGER PATERNALLY EXPRESSED ZN-FINGER PW1 PD017719: Y146-L385, G198-I436, E104-H336, C120-F351, K222-T442, K250-K451, G282-R452, R91-F267	BLAST_PRODOM

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					ZINC-FINGER NUCLEAR DNA-BINDING METAL-BINDING CLONE: 2610305D13 KRAZ1 FULL-LENGTH PD033163: M246-K368	BLAST_PRODOM
					KRAB BOX DOMAIN DM00605|I48689|11-85: Q5-E72 DM00605|P51523|5-79: Q5-P79 DM00605|P52736|1-72: V8-W67 DM00605|P52738|3-77: Q5-I68	BLAST_DOMO
					ATP/GTP-binding site motif A (P-loop): A399-S406	MOTIFS
					Zinc finger, C2H2 type, domain: C118-H140, C120-H140, C148-H168, C176-H196, C204-H224, C232-H252, C260-H280, C288-H308, C316-H336, C344-H364, C372-H392, C400-H420	MOTIFS
34	7524926CD1	410	S17, S79, S104, S150, S206, S211, S215, S228, T4, T118, T263, T270, Y345	N74, N116, N203, N254	Ets-domain: Q238-H321	HMMER_PFAM
					Sterile alpha motif (SAM)/Pointed domain: P114-S198	HMMER_PFAM
					erythroblast transformation specific domain: Q238-I323	HMMER_SMART
					SAM / Pointed domain: P114-S198	HMMER_SMART
					Ets-domain IPB000418: G235-G265, E266-F318	BLIMPS_BLOCKS
					Ets-domain signatures and profile: Y221-M269, G265-D319	PROFILESCAN
					ETS domain signature PR00454: I239-S252, T263-E281, R282-Y300, Y301-D319	BLIMPS_PRINTS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
					TRANSCRIPTION FACTOR DNA-BINDING NUCLEAR PROTO-ONCOGENE SPLICING REGULATION ALTERNATIVE PHOSPHORYLATION PD002757:P112-S228 D219-T227	BLAST_PRODOM
					TRANSCRIPTION FACTOR DNA-BINDING NUCLEAR REGULATION ACTIVATOR SPLICING ALTERNATIVE PROTO-ONCOGENE PD000803: I239-F318	BLAST_PRODOM
					TRANSCRIPTION DNA-BINDING NUCLEAR REGULATION ACTIVATOR FLI-1 PROTO-ONCOGENE ALTERNATIVE FACTOR PD010980: D319-Y410	BLAST_PRODOM
					TRANSCRIPTION DNA-BINDING NUCLEAR REGULATION ACTIVATOR FLI-1 PROTO-ONCOGENE FACTOR INTEGRATION PD013261: M1-G75	BLAST_PRODOM
					ETS-DOMAIN DM00281|P11308|286-384: L231-H330 DM00281|Q01543|273-371: L231-H330 DM00281|S60754|302-400: L231-H330	BLAST_DOMO
					ERG; TRANSFORMING; TRANSCRIPTION; DM05182|Q01543|76-271: S76-P220	BLAST_DOMO
					Ets-domain signature 1: L241-L249	MOTIFS
					Ets-domain signature 2: K285-Y300	MOTIFS

Table 3

SEQ ID NO:	Incyte Polypeptide ID	Amino Acid Residues	Potential Phosphorylation Sites	Potential Glycosylation Sites	Signature Sequences, Domains and Motifs	Analytical Methods and Databases
35	7509493CD1	835	S30, S41, S120, S121, S157, S196, S201, S209, S219, S263, S290, S305, S311, S402, S403, S447, S460, S577, S617, S659, S664, S794, S813, T54, T65, T182, T267, T276, T343, T451, T569, T621, T647, T705, T779	N216, N400, N811	Zinc finger, C2H2 type: I227-H249, I411-H433, T54-H76, A581-H603	HMMER_PFAM
					Zinc finger: I227-H249 I411-H433, T54-H76, A581-H603, L780-H806	HMMER_SMART
					Zinc finger, C2H2 type IPB000822: C56-E84	BLIMPS_BLOCKS
					ZINC-FINGER METAL-BINDING DNA-BINDING NUCLEAR WIZL WIZS FIS CDNA PLACE1006445 KIAA1221 PD156543:A50-P835 S52-G114, L6-G350	BLAST_PRODOM
					Zinc finger, C2H2 type, domain: C56-H76, C229-H249, C413-H433, C582-H603, C583-H603	MOTIFS

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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37/7511811CB1 3012	1-280, 1-773, 1-3008, 9-653, 10-734, 13-630, 15-355, 24-637, 30-367, 30-826, 31-274, 35-350, 37-669, 44-851, 45-293, 46-323, 58-277, 58-454, 59-604, 63-158, 101-249, 151-393, 160-554, 185-404, 185-427, 185-591, 210-762, 227-590, 234-657, 275-553, 290-669, 376-675, 391-964, 404-604, 412-780, 415-958, 432-615, 433-923, 497-938, 512-873, 528-1216, 528-1307, 528-1308, 560-1038, 604-1298, 621-891, 641-1329, 722-1596, 832-1204, 844-1421, 854-1082, 865-1373, 902-1381, 926-1408, 929-1366, 978-1251, 994-1606, 996-1595, 1002-1430, 1024-1548, 1048-1666, 1056-1326, 1097-1530, 1104-1646, 1126-1571, 1134-1614, 1139-1632, 1140-1617, 1192-1757, 1211-1728, 1214-1699, 1236-1759, 1255-1757, 1256-1704, 1271-1756, 1283-1527, 1367-1643, 1376-1572, 1402-1650, 1441-1611, 1465-1734, 1465-1752, 1692-1957, 1757-1992, 1757-2037, 1757-2178, 1757-2186, 1757-2396, 1762-2294, 1775-2347, 1792-2056, 1797-2075, 1810-2195, 1813-2074, 1828-2384, 1843-2457, 1863-2542, 1894-2447, 1894-2529, 1895-2059, 1904-2262, 1917-2547, 1949-2207, 1955-2145,
	1956-2193, 1963-2471, 1968-2166, 1978-2223, 1990-2484, 2006-2426, 2019-2419, 2020-2502, 2024-2729, 2028-2634, 2032-2636, 2037-2277, 2041-2307, 2072-2445, 2081-2173, 2093-2330, 2096-2683, 2102-2678, 2105-2323, 2111-2494, 2134-2763, 2143-2633, 2157-2779, 2161-2423, 2173-2452, 2187-2709, 2188-2429, 2188-2787, 2189-2421, 2190-2768, 2202-2612, 2203-2476, 2209-2443, 2209-2628, 2218-2732, 2220-2911, 2233-2454, 2236-2445, 2240-2498, 2241-2649, 2246-2511, 2256-2512, 2256-2988, 2261-2552, 2265-2434, 2266-2496, 2266-2873, 2270-2854, 2284-2989, 2285-2993, 2290-2403, 2313-2411, 2342-2790, 2344-2600, 2344-2610, 2344-2992, 2365-3008, 2374-2891, 2375-2952, 2381-2757, 2402-2889, 2412-2623, 2414-2612, 2418-2689, 2419-2619, 2419-2712, 2419-3010, 2430-2795, 2437-2868, 2440-2717, 2451-2660, 2454-2667, 2456-2715, 2467-2728, 2469-3010, 2472-2745, 2493-2889, 2497-3010, 2511-2929, 2517-3010, 2521-3010, 2525-3010, 2526-2993, 2527-3000, 2527-3010, 2528-3008, 2529-2991, 2531-3002, 2536-2994, 2545-3010, 2546-2994, 2552-2994, 2555-2994,
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
38/7511870CB1 2280	1-374, 1-449, 1-539, 1-674, 1-734, 6-232, 6-236, 6-237, 6-2269, 27-190, 27-245, 41-480, 246-716, 247-647, 247-870, 250-588, 258-555, 263-549, 263-624, 305-773, 403-692, 479-708, 479-739, 479-996, 484-916, 491-747, 607-1092, 653-1154, 663-1276, 664-918, 666-1236, 672-1162, 676-865, 702-1450, 711-979, 733-993, 733-1045, 744-1008, 818-1028, 849-1095, 872-1143, 877-1114, 877-1126, 877-1127, 877-1129, 877-1133, 877-1159, 960-1244, 996-1252, 1071-1266, 1133-1298, 1299-1563, 1303-1976, 1324-1449, 1324-1450, 1328-1905, 1355-1590, 1357-1638, 1370-1615, 1373-2048, 1377-2039, 1407-1676, 1423-1788, 1443-1957, 1455-1954, 1478-1783, 1491-1931, 1514-1974, 1515-1970, 1518-1798, 1518-1973, 1522-1974, 1526-1973, 1527-1971, 1549-1973, 1558-2046, 1561-1973, 1570-1803, 1575-1863, 1580-1835, 1585-1816, 1585-2186, 1596-1980, 1617-1974, 1621-1975, 1662-1891, 1662-1973, 1691-1928, 1693-1964, 1693-2039, 1694-1916, 1707-1999, 1712-1950, 1712-1961, 1720-2042, 1733-2265, 1768-2027, 1801-2268, 1837-2259, 1843-1973, 1846-1917, 1846-1973,
	1849-2105, 1850-2279, 1851-2280, 1876-2268, 1884-2131, 1886-2275, 1922-2039, 1935-2268, 1944-2277, 1946-2205, 2005-2261, 2039-2268, 2125-2280, 2184-2280
39/7510245CB1 2414	1-470, 27-638, 36-284, 36-477, 39-582, 40-320, 41-680, 43-534, 43-603, 43-604, 43-622, 44-710, 61-704, 73-677, 78-365, 78-543, 78-673, 86-721, 98-622, 98-2414, 115-398, 132-820, 138-401, 145-825, 203-811, 283-899, 321-718, 339-1009, 349-847, 349-968, 369-936, 376-723, 460-1161, 505-804, 518-1121, 530-1020, 556-1111, 569-1139, 585-1160, 597-1204, 660-1244, 677-1339, 679-1160, 695-1168, 695-1169, 738-1372, 741-1368, 782-995, 814-1436, 818-1296, 836-1365, 871-1051, 1058-1645, 1090-1396, 1174-1646, 1175-1322, 1195-1416, 1278-1772, 1408-1629, 1510-1769, 1545-1750, 1566-1865, 1616-1888, 1619-1865, 1635-1872, 1638-1877, 1649-2081, 1693-1892, 1695-1973, 1750-2038, 1750-2328, 1753-2414, 1768-2041
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	1089-1339, 1090-2027, 1092-1603, 1099-1389, 1101-1332, 1105-1304, 1106-1676, 1106-1756, 1113-1597, 1123-1864, 1127-1424, 1127-1652, 1128-1396, 1128-1632, 1130-1401, 1130-1478, 1130-1633, 1130-1925, 1131-1864, 1134-1393, 1148-1410, 1151-1540, 1153-1873, 1162-1376, 1166-1647, 1168-1406, 1169-1636, 1169-1690, 1178-1454, 1178-1480, 1185-1436, 1187-1397, 1193-1448, 1195-1794, 1196-1490, 1199-1800, 1200-1437, 1200-1440, 1200-1441, 1200-1456, 1200-1461, 1200-1472, 1200-1599, 1200-1714, 1200-1770, 1200-1822, 1200-1828, 1200-1861, 1201-1868, 1202-1485, 1206-1602, 1215-1530, 1216-1458, 1217-1532, 1218-1463, 1220-1416, 1223-1815, 1227-1356, 1229-1765, 1230-1523, 1231-2095, 1244-1894, 1254-1924, 1256-1544, 1267-1803, 1267-1862, 1270-1847, 1271-1404, 1274-1411, 1274-1496, 1274-1872, 1274-1938, 1277-1919, 1278-1547, 1281-1586, 1284-1576, 1285-1468, 1285-1544, 1285-1550, 1286-1596, 1292-1851, 1293-1477, 1294-2091, 1305-1553, 1306-1841, 1306-2094, 1309-2095, 1310-1720, 1311-1495, 1315-1429, 1315-1585, 1322-1602, 1325-1545,

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	1609-1921, 1614-1886, 1615-1860, 1617-1854, 1622-1858, 1625-1901, 1626-1878, 1631-1870, 1637-2012, 1641-1912, 1657-2012, 1659-2150, 1662-2206, 1663-1922, 1664-2165, 1665-1966, 1666-2012, 1671-2221, 1672-1927, 1672-1939, 1672-1982, 1672-2150, 1690-2233, 1691-2182, 1692-1965, 1693-1956, 1697-2228, 1698-2233, 1699-1908, 1699-2012, 1700-2233, 1704-2233, 1709-2233, 1711-1966, 1712-2233, 1715-2212, 1722-1826, 1736-2165, 1742-2175, 1744-2207, 1750-2228, 1751-2092, 1753-2003, 1753-2052, 1754-2219, 1754-2222, 1755-2227, 1766-2233, 1770-2221, 1770-2233, 1772-2221, 1774-2208, 1774-2233, 1782-2221, 1782-2224, 1782-2230, 1783-2233, 1785-2220, 1785-2230, 1786-2229, 1788-2225, 1789-2221, 1790-2226, 1792-2221, 1792-2233, 1793-2215, 1795-2233, 1796-2221, 1798-2083, 1798-2151, 1798-2171, 1799-2041, 1800-2083, 1801-2221, 1805-1953, 1805-2050, 1805-2225, 1806-2222, 1808-2220, 1808-2221, 1810-2233, 1813-2233, 1814-2062, 1815-2221, 1816-2226, 1820-2210, 1820-2224, 1821-2222, 1822-2215, 1824-2221, 1826-2221, 1830-2219, 1833-2092,
	1835-2026, 1837-2221, 1838-2124, 1838-2233, 1839-2086, 1842-2221, 1843-2221, 1848-2219, 1850-2219, 1851-2215, 1856-2097, 1856-2225, 1859-2222, 1861-2099, 1862-2142, 1867-2233, 1868-2093, 1868-2098, 1869-2233, 1877-2221, 1879-2163, 1879-2221, 1880-2131, 1880-2233, 1881-2124, 1885-2156, 1890-2132, 1897-2222, 1899-2106, 1907-2222, 1914-2221, 1922-2221, 1930-2222, 1931-2192, 1931-2233, 1932-2221, 1933-2222, 1936-2233, 1943-2209, 1955-2230, 1962-2223, 1964-2225, 1966-2230, 1967-2232, 1969-2219, 1974-2221, 1977-2146, 1977-2225, 1980-2217, 1988-2233, 1992-2225, 2000-2222, 2014-2233, 2020-2233, 2023-2221, 2024-2227, 2031-2215, 2031-2219, 2031-2220, 2038-2219, 2039-2220, 2043-2203, 2046-2222, 2047-2222, 2049-2233, 2058-2221, 2058-2222, 2062-2233, 2068-2219, 2079-2224, 2079-2233, 2086-2221, 2098-2222, 2116-2211
43/7512177CB1 1744	1-235, 1-392, 1-418, 1-458, 1-460, 1-496, 1-524, 1-535, 1-539, 1-574, 1-629, 1-645, 1-1744, 28-592, 28-639, 28-747, 95-582, 162-820, 487-966, 647-882, 647-1146, 699-1354, 720-1209, 723-1305, 728-1242, 758-1205, 946-1397, 952-1447, 995-1447, 997-1447, 1100-1398, 1144-1447

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	959-1196, 963-1268, 966-1209, 966-1226, 966-1244, 966-1260, 969-1190, 969-1195, 971-1239, 974-1158, 975-1256, 983-1241, 983-1337, 995-1337, 1007-1562, 1007-1585, 1008-1337, 1013-1279, 1013-1437, 1013-1559, 1018-1155, 1018-1271, 1018-1273, 1018-1285, 1025-1315, 1046-1295, 1052-1306, 1062-1694, 1068-1385, 1069-1324, 1070-1335, 1075-1265, 1077-1300, 1079-1196, 1079-1285, 1079-1334, 1081-1364, 1084-1358, 1086-1627, 1087-1350, 1087-1426, 1087-1587, 1088-1335, 1097-1356, 1099-1371, 1101-1590, 1102-1395, 1103-1375, 1105-1175, 1114-1354, 1114-1370, 1121-1392, 1121-1690, 1123-1408, 1124-1327, 1130-1938, 1135-1381, 1138-1442, 1146-1386, 1146-1392, 1146-1412, 1146-1444, 1147-1623, 1151-1408, 1151-1434, 1160-1404, 1171-1450, 1176-1457, 1185-1621, 1198-1593, 1203-1573, 1205-1696, 1215-1404, 1217-1917, 1218-1469, 1220-1503, 1221-1692, 1222-1455, 1224-1996, 1230-1365, 1230-1728, 1233-1471, 1235-1493, 1242-1522, 1250-1509, 1253-1613, 1255-1467, 1259-1458, 1259-1540, 1261-1438, 1264-1677, 1267-1691, 1284-1706, 1286-1553,
	1287-1995, 1288-1573, 1290-1585, 1293-1655, 1297-1443, 1304-1551, 1306-1915, 1315-1606, 1317-1558, 1319-1571, 1319-1574, 1319-1575, 1319-1576, 1319-1610, 1320-1557, 1331-1502, 1335-1737, 1336-1630, 1339-1641, 1348-1604, 1352-1605, 1354-1614, 1355-1608, 1356-1610, 1362-1537, 1368-1922, 1374-1630, 1377-1602, 1389-1635, 1391-1625, 1397-1603, 1401-1723, 1403-1629, 1415-1994, 1419-1675, 1421-1637, 1432-1658, 1432-1676, 1432-1942, 1432-1999, 1434-1546, 1438-2001, 1444-1657, 1447-2010, 1455-1684, 1459-1717, 1463-1984, 1465-1713, 1469-1682, 1469-1744, 1470-2010, 1475-1759, 1486-1753, 1492-2010, 1496-2008, 1499-1772, 1501-1793, 1501-2010, 1513-1769, 1518-1793, 1520-2010, 1523-2010, 1524-2010, 1527-1875, 1528-2010, 1533-2010, 1535-2010, 1536-2010, 1541-1799, 1542-1994, 1543-2010, 1545-1780, 1546-1842, 1548-1993, 1552-1980, 1554-1981, 1557-1992, 1564-1980, 1566-2010, 1567-1980, 1567-2010, 1569-2010, 1572-1997, 1572-1999, 1572-2010, 1576-1999, 1581-1986, 1582-1980, 1582-1994, 1583-1994, 1584-1994, 1587-1980, 1588-1991,

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	1752-1980, 1753-1991, 1753-2010, 1754-2005, 1754-2006, 1755-1980, 1756-1997, 1758-2008, 1760-2001, 1762-1991, 1770-1998, 1770-1999, 1771-1997, 1774-1999, 1779-1995, 1784-1997, 1796-1991, 1804-1994, 1808-2010, 1821-1980, 1826-1980, 1830-1980, 1832-1980, 1842-1980, 1848-2010, 1849-1994, 1851-1980, 1854-1980, 1862-1991, 1865-2009, 1874-1980, 1888-2010, 1890-1980, 1891-2010, 1897-2010, 1912-2010, 1922-1994
45/7512866CB1 2497	1-320, 1-395, 1-768, 1-1352, 2-232, 88-338, 132-931, 132-976, 223-653, 280-1073, 291-913, 542-1482, 570-1482, 579-1483, 595-1375, 652-1304, 653-1483, 655-1483, 667-951, 678-965, 718-980, 718-1317, 718-1353, 718-1528, 721-791, 731-1034, 731-1353, 734-791, 745-791, 745-848, 745-849, 745-871, 745-876, 752-881, 760-847, 760-881, 775-881, 823-878, 823-919, 835-932, 835-933, 835-955, 835-960, 835-1026, 837-965, 841-960, 842-878, 844-965, 920-1033, 948-1352, 1172-1285, 1184-1285, 1237-1285, 1249-1375, 1249-1380, 1250-1375, 1255-1375, 1255-1380, 1255-1446, 1280-1591, 1309-1379, 1322-1378, 1322-1379, 1334-1379, 1359-1605, 1366-1605, 1368-1743, 1411-1500, 1411-1507, 1424-1507, 1430-1507, 1431-1507, 1435-2028, 1526-2262, 1719-1922, 1983-2497
46/7512572CB1 1156	1-240, 1-259, 1-382, 1-1149, 21-252, 23-369, 29-322, 34-146, 34-387, 35-460, 45-318, 47-316, 47-320, 53-379, 76-460, 100-849, 100-862, 109-302, 160-424, 160-446, 164-444, 165-460, 166-420, 171-414, 177-322, 179-460, 180-428, 182-437, 186-407, 186-420, 186-422, 186-429, 186-430, 186-445, 186-447, 186-449, 186-457, 186-460, 187-420, 191-418, 191-454, 191-457, 192-418, 192-446, 193-445, 195-426, 195-427, 207-410, 242-392, 253-1108, 259-443, 261-358, 297-1108, 456-911, 461-662, 461-691, 461-710, 461-721, 461-738, 461-1022, 461-1108, 461-1115, 461-1139, 463-1077, 465-1047, 465-1110, 482-745, 483-765, 484-856, 485-746, 501-1070, 506-1154, 520-753, 523-1074, 530-1074, 542-770, 547-992, 550-786, 555-831, 559-1139, 563-833, 563-844, 563-1137, 566-1097, 577-819, 577-979, 577-1068, 577-1153, 578-1075, 583-843, 584-856, 587-857, 593-1154, 594-1118, 597-886, 597-934, 597-1095, 597-1111, 599-863, 599-889, 604-816, 604-881, 604-1053, 604-1081, 606-958, 607-1151, 613-760, 615-830, 615-888, 618-1064, 634-876, 641-1079,

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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47/7512518CB1 1460	1-278, 2-45, 62-191, 62-234, 65-290, 65-320, 65-1460, 81-360, 84-351, 85-357, 94-372, 95-305, 98-360, 105-204, 129-396, 131-396, 131-397, 136-396, 141-384, 178-389, 178-396, 307-573, 415-709, 434-690, 437-711, 511-697, 511-1086, 518-764, 547-969, 567-838, 570-1078, 607-807, 631-865, 631-979, 633-1264, 641-883, 656-1161, 659-969, 666-888, 668-1233, 687-927, 689-949, 698-945, 701-1328, 713-1335, 720-1169, 722-1216, 733-1083, 745-1179, 750-1371, 753-1314, 761-1344, 765-1416, 781-963, 817-1101, 823-1290, 831-1201, 836-1398, 837-1451, 841-1365, 870-1377, 877-1391, 877-1447, 887-1354, 890-1387, 908-1154, 913-1224, 918-1447, 927-1094, 927-1173, 929-1183, 931-1460, 938-1360, 938-1434, 956-1387, 959-1406, 961-1112, 977-1454, 987-1460, 988-1446, 990-1447, 992-1452, 998-1455, 1004-1257, 1012-1228, 1013-1266, 1013-1267, 1013-1281, 1013-1283, 1036-1452, 1050-1284, 1059-1446, 1060-1447, 1061-1455, 1074-1392, 1089-1453, 1097-1452, 1098-1457, 1100-1452, 1111-1439, 1112-1409, 1113-1445, 1129-1452, 1136-1313, 1155-1452,
	1157-1460, 1163-1454, 1166-1452, 1170-1455, 1173-1402, 1173-1452, 1184-1451, 1201-1452, 1209-1446, 1223-1452, 1224-1460, 1236-1424, 1245-1448, 1249-1447, 1263-1452, 1269-1460, 1294-1447, 1319-1452, 1329-1438, 1355-1446
48/7512669CB1 1168	1-269, 1-270, 3-216, 23-257, 23-503, 25-227, 25-259, 25-270, 25-271, 25-276, 25-304, 25-399, 26-341, 27-271, 27-285, 27-315, 27-342, 27-473, 28-93, 28-171, 28-247, 28-268, 28-284, 28-285, 28-288, 28-293, 28-308, 28-360, 28-497, 28-528, 28-563, 28-631, 28-1131, 29-286, 29-298, 29-301, 29-302, 29-309, 29-319, 29-325, 29-340, 29-388, 29-410, 29-416, 29-448, 29-473, 29-499, 29-517, 29-525, 29-531, 29-536, 29-543, 29-548, 29-582, 29-601, 30-278, 30-285, 30-316, 30-428, 30-611, 30-644, 31-104, 31-248, 31-257, 31-273, 31-283, 31-284, 31-286, 31-301, 31-313, 31-316, 31-320, 31-322, 31-336, 31-341, 31-350, 31-357, 31-558, 32-232, 32-238, 32-560, 32-607, 33-312, 34-278, 34-280, 34-334, 34-425, 34-475, 34-536, 35-146, 35-260, 36-263, 36-284, 36-318, 38-313, 38-320, 38-344, 40-330, 41-205, 41-269, 41-292, 41-293, 41-295, 41-303, 41-437, 41-471, 41-473, 41-489, 41-491, 41-493, 41-494, 41-499, 41-514, 41-522, 41-554, 42-512, 44-310, 44-325, 46-330, 47-454, 48-253, 48-283, 48-287, 48-291, 48-295, 48-297, 48-299, 48-316,

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Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	2094-2306, 2094-2613, 2095-2172, 2095-2323, 2095-2631, 2099-2774, 2106-2369, 2106-2674, 2107-2777, 2108-2361, 2109-2351, 2110-2356, 2110-2371, 2111-2794, 2112-2346, 2112-2406, 2112-2616, 2117-2393, 2119-2383, 2121-2681, 2121-2737, 2122-2696, 2125-2627, 2125-2654, 2125-2698, 2125-2783, 2126-2581, 2126-2641, 2126-2646, 2126-2676, 2126-2749, 2130-2763, 2131-2538, 2133-2786, 2135-2781, 2139-2703, 2144-2611, 2146-2626, 2146-2681, 2146-2794, 2155-2773, 2155-2785, 2156-2655, 2157-2388, 2158-2421, 2159-2794, 2168-2794, 2169-2794, 2173-2397, 2173-2755, 2173-2791, 2176-2389, 2176-2438, 2178-2456, 2179-2419, 2181-2794, 2185-2363, 2188-2780, 2189-2726, 2190-2430, 2192-2436, 2193-2442, 2195-2676, 2204-2794, 2205-2413, 2206-2447, 2206-2794, 2209-2413, 2209-2421, 2209-2429, 2209-2728, 2211-2463, 2212-2435, 2215-2794, 2216-2794, 2217-2794, 2218-2478, 2222-2404, 2228-2516, 2229-2381, 2229-2513, 2235-2461, 2235-2471, 2237-2472, 2237-2794, 2242-2428, 2250-2793, 2253-2657, 2253-2722, 2253-2739, 2255-2539, 2257-2534, 2257-2767,

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	3420-3614, 3420-3811, 3428-3798, 3428-3839, 3428-3840, 3429-3839, 3431-3668, 3431-3778, 3433-3693, 3436-3742, 3438-3840, 3440-3838, 3446-3812, 3458-3773, 3458-3840, 3460-3804, 3461-3804, 3475-3687, 3475-3840, 3476-3824, 3478-3839, 3483-3840, 3487-3840, 3492-3804, 3493-3774, 3493-3804, 3495-3804, 3497-3733, 3500-3760, 3501-3834, 3502-3834, 3502-3840, 3505-3803, 3510-3787, 3510-3838, 3510-3840, 3529-3766, 3537-3840, 3542-3761, 3542-3798, 3548-3781, 3548-3804, 3548-3833, 3551-3838, 3553-3804, 3563-3839, 3567-3838, 3570-3840, 3574-3771, 3598-3804, 3604-3840, 3614-3840, 3623-3840, 3654-3777, 3683-3840, 3694-3838, 3696-3798, 3696-3811, 3722-3838, 3728-3840
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	2055-2528, 2058-2472, 2060-2290, 2067-2320, 2070-2322, 2082-2345, 2083-2347, 2084-2324, 2084-2562, 2088-2318, 2097-2479, 2106-2361, 2106-2568, 2107-2568, 2110-2417, 2113-2281, 2113-2325, 2117-2361, 2117-2561, 2117-2568, 2119-2554, 2120-2397, 2121-2568, 2122-2568, 2123-2567, 2124-2485, 2130-2568, 2132-2568, 2133-2568, 2134-2568, 2139-2511, 2146-2555, 2150-2568, 2154-2568, 2155-2568, 2162-2426, 2162-2568, 2164-2568, 2167-2461, 2167-2568, 2168-2568, 2171-2568, 2176-2568, 2178-2568, 2183-2568, 2184-2558, 2184-2568, 2185-2568, 2187-2554, 2187-2568, 2188-2568, 2190-2568, 2191-2553, 2191-2568, 2192-2568, 2193-2568, 2195-2568, 2198-2568, 2204-2325, 2204-2568, 2211-2550, 2219-2568, 2220-2568, 2221-2384, 2223-2568, 2229-2568, 2252-2514, 2253-2568, 2255-2568, 2340-2434, 2403-2568, 2405-2568, 2414-2568, 2418-2501, 2418-2568, 2470-2568

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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	1570-1931, 1571-1933, 1576-1707, 1597-1869, 1608-1904, 1615-1900, 1624-1929, 1649-1933, 1655-1927, 1666-1927, 1744-1929, 1776-1920
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
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Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
	3034-3532, 3039-3511, 3040-3513, 3046-3448, 3052-3515, 3055-3512, 3056-3530, 3058-3515, 3059-3304, 3061-3286, 3061-3513, 3061-3515, 3064-3513, 3072-3322, 3075-3513, 3083-3323, 3083-3364, 3083-3447, 3083-3456, 3083-3469, 3083-3528, 3083-3529, 3089-3517, 3092-3512, 3093-3386, 3097-3512, 3101-3482, 3102-3503, 3106-3454, 3112-3513, 3114-3512, 3114-3529, 3122-3368, 3124-3513, 3132-3449, 3140-3512, 3146-3515, 3147-3512, 3147-3515, 3148-3514, 3149-3378, 3152-3512, 3152-3530, 3153-3507, 3153-3512, 3157-3515, 3160-3523, 3161-3465, 3163-3512, 3167-3520, 3193-3479, 3217-3477, 3223-3515, 3275-3437, 3287-3512, 3365-3509, 3384-3526, 3466-3529
57/7519735CB1 3732	1-850, 10-615, 24-3713, 26-131, 285-617, 307-368, 322-604, 368-582, 402-701, 402-861, 426-1103, 473-1157, 476-816, 525-909, 646-779, 665-926, 773-1355, 780-1239, 781-1248, 856-1247, 858-1409, 880-1409, 897-1096, 930-1158, 969-1247, 981-1832, 981-1836, 981-1874, 1011-1295, 1012-1321, 1036-1402, 1050-1356, 1119-1189, 1119-1317, 1119-1533, 1122-1533, 1158-1701, 1158-1841, 1244-1282, 1319-1985, 1322-1347, 1322-1348, 1322-1902, 1325-1348, 1486-1909, 1557-1994, 1558-2043, 1572-2154, 1575-1841, 1596-1639, 1605-2033, 1612-1639, 1613-1638, 1613-1639, 1615-1639, 1630-2100, 1660-1690, 1663-1690, 1667-1690, 1694-2089, 1702-1854, 1703-1854, 1709-2230, 1780-2379, 1797-2488, 1805-2080, 1890-2458, 1893-2095, 1896-2587, 1899-2501, 1915-2683, 1915-2686, 1944-2500, 1981-2249, 2027-2594, 2030-2171, 2036-2086, 2046-2170, 2062-2328, 2094-2171, 2098-2824, 2141-2890, 2155-2783, 2177-2793, 2194-2862, 2213-2797, 2215-2737, 2217-2636, 2220-2636, 2263-2938, 2268-2631, 2279-3066, 2280-2876, 2283-2663, 2285-2540, 2289-3044, 2323-2993,
	2332-2530, 2338-2621, 2342-2971, 2368-3066, 2372-3060, 2378-2969, 2383-2587, 2389-2982, 2401-2554, 2403-3115, 2406-3051, 2459-2974, 2460-3051, 2462-2523, 2464-3009, 2505-3126, 2510-2869, 2513-2743, 2513-2746, 2515-3155, 2523-3161, 2526-2979, 2532-2793, 2540-3178, 2542-3107, 2549-2798, 2549-3016, 2555-3133, 2559-3221, 2578-3137, 2589-2861, 2591-2853, 2596-2778, 2597-3195, 2611-3118, 2619-3041, 2654-3395, 2655-3165, 2655-3287, 2675-2937, 2697-3254, 2701-3137, 2766-3032, 2783-3034, 2790-3373, 2801-3307, 2822-3257, 2827-3103, 2853-3070, 2868-3228, 2870-3185, 2870-3313, 2871-3228, 2876-3228, 2935-3282, 2953-3504, 2989-3196, 3006-3420, 3016-3622, 3043-3711, 3054-3231, 3057-3240, 3092-3703, 3121-3700, 3134-3726, 3134-3727, 3135-3372, 3135-3388, 3135-3678, 3140-3726, 3143-3404, 3153-3720, 3186-3665, 3197-3698, 3216-3704, 3222-3695, 3230-3456, 3234-3732, 3239-3711, 3252-3715, 3255-3712, 3258-3715, 3259-3504, 3261-3486, 3261-3713, 3261-3715, 3264-3713, 3272-3522, 3275-3713, 3283-3523, 3283-3564, 3283-3728, 3283-3729,
	3283-3730, 3289-3717, 3292-3712, 3293-3586, 3297-3712, 3302-3703, 3306-3707, 3312-3713, 3314-3712, 3314-3729, 3322-3568, 3324-3713, 3340-3712, 3346-3715, 3347-3712, 3347-3715, 3348-3714, 3349-3578, 3352-3712, 3353-3707, 3353-3712, 3354-3712, 3357-3715, 3360-3711, 3363-3712, 3367-3712, 3376-3569, 3393-3679, 3417-3677, 3423-3715

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
58/7517880CB1 1535	1-627, 1-628, 1-640, 1-645, 2-1534, 196-984, 197-979, 533-1358, 838-1535
59/4524267CB1 794	1-253, 1-794, 25-584, 88-347, 146-408, 234-794, 340-584
60/7481640CB1 2446	1-226, 143-349, 143-387, 143-412, 143-413, 143-437, 143-481, 143-486, 143-491, 143-498, 143-527, 143-540, 143-552, 143-557, 143-562, 143-597, 143-605, 143-631, 143-639, 143-645, 143-648, 143-665, 143-687, 143-691, 143-695, 143-727, 143-739, 143-742, 143-743, 143-747, 143-749, 143-755, 143-756, 143-768, 143-782, 143-783, 143-787, 143-789, 143-798, 143-809, 143-816, 143-825, 143-829, 143-852, 143-873, 143-878, 143-884, 143-885, 143-887, 143-888, 143-948, 143-960, 143-985, 143-993, 143-1039, 144-787, 144-946, 145-777, 147-916, 149-757, 149-1083, 150-844, 150-879, 152-1038, 158-1059, 160-1105, 163-914, 185-994, 190-878, 197-1000, 197-1001, 220-875, 227-880, 229-848, 242-986, 258-1083, 262-866, 263-854, 277-966, 283-857, 288-887, 292-1083, 312-942, 320-972, 321-1118, 322-977, 324-1053, 330-1058, 333-1232, 336-1237, 337-941, 345-831, 359-941, 370-1069, 378-1116, 380-1099, 384-889, 393-1118, 395-851, 397-1150, 402-1202, 403-1048, 405-810, 405-851, 405-1006, 408-835, 409-1050, 410-1153, 411-1313, 418-1007, 430-958,
	433-961, 433-1038, 434-1053, 436-966, 438-969, 452-1195, 473-872, 473-1218, 485-1006, 486-1001, 486-1036, 497-871, 502-1040, 510-1205, 512-985, 520-980, 524-996, 524-1142, 525-1000, 527-916, 527-1398, 528-1168, 530-938, 533-923, 533-1048, 533-1167, 536-1023, 538-979, 539-1172, 540-1016, 540-1031, 540-1276, 552-1126, 552-1149, 563-1126, 571-1054, 583-1172, 592-1157, 593-1311, 594-1153, 595-1149, 599-1136, 600-1311, 603-1182, 604-1271, 605-1150, 609-1203, 612-1169, 620-1024, 621-1216, 622-1157, 630-1224, 638-975, 639-987, 643-1387, 659-1236, 659-1248, 662-1167, 664-1430, 670-1344, 674-1277, 682-1192, 683-1203, 683-1277, 692-1171, 723-1173, 728-1184, 729-1103, 729-1107, 737-1350, 739-1241, 741-1357, 744-1260, 751-1169, 751-1172, 754-1288, 758-1228, 762-1228, 782-1356, 785-1499, 803-1597, 804-1481, 805-1481, 806-1283, 812-1481, 820-1615, 821-1620, 822-1628, 824-1491, 824-1630, 869-1776, 909-1548, 915-1389, 920-1713, 922-1551, 938-1623, 941-1776, 945-1776, 949-1625, 973-1776, 975-1776, 976-1312, 978-1776,
	989-1770, 996-1776, 1001-1448, 1001-1454, 1008-1776, 1014-1776, 1015-1505, 1016-1513, 1018-1741, 1019-1776, 1022-1776, 1025-1776, 1031-1776, 1039-1776, 1042-1442, 1054-1776, 1056-1776, 1058-1776, 1062-1776, 1071-1776, 1085-1775, 1086-1776, 1090-1447, 1090-1701, 1090-1776, 1097-1776, 1103-1607, 1110-1499, 1113-1776, 1119-1613, 1119-1776, 1120-1776, 1121-1504, 1130-1776, 1137-1776, 1137-1826, 1138-1776, 1159-1776, 1165-1776, 1166-1591, 1169-1602, 1185-1776, 1195-1776, 1199-1505, 1203-1776, 1204-1776, 1217-1513, 1282-1776, 1301-1776, 1302-1773, 1305-1776, 1306-1776, 1314-1776, 1315-1776, 1321-1776, 1333-1776, 1335-1776, 1342-1776, 1379-1776, 1667-2446

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
61/7515465CB1 3748	1-437, 178-472, 178-894, 178-3741, 187-932, 190-589, 190-872, 193-604, 197-748, 213-686, 215-940, 246-779, 246-854, 257-281, 266-749, 266-810, 266-811, 266-1121, 270-874, 271-771, 345-915, 346-945, 391-846, 402-639, 402-3722, 438-642, 438-692, 438-875, 438-1088, 438-1127, 438-1317, 456-755, 468-1172, 527-1059, 660-836, 704-993, 749-942, 1012-1296, 1086-1277, 1137-1370, 1297-1584, 1297-1749, 1300-1379, 1300-1686, 1361-1667, 1361-1696, 1430-1492, 1435-1671, 1472-1752, 1483-1756, 1488-1740, 1506-1909, 1521-1793, 1521-2040, 1536-1803, 1574-1827, 1574-2115, 1583-1821, 1626-1926, 1632-1830, 1683-1983, 1696-1950, 1728-1980, 1728-2220, 1758-2021, 1765-2099, 1772-2011, 1772-2198, 1796-2034, 1818-2062, 1841-2146, 1846-2081, 1864-2137, 1900-2207, 1910-2159, 1958-2225, 2016-2517, 2024-2299, 2083-2612, 2108-2385, 2109-2363, 2181-2433, 2181-2608, 2183-2428, 2194-2464, 2196-2474, 2214-2678, 2224-2523, 2267-2540, 2276-2511, 2276-2522, 2276-2790, 2277-2574, 2280-2548, 2288-2527, 2303-2546, 2303-2595, 2306-2567, 2335-2583,
	2349-2444, 2358-2616, 2396-2676, 2397-2644, 2421-2680, 2442-2681, 2448-2752, 2449-2728, 2456-2688, 2470-2727, 2493-2857, 2497-2700, 2497-2796, 2522-2670, 2530-2761, 2530-3172, 2532-2852, 2548-2694, 2548-2982, 2564-2856, 2564-3161, 2602-2845, 2604-2839, 2610-2857, 2610-3161, 2613-2852, 2616-2877, 2628-2939, 2655-2872, 2656-2904, 2658-2937, 2666-3129, 2671-2985, 2672-2929, 2673-2945, 2687-2961, 2696-2968, 2701-2934, 2707-2942, 2707-2966, 2709-2944, 2719-2964, 2719-3345, 2720-3017, 2726-3039, 2743-2966, 2743-2983, 2754-3078, 2755-3038, 2773-3025, 2785-2994, 2785-3068, 2791-3066, 2801-3104, 2801-3228, 2802-3081, 2815-3056, 2817-3081, 2818-3072, 2826-3095, 2827-3085, 2828-2979, 2828-3107, 2836-3084, 2837-3084, 2850-3093, 2853-3118, 2857-3442, 2868-3176, 2871-3065, 2875-3075, 2875-3187, 2885-3353, 2912-3109, 2912-3194, 2913-3167, 2948-3217, 2948-3557, 2949-3232, 2956-3200, 2957-3241, 2963-3194, 2964-3168, 2965-3266, 2967-3245, 2997-3330, 3001-3635, 3006-3246, 3006-3283, 3007-3247, 3007-3250, 3007-3262, 3007-3281,
	3007-3348, 3009-3253, 3012-3330, 3019-3637, 3021-3664, 3024-3271, 3024-3274, 3044-3348, 3048-3315, 3054-3637, 3058-3218, 3059-3292, 3072-3735, 3075-3311, 3075-3637, 3085-3330, 3093-3267, 3093-3379, 3093-3391, 3093-3404, 3101-3707, 3109-3639, 3110-3241, 3111-3236, 3115-3428, 3123-3637, 3129-3748, 3130-3664, 3132-3632, 3151-3559, 3152-3392, 3155-3412, 3155-3637, 3156-3399, 3156-3637, 3161-3435, 3168-3637, 3171-3637, 3182-3650, 3188-3664, 3193-3649, 3199-3637, 3204-3454, 3204-3664, 3209-3652, 3220-3478, 3220-3486, 3220-3503, 3221-3637, 3231-3648, 3233-3535, 3243-3637, 3249-3664, 3249-3741, 3252-3741, 3255-3741, 3256-3556, 3256-3664, 3258-3729, 3259-3732, 3259-3735, 3269-3741, 3270-3741, 3272-3726, 3282-3659, 3282-3741, 3284-3570, 3286-3729, 3290-3478, 3290-3740, 3297-3597, 3297-3664, 3297-3733, 3299-3714, 3300-3555, 3306-3559, 3306-3635, 3307-3605, 3311-3726, 3312-3637, 3324-3632, 3324-3722, 3325-3535, 3333-3728, 3352-3569, 3352-3726, 3359-3639, 3362-3637, 3364-3630, 3364-3649, 3371-3728, 3378-3748, 3381-3637,
	3396-3648, 3423-3734, 3427-3637, 3462-3716

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
62/4912891CB1 1808	1-280, 1-543, 2-293, 27-507, 29-300, 34-298, 37-501, 39-631, 60-365, 174-323, 174-357, 174-766, 174-792, 175-776, 177-772, 191-456, 216-813, 220-811, 292-974, 309-643, 325-597, 325-900, 334-611, 407-1009, 425-1012, 612-927, 628-919, 628-930, 628-1096, 685-896, 702-887, 708-957, 711-1373, 724-1258, 747-1039, 769-803, 775-1372, 785-1027, 813-1475, 823-1337, 830-1469, 867-1445, 885-1499, 922-1543, 933-1338, 957-1045, 957-1147, 957-1202, 957-1213, 977-1389, 1000-1077, 1000-1082, 1000-1207, 1004-1045, 1036-1653, 1048-1662, 1078-1658, 1079-1513, 1080-1202, 1080-1213, 1091-1213, 1096-1392, 1104-1346, 1114-1743, 1125-1147, 1125-1381, 1136-1664, 1162-1369, 1163-1201, 1163-1207, 1167-1714, 1168-1207, 1171-1213, 1171-1222, 1171-1793, 1175-1222, 1176-1222, 1190-1695, 1206-1801, 1211-1411, 1248-1380, 1248-1381, 1255-1306, 1258-1795, 1259-1296, 1259-1381, 1274-1306, 1284-1369, 1284-1667, 1293-1381, 1337-1369, 1348-1609, 1348-1793, 1349-1763, 1354-1795, 1363-1795, 1365-1796, 1367-1793, 1383-1793, 1392-1793, 1414-1621,
	1420-1502, 1420-1627, 1422-1795, 1423-1497, 1424-1492, 1424-1621, 1434-1502, 1437-1730, 1460-1492, 1485-1738, 1485-1793, 1508-1549, 1508-1559, 1508-1576, 1527-1576, 1533-1801, 1536-1621, 1542-1791, 1583-1627, 1589-1787
63/70552759CB1 1706	1-579, 1-590, 1-670, 1-846, 57-351, 61-349, 70-627, 71-332, 71-453, 71-454, 73-453, 74-700, 75-224, 75-251, 75-301, 75-620, 75-641, 75-649, 75-674, 75-680, 75-704, 75-711, 75-813, 75-845, 83-687, 86-210, 86-225, 86-291, 86-347, 86-351, 86-584, 86-612, 86-614, 93-836, 94-739, 95-578, 102-814, 104-570, 104-760, 104-794, 113-757, 114-452, 123-753, 127-409, 127-685, 197-453, 209-862, 218-351, 218-514, 330-453, 360-862, 368-453, 467-1146, 489-890, 510-1196, 530-1197, 534-1196, 542-1126, 552-1117, 554-1264, 590-1222, 592-860, 599-1187, 655-1267, 656-983, 873-941, 873-947, 873-983, 873-1047, 873-1066, 873-1109, 873-1153, 873-1155, 874-931, 874-949, 874-982, 874-1067, 876-937, 876-948, 882-962, 882-984, 882-1067, 884-1067, 885-1155, 889-1155, 893-1319, 906-1067, 915-1155, 924-984, 928-984, 928-1155, 966-1067, 966-1068, 966-1235, 968-1155, 990-1652, 992-1415, 1013-1045, 1013-1070, 1013-1273, 1026-1155, 1033-1652, 1035-1067, 1041-1151, 1041-1235, 1041-1318, 1041-1323, 1053-1579, 1069-1656, 1070-1704, 1096-1429,
	1096-1706, 1123-1593, 1126-1235, 1126-1319, 1154-1504, 1154-1507, 1179-1269, 1192-1323, 1193-1582, 1194-1235, 1209-1319, 1251-1323, 1259-1316, 1259-1319, 1259-1357, 1281-1319, 1284-1319, 1284-1656, 1296-1323, 1296-1357, 1296-1490, 1353-1459, 1426-1702, 1427-1461, 1433-1490, 1437-1488, 1437-1543, 1461-1697, 1517-1572, 1595-1629, 1595-1696, 1607-1697, 1632-1704, 1641-1697, 1671-1697

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
64/7524194CB1 2995	1-722, 1-2976, 149-704, 395-860, 399-1038, 595-784, 595-844, 595-954, 595-1029, 595-1030, 595-1082, 595-1179, 595-1261, 595-1279, 595-1283, 595-1341, 595-1351, 598-1030, 604-1038, 698-979, 698-1517, 735-1289, 749-1228, 832-1294, 841-1866, 858-1302, 866-1409, 868-1832, 981-1372, 1123-1652, 1124-1410, 1148-1443, 1254-1707, 1254-1789, 1403-1749, 1414-1834, 1460-2334, 1594-2391, 1594-2485, 1616-2170, 1645-1826, 1645-1885, 1645-2228, 1695-1867, 1695-2271, 1713-1960, 1734-2028, 1826-1930, 1960-2410, 1962-2224, 1993-2227, 1994-2098, 2060-2588, 2069-2328, 2070-2311, 2076-2612, 2083-2588, 2094-2596, 2110-2731, 2125-2722, 2145-2415, 2145-2687, 2153-2368, 2154-2696, 2165-2676, 2170-2939, 2177-2565, 2177-2650, 2179-2464, 2196-2904, 2196-2987, 2196-2988, 2210-2849, 2214-2490, 2216-2547, 2226-2809, 2252-2951, 2268-2905, 2269-2899, 2275-2553, 2290-2921, 2309-2924, 2315-2940, 2323-2820, 2325-2703, 2326-2691, 2328-2756, 2332-2975, 2347-2615, 2347-2918, 2392-2915, 2410-2913, 2410-2930, 2425-2930, 2427-2952, 2434-2734,
	2481-2713, 2481-2744, 2489-2918, 2489-2919, 2497-2975, 2498-2975, 2499-2994, 2506-2978, 2510-2978, 2512-2976, 2516-2976, 2520-2993, 2524-2993, 2535-2995, 2537-2975, 2538-2982, 2547-2975, 2549-2981, 2552-2975, 2556-2975, 2557-2975, 2560-2812, 2560-2981, 2570-2975, 2591-2981, 2593-2975, 2614-2979, 2624-2976, 2638-2976, 2652-2939, 2673-2896, 2683-2975, 2691-2975, 2692-2980, 2729-2975
65/7523973CB1 3597	1-342, 1-455, 22-455, 51-2570, 52-451, 52-455, 91-451, 91-455, 92-441, 92-455, 94-455, 96-455, 97-455, 113-455, 245-455, 579-831, 1183-1470, 1183-1979, 1611-1836, 1751-2039, 1999-2252, 2151-3323, 2325-2576, 2605-2918, 2796-3071, 2809-3272, 2811-3235, 2820-3301, 2826-3137, 2830-3167, 2837-3072, 2837-3192, 2837-3296, 2846-3297, 2862-3275, 3004-3259, 3016-3273, 3049-3319, 3049-3321, 3049-3322, 3049-3597, 3050-3322, 3051-3322, 3055-3322, 3301-3597, 3324-3411, 3324-3441, 3324-3466, 3324-3509, 3324-3533, 3324-3547, 3324-3549, 3324-3570, 3324-3571, 3324-3575, 3324-3593, 3324-3595, 3324-3596, 3324-3597, 3325-3597, 3327-3546, 3327-3597, 3329-3597
66/7524553CB1 2226	1-208, 1-526, 8-255, 18-493, 25-309, 53-265, 59-336, 88-361, 178-483, 197-656, 197-875, 198-500, 198-704, 198-727, 222-614, 397-640, 500-725, 523-768, 523-801, 544-766, 544-1020, 564-889, 602-1350, 680-1279, 925-1238, 949-1188, 949-1380, 975-1332, 975-1770, 1094-1556, 1156-1384, 1244-1525, 1251-1499, 1252-1401, 1316-1536, 1376-1616, 1376-1982, 1403-1731, 1425-1550, 1430-1660, 1441-1700, 1441-1730, 1481-1751, 1512-1782, 1512-2076, 1590-1846, 1596-1859, 1615-2226, 1626-2108, 1633-1876, 1633-2023, 1645-1972, 1674-1929, 1676-1898
67/2666675CB1 1215	1-259, 1-588, 191-328, 191-349, 191-352, 191-395, 191-410, 191-517, 191-762, 192-475, 415-758, 652-920, 654-907, 667-1031, 673-932, 694-767

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
68/4032169CB1 3803	1-197, 1-218, 1-243, 1-395, 5-270, 65-498, 69-478, 95-322, 95-708, 191-419, 196-437, 274-773, 323-362, 327-737, 369-622, 384-857, 398-889, 460-821, 494-563, 494-792, 494-822, 494-878, 502-573, 503-631, 503-687, 503-718, 503-864, 523-573, 523-864, 523-963, 524-563, 524-573, 524-948, 526-888, 526-948, 531-573, 535-718, 559-926, 565-1032, 611-1032, 615-995, 619-900, 623-940, 653-718, 670-982, 690-996, 695-893, 696-1057, 706-1219, 746-940, 754-942, 754-956, 754-992, 754-1047, 754-1060, 754-1116, 754-1131, 754-1200, 754-1213, 755-983, 755-1116, 755-1138, 771-877, 771-940, 771-1026, 771-1067, 771-1162, 774-1080, 775-970, 775-1067, 775-1138, 776-1200, 778-960, 778-1138, 786-1160, 791-1060, 817-1284, 825-1293, 830-928, 830-976, 830-1054, 830-1138, 830-1157, 830-1178, 830-1193, 830-1225, 838-877, 838-941, 838-973, 838-975, 838-1013, 838-1039, 838-1253, 838-1296, 841-1287, 846-1112, 851-1013, 853-1230, 860-1162, 862-1061, 867-1371, 881-1013, 897-1343, 904-1318, 913-1213, 913-1227, 920-1128, 920-1393, 922-989,
	922-1025, 922-1057, 922-1067, 922-1180, 922-1228, 922-1308, 922-1380, 930-1194, 937-1293, 944-1227, 946-1138, 954-1296, 968-1276, 983-1452, 985-1308, 991-1479, 997-1293, 1003-1212, 1004-1392, 1005-1276, 1006-1465, 1007-1415, 1019-1278, 1019-1455, 1028-1067, 1028-1237, 1028-1268, 1028-1296, 1028-1318, 1043-1293, 1072-1500, 1086-1499, 1091-1189, 1091-1265, 1091-1293, 1091-1305, 1091-1306, 1091-1376, 1091-1461, 1091-1465, 1091-1500, 1091-1516, 1103-1364, 1103-1412, 1103-1461, 1110-1423, 1111-1445, 1114-1338, 1122-1469, 1127-1444, 1147-1465, 1153-1461, 1165-1487, 1171-1417, 1172-1517, 1173-1444, 1175-1528, 1183-1446, 1183-1517, 1196-1500, 1198-1402, 1207-1477, 1210-1709, 1211-1758, 1244-1444, 1244-1517, 1251-1757, 1258-1293, 1258-1460, 1258-1500, 1258-1527, 1259-1461, 1280-1461, 1282-1416, 1302-1500, 1324-1464, 1333-1461, 1340-1500, 1341-1461, 1342-1517, 1355-1517, 1363-1413, 1365-1500, 1366-1665, 1425-1459, 1425-1517, 1426-1461, 1426-1477, 1427-1460, 1433-1517, 1457-1829, 1618-2112, 1637-1916, 1650-2047,
	1686-1934, 1777-2203, 1842-2139, 1856-2126, 1856-2294, 1937-2168, 2147-2429, 2151-2418, 2161-2458, 2182-2658, 2231-2537, 2234-2507, 2286-2856, 2292-2528, 2333-2891, 2404-2801, 2426-2697, 2439-2675, 2440-2796, 2470-2999, 2532-2950, 2541-2764, 2599-2806, 2663-2925, 2663-2951, 2732-3140, 2732-3282, 2761-2950, 2791-2950, 2791-3157, 2800-2950, 2995-3187, 3010-3225, 3010-3332, 3038-3283, 3087-3326, 3103-3361, 3108-3408, 3117-3345, 3212-3773, 3213-3348, 3213-3767, 3224-3408, 3254-3797, 3260-3540, 3307-3567, 3439-3634, 3559-3794, 3590-3777
69/7524926CB1 1263	1-615, 539-1263, 540-1262, 565-1263

Table 4

Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length	Sequence Fragments
70/7509493CB1 2880	1-447, 1-469, 1-517, 1-646, 204-417, 204-710, 237-959, 346-485, 355-617, 392-480, 407-713, 413-480, 417-480, 418-480, 422-480, 429-480, 438-480, 590-953, 606-894, 606-2880, 691-1278, 756-1403, 854-1363, 854-1556, 854-1594, 860-1540, 889-1009, 889-1010, 889-1031, 889-1042, 889-1047, 890-1047, 894-1461, 898-1031, 902-1047, 903-988, 903-1022, 903-1047, 903-1763, 905-1002, 909-1391, 910-1002, 910-1022, 910-1023, 910-1031, 911-982, 911-999, 931-1042, 932-999, 932-1031, 934-1689, 936-1031, 937-1147, 938-1047, 939-1031, 948-1202, 948-1238, 953-1269, 961-1031, 961-1047, 974-1240, 975-1031, 993-1031, 994-1254, 994-1652, 1005-1811, 1043-1336, 1057-1561, 1082-1190, 1090-1342, 1132-1784, 1164-1285, 1206-1480, 1206-1491, 1224-1436, 1301-1605, 1303-1690, 1359-2074, 1360-2150, 1365-2085, 1373-1583, 1391-1583, 1394-1802, 1413-1787, 1413-2109, 1432-1574, 1432-1952, 1432-1960, 1432-2042, 1435-1941, 1436-1562, 1436-2132, 1440-2150, 1441-1504, 1441-1510, 1441-1594, 1441-1599, 1441-1602, 1441-1691, 1442-2151, 1455-1599,
	1461-1599, 1477-2151, 1484-1553, 1484-1554, 1484-1575, 1484-1583, 1486-1599, 1488-1583, 1489-1599, 1490-2067, 1491-1583, 1500-1599, 1502-1939, 1506-1599, 1512-1583, 1513-1583, 1527-1599, 1546-1599, 1557-1599, 1571-1719, 1593-1792, 1593-1803, 1610-1791, 1613-2214, 1614-2324, 1617-2408, 1636-2357, 1643-2115, 1644-2263, 1656-1916, 1657-2151, 1664-2141, 1671-2141, 1676-2104, 1678-2104, 1679-2104, 1679-2141, 1681-2104, 1683-2141, 1686-2104, 1699-2104, 1701-2141, 1706-2104, 1706-2141, 1707-2104, 1711-2104, 1729-2104, 1729-2141, 1757-2017, 1776-2409, 1784-2104, 1787-2124, 1788-2104, 1794-1943, 1794-2104, 1798-2151, 1807-2104, 1808-2063, 1824-2104, 1839-2426, 1842-2151, 1844-2104, 1847-2156, 1875-2151, 1879-2104, 1885-2141, 1890-2466, 1894-2394, 1898-2395, 1914-2104, 1924-2028, 1926-2141, 1939-2150, 1944-2105, 1972-2084, 1972-2093, 1988-2104, 1994-2064, 1994-2071, 1994-2072, 1994-2093, 1995-2234, 2001-2093, 2019-2093, 2022-2093, 2033-2093, 2037-2093, 2055-2114, 2056-2093, 2067-2093, 2106-2678, 2151-2483, 2152-2463,
	2153-2660, 2157-2766, 2160-2409, 2160-2454, 2160-2482, 2160-2536, 2160-2701, 2160-2833, 2162-2542, 2184-2446, 2191-2494, 2197-2463, 2201-2587, 2273-2464, 2273-2473, 2273-2481, 2273-2581, 2273-2756, 2273-2792, 2274-2518, 2274-2720, 2276-2478, 2276-2880

Table 5

Polynucleotide SEQ ID NO:	Incyte Project ID:	Representative Library
36	7511189CB1	TLYJINT01
37	7511811CB1	COLNNOT07
38	7511870CB1	KERANOT01
39	7510245CB1	SINTFER02
40	7511072CB1	PROSTUT12
41	7512247CB1	PROSTUS23
42	7512244CB1	BRSTNOT04
43	7512177CB1	BRAVUNT02
44	7511867CB1	SPLNNOT04
45	7512866CB1	PLACNOT02
46	7512572CB1	SPLNNOT04
47	7512518CB1	NERDTDN03
48	7512669CB1	BRAIUNT01
49	7512706CB1	LUNGNOT23
50	7512872CB1	BRAITUT08
51	7512925CB1	SPLNTUE01
52	7512927CB1	PANCTUT02
53	7512943CB1	THYMNOT04
54	7512955CB1	FIBPFEN06
55	7512950CB1	BRACNOK02
56	7517524CB1	ESOGTME01
57	7519735CB1	CORPNOT02
59	4524267CB1	HNT2TXT01
60	7481640CB1	SMCANOT01
61	7515465CB1	BRSTNOT02
62	4912891CB1	THYMDIT01
63	70552759CB1	BMARTXE01
64	7524194CB1	BLADNOT03
65	7523973CB1	SCORNON02
66	7524553CB1	SPLNNOT04
67	2666675CB1	TLYMNOT08
68	4032169CB1	BRAINOT23
70	7509493CB1	OVARNOE02

Table 6

Library	Vector	Library Description
BLADNOT03	pINCY	Library was constructed using RNA isolated from bladder tissue removed from an 80-year-old Caucasian female during a radical cystectomy and lymph node excision. Pathology for the associated tumor tissue indicated grade 3 invasive transitional cell carcinoma. Patient history included malignant neoplasm of the uterus, atherosclerosis, and atrial fibrillation. Family history included acute renal failure, osteoarthritis, and atherosclerosis.
BMARTXE01	pINCY	This 5' biased random primed library was constructed using RNA isolated from treated SH-SY5Y cells derived from a metastatic bone marrow neuroblastoma, removed from a 4-year-old Caucasian female (Schering AG). The medium was MEM/HAM'S F12 with 10% fetal calf serum. After reaching about 80% confluency cells were treated with 6-Hydroxydopamine (6-OHDA) at 100 microM for 8 hours.
BRACNOK02	PSPORT1	This amplified and normalized library was constructed using RNA isolated from posterior cingulate tissue removed from an 85-year-old Caucasian female who died from myocardial infarction and retroperitoneal hemorrhage. Pathology indicated atherosclerosis, moderate to severe, involving the circle of Willis, middle cerebral, basilar and vertebral arteries; infarction, remote, left dentate nucleus; and amyloid plaque deposition consistent with age. There was mild to moderate leptomeningeal fibrosis, especially over the convexity of the frontal lobe. There was mild generalized atrophy involving all lobes. The white matter was mildly thinned. Cortical thickness in the temporal lobes, both maximal and minimal, was slightly reduced. The substantia nigra pars compacta appeared mildly depigmented. Patient history included COPD, hypertension, and recurrent deep venous thrombosis. 6.4 million independent clones from this amplified library were normalized in one round using conditions adapted from Soares et al., PNAS
		(1994) 91:9228-9232 and Bonaldo et al., Genome Research 6 (1996):791.
BRAINOT23	pINCY	Library was constructed using RNA isolated from right temporal lobe tissue removed from a 45-year-old Black male during a brain lobectomy. Pathology for the associated tumor tissue indicated dysembryoplastic neuroepithelial tumor of the right temporal lobe. The right temporal region dura was consistent with calcifying pseudotumor of the neuraxis. The patient presented with convulsive intractable epilepsy, partial epilepsy, and memory disturbance. Patient history included obesity, meningitis, backache, unspecified sleep apnea, acute stressreaction, acquired knee deformity, and chronic sinusitis. Family history included obesity, benign hypertension, cirrhosis of the liver, alcohol abuse, hyperlipidemia,cerebrovascular disease, and type II diabetes.
BRAITUT08	pINCY	Library was constructed using RNA isolated from brain tumor tissue removed from the left frontal lobe of a 47-year-old Caucasian male during excision of cerebral meningeal tissue. Pathology indicated grade 4 fibrillary astrocytoma with focal tumoral radionecrosis. Patient history included cerebrovascular disease, deficiency anemia, hyperlipidemia, epilepsy, and tobacco use. Family history included cerebrovascular disease and a malignant prostate neoplasm.

Table 6

Library	Vector	Library Description
BRAIUNT01	pINCY	Library was constructed using RNA isolated from SK-N-MC, a neuroepithelioma cell line (ATCC HTB-10) derived from a 14-year-old Caucasian female with neuroepithelioma, with metastasis to the supra-orbital area.
BRAVUNT02	PSPORT1	Library was constructed using pooled RNA isolated from separate populations of unstimulated astrocytes.
BRSTNOT02	PSPORT1	Library was constructed using RNA isolated from diseased breast tissue removed from a 55-year-old Caucasian female during a unilateral extended simple mastectomy. Pathology indicated proliferative fibrocysytic changes characterized by apocrine metaplasia, sclerosing adenosis, cyst formation, and ductal hyperplasia without atypia. Pathology for the associated tumor tissue indicated an invasive grade 4 mammary adenocarcinoma. Patient history included atrial tachycardia and a benign neoplasm. Family history included cardiovascular and cerebrovascular disease.
BRSTNOT04	PSPORT1	Library was constructed using RNA isolated from breast tissue removed from a 62-year-old East Indian female during a unilateral extended simple mastectomy. Pathology for the associated tumor tissue indicated an invasive grade 3 ductal carcinoma. Patient history included benign hypertension, hyperlipidemia, and hematuria. Family history included cerebrovascular and cardiovascular disease, hyperlipidemia, and liver cancer.
COLNNOT07	PSPORT1	Library was constructed using RNA isolated from colon tissue removed from a 60-year-old Caucasian male during a left hemicolectomy.
CORPNOT02	pINCY	Library was constructed using RNA isolated from diseased corpus callosum tissue removed from the brain of a 74-year-old Caucasian male who died from Alzheimer's disease.
ESOGTME01	PSPORT	This 5' biased random primed library was constructed using RNA isolated from esophageal tissue removed from a 53-year-old Caucasian male during a partial esophagectomy, proximal gastrectomy, and regional lymph node biopsy. Pathology indicated no significant abnormality in the non-neoplastic esophagus. Pathology for the matched tumor tissue indicated invasive grade 4 (of 4) adenocarcinoma, forming a sessile mass situated in the lower esophagus, 2 cm from the gastroesophageal junction and 7 cm from the proximal margin. The tumor invaded through the muscularis propria into the adventitial soft tissue. Metastatic carcinoma was identified in 2 of 5 paragastric lymph nodes with perinodal extension. The patient presented with dysphagia. Patient history included membranous nephritis, hyperlipidemia, benign hypertension, and anxiety state. Previous surgeries included an adenotonsillectomy, appendectomy, and inguinal hernia repair. The patient was not taking any medications. Family history included atherosclerotic
		coronary artery disease, alcoholic cirrhosis, alcohol abuse, and an abdominal aortic aneurysm rupture in the father; breast cancer in the mother; a myocardial infarction and atherosclerotic coronary artery disease in the sibling(s); and myocardial infarction and atherosclerotic coronary artery disease in the grandparent(s).

Table 6

Library	Vector	Library Description
FIBPFEN06	pINCY	The normalized prostate stromal fibroblast tissue libraries were constructed from 1.56 million independent clones from a prostate fibroblast library. Starting RNA was made from fibroblasts of prostate stroma removed from a male fetus, who died after 26 weeks' gestation. The libraries were normalized in two rounds using conditions adapted from Soares et al., PNAS (1994) 91:9228 and Bonaldo et al., Genome Research (1996) 6:791, except that a significantly longer (48-hours/round)reannealing hybridization was used. The library was then linearized and recircularized to select for insert containing clones as follows: plasmid DNA was prepped from approximately 1 million clones from the normalized prostate stromal fibroblast tissue libraries following soft agar transformation.
HNT2TXT01	pINCY	Library was constructed using RNA isolated from the hNT2 cell line (derived from a human teratocarcinoma that exhibited properties characteristic of a committed neuronal precursor). Cells were treated with mouse leptin at 1 ug/ml and 9 cis retinoic acid at 3.3 uM for 6 days.
KERANOT01	PBLUESCRIPT	Library was constructed using RNA isolated from neonatal keratinocytes obtained from the leg skin of a spontaneously aborted black male.
LUNGNOT23	pINCY	Library was constructed using RNA isolated from left lobe lung tissue removed from a 58-year-old Caucasian male. Pathology for the associated tumor tissue indicated metastatic grade 3 (of 4) osteosarcoma. Patient history included soft tissue cancer, secondary cancer of the lung, prostate cancer, and an acute duodenal ulcer with hemorrhage. Family history included prostate cancer, breast cancer, and acute leukemia.
NERDTDN03	pINCY	This normalized dorsal root ganglion tissue library was constructed from 1.05 million independent clones from a dorsal root ganglion tissue library. Starting RNA was made from dorsal root ganglion tissue removed from the cervical spine of a 32-year-old Caucasian male who died from acute pulmonary edema, acute bronchopneumonia, bilateral pleural effusions, pericardial effusion, and malignant lymphoma (natural killer cell type). The patient presented with pyrexia of unknown origin, malaise, fatigue, and gastrointestinal bleeding. Patient history included probable cytomegalovirus infection, liver congestion, and steatosis, splenomegaly, hemorrhagic cystitis, thyroid hemorrhage, respiratory failure, pneumonia of the left lung, natural killer cell lymphoma of the pharynx, Bell's palsy, and tobacco and alcohol abuse. Previous surgeries included colonoscopy, closed colon biopsy, adenotonsillectomy, and nasopharyngeal endoscopy and biopsy. Patient medications included Diflucan (fluconazole), Deltasone (prednisone),
		hydrocodone, Lortab, Alprazolam, Reazodone, ProMace-Cytabom, Etoposide, Cisplatin, Cytarabine, and dexamethasone. The patient received radiation therapy and multiple blood transfusions. The library was normalized in 2 rounds using conditions adapted from Soares et al., PNAS (1994) 91:9228-9232 and Bonaldo et al., Genome Research 6 (1996):791, except that a significantly longer (48 hours/round) reannealing hybridization was used.

Table 6

Library	Vector	Library Description
OVARNOE02	PCDNA2.1	This 5' biased random primed library was constructed using RNA isolated from right ovary tissue removed from a 47-year-old Caucasian female during total abdominal hysterectomy, bilateral salpingo-oophorectomy, incisional hernia repair, and panniculectomy. The patient presented with premenopausal menorrhagia. Patient history included osteoarthritis, tubal pregnancy, and polio osteopathy of the left leg. Previous surgeries included gastroenterostomy, plastic repair of the palate, adenotonsillectomy, dilation and curettage, cholecystectomy, and bladder reconstruction. Patient medications included vitamins, iron, and zinc. Family history included benign hypertension and type II diabetes in the father; and type II diabetes in the sibling(s).
PANCTUT02	pINCY	Library was constructed using RNA isolated from pancreatic tumor tissue removed from a 45-year-old Caucasian female during radical pancreaticoduodenectomy. Pathology indicated a grade 4 anaplastic carcinoma. Family history included benign hypertension, hyperlipidemia and atherosclerotic coronary artery disease.
PLACNOT02	pINCY	Library was constructed using RNA isolated from the placental tissue of a Hispanic female fetus, who was prematurely delivered at 21 weeks' gestation. Serologies of the mother's blood were positive for CMV (cytomegalovirus).
PROSTUS23	pINCY	This subtracted prostate tumor library was constructed using 10 million clones from a pooled prostate tumor library that was subjected to 2 rounds of subtractive hybridization with 10 million clones from a pooled prostate tissue library. The starting library for subtraction was constructed by pooling equal numbers of clones from 4 prostate tumor libraries using mRNA isolated from prostate tumor removed from Caucasian males at ages 58 (A), 61 (B), 66 (C), and 68 (D) during prostatectomy with lymph node excision. Pathology indicated adenocarcinoma in all donors. History included elevated PSA, induration and tobacco abuse in donor A; elevated PSA, induration, prostate hyperplasia, renal failure, osteoarthritis, renal artery stenosis, benign HTN, thrombocytopenia, hyperlipidemia, tobacco/alcohol abuse and hepatitis C (carrier) in donor B; elevated PSA, induration, and tobacco abuse in donor C; and elevated PSA, induration, hypercholesterolemia, and kidney calculus in donor D. The hybridization probe for
		subtraction was constructed by pooling equal numbers of cDNA clones from 3 prostate tissue libraries derived from prostate tissue, prostate epithelial cells, and fibroblasts from prostate stroma from 3 different donors. Subtractive hybridization conditions were based on the methodologies of Swaroop et al., NAR 19 (1991):1954 and Bonaldo, et al. Genome Research 6 (1996):791.
PROSTUT12	pINCY	Library was constructed using RNA isolated from prostate tumor tissue removed from a 65-year-old Caucasian male during a radical prostatectomy. Pathology indicated an adenocarcinoma (Gleason grade 2+2). Adenofibromatous hyperplasia was also present. The patient presented with elevated prostate specific antigen (PSA).

Table 6

Library	Vector	Library Description
SCORNON02	PSPORT1	This normalized spinal cord library was constructed from 3.24M independent clones from the a spinal cord tissue library. RNA was isolated from the spinal cord tissue removed from a 71-year-old Caucasian male who died from respiratory arrest. Patient history included myocardial infarction, gangrene, and end stage renal disease. The normalization and hybridization conditions were adapted from Soares et al.(PNAS (1994) 91:9228).
SINTFER02	pINCY	This random primed library was constructed using RNA isolated from small intestine tissue removed from a Caucasian male fetus who died from fetal demise.
SMCANOT01	pINCY	Library was constructed using RNA isolated from an aortic smooth muscle cell line derived from the explanted heart of a male during a heart transplant.
SPLNNOT04	pINCY	Library was constructed using RNA isolated from the spleen tissue of a 2-year-old Hispanic male, who died from cerebral anoxia. Past medical history and serologies were negative.
SPLNTUE01	PCDNA2.1	This 5' biased random primed library was constructed using RNA isolated from spleen tumor tissue removed from a 28-year-old male during total splenectomy. Pathology indicated malignant lymphoma, diffuse large cell type, B-cell phenotype with abundant reactive T-cells and marked granulomatous response involving the spleen, where it formed approximately 45 nodules, liver, and multiple lymph nodes.
THYMDIT01	pINCY	The library was constructed using RNA isolated from diseased thymus tissue removed from a 16-year-old Caucasian female during a total excision of thymus and regional lymph node excision. Pathology indicated thymic follicular hyperplasia. The right lateral thymus showed reactive lymph nodes. A single reactive lymph node was also identified at the inferior thymus margin. The patient presented with myasthenia gravis, malaise, fatigue, dysphagia, severe muscle weakness, and prominent eyes. Patient history included frozen face muscles. Family history included depressive disorder, hepatitis B, myocardial infarction, atherosclerotic coronary artery disease, leukemia, multiple sclerosis, and lupus.
THYMNOT04	pINCY	Library was constructed using RNA isolated from thymus tissue removed from a 3-year-old Caucasian male, who died from anoxia. Serologies were negative. The patient was not taking any medications.
TLYJINT01	pINCY	Library was constructed using RNA isolated from a Jurkat cell line derived from the T cells of a male. The cells were treated for 18 hours with 50 ng/ml phorbol ester (PMA) and 1 micromolar calcium ionophore. Patient history included acute T-cell leukemia.
TLYMNOT08	pINCY	The library was constructed using RNA isolated from anergicallogenic T-lymphocyte tissue removed from an adult (40-50-year old) Caucasian male.The cells were incubated for 3 days in the presence of 1 microgram/ml OKT3 mAb and 5% human serum.

Table 7

Program	Description	Reference	Parameter Threshold
ABI FACTURA	A program that removes vector sequences and masks ambiguous bases in nucleic acid sequences.	Applied Biosystems, Foster City, CA.	
ABI/PARACEL FDF	A Fast Data Finder useful in comparing and annotating amino acid or nucleic acid sequences.	Applied Biosystems, Foster City, CA; Paracel Inc., Pasadena, CA.	Mismatch <50%
ABI AutoAssembler	A program that assembles nucleic acid sequences.	Applied Biosystems, Foster City, CA.	
BLAST	A Basic Local Alignment Search Tool useful in sequence similarity search for amino acid and nucleic acid sequences. BLAST includes five functions: blastp, blastn, blastx, tblastn, and tblastx.	Altschul, S.F. et al. (1990) J. Mol. Biol. 215:403-410; Altschul, S.F. et al. (1997) Nucleic Acids Res. 25:3389-3402.	ESTs: Probability value = 1.0E-8 or less; Full Length sequences: Probability value = 1.0E-10 or less
FASTA	A Pearson and Lipman algorithm that searches for similarity between a query sequence and a group of sequences of the same type. FASTA comprises as least five functions: fasta, tfasta, fastx, tfastx, and ssearch.	Pearson, W.R. and D.J. Lipman (1988) Proc. Natl. Acad Sci. USA 85:2444-2448; Pearson, W.R. (1990) Methods Enzymol. 183:63-98; and Smith, T.F. and M.S. Waterman (1981) Adv. Appl. Math. 2:482-489.	ESTs: fasta E value = 1.06E-6; Assembled ESTs: fasta Identity = 95% or greater and Match length = 200 bases or greater; fastx E value = 1.0E-8 or less; Full Length sequences: fastx score = 100 or greater
BLIMPS	A BLocks IMProved Searcher that matches a sequence against those in BLOCKS, PRINTS, DOMO, PRODOM, and PFAM databases to search for gene families, sequence homology, and structural fingerprint regions.	Henikoff, S. and J.G. Henikoff (1991) Nucleic Acids Res. 19:6565-6572; Henikoff, J.G. and S. Henikoff (1996) Methods Enzymol. 266:88-105; and Attwood, T.K. et al. (1997) J. Chem. Inf. Comput. Sci. 37:417-424.	Probability value = 1.0E-3 or less

Table 7

Program	Description	Reference	Parameter Threshold
HMMER	An algorithm for searching a query sequence against hidden Markov model (HMM)-based databases of protein family consensus sequences, such as PFAM, INCY, SMART and TIGRFAM.	Krogh, A. et al. (1994) J. Mol. Biol. 235:1501-1531; Sonnhammer, E.L.L. et al. (1988) Nucleic Acids Res. 26:320-322; Durbin, R. et al. (1998) Our World View, in a Nutshell, Cambridge Univ. Press, pp. 1-350.	PFAM, INCY, SMART or TIGRFAM hits: Probability value = 1.0E-3 or less; Signal peptide hits: Score = 0 or greater
ProfileScan	An algorithm that searches for structural and sequence motifs in protein sequences that match sequence patterns defined in Prosite.	Gribskov, M. et al. (1988) CABIOS 4:61-66; Gribskov, M. et al. (1989) Methods Enzymol. 183:146-159; Bairoch, A. et al. (1997) Nucleic Acids Res. 25:217-221.	Normalized quality score ≥ GCG specified "HIGH" value for that particular Prosite motif. Generally, score = 1.4-2.1.
Phred	A base-calling algorithm that examines automated sequencer traces with high sensitivity and probability.	Ewing, B. et al. (1998) Genome Res. 8:175-185; Ewing, B. and P. Green (1998) Genome Res. 8:186-194.	
Phrap	A Phils Revised Assembly Program including SWAT and CrossMatch, programs based on efficient implementation of the Smith-Waterman algorithm, useful in searching sequence homology and assembling DNA sequences.	Smith, T.F. and M.S. Waterman (1981) Adv. Appl. Math. 2:482-489; Smith, T.F. and M.S. Waterman (1981) J. Mol. Biol. 147:195-197; and Green, P., University of Washington, Seattle, WA.	Score = 120 or greater; Match length = 56 or greater
Consed	A graphical tool for viewing and editing Phrap assemblies.	Gordon, D. et al. (1998) Genome Res. 8:195-202.	
SPScan	A weight matrix analysis program that scans protein sequences for the presence of secretory signal peptides.	Nielson, H. et al. (1997) Protein Engineering 10:1-6; Claverie, J.M. and S. Audic (1997) CABIOS 12:431-439.	Score = 3.5 or greater
TMAP	A program that uses weight matrices to delineate transmembrane segments on protein sequences and determine orientation.	Persson, B. and P. Argos (1994) J. Mol. Biol. 237:182-192; Persson, B. and P. Argos (1996) Protein Sci. 5:363-371.	

Table 7

Program	Description	Reference	Parameter Threshold
TMHMMER	A program that uses a hidden Markov model (HMM) to delineate transmembrane segments on protein sequences and determine orientation.	Sonnhammer, E.L. et al. (1998) Proc. Sixth Intl. Conf. On Intelligent Systems for Mol. Biol., Glasgow et al., eds., The Am. Assoc. for Artificial Intelligence (AAAI) Press, Menlo Park, CA, and MIT Press, Cambridge, MA, pp. 175-182.	
Motifs	A program that searches amino acid sequences for patterns that matched those defined in Prosite.	Bairoch, A. et al. (1997) Nucleic Acids Res. 25:217-221; Wisconsin Package Program Manual, version 9, page M51-59, Genetics Computer Group, Madison, WI.	

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
36	7511189	1493603H1	SNP00104859	40	43	A	G	A	noncoding	n/a	n/a	n/a	n/a
37	7511811	1451677H1	SNP00020092	51	2686	G	G	A	noncoding	n/a	n/a	n/a	n/a
37	7511811	1841920R6	SNP00026715	232	2440	C	T	C	noncoding	n/a	n/a	n/a	n/a
37	7511811	1841920T6	SNP00020092	227	2703	G	G	A	noncoding	n/a	n/a	n/a	n/a
37	7511811	2945475H1	SNP00002207	225	2476	G	C	G	noncoding	n/a	n/a	n/a	n/a
38	7511870	1805285F6	SNP00098636	21	753	C	T	C	Y138	n/a	n/a	n/a	n/a
38	7511870	1805285F6	SNP00098637	217	949	A	A	G	K204	n/a	n/a	n/a	n/a
38	7511870	1805285H1	SNP00098637	217	950	A	A	G	K204	n/a	n/a	n/a	n/a
38	7511870	1805285T6	SNP00044580	214	1767	C	C	T	noncoding	n/a	n/a	n/a	n/a
38	7511870	1805285T6	SNP00044581	71	1910	A	A	G	noncoding	n/a	n/a	n/a	n/a
38	7511870	1921210H1	SNP00044582	4	2218	T	T	G	noncoding	n/a	n/a	n/a	n/a
38	7511870	460707T6	SNP00044580	131	1801	T	C	T	noncoding	n/a	n/a	n/a	n/a
38	7511870	7410986H1	SNP00098636	323	755	T	T	C	L139	n/a	n/a	n/a	n/a
38	7511870	7410986H1	SNP00098637	127	952	G	A	G	V205	n/a	n/a	n/a	n/a
39	7510245	1441953F6	SNP00007123	74	2159	C	C	A	noncoding	n/a	n/a	n/a	n/a
39	7510245	1441953F6	SNP00032259	144	2229	C	C	G	noncoding	n/a	n/a	n/a	n/a
39	7510245	1995474H1	SNP00014672	47	1471	C	C	T	noncoding	n/a	n/a	n/a	n/a
39	7510245	3812757T8	SNP00112082	51	641	C	C	A	D98	n/d	n/a	n/a	n/a
39	7510245	7361117H1	SNP00007123	339	2142	C	C	A	noncoding	n/a	n/a	n/a	n/a
39	7510245	7361117H1	SNP00032259	409	2212	C	C	G	noncoding	n/a	n/a	n/a	n/a
39	7510245	7716125J1	SNP00007123	416	2149	C	C	A	noncoding	n/a	n/a	n/a	n/a
40	7511072	000814H1	SNP00144385	25	988	G	C	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	001887H1	SNP00003211	20	1016	T	C	T	noncoding	n/a	n/a	n/a	n/a
40	7511072	001887H1	SNP00058000	229	1225	T	T	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	038341H1	SNP00063071	109	911	C	C	T	noncoding	n/a	n/a	n/a	n/a
40	7511072	1002401H1	SNP00047378	24	1595	C	C	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	1290827H1	SNP00132379	24	666	C	C	T	noncoding	n/a	n/a	n/a	n/a
40	7511072	1390433H1	SNP00143328	203	775	G	G	C	noncoding	n/a	n/a	n/a	n/a

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
40	7511072	1810565F6	SNP00038952	158	179	C	C	G	noncoding	0.48	n/a	n/a	n/a
40	7511072	2410685H1	SNP00058000	170	1214	T	T	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	2522735H1	SNP00144386	21	1715	T	T	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	2728330H1	SNP00022964	21	695	G	G	C	noncoding	n/d	n/a	n/a	n/a
40	7511072	2836674T6	SNP00058000	480	1227	T	T	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	3163333H1	SNP00146056	206	1230	C	C	T	noncoding	n/a	n/a	n/a	n/a
40	7511072	3327356H1	SNP00063001	27	227	A	A	C	noncoding	n/a	n/a	n/a	n/a
40	7511072	3691238H1	SNP00058000	169	1215	T	T	G	noncoding	n/a	n/a	n/a	n/a
40	7511072	4806111H1	SNP00143328	141	777	G	G	C	noncoding	n/a	n/a	n/a	n/a
40	7511072	6287709H2	SNP00143963	389	1197	C	C	T	noncoding	n/a	n/a	n/a	n/a
41	7512247	154776H1	SNP00060368	145	1685	T	C	T	noncoding	n/a	n/a	n/a	n/a
41	7512247	2993050F6	SNP00145896	124	141	T	T	G	noncoding	n/a	n/a	n/a	n/a
41	7512247	7589130H1	SNP00145896	107	142	T	T	G	noncoding	n/a	n/a	n/a	n/a
41	7512247	7709205J1	SNP00145896	151	124	T	T	G	noncoding	n/a	n/a	n/a	n/a
41	7512247	7729503J1	SNP00145896	109	121	G	T	G	noncoding	n/a	n/a	n/a	n/a
42	7512244	1319231H1	SNP00006366	222	1322	C	C	A	P405	n/a	n/a	n/a	n/a
42	7512244	1440393H1	SNP00029688	25	60	C	C	T	noncoding	n/a	n/a	n/a	n/a
42	7512244	1602271H1	SNP00029689	188	2177	C	C	G	noncoding	0.88	n/a	n/a	n/a
42	7512244	7660990H1	SNP00006366	140	1321	C	C	A	P405	n/a	n/a	n/a	n/a
42	7512244	7700539J1	SNP00029688	25	44	C	C	T	noncoding	n/a	n/a	n/a	n/a
44	7511867	1431678H1	SNP00042039	89	1819	C	C	G	noncoding	n/a	n/a	n/a	n/a
44	7511867	1431678H1	SNP00042040	158	1888	G	G	A	noncoding	n/a	n/a	n/a	n/a
44	7511867	1438027H1	SNP00069656	89	276	A	A	G	R25	n/d	n/a	n/a	n/a
44	7511867	1522986F6	SNP00069656	86	277	A	A	G	S26	n/d	n/a	n/a	n/a
44	7511867	3907738H1	SNP00042039	113	1820	C	C	G	noncoding	n/a	n/a	n/a	n/a
44	7511867	3907738H1	SNP00042040	182	1889	G	G	A	noncoding	n/a	n/a	n/a	n/a
44	7511867	4807479H1	SNP00069656	89	280	A	A	G	M27	n/d	n/a	n/a	n/a
44	7511867	6362428H1	SNP00134258	95	1981	G	G	T	noncoding	n/a	n/a	n/a	n/a

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
46	7512572	1238605H1	SNP00074591	30	642	C	C	T	R153	n/a	n/a	n/a	n/a
46	7512572	1649445F6	SNP00099007	228	603	T	T	C	H140	n/a	n/a	n/a	n/a
46	7512572	1649445T6	SNP00136067	4	1063	G	G	C	noncoding	n/a	n/a	n/a	n/a
46	7512572	2256552H1	SNP00136067	101	975	G	G	C	K264	n/a	n/a	n/a	n/a
46	7512572	2362640T6	SNP00099007	418	661	T	T	C	S160	n/a	n/a	n/a	n/a
46	7512572	3321944T6	SNP00074591	435	640	C	C	T	R153	n/a	n/a	n/a	n/a
46	7512572	7630861J1	SNP00136067	98	976	G	G	C	G265	n/a	n/a	n/a	n/a
48	7512669	1429110F6	SNP00137212	292	332	C	C	T	A91	n/a	n/a	n/a	n/a
48	7512669	1511111T6	SNP00034940	209	857	C	C	T	D266	n/d	n/d	n/d	n/d
48	7512669	1906260H1	SNP00137212	237	339	C	C	T	R94	n/a	n/a	n/a	n/a
48	7512669	2618271T6	SNP00034940	162	899	C	C	T	T280	n/d	n/d	n/d	n/d
48	7512669	2717868T6	SNP00034940	186	868	C	C	T	T270	n/d	n/d	n/d	n/d
48	7512669	7373718H1	SNP00137212	275	341	C	C	T	R94	n/a	n/a	n/a	n/a
48	7512669	7753122J1	SNP00137212	274	318	C	C	T	L87	n/a	n/a	n/a	n/a
49	7512706	001388H1	SNP00042084	258	2742	A	A	G	noncoding	n/a	n/a	n/a	n/a
49	7512706	099627H1	SNP00061682	53	1290	T	T	C	noncoding	n/a	n/a	n/a	n/a
49	7512706	127435H1	SNP00132104	135	2229	T	T	G	noncoding	n/a	n/a	n/a	n/a
49	7512706	127435R1	SNP00132832	124	2218	G	G	T	noncoding	n/a	n/a	n/a	n/a
49	7512706	2055708R6	SNP00132832	429	2219	G	G	T	noncoding	n/a	n/a	n/a	n/a
49	7512706	3139805T6	SNP00132832	498	2231	G	G	T	noncoding	n/a	n/a	n/a	n/a
49	7512706	7019246H1	SNP00042084	386	2730	A	A	G	noncoding	n/a	n/a	n/a	n/a
49	7512706	7339457H1	SNP00042084	329	2724	A	A	G	noncoding	n/a	n/a	n/a	n/a
50	7512872	1234232H1	SNP00147591	98	1093	G	G	A	K273	n/a	n/a	n/a	n/a
50	7512872	1272696H1	SNP00020360	120	1206	G	G	A	noncoding	n/a	n/a	n/a	n/a
50	7512872	1305862H1	SNP00033354	168	577	T	T	C	F101	n/a	n/a	n/a	n/a
50	7512872	1851066F6	SNP00033354	156	576	T	T	C	F101	n/a	n/a	n/a	n/a
50	7512872	1851429T6	SNP00020360	229	1218	G	G	A	noncoding	n/a	n/a	n/a	n/a
50	7512872	1879385T6	SNP00020360	213	1213	G	G	A	noncoding	n/a	n/a	n/a	n/a

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
50	7512872	2765410F6	SNP00033354	118	582	T	T	C	V103	n/a	n/a	n/a	n/a
50	7512872	2765410T6	SNP00020360	222	1209	G	G	A	noncoding	n/a	n/a	n/a	n/a
50	7512872	2917293H1	SNP00107375	140	1109	T	G	T	C279	n/d	n/d	n/d	n/d
51	7512925	1592866H1	SNP00108384	155	155	G	G	C	G16	n/a	n/a	n/a	n/a
51	7512925	2131847H1	SNP00155163	70	2481	G	G	A	E792	n/a	n/a	n/a	n/a
51	7512925	2131847R6	SNP00155163	70	2485	G	G	A	R793	n/a	n/a	n/a	n/a
51	7512925	2385359H1	SNP00004391	102	3126	C	C	T	noncoding	0.83	0.84	n/d	0.9
51	7512925	2422460H1	SNP00051226	22	2768	T	C	T	I887	n/a	n/a	n/a	n/a
51	7512925	6780816J1	SNP00155163	385	2464	G	G	A	S786	n/a	n/a	n/a	n/a
51	7512925	6955796H1	SNP00108384	146	124	G	G	C	S6	n/a	n/a	n/a	n/a
51	7512925	7758466J1	SNP00108384	53	132	C	G	C	P9	n/a	n/a	n/a	n/a
52	7512927	052735H1	SNP00002883	42	1534	G	G	A	noncoding	0.78	0.79	0.85	0.79
52	7512927	1337323H1	SNP00021203	124	718	T	T	C	noncoding	n/a	n/a	n/a	n/a
53	7512943	059835H1	SNP00020653	89	1832	A	A	G	noncoding	n/a	n/a	n/a	n/a
53	7512943	1338891T6	SNP00020653	46	1833	A	A	G	noncoding	n/a	n/a	n/a	n/a
53	7512943	2706292T6	SNP00020653	42	1836	A	A	G	noncoding	n/a	n/a	n/a	n/a
53	7512943	5301370H1	SNP00099887	90	211	C	A	C	L44	0.13	0.05	0.07	n/d
54	7512955	2293452H1	SNP00033331	221	737	A	A	G	noncoding	n/a	n/a	n/a	n/a
54	7512955	2353663H1	SNP00034254	20	457	T	C	T	noncoding	0.98	n/a	n/a	n/a
54	7512955	3632493T9	SNP00141890	12	553	A	C	A	noncoding	n/a	n/a	n/a	n/a
54	7512955	7625489J1	SNP00007665	62	80	C	C	A	noncoding	n/a	n/a	n/a	n/a
54	7512955	8625806J1	SNP00007665	30	65	C	C	A	noncoding	n/a	n/a	n/a	n/a
55	7512950	1236609F7	SNP00140324	177	3267	A	A	C	Y1072	n/a	n/a	n/a	n/a
55	7512950	1301396H1	SNP00011118	64	1756	C	T	C	D568	0.57	0.32	0.65	0.56
55	7512950	1830685H1	SNP00100988	5	2914	T	T	C	P954	n/a	n/a	n/a	n/a
55	7512950	2555430H1	SNP00061527	183	683	G	G	A	G211	n/d	n/d	n/d	n/d
55	7512950	3090243H1	SNP00103490	151	2273	G	G	C	D741	n/d	n/a	n/a	n/a
55	7512950	5425662F6	SNP00103490	377	2292	G	G	C	R747	n/d	n/a	n/a	n/a

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
55	7512950	6892693H1	SNP00100987	173	2469	T	T	C	F806	n/d	n/d	n/d	n/d
55	7512950	7623345H1	SNP00061527	18	685	G	G	A	G211	n/d	n/d	n/d	n/d
55	7512950	7700161H1	SNP00103489	449	2209	C	C	T	D719	n/a	n/a	n/a	n/a
55	7512950	7700161H1	SNP00103490	483	2243	G	G	C	G731	n/d	n/a	n/a	n/a
55	7512950	7735074J1	SNP00103489	338	2239	T	C	T	A729	n/a	n/a	n/a	n/a
56	7517524	1610854H1	SNP00010367	28	2881	A	A	G	noncoding	n/d	n/a	n/a	n/a
56	7517524	3003120H1	SNP00040422	156	1271	C	C	A	noncoding	n/a	n/a	n/a	n/a
57	7519735	1610854H1	SNP00010367	28	3081	A	A	G	M1013	n/d	n/a	n/a	n/a
57	7519735	3003120H1	SNP00040422	156	1170	C	C	A	H376	n/a	n/a	n/a	n/a
58	7517880	1511916F6	SNP00000813	137	41	G	A	G	E6	0.13	0.04	n/d	0.12
58	7517880	1797396R6	SNP00016369	464	343	T	T	C	S106	n/a	n/a	n/a	n/a
60	7481640	2478134H1	SNP00029688	60	60	C	C	T	noncoding	n/a	n/a	n/a	n/a
61	7515465	1210173R2	SNP00037783	188	2368	C	C	T	F696	n/a	n/a	n/a	n/a
61	7515465	1532060F6	SNP00141249	58	806	G	G	A	E176	n/a	n/a	n/a	n/a
61	7515465	1532060F6	SNP00141250	333	1081	G	G	A	E267	n/a	n/a	n/a	n/a
61	7515465	1746650F6	SNP00037782	128	1648	C	C	T	L456	n/a	n/a	n/a	n/a
61	7515465	1746650F6	SNP00061536	54	1574	A	A	C	I432	n/a	n/a	n/a	n/a
61	7515465	1746650F6	SNP00141612	226	1746	A	A	C	K489	n/a	n/a	n/a	n/a
61	7515465	1746650H1	SNP00061536	84	1604	A	A	C	T442	n/a	n/a	n/a	n/a
61	7515465	1866127F6	SNP00127034	278	2049	T	T	C	V590	n/a	n/a	n/a	n/a
61	7515465	2500283F6	SNP00127034	315	2042	T	T	C	F588	n/a	n/a	n/a	n/a
61	7515465	2544120H1	SNP00127034	141	2050	T	T	C	V590	n/a	n/a	n/a	n/a
61	7515465	2874433H1	SNP00141612	61	1776	A	A	C	E499	n/a	n/a	n/a	n/a
61	7515465	458868R6	SNP00037783	210	2292	C	C	T	A671	n/a	n/a	n/a	n/a
61	7515465	7712425J1	SNP00141249	450	816	G	G	A	G179	n/a	n/a	n/a	n/a
62	4912891	1524859T6	SNP00105455	217	1408	G	G	A	R408	0.67	0.69	0.69	0.7
62	4912891	4306648H2	SNP00103681	16	47	C	C	T	noncoding	n/a	n/a	n/a	n/a
62	4912891	6006483F6	SNP00103681	47	53	T	C	T	noncoding	n/a	n/a	n/a	n/a

Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
63	70552759	5499536R9	SNP00114657	427	773	T	C	T	V215	n/a	n/a	n/a	n/a
63	70552759	7174473H1	SNP00146781	142	262	C	T	C	T45	n/a	n/a	n/a	n/a
64	7524194	1401538F6	SNP00105969	72	2216	C	C	T	noncoding	n/d	n/d	n/d	n/d
64	7524194	1401538T6	SNP00001958	262	2669	G	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	1401538T6	SNP00019450	65	2866	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	1401538T6	SNP00149655	296	2635	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	1401538T6	SNP00149656	149	2782	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	1601053T6	SNP00001958	212	2704	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	1601053T6	SNP00019450	15	2901	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	1601053T6	SNP00149655	246	2670	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	1601053T6	SNP00149656	99	2817	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	1929389F6	SNP00019450	71	2845	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	2496984T6	SNP00001958	271	2648	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	2496984T6	SNP00149655	305	2614	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	2496984T6	SNP00149656	158	2761	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	2917582T6	SNP00001958	269	2650	G	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	2917582T6	SNP00019450	72	2847	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	2917582T6	SNP00149655	303	2616	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	2917582T6	SNP00149656	156	2763	T	T	C	noncoding	n/a	n/a	n/a	n/a
64	7524194	2960640T6	SNP00001958	206	2708	A	A	G	noncoding	n/a	n/a	n/a	n/a
64	7524194	2960640T6	SNP00019450	9	2905	A	A	G	noncoding	n/a	n/a	n/a	n/a
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Table 8

SEQ ID NO:	PID	EST ID	SNP ID	EST SNP	CB1 SNP	EST Allele	Allele 1	Allele 2	Amino Acid	Caucasian Allele 1 frequency	African Allele 1 frequency	Asian Allele 1 frequency	Hispanic Allele 1 frequency
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68	4032169	1359580T6	SNP00008806	142	2765	T	T	G	noncoding	n/a	n/a	n/a	n/a
68	4032169	1359580T6	SNP00146462	53	2854	C	C	T	noncoding	n/a	n/a	n/a	n/a
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68	4032169	2808136H1	SNP00096700	131	1497	C	C	T	S439	0.99	0.98	0.94	0.92
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70	7509493	2346414F6	SNP00020237	267	2802	T	C	T	noncoding	0.67	0.84	n/d	0.76

What is claimed is:

1. An isolated polypeptide selected from the group consisting of:

a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-27 and SEQ ID NO:29-35,

b) a polypeptide consisting essentially of the amino acid sequence of SEQ ID NO:28,

c) a polypeptide consisting essentially of a naturally occurring amino acid sequence at least 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1, SEQ ID NO:7, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:20, SEQ ID NO:26, and SEQ ID NO:34,

d) a polypeptide comprising a naturally occurring amino acid sequence at least 95% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:2, SEQ ID NO:4, and SEQ ID NO:10,

e) a polypeptide comprising a naturally occurring amino acid sequence at least 93% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:3 and SEQ ID NO:23,

f) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:5, SEQ ID NO:8-9, SEQ ID NO:17, SEQ ID NO:21-22, SEQ ID NO:24, SEQ ID NO:27, SEQ ID NO:29, and SEQ ID NO:31-32,

g) a polypeptide comprising a naturally occurring amino acid sequence at least 94% identical to the amino acid sequence of SEQ ID NO:12,

h) a polypeptide comprising a naturally occurring amino acid sequence at least 91% identical to the amino acid sequence of SEQ ID NO:14,

i) a polypeptide comprising a naturally occurring amino acid sequence at least 98% identical to the amino acid sequence of SEQ ID NO:18,

j) a polypeptide comprising a naturally occurring amino acid sequence at least 97% identical to the amino acid sequence of SEQ ID NO:19,

k) a polypeptide comprising a naturally occurring amino acid sequence at least 96% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:25 and SEQ ID NO:35,

l) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, and

m) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

2. An isolated polypeptide of claim 1 selected from the group consisting of:

a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-27 and SEQ ID NO:29-35, and

b) a polypeptide consisting essentially of the amino acid sequence of SEQ ID NO:28.

3. An isolated polynucleotide encoding a polypeptide of claim 1.

4. An isolated polynucleotide encoding a polypeptide of claim 2.

5. An isolated polynucleotide of claim 4 comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70.

6. A recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide of claim 3.

7. A cell transformed with a recombinant polynucleotide of claim 6.

8. A transgenic organism comprising a recombinant polynucleotide of claim 6.

9. A method of producing a polypeptide of claim 1, the method comprising:

a) culturing a cell under conditions suitable for expression of the polypeptide, wherein said cell is transformed with a recombinant polynucleotide, and said recombinant polynucleotide comprises a promoter sequence operably linked to a polynucleotide encoding the polypeptide of claim 1, and

b) recovering the polypeptide so expressed.

10. A method of claim 9, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

11. An isolated antibody which specifically binds to a polypeptide of claim 1.

12. An isolated polynucleotide selected from the group consisting of:

a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:36-70,

b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least

90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:36, SEQ ID NO:40, SEQ ID NO:45, SEQ ID NO:57, SEQ ID NO:59-60, and SEQ ID NO:66-67,

c) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 99% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:37 and SEQ ID NO:54,

d) a polynucleotide consisting essentially of a naturally occurring polynucleotide sequence at least 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:38, SEQ ID NO:41, SEQ ID NO:46, SEQ ID NO:49-53, SEQ ID NO:55, SEQ ID NO:62, and SEQ ID NO:69,

e) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 98% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:39, SEQ ID NO:43, SEQ ID NO:47-48, and SEQ ID NO:68,

f) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 95% identical to the polynucleotide sequence of SEQ ID NO:42,

g) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 92% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:44, SEQ ID NO:61, and SEQ ID NO:65,

h) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 97% identical to the polynucleotide sequence of SEQ ID NO:56,

i) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 94% identical to the polynucleotide sequence of SEQ ID NO:58,

j) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 96% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:63 and SEQ ID NO:70,

k) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 93% identical to the polynucleotide sequence of SEQ ID NO:64,

l) a polynucleotide complementary to a polynucleotide of a),

m) a polynucleotide complementary to a polynucleotide of b),

n) a polynucleotide complementary to a polynucleotide of c),

o) a polynucleotide complementary to a polynucleotide of d),

p) a polynucleotide complementary to a polynucleotide of e),

q) a polynucleotide complementary to a polynucleotide of f),

r) a polynucleotide complementary to a polynucleotide of g),

s) a polynucleotide complementary to a polynucleotide of h),

t) a polynucleotide complementary to a polynucleotide of i),

u) a polynucleotide complementary to a polynucleotide of j),

v) a polynucleotide complementary to a polynucleotide of k), and

w) an RNA equivalent of a)-v).

13. An isolated polynucleotide comprising at least 60 contiguous nucleotides of a polynucleotide of claim 12.

14. A method of detecting a target polynucleotide in a sample, said target polynucleotide having a sequence of a polynucleotide of claim 12, the method comprising:

a) hybridizing the sample with a probe comprising at least 20 contiguous nucleotides comprising a sequence complementary to said target polynucleotide in the sample, and which probe specifically hybridizes to said target polynucleotide, under conditions whereby a hybridization complex is formed between said probe and said target polynucleotide or fragments thereof, and

b) detecting the presence or absence of said hybridization complex, and, optionally, if present, the amount thereof.

15. A method of claim 14, wherein the probe comprises at least 60 contiguous nucleotides.

16. A method of detecting a target polynucleotide in a sample, said target polynucleotide having a sequence of a polynucleotide of claim 12, the method comprising:

a) amplifying said target polynucleotide or fragment thereof using polymerase chain reaction amplification, and

b) detecting the presence or absence of said amplified target polynucleotide or fragment thereof, and, optionally, if present, the amount thereof.

17. A composition comprising a polypeptide of claim 1 and a pharmaceutically acceptable excipient.

18. A composition of claim 17, wherein the polypeptide is selected from the group consisting of:

a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-27 and SEQ ID NO:29-35, and

b) a polypeptide consisting essentially of the amino acid sequence of SEQ ID NO:28.

19. A method for treating a disease or condition associated with decreased expression of functional NAAP, comprising administering to a patient in need of such treatment the composition of claim 17.

20. A method of screening a compound for effectiveness as an agonist of a polypeptide of claim 1, the method comprising:

a) contacting a sample comprising a polypeptide of claim 1 with a compound, and
b) detecting agonist activity in the sample.

21. A composition comprising an agonist compound identified by a method of claim 20 and a pharmaceutically acceptable excipient.

22. A method for treating a disease or condition associated with decreased expression of functional NAAP, comprising administering to a patient in need of such treatment a composition of claim 21.

23. A method of screening a compound for effectiveness as an antagonist of a polypeptide of claim 1, the method comprising:

a) contacting a sample comprising a polypeptide of claim 1 with a compound, and
b) detecting antagonist activity in the sample.

24. A composition comprising an antagonist compound identified by a method of claim 23 and a pharmaceutically acceptable excipient.

25. A method for treating a disease or condition associated with overexpression of functional NAAP, comprising administering to a patient in need of such treatment a composition of claim 24.

26. A method of screening for a compound that specifically binds to the polypeptide of claim 1, the method comprising:

a) combining the polypeptide of claim 1 with at least one test compound under suitable conditions, and
b) detecting binding of the polypeptide of claim 1 to the test compound, thereby identifying a compound that specifically binds to the polypeptide of claim 1.

27. A method of screening for a compound that modulates the activity of the polypeptide of claim 1, the method comprising:

a) combining the polypeptide of claim 1 with at least one test compound under conditions permissive for the activity of the polypeptide of claim 1,

b) assessing the activity of the polypeptide of claim 1 in the presence of the test compound, and

c) comparing the activity of the polypeptide of claim 1 in the presence of the test compound with the activity of the polypeptide of claim 1 in the absence of the test compound, wherein a change in the activity of the polypeptide of claim 1 in the presence of the test compound is indicative of a compound that modulates the activity of the polypeptide of claim 1.

28. A method of screening a compound for effectiveness in altering expression of a target polynucleotide, wherein said target polynucleotide comprises a sequence of claim 5, the method comprising:

a) contacting a sample comprising the target polynucleotide with a compound, under conditions suitable for the expression of the target polynucleotide,

b) detecting altered expression of the target polynucleotide, and

c) comparing the expression of the target polynucleotide in the presence of varying amounts of the compound and in the absence of the compound.

29. A method of screening for potential toxicity of a test compound, the method comprising:

a) treating a biological sample containing nucleic acids with the test compound,

b) hybridizing the nucleic acids of the treated biological sample with a probe comprising at least 20 contiguous nucleotides of a polynucleotide of claim 12 under conditions whereby a specific hybridization complex is formed between said probe and a target polynucleotide in the biological sample, said target polynucleotide comprising a polynucleotide sequence of a polynucleotide of claim 12 or fragment thereof,

c) quantifying the amount of hybridization complex, and

d) comparing the amount of hybridization complex in the treated biological sample with the amount of hybridization complex in an untreated biological sample, wherein a difference in the amount of hybridization complex in the treated biological sample indicates potential toxicity of the test compound.

30. A method for a diagnostic test for a condition or disease associated with the expression of NAAP in a biological sample, the method comprising:

a) combining the biological sample with an antibody of claim 11, under conditions suitable for the antibody to bind the polypeptide and form an antibody:polypeptide complex, and

b) detecting the complex, wherein the presence of the complex correlates with the presence of the polypeptide in the biological sample.

31. The antibody of claim 11, wherein the antibody is:

a) a chimeric antibody,

b) a single chain antibody,

c) a Fab fragment,

d) a $F(ab')_2$ fragment, or

e) a humanized antibody.

32. A composition comprising an antibody of claim 11 and an acceptable excipient.

33. A method of diagnosing a condition or disease associated with the expression of NAAP in a subject, comprising administering to said subject an effective amount of the composition of claim 32.

34. A composition of claim 32, further comprising a label.

35. A method of diagnosing a condition or disease associated with the expression of NAAP in a subject, comprising administering to said subject an effective amount of the composition of claim 34.

36. A method of preparing a polyclonal antibody with the specificity of the antibody of claim 11, the method comprising:

a) immunizing an animal with a polypeptide consisting of an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, or an immunogenic fragment thereof, under conditions to elicit an antibody response,

b) isolating antibodies from the animal, and

c) screening the isolated antibodies with the polypeptide, thereby identifying a polyclonal antibody which specifically binds to a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

37. A polyclonal antibody produced by a method of claim 36.

38. A composition comprising the polyclonal antibody of claim 37 and a suitable carrier.

39. A method of making a monoclonal antibody with the specificity of the antibody of claim 11, the method comprising:

a) immunizing an animal with a polypeptide consisting of an amino acid sequence selected from the group consisting of SEQ ID NO:1-35, or an immunogenic fragment thereof, under conditions to elicit an antibody response,

b) isolating antibody producing cells from the animal,

c) fusing the antibody producing cells with immortalized cells to form monoclonal antibody-producing hybridoma cells,

d) culturing the hybridoma cells, and

e) isolating from the culture monoclonal antibody which specifically binds to a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

40. A monoclonal antibody produced by a method of claim 39.

41. A composition comprising the monoclonal antibody of claim 40 and a suitable carrier.

42. The antibody of claim 11, wherein the antibody is produced by screening a Fab expression library.

43. The antibody of claim 11, wherein the antibody is produced by screening a recombinant immunoglobulin library.

44. A method of detecting a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35 in a sample, the method comprising:

a) incubating the antibody of claim 11 with the sample under conditions to allow specific binding of the antibody and the polypeptide, and

b) detecting specific binding, wherein specific binding indicates the presence of a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35 in the sample.

45. A method of purifying a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35 from a sample, the method comprising:

a) incubating the antibody of claim 11 with the sample under conditions to allow specific binding of the antibody and the polypeptide, and

b) separating the antibody from the sample and obtaining the purified polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-35.

46. A microarray wherein at least one element of the microarray is a polynucleotide of claim 13.

47. A method of generating an expression profile of a sample which contains polynucleotides, the method comprising:

a) labeling the polynucleotides of the sample,

b) contacting the elements of the microarray of claim 46 with the labeled polynucleotides of the sample under conditions suitable for the formation of a hybridization complex, and

c) quantifying the expression of the polynucleotides in the sample.

48. An array comprising different nucleotide molecules affixed in distinct physical locations on a solid substrate, wherein at least one of said nucleotide molecules comprises a first oligonucleotide or polynucleotide sequence specifically hybridizable with at least 30 contiguous nucleotides of a target polynucleotide, and wherein said target polynucleotide is a polynucleotide of claim 12.

49. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to at least 30 contiguous nucleotides of said target polynucleotide.

50. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to at least 60 contiguous nucleotides of said target polynucleotide.

51. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to said target polynucleotide.

52. An array of claim 48, which is a microarray.

53. An array of claim 48, further comprising said target polynucleotide hybridized to a nucleotide molecule comprising said first oligonucleotide or polynucleotide sequence.

54. An array of claim 48, wherein a linker joins at least one of said nucleotide molecules to said solid substrate.

55. An array of claim 48, wherein each distinct physical location on the substrate contains multiple nucleotide molecules, and the multiple nucleotide molecules at any single distinct physical location have the same sequence, and each distinct physical location on the substrate contains nucleotide molecules having a sequence which differs from the sequence of nucleotide molecules at another distinct physical location on the substrate.

56. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:1.

57. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:2.

58. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:3.

59. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:4.

60. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:5.

61. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:6.

62. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:7.

63. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:8.

64. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:9.

65. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:10.

66. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:11.

67. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:12.

68. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:13.

69. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:14.

70. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:15.

71. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:16.

72. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:17.

73. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:18.

74. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:19.

75. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:20.

76. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:21.

77. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:22.

78. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:23.

79. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:24.

80. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:25.

81. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:26.

82. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:27.

83. A polypeptide of claim 1, consisting essentially of the amino acid sequence of SEQ ID NO:28.

84. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:29.

85. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:30.

86. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:31.

87. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:32.

88. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:33.

89. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:34.

90. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:35.

91. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:36.

92. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:37.

93. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:38.

94. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:39.

95. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:40.

96. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:41.

97. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:42.

98. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:43.

99. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:44.

100. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:45.

101. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:46.

102. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:47.

103. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:48.

104. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:49.

105. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:50.

106. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:51.

107. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:52.

108. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:53.

109. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:54.

110. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:55.

111. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:56.

112. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:57.

113. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:58.

114. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:59.

115. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:60.

116. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:61.

117. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:62.

118. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:63.

119. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:64.

120. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:65.

121. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:66.

122. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:67.

123. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:68.

124. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:69.

125. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:70.

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HAFALIA, April J.A.; KHARE, Reena;
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LEE, Sally; YUE, Henry;
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Ala Arg Val Leu Gly Ser Glu Gly Glu Glu Glu Asp Glu Ala Leu
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Ser Pro Ala Lys Gly Gln Lys Pro Ala Leu Asp Cys Ser Gln Val
80 85 90
Ser Pro Pro Arg Pro Ala Thr Ser Pro Glu Asn Asn Ala Ser Leu
95 100 105
Ser Asp Thr Ser Pro Met Asp Ser Ser Pro Ser Gly Ile Pro Lys
110 115 120
Arg Arg Thr Ala Arg Lys Gln Leu Pro Lys Arg Thr Ile Gln Glu
125 130 135
Val Leu Glu Glu Gln Ser Glu Asp Glu Asp Arg Glu Ala Lys Arg
140 145 150
Lys Lys Glu Glu Glu Glu Glu Glu Thr Pro Lys Glu Ser Leu Thr
155 160 165
Glu Ala Glu Val Ala Thr Glu Lys Glu Gly Glu Asp Gly Asp Gln
170 175 180
Pro Thr Thr Pro Pro Lys Pro Leu Lys Thr Ser Lys Ala Glu Thr
185 190 195
Pro Thr Glu Ser Val Ser Glu Pro Glu Val Ala Thr Lys Gln Glu
200 205 210
Leu Gln Glu Glu Glu Glu Gln Thr Lys Pro Pro Arg Arg Ala Pro
215 220 225
Lys Thr Leu Ser Ser Phe Phe Thr Pro Arg Lys Pro Ala Val Lys
230 235 240
Lys Glu Val Lys Glu Glu Glu Pro Gly Ala Pro Gly Lys Glu Gly
245 250 255
Ala Ala Glu Gly Pro Leu Asp Pro Ser Gly Tyr Asn Pro Ala Lys
260 265 270
Asn Asn Tyr His Pro Val Glu Asp Ala Cys Trp Lys Pro Gly Gln
275 280 285
Lys Val Pro Tyr Leu Ala Val Ala Arg Thr Phe Glu Lys Ile Glu
290 295 300
Glu Val Ser Ala Arg Leu Arg Met Val Glu Thr Leu Ser Asn Leu
305 310 315
Leu Arg Ser Val Val Ala Leu Ser Pro Pro Asp Leu Leu Pro Val
320 325 330
Leu Tyr Leu Ser Leu Asn His Leu Gly Pro Pro Gln Gln Gly Leu
335 340 345
Glu Leu Gly Val Gly Asp Gly Val Leu Leu Lys Ala Val Ala Gln

350 355 360
Ala Thr Gly Arg Gln Leu Glu Ser Val Arg Ala Glu Ala Ala Glu
365 370 375
Lys Gly Asp Val Gly Leu Val Ala Glu Asn Ser Arg Ser Thr Gln
380 385 390
Arg Leu Met Leu Pro Pro Pro Pro Leu Thr Ala Ser Gly Val Phe
395 400 405
Ser Lys Phe Arg Asp Ile Ala Arg Leu Thr Gly Ser Ala Ser Thr
410 415 420
Ala Lys Lys Ile Asp Ile Ile Lys Gly Leu Phe Val Ala Cys Arg
425 430 435
His Ser Glu Ala Arg Phe Ile Ala Arg Ser Leu Ser Gly Arg Leu
440 445 450
Arg Leu Gly Leu Ala Glu Gln Ser Val Leu Ala Ala Leu Ser Gln
455 460 465
Ala Val Ser Leu Thr Pro Pro Gly Gln Glu Phe Pro Pro Ala Met
470 475 480
Val Asp Ala Gly Lys Gly Lys Thr Ala Glu Ala Arg Lys Thr Trp
485 490 495
Leu Glu Glu Gln Gly Met Ile Leu Lys Gln Thr Phe Cys Glu Val
500 505 510
Pro Asp Leu Asp Arg Ile Ile Pro Val Leu Leu Glu His Gly Leu
515 520 525
Glu Arg Leu Pro Glu His Cys Lys Leu Ser Pro Asp Pro Arg Pro
530 535 540
Gly Arg Arg Gly Gly Glu Asp Leu Gln Gln Glu Ser Gly Arg Gln
545 550 555
His Trp Glu Val Pro Gly His His Gln Pro His Pro Gln Asp
560 565

<210> 3
<211> 346
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7511870CD1

<400> 3
Met Leu Gly Val Arg Cys Leu Leu Arg Ser Val Arg Phe Cys Ser
1 5 10 15
Ser Ala Pro Phe Pro Lys His Lys Pro Ser Ala Lys Leu Ser Val
20 25 30
Arg Asp Ala Leu Gly Ala Gln Asn Ala Ser Gly Glu Arg Ile Lys
35 40 45
Ile Gln Gly Trp Ile Arg Ser Val Arg Ser Gln Lys Glu Val Leu
50 55 60
Phe Leu His Val Asn Asp Gly Ser Ser Leu Glu Ser Leu Gln Val
65 70 75
Val Ala Asp Ser Gly Leu Asp Ser Arg Glu Leu Thr Phe Gly Ser
80 85 90
Ser Val Glu Val Gln Gly Gln Leu Ile Lys Ser Pro Ser Lys Arg
95 100 105
Gln Asn Val Glu Leu Lys Ala Glu Lys Ile Lys Val Ile Gly Asn
110 115 120
Cys Asp Ala Lys Asp Phe Pro Ile Lys Tyr Lys Glu Arg His Pro
125 130 135
Leu Glu Tyr Leu Arg Gln Tyr Pro His Phe Arg Cys Arg Thr Asn
140 145 150
Val Leu Gly Ser Ile Leu Arg Ile Arg Ser Glu Ala Thr Ala Ala
155 160 165
Ile His Ser Phe Phe Lys Asp Ser Gly Phe Val His Ile His Thr

```
                170                 175                 180
Pro Ile Ile Thr Ser Asn Asp Ser Glu Gly Ala Gly Glu Leu Phe
                185                 190                 195
Gln Leu Glu Pro Ser Gly Lys Leu Lys Val Pro Glu Glu Asn Phe
                200                 205                 210
Phe Asn Val Pro Ala Phe Leu Thr Val Ser Gly Gln Leu His Leu
                215                 220                 225
Glu Val Met Ser Gly Ala Phe Thr Gln Val Phe Thr Phe Gly Pro
                230                 235                 240
Thr Phe Arg Ala Glu Asn Ser Gln Ser Arg Arg His Leu Ala Glu
                245                 250                 255
Phe Tyr Met Ile Glu Ala Glu Ile Ser Phe Val Asp Ser Leu Gln
                260                 265                 270
Asp Leu Met Gln Val Ile Glu Glu Leu Phe Lys Ala Thr Thr Met
                275                 280                 285
Met Val Leu Ser Lys Cys Pro Glu Asp Val Glu Leu Cys His Lys
                290                 295                 300
Phe Ile Ala Pro Gly Gln Lys Asp Arg Leu Glu His Met Leu Lys
                305                 310                 315
Asn Asn Phe Leu Met Leu Leu Leu Leu Ile Phe Trp Phe Leu Glu
                320                 325                 330
Leu Gly Asn Ser Leu Glu Glu Ala Ser Glu Lys Asn Asp Thr Ile
                335                 340                 345
Ser


<210> 4
<211> 287
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7510245CD1

<400> 4
Met Ala Gly Tyr Ala Thr Thr Pro Ser Pro Met Gln Thr Leu Gln
  1             5                  10                  15
Glu Glu Ala Val Cys Ala Ile Cys Leu Asp Tyr Phe Lys Asp Pro
                20                  25                  30
Val Ser Ile Ser Cys Gly His Asn Phe Cys Arg Gly Cys Val Thr
                35                  40                  45
Gln Leu Trp Ser Lys Glu Asp Glu Glu Asp Gln Asn Glu Glu Glu
                50                  55                  60
Asp Glu Trp Glu Glu Glu Glu Asp Glu Glu Ala Val Gly Ala Met
                65                  70                  75
Asp Gly Trp Asp Gly Ser Ile Arg Glu Val Leu Tyr Arg Gly Asn
                80                  85                  90
Ala Asp Glu Glu Leu Phe Gln Asp Gln Asp Asp Asp Glu Leu Trp
                95                 100                 105
Leu Gly Asp Ser Gly Ile Thr Asn Trp Asp Asn Val Asp Tyr Met
               110                 115                 120
Trp Asp Glu Glu Glu Glu Glu Glu Glu Glu Asp Gln Asp Tyr Tyr
               125                 130                 135
Leu Gly Gly Leu Arg Pro Asp Leu Arg Ile Asp Val Tyr Arg Glu
               140                 145                 150
Glu Glu Ile Leu Glu Ala Tyr Asp Glu Asp Glu Asp Glu Glu Leu
               155                 160                 165
Tyr Pro Asp Ile His Pro Pro Pro Ser Leu Pro Leu Pro Gly Gln
               170                 175                 180
Phe Thr Cys Pro Gln Cys Arg Lys Ser Phe Thr Arg Arg Ser Phe
               185                 190                 195
Arg Pro Asn Leu Gln Leu Ala Asn Met Val Gln Ile Ile Arg Gln
```

```
                200                     205                     210
Met Cys Pro Thr Pro Tyr Arg Gly Asn Arg Ser Asn Asp Gln Gly
                215                     220                     225
Met Cys Phe Lys His Gln Glu Ala Leu Lys Leu Phe Cys Glu Val
                230                     235                     240
Asp Lys Glu Ala Ile Cys Val Val Cys Arg Glu Ser Arg Ser His
                245                     250                     255
Lys Gln His Ser Val Val Pro Leu Glu Glu Val Val Gln Glu Tyr
                260                     265                     270
Gln Lys Ile Gly Ser Thr Ser Ser Val Glu Leu Ser Glu Ser Val
                275                     280                     285
Gly Gln


<210> 5
<211> 90
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7511072CD1

<400> 5
Met Ser Ser Glu Glu Gly Lys Leu Phe Val Gly Gly Leu Asn Phe
  1               5                  10                  15
Asn Thr Asp Glu Gln Ala Leu Glu Asp His Phe Ser Ser Phe Gly
                 20                  25                  30
Pro Ile Ser Glu Val Val Val Val Lys Asp Arg Glu Thr Gln Arg
                 35                  40                  45
Ser Arg Gly Phe Gly Phe Ile Thr Phe Thr Asn Pro Glu His Ala
                 50                  55                  60
Ser Val Ala Met Arg Ala Met Asn Gly Glu Gly Val Glu Gly Ser
                 65                  70                  75
Gly Arg Gln Ser Leu Ala Ala Trp Glu Ala Thr Thr Gly Pro Ala
                 80                  85                  90


<210> 6
<211> 470
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512247CD1

<400> 6
Met Pro Leu Leu Gly Leu Leu Pro Arg Arg Ala Trp Ala Ser Leu
  1               5                  10                  15
Leu Ser Gln Leu Leu Arg Pro Pro Cys Ala Ser Cys Thr Gly Ala
                 20                  25                  30
Val Arg Cys Gln Ser Gln Val Ala Glu Ala Val Leu Thr Ser Gln
                 35                  40                  45
Leu Lys Ala His Gln Glu Lys Pro Asn Phe Ile Ile Lys Thr Pro
                 50                  55                  60
Lys Gly Thr Arg Asp Leu Ser Pro Gln His Met Val Val Arg Glu
                 65                  70                  75
Lys Ile Leu Asp Leu Val Ile Ser Cys Phe Lys Arg His Gly Ala
                 80                  85                  90
Lys Gly Met Asp Thr Pro Ala Phe Glu Leu Lys Glu Thr Leu Thr
                 95                 100                 105
Glu Lys Tyr Gly Glu Asp Ser Gly Leu Met Tyr Asp Leu Lys Asp
```

110 115 120
Gln Gly Gly Glu Leu Leu Ser Leu Arg Tyr Asp Leu Thr Val Pro
125 130 135
Phe Ala Arg Tyr Leu Ala Met Asn Lys Val Lys Lys Met Lys Arg
140 145 150
Tyr His Val Gly Lys Val Trp Arg Arg Glu Ser Pro Thr Ile Val
155 160 165
Gln Gly Arg Tyr Arg Glu Phe Cys Gln Cys Asp Phe Asp Ile Ala
170 175 180
Gly Gln Phe Asp Pro Met Ile Pro Asp Ala Glu Cys Leu Lys Ile
185 190 195
Met Cys Glu Ile Leu Ser Gly Leu Gln Leu Gly Asp Phe Leu Ile
200 205 210
Lys Val Asn Asp Arg Arg Ile Val Asp Gly Met Phe Ala Val Cys
215 220 225
Gly Val Pro Glu Ser Lys Phe Arg Ala Ile Cys Ser Ser Ile Asp
230 235 240
Lys Leu Asp Lys Met Ala Trp Lys Asp Val Arg His Glu Met Val
245 250 255
Val Lys Lys Gly Leu Ala Pro Glu Val Ala Asp Arg Ile Gly Asp
260 265 270
Tyr Val Gln Cys His Gly Gly Val Ser Leu Val Glu Gln Met Phe
275 280 285
Gln Asp Pro Arg Leu Ser Gln Asn Lys Gln Ala Leu Glu Gly Leu
290 295 300
Gly Asp Leu Lys Leu Leu Phe Glu Tyr Leu Thr Leu Phe Gly Ile
305 310 315
Ala Asp Lys Ile Ser Phe Asp Leu Ser Leu Ala Arg Gly Leu Asp
320 325 330
Tyr Tyr Thr Gly Val Ile Tyr Glu Ala Val Leu Leu Gln Thr Pro
335 340 345
Thr Gln Ala Gly Glu Glu Pro Leu Asn Val Gly Ser Val Ala Ala
350 355 360
Gly Gly Arg Tyr Asp Gly Leu Val Gly Met Phe Asp Pro Lys Gly
365 370 375
His Lys Val Pro Cys Val Gly Leu Ser Ile Gly Val Glu Arg Ile
380 385 390
Phe Tyr Ile Val Glu Gln Arg Met Lys Thr Lys Gly Glu Lys Val
395 400 405
Arg Thr Thr Glu Thr Gln Val Phe Val Ala Thr Pro Gln Lys Asn
410 415 420
Phe Leu Gln Glu Arg Leu Lys Leu Ile Ala Glu Leu Trp Asp Ser
425 430 435
Gly Ile Lys Ala Glu Met Leu Tyr Lys Asn Asn Pro Lys Leu Leu
440 445 450
Thr Gln Leu His Tyr Cys Glu Ser Thr Gly Ile Pro Leu Val Val
455 460 465
Ile Ile Gly Gly His
470

<210> 7
<211> 445
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512244CD1

<400> 7
Met Asp Glu Leu Phe Pro Leu Ile Phe Pro Ala Glu Pro Ala Gln
1 5 10 15
Ala Ser Gly Pro Tyr Val Glu Ile Ile Glu Gln Pro Lys Gln Arg

```
                    20                       25                       30
Gly Met Arg Phe Arg Tyr Lys Cys Glu Gly Arg Ser Ala Gly Ser
                    35                       40                       45
Ile Pro Gly Glu Arg Ser Thr Asp Thr Thr Lys Thr His Pro Thr
                    50                       55                       60
Ile Lys Ile Asn Gly Tyr Thr Gly Pro Gly Thr Val Arg Ile Ser
                    65                       70                       75
Leu Val Thr Lys Asp Pro Pro His Arg Pro His Pro His Glu Leu
                    80                       85                       90
Val Gly Lys Asp Cys Arg Asp Gly Phe Tyr Glu Ala Glu Leu Cys
                    95                       100                      105
Pro Asp Arg Cys Ile His Ser Phe Gln Asn Leu Gly Ile Gln Cys
                    110                      115                      120
Val Lys Lys Arg Asp Leu Glu Gln Ala Ile Ser Gln Arg Ile Gln
                    125                      130                      135
Thr Asn Asn Asn Pro Phe Gln Val Pro Ile Glu Glu Gln Arg Gly
                    140                      145                      150
Asp Tyr Asp Leu Asn Ala Val Arg Leu Cys Phe Gln Val Thr Val
                    155                      160                      165
Arg Asp Pro Ser Gly Arg Pro Leu Arg Leu Pro Pro Val Leu Ser
                    170                      175                      180
His Pro Ile Phe Asp Asn His Asp Arg His Arg Ile Glu Glu Lys
                    185                      190                      195
Arg Lys Arg Thr Tyr Glu Thr Phe Lys Ser Ile Met Lys Lys Ser
                    200                      205                      210
Pro Phe Ser Gly Pro Thr Asp Pro Arg Pro Pro Pro Arg Arg Ile
                    215                      220                      225
Ala Val Pro Ser Arg Ser Ser Ala Ser Val Pro Lys Pro Ala Pro
                    230                      235                      240
Gln Pro Tyr Pro Phe Thr Ser Ser Leu Ser Thr Ile Asn Tyr Asp
                    245                      250                      255
Glu Phe Pro Thr Met Val Phe Pro Ser Gly Gln Ile Ser Gln Ala
                    260                      265                      270
Ser Ala Leu Ala Pro Ala Pro Pro Gln Val Leu Pro Gln Ala Pro
                    275                      280                      285
Ala Pro Ala Pro Ala Pro Ala Met Val Ser Ala Leu Ala Gln Ala
                    290                      295                      300
Pro Ala Pro Val Pro Val Leu Ala Pro Gly Pro Pro Gln Ala Val
                    305                      310                      315
Ala Pro Pro Ala Pro Lys Pro Thr Gln Ala Gly Glu Gly Thr Leu
                    320                      325                      330
Ser Glu Ala Leu Leu Gln Leu Gln Phe Asp Asp Glu Asp Leu Gly
                    335                      340                      345
Ala Leu Leu Gly Asn Ser Thr Asp Pro Ala Val Phe Thr Asp Leu
                    350                      355                      360
Ala Ser Val Asp Asn Ser Glu Phe Gln Gln Leu Leu Asn Gln Gly
                    365                      370                      375
Ile Pro Val Ala Pro His Thr Thr Glu Pro Met Leu Met Glu Tyr
                    380                      385                      390
Pro Glu Ala Ile Thr Arg Leu Val Thr Gly Ala Gln Arg Pro Pro
                    395                      400                      405
Asp Pro Ala Pro Ala Pro Leu Gly Ala Pro Gly Leu Pro Asn Gly
                    410                      415                      420
Leu Leu Ser Gly Asp Glu Asp Phe Ser Ser Ile Ala Asp Met Asp
                    425                      430                      435
Phe Ser Ala Leu Leu Ser Gln Ile Ser Ser
                    440                      445
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<210> 8
<211> 375
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512177CD1

<400> 8
Met Asn Ser Gly Ile Ser Gln Val Phe Gln Arg Glu Leu Thr Cys
1 5 10 15
Pro Ile Cys Leu Asn Tyr Phe Ile Asp Pro Val Thr Ile Asp Cys
20 25 30
Gly His Ser Phe Cys Arg Pro Cys Phe Tyr Leu Asn Trp Gln Asp
35 40 45
Ile Pro Ile Leu Thr Gln Cys Phe Glu Cys Leu Lys Thr Thr Gln
50 55 60
Gln Arg Asn Leu Lys Thr Asn Ile Arg Leu Lys Lys Met Ala Ser
65 70 75
Arg Ala Arg Lys Ala Ser Leu Trp Leu Phe Leu Ser Ser Glu Glu
80 85 90
Gln Met Cys Gly Thr His Arg Glu Thr Lys Lys Ile Phe Cys Glu
95 100 105
Val Asp Arg Ser Leu Leu Cys Leu Leu Cys Ser Ser Ser Leu Glu
110 115 120
His Arg Tyr His Arg His Cys Pro Ala Glu Trp Ala Ala Glu Glu
125 130 135
His Arg Glu Lys Leu Leu Lys Lys Met Gln Ser Leu Trp Glu Lys
140 145 150
Val Cys Glu Asn Gln Arg Asn Leu Asn Val Glu Thr Thr Arg Ile
155 160 165
Ser His Trp Lys Ala Phe Gly Asp Ile Leu His Arg Ser Glu Ser
170 175 180
Val Leu Leu His Met Pro Gln Pro Leu Asn Leu Glu Leu Arg Ala
185 190 195
Gly Pro Ile Thr Gly Leu Arg Asp Arg Leu Asn Gln Phe Arg Val
200 205 210
Asp Ile Thr Leu Pro His Asn Glu Ala Asn Ser His Ile Phe Arg
215 220 225
Arg Gly Asp Leu Arg Ser Ile Cys Ile Gly Cys Asp Arg Gln Asn
230 235 240
Ala Pro His Ile Thr Ala Thr Pro Thr Ser Phe Leu Ala Trp Gly
245 250 255
Ala Gln Thr Phe Thr Ser Gly Lys Tyr Tyr Trp Glu Val His Val
260 265 270
Gly Asp Ser Trp Asn Trp Ala Phe Gly Val Cys Asn Lys Tyr Trp
275 280 285
Lys Gly Thr Asn Gln Asn Gly Asn Ile His Gly Glu Glu Gly Leu
290 295 300
Phe Ser Leu Gly Cys Val Lys Asn Asp Ile Gln Cys Ser Leu Phe
305 310 315
Thr Thr Ser Pro Leu Thr Leu Gln Tyr Val Pro Arg Pro Thr Asn
320 325 330
His Val Gly Leu Phe Leu Asp Cys Glu Ala Arg Thr Val Ser Phe
335 340 345
Val Asp Val Asn Gln Ser Ser Pro Ile His Thr Ile Pro Asn Cys
350 355 360
Ser Phe Ser Pro Pro Leu Arg Pro Ile Phe Cys Cys Val His Leu
365 370 375

<210> 9
<211> 96
<212> PRT
<213> Homo sapiens

<220>

<221> misc_feature
<223> Incyte ID No: 7511867CD1

<400> 9
Met Ala Ala Val Gln Ala Ala Glu Val Lys Val Asp Gly Ser Glu
1 5 10 15
Pro Lys Leu Ser Lys Asn Glu Leu Lys Arg Arg Leu Lys Ala Glu
20 25 30
Lys Lys Val Ala Glu Lys Glu Ala Lys Gln Lys Glu Leu Ser Glu
35 40 45
Lys Gln Leu Ser Gln Ala Thr Ala Ala Ala Thr Asn His Thr Thr
50 55 60
Asp Asn Gly Val Gly Pro Glu Glu Glu Ser Val Asp Pro Asn Val
65 70 75
Gly Ser Met Pro Lys Glu Leu Leu Gly Glu Ser Ser Ser Ser Met
80 85 90
Ile Phe Glu Glu Arg Gly
95

<210> 10
<211> 430
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512866CD1

<400> 10
Met Ala Ala Val Val Leu Pro Pro Thr Ala Ala Leu Ser Ser Leu
1 5 10 15
Phe Pro Ala Ser Gln Arg Glu Gly His Thr Glu Gly Gly Glu Leu
20 25 30
Val Asn Glu Leu Leu Lys Ser Trp Leu Lys Gly Leu Val Thr Phe
35 40 45
Glu Asp Val Ala Val Glu Phe Thr Gln Glu Glu Trp Ala Leu Leu
50 55 60
Asp Pro Ala Gln Arg Thr Leu Tyr Arg Asp Val Met Leu Glu Asn
65 70 75
Cys Arg Asn Leu Ala Ser Leu Gly Asn Gln Val Asp Lys Pro Arg
80 85 90
Leu Ile Ser Gln Leu Glu Gln Glu Asp Lys Val Met Thr Glu Glu
95 100 105
Arg Gly Ile Leu Ser Gly Thr Cys Pro Asp Val Glu Asn Pro Phe
110 115 120
Lys Ala Lys Gly Leu Thr Pro Lys Leu His Val Phe Arg Lys Glu
125 130 135
Gln Ser Arg Asn Met Lys Met Glu Arg Asn His Leu Gly Ala Thr
140 145 150
Leu Asn Glu Cys Asn Gln Cys Phe Lys Val Phe Ser Thr Lys Ser
155 160 165
Ser Leu Thr Arg His Arg Lys Ile His Thr Gly Glu Arg Pro Tyr
170 175 180
Gly Cys Ser Glu Cys Gly Lys Ser Tyr Ser Ser Arg Ser Tyr Leu
185 190 195
Ala Val His Lys Arg Ile His Asn Gly Glu Lys Pro Tyr Glu Cys
200 205 210
Asn Asp Cys Gly Lys Thr Phe Ser Ser Arg Ser Tyr Leu Thr Val
215 220 225
His Lys Arg Ile His Asn Gly Glu Lys Pro Tyr Glu Cys Ser Asp
230 235 240
Cys Gly Lys Thr Phe Ser Asn Ser Ser Tyr Leu Arg Pro His Leu
245 250 255

```
Arg Ile His Thr Gly Glu Lys Pro Tyr Lys Cys Asn Gln Cys Phe
                260                 265                 270
Arg Glu Phe Arg Thr Gln Ser Ile Phe Thr Arg His Lys Arg Val
                275                 280                 285
His Thr Gly Glu Gly His Tyr Val Cys Asn Gln Cys Gly Lys Ala
                290                 295                 300
Phe Gly Thr Arg Ser Ser Leu Ser Ser His Tyr Ser Ile His Thr
                305                 310                 315
Gly Glu Tyr Pro Tyr Glu Cys His Asp Cys Gly Arg Thr Phe Arg
                320                 325                 330
Arg Arg Ser Asn Leu Thr Gln His Ile Arg Thr His Thr Gly Glu
                335                 340                 345
Lys Pro Tyr Thr Cys Asn Glu Cys Gly Lys Ser Phe Thr Asn Ser
                350                 355                 360
Phe Ser Leu Thr Ile His Arg Arg Ile His Asn Gly Glu Lys Ser
                365                 370                 375
Tyr Glu Cys Ser Asp Cys Gly Lys Ser Phe Asn Val Leu Ser Ser
                380                 385                 390
Val Lys Lys His Met Arg Thr His Thr Gly Lys Lys Pro Tyr Glu
                395                 400                 405
Cys Asn Tyr Cys Gly Lys Ser Phe Thr Ser Asn Ser Tyr Leu Ser
                410                 415                 420
Val His Thr Arg Met His Asn Arg Gln Met
                425                 430

<210> 11
<211> 268
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512572CD1

<400> 11
Met Ala Thr Ala Val Arg Ala Val Gly Cys Leu Pro Val Leu Cys
  1               5                  10                  15
Ser Gly Thr Ala Gly His Leu Leu Gly Arg Gln Cys Ser Leu Asn
                 20                  25                  30
Thr Leu Pro Ala Ala Ser Ile Leu Ala Trp Lys Ser Val Leu Gly
                 35                  40                  45
Asn Gly His Leu Ser Ser Leu Gly Thr Arg Asp Thr His Pro Tyr
                 50                  55                  60
Ala Ser Leu Ser Arg Ala Leu Gln Thr Gln Cys Cys Ile Ser Ser
                 65                  70                  75
Pro Ser His Leu Met Ser Gln Gln Tyr Arg Pro Tyr Ser Phe Phe
                 80                  85                  90
Thr Lys Cys Phe Ser Ile Gly Lys Ala Thr Asp Arg Met Asp Ala
                 95                 100                 105
Phe Arg Lys Ala Lys Asn Arg Ala Val His His Leu His Tyr Ile
                110                 115                 120
Glu Arg Tyr Glu Asp His Thr Ile Phe His Asp Ile Ser Leu Arg
                125                 130                 135
Phe Lys Arg Thr His Ile Lys Met Lys Lys Gln Pro Lys Gly Tyr
                140                 145                 150
Gly Leu Arg Cys His Arg Ala Ile Ile Thr Ile Cys Arg Leu Ile
                155                 160                 165
Gly Ile Lys Asp Met Tyr Ala Lys Val Ser Gly Ser Ile Asn Met
                170                 175                 180
Leu Ser Leu Thr Gln Gly Leu Phe Arg Gly Leu Ser Arg Gln Glu
                185                 190                 195
Thr His Gln Gln Leu Ala Asp Lys Lys Gly Leu His Val Val Glu
                200                 205                 210
```

Ile Arg Glu Glu Cys Gly Pro Leu Pro Ile Val Val Ala Ser Pro
215 220 225
Arg Gly Pro Leu Arg Lys Asp Pro Glu Pro Glu Asp Glu Val Pro
230 235 240
Asp Val Lys Leu Asp Trp Glu Asp Val Lys Thr Ala Gln Gly Met
245 250 255
Lys Arg Ser Val Trp Ser Asn Leu Lys Arg Ala Ala Thr
260 265

<210> 12
<211> 79
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512518CD1

<400> 12
Met Ser Glu Glu Glu Gln Gly Ser Gly Thr Thr Thr Gly Cys Gly
1 5 10 15
Leu Pro Ser Ile Glu Gln Met Leu Ala Ala Asn Pro Gly Lys Thr
20 25 30
Pro Ile Ser Leu Leu Gln Glu Tyr Gly Thr Arg Ile Gly Lys Thr
35 40 45
Pro Val Tyr Asp Leu Leu Lys Ala Glu Gly Gln Ala His Gln Pro
50 55 60
Asn Phe Thr Phe Arg Val Thr Val Gly Asp Thr Ser Cys Thr Val
65 70 75
Leu Phe Leu Pro

<210> 13
<211> 285
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512669CD1

<400> 13
Met Ala Ala Ala Glu Ala Ala Asn Cys Ile Met Glu Val Ser Cys
1 5 10 15
Gly Gln Ala Glu Ser Ser Glu Lys Pro Asn Ala Glu Asp Met Thr
20 25 30
Ser Lys Asp Tyr Tyr Phe Asp Ser Tyr Ala His Phe Gly Ile His
35 40 45
Glu Glu Met Leu Lys Asp Glu Val Arg Thr Leu Thr Tyr Arg Asn
50 55 60
Ser Met Phe His Asn Arg His Leu Phe Lys Asp Lys Val Val Leu
65 70 75
Asp Val Gly Ser Gly Thr Gly Ile Leu Cys Met Phe Ala Ala Lys
80 85 90
Ala Gly Ala Arg Lys Val Ile Gly Ile Glu Cys Ser Ser Ile Ser
95 100 105
Asp Tyr Ala Val Lys Ile Val Lys Ala Asn Lys Leu Asp His Val
110 115 120
Val Thr Ile Ile Lys Gly Lys Val Glu Glu Val Glu Leu Pro Val
125 130 135
Glu Lys Val Asp Ile Ile Ile Ser Glu Trp Met Gly Tyr Cys Leu
140 145 150
Phe Tyr Glu Ser Met Leu Asn Thr Val Leu Tyr Ala Arg Asp Lys

```
                155                 160                 165
Trp Leu Glu Val Asp Ile Tyr Thr Val Lys Val Glu Asp Leu Thr
                170                 175                 180
Phe Thr Ser Pro Phe Cys Leu Gln Val Lys Arg Asn Asp Tyr Val
                185                 190                 195
His Ala Leu Val Ala Tyr Phe Asn Ile Glu Phe Thr Arg Cys His
                200                 205                 210
Lys Arg Thr Gly Phe Ser Thr Ser Pro Glu Ser Pro Tyr Thr His
                215                 220                 225
Trp Lys Gln Thr Val Phe Tyr Met Glu Asp Tyr Leu Thr Val Lys
                230                 235                 240
Thr Gly Glu Glu Ile Phe Gly Thr Ile Gly Met Arg Pro Asn Ala
                245                 250                 255
Lys Asn Asn Arg Asp Leu Asp Phe Thr Ile Asp Leu Asp Phe Lys
                260                 265                 270
Gly Gln Leu Cys Glu Leu Ser Cys Ser Thr Asp Tyr Arg Met Arg
                275                 280                 285
```

<210> 14
<211> 110
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512706CD1

<400> 14

```
Met Arg Leu Phe Val Ser Asp Gly Val Pro Gly Cys Leu Pro Val
  1               5                  10                  15
Leu Ala Ala Ala Gly Arg Ala Arg Gly Arg Ala Glu Val Leu Ile
                 20                  25                  30
Ser Thr Val Gly Pro Glu Asp Cys Val Val Pro Phe Leu Thr Arg
                 35                  40                  45
Pro Lys Val Pro Val Leu Gln Leu Asp Ser Gly Asn Tyr Leu Phe
                 50                  55                  60
Ser Thr Ser Ala Ile Cys Arg Tyr Phe Phe Leu Leu Ser Gly Trp
                 65                  70                  75
Glu Gln Asp Asp Leu Thr Asn Gln Trp Leu Glu Trp Glu Ala Thr
                 80                  85                  90
Glu Leu Gln Trp Ser Lys Ala Arg Arg Gly Lys Met Phe Leu Val
                 95                 100                 105
Gln Cys Gly Glu Pro
                110
```

<210> 15
<211> 287
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512872CD1

<400> 15

```
Met Ser Glu Glu Gln Phe Gly Gly Asp Gly Ala Ala Ala Ala Ala
  1               5                  10                  15
Thr Ala Ala Val Gly Gly Ser Ala Gly Glu Gln Glu Gly Ala Met
                 20                  25                  30
Val Ala Ala Thr Gln Gly Ala Ala Ala Ala Ala Gly Ser Gly Ala
                 35                  40                  45
Gly Thr Gly Gly Gly Thr Ala Ser Gly Gly Thr Glu Gly Gly Ser
```

50 55 60
Ala Glu Ser Glu Gly Ala Lys Ile Asp Ala Ser Lys Asn Glu Glu
65 70 75
Asp Glu Gly Lys Met Phe Ile Gly Gly Leu Ser Trp Asp Thr Thr
80 85 90
Lys Lys Asp Leu Lys Asp Tyr Phe Ser Lys Phe Gly Glu Val Val
95 100 105
Asp Cys Thr Leu Lys Leu Asp Pro Ile Thr Gly Arg Ser Arg Gly
110 115 120
Phe Gly Phe Val Leu Phe Lys Glu Ser Glu Ser Val Asp Lys Val
125 130 135
Met Asp Gln Lys Glu His Lys Leu Asn Gly Lys Val Ile Asp Pro
140 145 150
Lys Arg Ala Lys Ala Met Lys Thr Lys Glu Pro Val Lys Lys Ile
155 160 165
Phe Val Gly Gly Leu Ser Pro Asp Thr Pro Glu Glu Lys Ile Arg
170 175 180
Glu Tyr Phe Gly Gly Phe Gly Glu Val Glu Ser Ile Glu Leu Pro
185 190 195
Met Asp Asn Lys Thr Asn Lys Arg Arg Gly Phe Cys Phe Ile Thr
200 205 210
Phe Lys Glu Glu Glu Pro Val Lys Lys Ile Met Glu Lys Lys Tyr
215 220 225
His Asn Val Gly Leu Ser Lys Cys Glu Ile Lys Val Ala Met Ser
230 235 240
Lys Glu Gln Tyr Gln Gln Gln Gln Gln Trp Gly Ser Arg Gly Gly
245 250 255
Phe Ala Gly Arg Ala Arg Gly Arg Gly Gly Asp Gln Gln Ser Gly
260 265 270
Tyr Gly Lys Val Ser Arg Arg Gly Gly His Gln Asn Ser Tyr Lys
275 280 285
Pro Tyr

<210> 16
<211> 948
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512925CD1

<400> 16
Met Ser Gly Pro Gly Asn Lys Arg Ala Ala Gly Asp Gly Gly Ser
1 5 10 15
Gly Pro Pro Glu Lys Lys Leu Ser Arg Glu Glu Lys Thr Thr Thr
20 25 30
Thr Leu Ile Glu Pro Ile Arg Leu Gly Gly Ile Ser Ser Thr Glu
35 40 45
Glu Met Asp Leu Lys Val Leu Gln Phe Lys Asn Lys Lys Leu Ala
50 55 60
Glu Arg Leu Glu Gln Arg Gln Ala Cys Glu Asp Glu Leu Arg Glu
65 70 75
Arg Ile Glu Lys Leu Glu Lys Arg Gln Ala Thr Asp Asp Ala Thr
80 85 90
Leu Leu Ile Val Asn Arg Tyr Trp Ala Gln Leu Asp Glu Thr Val
95 100 105
Glu Ala Leu Leu Arg Cys His Glu Ser Gln Gly Glu Leu Ser Ser
110 115 120
Ala Pro Glu Ala Pro Gly Thr Gln Glu Gly Pro Thr Cys Asp Gly
125 130 135
Thr Pro Leu Pro Glu Pro Gly Thr Ser Glu Leu Arg Asp Pro Leu

140 145 150
Leu Met Gln Leu Arg Pro Pro Leu Ser Glu Pro Ala Leu Ala Phe
155 160 165
Val Val Ala Leu Gly Ala Ser Ser Ser Glu Glu Val Glu Leu Glu
170 175 180
Leu Gln Gly Arg Met Glu Phe Ser Lys Ala Ala Val Ser Arg Val
185 190 195
Val Glu Ala Ser Asp Arg Leu Gln Arg Arg Val Glu Glu Leu Cys
200 205 210
Gln Arg Val Tyr Ser Arg Gly Asp Ser Glu Pro Leu Ser Glu Ala
215 220 225
Ala Gln Ala His Thr Arg Glu Leu Gly Arg Glu Asn Arg Arg Leu
230 235 240
Gln Asp Leu Ala Thr Gln Leu Gln Glu Lys His His Arg Ile Ser
245 250 255
Leu Glu Tyr Ser Glu Leu Gln Asp Lys Val Thr Ser Ala Glu Thr
260 265 270
Lys Val Leu Glu Met Glu Thr Thr Val Glu Asp Leu Gln Trp Asp
275 280 285
Ile Glu Lys Leu Arg Lys Arg Glu Gln Lys Leu Asn Lys His Leu
290 295 300
Ala Glu Ala Leu Glu Gln Leu Asn Ser Gly Tyr Tyr Val Ser Gly
305 310 315
Ser Ser Ser Gly Phe Gln Gly Gly Gln Ile Thr Leu Ser Met Gln
320 325 330
Lys Phe Glu Met Leu Asn Ala Glu Leu Glu Glu Asn Gln Glu Leu
335 340 345
Ala Asn Ser Arg Met Ala Glu Leu Glu Lys Leu Gln Ala Glu Leu
350 355 360
Gln Gly Ala Val Arg Thr Asn Glu Arg Leu Lys Val Ala Leu Arg
365 370 375
Ser Leu Pro Glu Glu Val Val Arg Glu Thr Gly Glu Tyr Arg Met
380 385 390
Leu Gln Ala Gln Phe Ser Leu Leu Tyr Asn Glu Ser Leu Gln Val
395 400 405
Lys Thr Gln Leu Asp Glu Ala Arg Gly Leu Leu Leu Ala Thr Lys
410 415 420
Asn Ser His Leu Arg His Ile Glu His Met Glu Ser Asp Glu Leu
425 430 435
Gly Leu Gln Lys Lys Leu Arg Thr Glu Val Ile Gln Leu Glu Asp
440 445 450
Thr Leu Ala Gln Val Arg Lys Glu Tyr Glu Met Leu Arg Ile Glu
455 460 465
Phe Glu Gln Asn Leu Ala Ala Asn Glu Gln Ala Gly Pro Ile Asn
470 475 480
Arg Glu Met Arg His Leu Ile Ser Ser Leu Gln Asn His Asn His
485 490 495
Gln Leu Lys Gly Asp Ala Gln Arg Tyr Lys Arg Lys Leu Arg Glu
500 505 510
Val Gln Ala Glu Ile Gly Lys Leu Arg Ala Gln Ala Ser Gly Ser
515 520 525
Ala His Ser Thr Pro Asn Leu Gly His Pro Glu Asp Ser Gly Val
530 535 540
Ser Ala Pro Ala Pro Gly Lys Glu Glu Gly Gly Pro Gly Pro Val
545 550 555
Ser Thr Pro Asp Asn Arg Lys Glu Met Ala Pro Val Pro Gly Thr
560 565 570
Thr Thr Thr Thr Thr Ser Val Lys Lys Glu Glu Leu Val Pro Ser
575 580 585
Glu Glu Asp Phe Gln Gly Ile Thr Pro Gly Ala Gln Gly Pro Ser
590 595 600
Ser Arg Gly Arg Glu Pro Glu Ala Arg Pro Lys Arg Glu Leu Arg
605 610 615

Glu Arg Glu Gly Pro Ser Leu Gly Pro Pro Pro Val Ala Ser Ala
620 625 630
Leu Ser Arg Ala Asp Arg Glu Lys Ala Lys Val Glu Glu Thr Lys
635 640 645
Arg Lys Glu Ser Glu Leu Leu Lys Gly Leu Arg Ala Glu Leu Lys
650 655 660
Lys Ala Gln Glu Ser Gln Lys Glu Met Lys Leu Leu Leu Asp Met
665 670 675
Tyr Lys Ser Ala Pro Lys Glu Gln Arg Asp Lys Val Gln Leu Met
680 685 690
Ala Ala Glu Arg Lys Ala Lys Ala Glu Val Asp Glu Leu Arg Ser
695 700 705
Arg Ile Arg Glu Leu Glu Glu Arg Asp Arg Arg Glu Ser Lys Lys
710 715 720
Ile Ala Asp Glu Asp Ala Leu Arg Arg Ile Arg Gln Ala Glu Glu
725 730 735
Gln Ile Glu His Leu Gln Arg Lys Leu Gly Ala Thr Lys Gln Glu
740 745 750
Glu Glu Ala Leu Leu Ser Glu Met Asp Val Thr Gly Gln Ala Phe
755 760 765
Glu Asp Met Gln Glu Gln Asn Gly Arg Leu Leu Gln Gln Leu Arg
770 775 780
Glu Lys Asp Asp Ala Asn Phe Lys Leu Met Ser Glu Arg Ile Lys
785 790 795
Ala Asn Gln Ile His Lys Leu Leu Arg Glu Glu Lys Asp Glu Leu
800 805 810
Gly Glu Gln Val Leu Gly Leu Lys Ser Gln Val Asp Ala Gln Leu
815 820 825
Leu Thr Val Gln Lys Leu Glu Glu Lys Glu Arg Ala Leu Gln Gly
830 835 840
Ser Leu Gly Gly Val Glu Lys Glu Leu Thr Leu Arg Ser Gln Ala
845 850 855
Leu Glu Leu Asn Lys Arg Lys Ala Val Glu Ala Ala Gln Leu Ala
860 865 870
Glu Asp Leu Lys Val Gln Leu Glu His Val Gln Thr Arg Leu Arg
875 880 885
Glu Ile Gln Pro Cys Leu Ala Glu Ser Arg Ala Ala Arg Glu Lys
890 895 900
Glu Ser Phe Asn Leu Lys Arg Ala Gln Glu Asp Ile Ser Arg Leu
905 910 915
Arg Arg Lys Leu Glu Lys Gln Arg Lys Val Glu Val Tyr Ala Asp
920 925 930
Ala Asp Glu Ile Leu Gln Glu Glu Ile Lys Glu Tyr Lys Val Arg
935 940 945
Ala Ala Ala

<210> 17
<211> 94
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512927CD1

<400> 17
Met Phe Glu Glu Lys Ala Ser Ser Pro Ser Gly Lys Met Gly Gly
1 5 10 15
Glu Glu Lys Pro Ile Gly Ala Gly Glu Glu Lys Gln Lys Glu Gly
20 25 30
Gly Lys Lys Lys Asn Lys Glu Gly Ser Gly Asp Gly Gly Arg Ala
35 40 45

Glu Lys Arg Gln Lys Lys Ile Ala Ser Gln Leu Lys Ser Leu Cys
50 55 60
Leu Met Val Asn Arg Leu Met Arg Asn Leu Gly Lys Leu His His
65 70 75
Ile Lys Leu Pro Val Glu Leu Val Lys Ala Trp Pro Thr Thr Pro
80 85 90
Leu Leu Leu Lys

<210> 18
<211> 189
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512943CD1

<400> 18
Met Arg Leu Ala Ala Ala Ala Asn Glu Ala Tyr Thr Ala Pro Leu
1 5 10 15
Ala Val Ser Gly Leu Leu Gly Cys Lys Gln Cys Gly Gly Gly Arg
20 25 30
Asp Gln Asp Glu Glu Leu Gly Ile Arg Ile Pro Arg Pro Leu Gly
35 40 45
Gln Gly Pro Ser Arg Phe Ile Pro Glu Lys Glu Ile Leu Gln Val
50 55 60
Gly Ser Glu Asp Ala Gln Met His Ala Leu Phe Ala Asp Ser Phe
65 70 75
Ala Ala Leu Gly Arg Leu Asp Asn Ile Thr Leu Val Met Val Phe
80 85 90
His Pro Gln Tyr Leu Glu Ser Phe Leu Lys Thr Gln His Tyr Leu
95 100 105
Leu Gln Met Asp Gly Pro Leu Pro Leu His Tyr Arg His Tyr Ile
110 115 120
Gly Ile Met Ala Ala Ala Arg His Gln Cys Ser Tyr Leu Val Asn
125 130 135
Leu His Val Asn Asp Phe Leu His Val Gly Gly Asp Pro Lys Trp
140 145 150
Leu Asn Gly Leu Glu Asn Ala Pro Gln Lys Leu Gln Asn Leu Gly
155 160 165
Glu Leu Asn Lys Val Leu Ala His Arg Pro Trp Leu Ile Thr Lys
170 175 180
Glu His Ile Glu Ser Arg Asn Ser Leu
185

<210> 19
<211> 96
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512955CD1

<400> 19
Met Arg Leu Ser Ala Leu Leu Ala Leu Ala Ser Lys Val Thr Leu
1 5 10 15
Pro Pro His Tyr Arg Tyr Gly Met Ser Pro Pro Gly Ser Val Ala
20 25 30
Asp Lys Arg Lys Asn Pro Pro Trp Ile Arg Arg Arg Pro Val Val
35 40 45
Val Glu Pro Ile Ser Asp Glu Asp Trp Tyr Leu Phe Cys Gly Asp

50 55 60
Thr Val Glu Ile Leu Glu Gly Lys Asp Ala Gly Lys Gln Gly Lys
65 70 75
Val Val Gln Val Ile Arg Gln Arg Asn Trp Val Val Val Gly Gly
80 85 90
Leu Asn Thr Glu Thr His
95

<210> 20
<211> 1395
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7512950CD1

<400> 20
Met Tyr Ala Val Tyr Lys Gln Ala His Pro Pro Thr Gly Leu Glu
1 5 10 15
Phe Ser Met Tyr Cys Asn Phe Phe Asn Asn Ser Glu Arg Asn Leu
20 25 30
Val Val Ala Gly Thr Ser Gln Leu Tyr Val Tyr Arg Leu Asn Arg
35 40 45
Asp Ala Glu Ala Leu Thr Lys Asn Asp Arg Ser Thr Glu Gly Lys
50 55 60
Ala His Arg Glu Lys Leu Glu Leu Ala Ala Ser Phe Ser Phe Phe
65 70 75
Gly Asn Val Met Ser Met Ala Ser Val Gln Leu Ala Gly Ala Lys
80 85 90
Arg Asp Ala Leu Leu Leu Ser Phe Lys Asp Ala Lys Leu Ser Val
95 100 105
Val Glu Tyr Asp Pro Gly Thr His Asp Leu Lys Thr Leu Ser Leu
110 115 120
His Tyr Phe Glu Glu Pro Glu Leu Arg Asp Gly Phe Val Gln Asn
125 130 135
Val His Thr Pro Arg Val Arg Val Asp Pro Asp Gly Arg Cys Ala
140 145 150
Ala Met Leu Val Tyr Gly Thr Arg Leu Val Val Leu Pro Phe Arg
155 160 165
Arg Glu Ser Leu Ala Glu Glu His Glu Gly Leu Val Gly Glu Gly
170 175 180
Gln Arg Ser Ser Phe Leu Pro Ser Tyr Ile Ile Asp Val Arg Ala
185 190 195
Leu Asp Glu Lys Leu Leu Asn Ile Ile Asp Leu Gln Phe Leu His
200 205 210
Gly Tyr Tyr Glu Pro Thr Leu Leu Ile Leu Phe Glu Pro Asn Gln
215 220 225
Thr Trp Pro Gly Arg Val Ala Val Arg Gln Asp Thr Cys Ser Ile
230 235 240
Val Ala Ile Ser Leu Asn Ile Thr Gln Lys Val His Pro Val Ile
245 250 255
Trp Ser Leu Thr Ser Leu Pro Phe Asp Cys Thr Gln Ala Leu Ala
260 265 270
Val Pro Lys Pro Ile Gly Gly Val Val Val Phe Ala Val Asn Ser
275 280 285
Leu Leu Tyr Leu Asn Gln Ser Val Pro Pro Tyr Gly Val Ala Leu
290 295 300
Asn Ser Leu Thr Thr Gly Thr Thr Ala Phe Pro Leu Arg Thr Gln
305 310 315
Glu Gly Val Arg Ile Thr Leu Asp Cys Ala Gln Ala Thr Phe Ile
320 325 330
Ser Tyr Asp Lys Met Val Ile Ser Leu Lys Gly Gly Glu Ile Tyr

335 340 345
Val Leu Thr Leu Ile Thr Asp Gly Met Arg Ser Val Arg Ala Phe
350 355 360
His Phe Asp Lys Ala Ala Ala Ser Val Leu Thr Thr Ser Met Val
365 370 375
Thr Met Glu Pro Gly Tyr Leu Phe Leu Gly Ser Arg Leu Gly Asn
380 385 390
Ser Leu Leu Leu Lys Tyr Thr Glu Lys Leu Gln Glu Pro Pro Ala
395 400 405
Ser Ala Val Arg Glu Ala Ala Asp Lys Glu Glu Pro Pro Ser Lys
410 415 420
Lys Lys Arg Val Asp Ala Thr Ala Gly Trp Ser Ala Ala Gly Lys
425 430 435
Ser Val Pro Gln Asp Glu Val Asp Glu Ile Glu Val Tyr Gly Ser
440 445 450
Glu Ala Gln Ser Gly Thr Gln Leu Ala Thr Tyr Ser Phe Glu Val
455 460 465
Cys Asp Ser Ile Leu Asn Ile Gly Pro Cys Ala Asn Ala Ala Val
470 475 480
Gly Glu Pro Ala Phe Leu Ser Glu Glu Phe Gln Asn Ser Pro Glu
485 490 495
Pro Asp Leu Glu Ile Val Val Cys Ser Gly His Gly Lys Asn Gly
500 505 510
Ala Leu Ser Val Leu Gln Lys Ser Ile Arg Pro Gln Val Val Thr
515 520 525
Thr Phe Glu Leu Pro Gly Cys Tyr Asp Met Trp Thr Val Ile Ala
530 535 540
Pro Val Arg Lys Glu Glu Glu Asp Asn Pro Lys Gly Glu Gly Thr
545 550 555
Glu Gln Glu Pro Ser Thr Thr Pro Glu Ala Asp Asp Asp Gly Arg
560 565 570
Arg His Gly Phe Leu Ile Leu Ser Arg Glu Asp Ser Thr Met Ile
575 580 585
Leu Gln Thr Gly Gln Glu Ile Met Glu Leu Asp Thr Ser Gly Phe
590 595 600
Ala Thr Gln Gly Pro Thr Val Phe Ala Gly Asn Ile Gly Asp Asn
605 610 615
Arg Tyr Ile Val Gln Val Ser Pro Leu Gly Ile Arg Leu Leu Glu
620 625 630
Gly Val Asn Gln Leu His Phe Ile Pro Val Asp Leu Gly Ala Pro
635 640 645
Ile Val Gln Cys Ala Val Ala Asp Pro Tyr Val Val Ile Met Ser
650 655 660
Ala Glu Gly His Val Thr Met Phe Leu Leu Lys Ser Asp Ser Tyr
665 670 675
Gly Gly Arg His His Arg Leu Ala Leu His Lys Pro Pro Leu His
680 685 690
His Gln Ser Lys Val Ile Thr Leu Cys Leu Tyr Arg Asp Leu Ser
695 700 705
Gly Met Phe Thr Thr Glu Ser Arg Leu Gly Gly Ala Arg Asp Glu
710 715 720
Leu Gly Gly Arg Ser Gly Pro Glu Ala Glu Gly Leu Gly Ser Glu
725 730 735
Thr Ser Pro Thr Val Asp Asp Glu Glu Glu Met Leu Tyr Gly Asp
740 745 750
Ser Gly Ser Leu Phe Ser Pro Ser Lys Glu Glu Ala Arg Arg Ser
755 760 765
Ser Gln Pro Pro Ala Asp Arg Asp Pro Ala Pro Phe Arg Ala Glu
770 775 780
Pro Thr His Trp Cys Leu Leu Val Arg Glu Asn Gly Thr Met Glu
785 790 795
Ile Tyr Gln Leu Pro Asp Trp Arg Leu Val Phe Leu Val Lys Asn
800 805 810

Phe Pro Val Gly Gln Arg Val Leu Val Asp Ser Ser Phe Gly Gln
815 820 825
Pro Thr Thr Gln Gly Glu Ala Arg Arg Glu Glu Ala Thr Arg Gln
830 835 840
Gly Glu Leu Pro Leu Val Lys Glu Val Leu Leu Val Ala Leu Gly
845 850 855
Ser Arg Gln Ser Arg Pro Tyr Leu Leu Val His Val Asp Gln Glu
860 865 870
Leu Leu Ile Tyr Glu Ala Phe Pro His Asp Ser Gln Leu Gly Gln
875 880 885
Gly Asn Leu Lys Val Arg Phe Lys Lys Val Phe Ile Cys Gly Pro
890 895 900
Ser Pro His Trp Leu Leu Val Thr Gly Arg Gly Ala Leu Arg Leu
905 910 915
His Pro Met Ala Ile Asp Gly Pro Val Asp Ser Phe Ala Pro Phe
920 925 930
His Asn Val Asn Cys Pro Arg Gly Phe Leu Tyr Phe Asn Arg Gln
935 940 945
Gly Glu Leu Arg Ile Ser Val Leu Pro Ala Tyr Leu Ser Tyr Asp
950 955 960
Ala Pro Trp Pro Val Arg Lys Ile Pro Leu Arg Cys Thr Ala His
965 970 975
Tyr Val Ala Tyr His Val Glu Ser Lys Val Tyr Ala Val Ala Thr
980 985 990
Ser Thr Asn Thr Pro Cys Ala Arg Ile Pro Arg Met Thr Gly Glu
995 1000 1005
Glu Lys Glu Phe Glu Thr Ile Glu Arg Asp Glu Arg Tyr Ile His
1010 1015 1020
Pro Gln Gln Glu Ala Phe Ser Ile Gln Leu Ile Ser Pro Val Ser
1025 1030 1035
Trp Glu Ala Ile Pro Asn Ala Arg Ile Glu Leu Gln Glu Trp Glu
1040 1045 1050
His Val Thr Cys Met Lys Thr Val Ser Leu Arg Ser Glu Glu Thr
1055 1060 1065
Val Ser Gly Leu Lys Gly Tyr Val Ala Ala Gly Thr Cys Leu Met
1070 1075 1080
Gln Gly Glu Glu Val Thr Cys Arg Gly Arg Ile Leu Ile Met Asp
1085 1090 1095
Val Ile Glu Val Val Pro Glu Pro Gly Gln Pro Leu Thr Lys Asn
1100 1105 1110
Lys Phe Lys Val Leu Tyr Glu Lys Glu Gln Lys Gly Pro Val Thr
1115 1120 1125
Ala Leu Cys His Cys Asn Gly His Leu Val Ser Ala Ile Gly Gln
1130 1135 1140
Lys Ile Phe Leu Trp Ser Leu Arg Ala Ser Glu Leu Thr Gly Met
1145 1150 1155
Ala Phe Ile Asp Thr Gln Leu Tyr Ile His Gln Met Ile Ser Val
1160 1165 1170
Lys Asn Phe Ile Leu Ala Ala Asp Val Met Lys Ser Ile Ser Leu
1175 1180 1185
Leu Arg Tyr Gln Glu Glu Ser Lys Thr Leu Ser Leu Val Ser Arg
1190 1195 1200
Asp Ala Lys Pro Leu Glu Val Tyr Ser Val Asp Phe Met Val Asp
1205 1210 1215
Asn Ala Gln Leu Gly Phe Leu Val Ser Asp Arg Asp Arg Asn Leu
1220 1225 1230
Met Val Tyr Met Tyr Leu Pro Glu Ala Lys Glu Ser Phe Gly Gly
1235 1240 1245
Met Arg Leu Leu Arg Arg Ala Asp Phe His Val Gly Ala His Val
1250 1255 1260
Asn Thr Phe Trp Arg Thr Pro Cys Arg Gly Ala Thr Glu Gly Leu
1265 1270 1275
Ser Lys Lys Ser Val Val Trp Glu Asn Lys His Ile Thr Trp Phe

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                1280                1285                1290
Ala Thr Leu Asp Gly Gly Ile Gly Leu Leu Leu Pro Met Gln Glu
                1295                1300                1305
Lys Thr Tyr Arg Arg Leu Leu Met Leu Gln Asn Ala Leu Thr Thr
                1310                1315                1320
Met Leu Pro His His Ala Gly Leu Asn Pro Arg Ala Phe Arg Met
                1325                1330                1335
Leu His Val Asp Arg Arg Thr Leu Gln Asn Ala Val Arg Asn Val
                1340                1345                1350
Leu Asp Gly Glu Leu Leu Asn Arg Tyr Leu Tyr Leu Ser Thr Met
                1355                1360                1365
Glu Arg Ser Glu Leu Ala Lys Lys Ile Gly Thr Thr Pro Asp Ile
                1370                1375                1380
Ile Leu Asp Asp Leu Leu Glu Thr Asp Arg Val Thr Ala His Phe
                1385                1390                1395


<210> 21
<211> 147
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517524CD1

<400> 21
Met Ser Asp Gln Ala Pro Lys Val Pro Glu Glu Met Phe Arg Glu
  1               5                  10                  15
Val Lys Tyr Tyr Ala Val Gly Asp Ile Asp Pro Gln Val Ile Gln
                 20                  25                  30
Leu Leu Lys Ala Gly Lys Ala Lys Glu Val Ser Tyr Asn Ala Leu
                 35                  40                  45
Ala Ser His Ile Ile Ser Glu Asp Gly Asp Asn Pro Glu Val Gly
                 50                  55                  60
Glu Ala Arg Glu Val Phe Asp Leu Pro Val Val Lys Pro Ser Trp
                 65                  70                  75
Val Ile Leu Ser Val Gln Cys Gly Thr Leu Leu Pro Val Asn Gly
                 80                  85                  90
Phe Ser Pro Glu Ser Cys Gln Ile Phe Phe Gly Ile Thr Ala Cys
                 95                 100                 105
Leu Ser Gln Gly Val Asp Thr Ser Trp Ser Ser Leu Leu Glu Ser
                110                 115                 120
Ser Arg Ala Leu Pro Gly Arg Gly Arg Glu Gly Ser Leu Ser Ser
                125                 130                 135
Arg Ser Trp Glu Ala Gln Arg Ser Ser Ala Phe Phe
                140                 145

<210> 22
<211> 1069
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519735CD1

<400> 22
Met Ser Asp Gln Ala Pro Lys Val Pro Glu Glu Met Phe Arg Glu
  1               5                  10                  15
Val Lys Tyr Tyr Ala Val Gly Asp Ile Asp Pro Gln Val Ile Gln
                 20                  25                  30
Leu Leu Lys Ala Gly Lys Ala Lys Glu Val Ser Tyr Asn Ala Leu
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                35                  40                  45
Ala Ser His Ile Ile Ser Glu Asp Gly Asp Asn Pro Glu Val Gly
                50                  55                  60
Glu Ala Arg Glu Val Phe Asp Leu Pro Val Val Lys Pro Ser Trp
                65                  70                  75
Val Ile Leu Ser Val Gln Cys Gly Thr Leu Leu Pro Val Asn Gly
                80                  85                  90
Phe Ser Pro Glu Ser Cys Gln Ile Phe Phe Gly Ile Thr Ala Cys
                95                  100                 105
Leu Ser Gln Val Ser Ser Glu Asp Arg Ser Ala Leu Trp Ala Leu
                110                 115                 120
Val Thr Phe Tyr Gly Gly Asp Cys Gln Leu Thr Leu Asn Lys Lys
                125                 130                 135
Cys Thr His Leu Ile Val Pro Glu Pro Lys Gly Glu Lys Tyr Glu
                140                 145                 150
Cys Ala Leu Lys Arg Ala Ser Ile Lys Ile Val Thr Pro Asp Trp
                155                 160                 165
Val Leu Asp Cys Val Ser Glu Lys Thr Lys Lys Asp Glu Ala Phe
                170                 175                 180
Tyr His Pro Arg Leu Ile Ile Tyr Glu Glu Glu Glu Glu Glu Glu
                185                 190                 195
Glu Glu Glu Glu Glu Val Glu Asn Glu Glu Gln Asp Ser Gln Asn
                200                 205                 210
Glu Gly Ser Thr Asp Glu Lys Ser Ser Pro Ala Ser Ser Gln Glu
                215                 220                 225
Gly Ser Pro Ser Gly Asp Gln Gln Phe Ser Pro Lys Ser Asn Thr
                230                 235                 240
Glu Lys Ser Lys Gly Glu Leu Met Phe Asp Asp Ser Ser Asp Ser
                245                 250                 255
Ser Pro Glu Lys Gln Glu Arg Asn Leu Asn Trp Thr Pro Ala Glu
                260                 265                 270
Val Pro Gln Leu Ala Ala Ala Lys Arg Arg Leu Pro Gln Gly Lys
                275                 280                 285
Glu Pro Gly Leu Ile Asn Leu Cys Ala Asn Val Pro Pro Val Pro
                290                 295                 300
Gly Asn Ile Leu Pro Pro Glu Val Arg Gly Asn Leu Met Ala Ala
                305                 310                 315
Gly Gln Asn Leu Gln Ser Ser Glu Arg Ser Glu Met Ile Ala Thr
                320                 325                 330
Trp Ser Pro Ala Val Arg Thr Leu Arg Asn Ile Thr Asn Asn Ala
                335                 340                 345
Asp Ile Gln Gln Met Asn Arg Pro Ser Asn Val Ala His Ile Leu
                350                 355                 360
Gln Thr Leu Ser Ala Pro Thr Lys Asn Leu Glu Gln Gln Val Asn
                365                 370                 375
His Ser Gln Gln Gly His Thr Asn Ala Asn Ala Val Leu Phe Ser
                380                 385                 390
Gln Val Lys Val Thr Pro Glu Thr His Met Leu Gln Gln Gln Gln
                395                 400                 405
Gln Ala Gln Gln Gln Gln Gln Gln His Pro Val Leu His Leu Gln
                410                 415                 420
Pro Gln Gln Ile Met Gln Leu Gln Gln Gln Gln Gln Gln Gln Ile
                425                 430                 435
Ser Gln Gln Pro Tyr Pro Gln Gln Pro Pro His Pro Phe Ser Gln
                440                 445                 450
Gln Gln Gln Gln Gln Gln Gln Ala His Pro His Gln Phe Ser Gln
                455                 460                 465
Gln Gln Leu Gln Phe Pro Gln Gln Gln Leu His Pro Pro Gln Gln
                470                 475                 480
Leu His Arg Pro Gln Gln Gln Leu Gln Pro Phe Gln Gln Gln His
                485                 490                 495
Ala Leu Gln Gln Gln Phe His Gln Leu Gln Gln His Gln Leu Gln
                500                 505                 510
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Gln Gln Gln Leu Ala Gln Leu Gln Gln Gln His Ser Leu Leu Gln
515 520 525
Gln Gln Gln Gln Gln Gln Ile Gln Gln Gln Gln Leu Gln Arg Met
530 535 540
His Gln Gln Gln Gln Gln Gln Gln Met Gln Ser Gln Thr Ala Pro
545 550 555
His Leu Ser Gln Thr Ser Gln Ala Leu Gln His Gln Val Pro Pro
560 565 570
Gln Gln Pro Pro Gln Gln Gln Gln Gln Gln Gln Pro Pro Pro Ser
575 580 585
Pro Gln Gln His Gln Leu Phe Gly His Asp Pro Ala Val Glu Ile
590 595 600
Pro Glu Glu Gly Phe Leu Leu Gly Cys Val Phe Ala Ile Ala Asp
605 610 615
Tyr Pro Glu Gln Met Ser Asp Lys Gln Leu Leu Ala Thr Trp Lys
620 625 630
Arg Ile Ile Gln Ala His Gly Gly Thr Val Asp Pro Thr Phe Thr
635 640 645
Ser Arg Cys Thr His Leu Leu Cys Glu Ser Gln Val Ser Ser Ala
650 655 660
Tyr Ala Gln Ala Ile Arg Glu Arg Lys Arg Cys Val Thr Ala His
665 670 675
Trp Leu Asn Thr Val Leu Lys Lys Lys Lys Met Val Pro Pro His
680 685 690
Arg Ala Leu His Phe Pro Val Ala Phe Pro Pro Gly Gly Lys Pro
695 700 705
Cys Ser Gln His Ile Ile Ser Val Thr Gly Phe Val Asp Ser Asp
710 715 720
Arg Asp Asp Leu Lys Leu Met Ala Tyr Leu Ala Gly Ala Lys Tyr
725 730 735
Thr Gly Tyr Leu Cys Arg Ser Asn Thr Val Leu Ile Cys Lys Glu
740 745 750
Pro Thr Gly Leu Lys Tyr Glu Lys Ala Lys Glu Trp Arg Ile Pro
755 760 765
Cys Val Asn Ala Gln Trp Leu Gly Asp Ile Leu Leu Gly Asn Phe
770 775 780
Glu Ala Leu Arg Gln Ile Gln Tyr Ser Arg Tyr Thr Ala Phe Ser
785 790 795
Leu Gln Asp Pro Phe Ala Pro Thr Gln His Leu Val Leu Asn Leu
800 805 810
Leu Asp Ala Trp Arg Val Pro Leu Lys Val Ser Ala Glu Leu Leu
815 820 825
Met Ser Ile Arg Leu Pro Pro Lys Leu Lys Gln Asn Glu Val Ala
830 835 840
Asn Val Gln Pro Ser Ser Lys Arg Ala Arg Ile Glu Asp Val Pro
845 850 855
Pro Pro Thr Lys Lys Leu Thr Pro Glu Leu Thr Pro Phe Val Leu
860 865 870
Phe Thr Gly Phe Glu Pro Val Gln Val Gln Gln Tyr Ile Lys Lys
875 880 885
Leu Tyr Ile Leu Gly Gly Glu Val Ala Glu Ser Ala Gln Lys Cys
890 895 900
Thr His Leu Ile Ala Ser Lys Val Thr Arg Thr Val Lys Phe Leu
905 910 915
Thr Ala Ile Ser Val Val Lys His Ile Val Thr Pro Glu Trp Leu
920 925 930
Glu Glu Cys Phe Arg Cys Gln Lys Phe Ile Asp Glu Gln Asn Tyr
935 940 945
Ile Leu Arg Asp Ala Glu Ala Glu Val Leu Phe Ser Phe Ser Leu
950 955 960
Glu Glu Ser Leu Lys Arg Ala His Val Ser Pro Leu Phe Lys Ala
965 970 975
Lys Tyr Phe Tyr Ile Thr Pro Gly Ile Cys Pro Ser Leu Ser Thr

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                980                 985                 990
Met Lys Ala Ile Val Glu Cys Ala Gly Gly Lys Val Leu Ser Lys
                995                 1000                1005
Gln Pro Ser Phe Arg Lys Leu Met Glu His Lys Gln Asn Ser Ser
                1010                1015                1020
Leu Ser Glu Ile Ile Leu Ile Ser Cys Glu Asn Asp Leu His Leu
                1025                1030                1035
Cys Arg Glu Tyr Phe Ala Arg Gly Ile Asp Val His Asn Ala Glu
                1040                1045                1050
Phe Val Leu Thr Gly Val Leu Thr Gln Thr Leu Asp Tyr Glu Ser
                1055                1060                1065
Tyr Lys Phe Asn


<210> 23
<211> 261
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517880CD1

<400> 23
Met Thr Arg Leu Gln Glu Glu Met Ser Leu Trp Leu Gly Ala Pro
 1               5                   10                  15
Val Pro Asp Ile Pro Pro Asp Ser Ala Val Glu Leu Trp Lys Pro
                20                  25                  30
Gly Ala Gln Asp Ala Ser Ser Gln Ala Gln Gly Gly Ser Ser Cys
                35                  40                  45
Ile Leu Arg Glu Glu Ala Arg Met Pro His Ser Ala Gly Gly Thr
                50                  55                  60
Ala Gly Val Gly Leu Glu Ala Ala Glu Pro Thr Ala Leu Leu Thr
                65                  70                  75
Arg Ala Glu Pro Pro Ser Glu Pro Thr Glu Ile Arg Pro Gln Lys
                80                  85                  90
Arg Lys Lys Gly Pro Ala Pro Lys Met Leu Gly Asn Glu Leu Cys
                95                  100                 105
Ser Val Cys Gly Asp Lys Ala Ser Gly Phe His Tyr Asn Val Leu
                110                 115                 120
Ser Cys Glu Gly Cys Lys Gly Phe Phe Arg Arg Ser Val Ile Lys
                125                 130                 135
Gly Ala His Tyr Ile Cys His Ser Gly Gly His Cys Pro Met Asp
                140                 145                 150
Thr Tyr Met Arg Arg Lys Cys Gln Glu Cys Arg Leu Arg Lys Cys
                155                 160                 165
Arg Gln Ala Gly Met Arg Glu Glu Cys Val Leu Ser Glu Glu Gln
                170                 175                 180
Ile Arg Leu Lys Lys Leu Lys Arg Gln Glu Glu Glu Gln Ala His
                185                 190                 195
Ala Thr Ser Leu Pro Pro Arg Ala Ser Ser Pro Pro Gln Ile Leu
                200                 205                 210
Pro Gln Leu Ser Pro Glu Gln Leu Gly Met Ile Glu Lys Leu Val
                215                 220                 225
Ala Ala Gln Gln Gln Cys Asn Arg Arg Ser Phe Ser Asp Arg Leu
                230                 235                 240
Arg Val Thr Val Leu Asp Thr Pro Gly Glu Arg Arg Leu Arg Pro
                245                 250                 255
Asp His Gln Trp Ala Ser
                260

<210> 24
<211> 125
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<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 4524267CD1

<400> 24
Met Asp Leu Asp Phe Pro Gln Ala Phe Gln Lys Glu Leu Thr Cys
1 5 10 15
Leu Ile Cys Leu Asn Tyr Leu Thr Asp Pro Ile Thr Ile Gly Cys
20 25 30
Gly His Ser Phe Cys Arg Pro Cys Leu Cys Leu Cys Trp Glu Glu
35 40 45
Ala His Thr Pro Ala Leu Arg Ala Gly Asn Cys His Ser Arg Lys
50 55 60
Ile Ser Thr Asn Ile Leu Leu Lys Asn Leu Val Ser Ile Ala Arg
65 70 75
Lys Ala Ser Leu Trp Gln Phe Leu Ser Ser Asn Glu Gln Met Cys
80 85 90
Gly Ile His Arg Glu Thr Lys Met Phe Cys Asp Val Gly Lys Ser
95 100 105
Leu Leu Cys Phe Leu Cys Ser Asn Ser Gln Glu His Trp Gly Thr
110 115 120
Glu Thr Leu Ala His
125

<210> 25
<211> 550
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7481640CD1

<400> 25
Met Asp Glu Leu Phe Pro Leu Ile Phe Pro Ala Glu Pro Ala Gln
1 5 10 15
Ala Ser Gly Pro Tyr Val Glu Ile Ile Glu Gln Pro Lys Gln Arg
20 25 30
Gly Met Arg Phe Arg Tyr Lys Cys Glu Gly Arg Ser Ala Gly Ser
35 40 45
Ile Pro Gly Glu Arg Ser Thr Asp Thr Thr Lys Thr His Pro Thr
50 55 60
Ile Lys Ile Asn Gly Tyr Thr Gly Pro Gly Thr Val Arg Ile Ser
65 70 75
Leu Val Thr Lys Asp Pro Pro His Arg Pro His Pro His Glu Leu
80 85 90
Val Gly Lys Asp Cys Arg Asp Gly Phe Tyr Glu Ala Glu Leu Cys
95 100 105
Pro Asp Arg Cys Ile His Ser Phe Gln Asn Leu Gly Ile Gln Cys
110 115 120
Val Lys Lys Arg Asp Leu Glu Gln Ala Ile Ser Gln Arg Ile Gln
125 130 135
Thr Asn Asn Asn Pro Phe Gln Val Pro Ile Glu Glu Gln Arg Gly
140 145 150
Asp Tyr Asp Leu Asn Ala Val Arg Leu Cys Phe Gln Val Thr Val
155 160 165
Arg Asp Pro Ser Gly Arg Pro Leu Arg Leu Thr Pro Val Leu Ser
170 175 180
His Pro Ile Phe Asp Asn Arg Ala Pro Asn Thr Ala Glu Leu Lys
185 190 195

```
Ile Cys Arg Val Asn Arg Asn Ser Gly Ser Cys Leu Gly Gly Asp
                200                 205                 210
Glu Ile Phe Leu Leu Cys Asp Lys Val Gln Lys Glu Asp Ile Glu
                215                 220                 225
Val Tyr Phe Thr Gly Pro Gly Trp Glu Ala Arg Gly Ser Phe Ser
                230                 235                 240
Gln Ala Asp Val His Arg Gln Val Ala Ile Val Phe Arg Thr Pro
                245                 250                 255
Pro Tyr Ala Asp Pro Ser Leu Gln Ala Pro Val Arg Val Ser Met
                260                 265                 270
Gln Leu Arg Arg Pro Ser Asp Arg Glu Leu Ser Glu Pro Met Glu
                275                 280                 285
Phe Gln Tyr Leu Pro Asp Thr Asp Asp Arg His Arg Ile Glu Glu
                290                 295                 300
Lys Arg Lys Arg Thr Tyr Glu Thr Phe Lys Ser Ile Met Lys Lys
                305                 310                 315
Ser Pro Phe Asn Gly Pro Thr Glu Pro Arg Pro Pro Pro Arg Arg
                320                 325                 330
Ile Ala Val Pro Ser Arg Gly Pro Thr Ser Val Pro Lys Pro Ala
                335                 340                 345
Pro Gln Pro Tyr Ala Phe Ser Thr Ser Leu Ser Thr Ile Asn Phe
                350                 355                 360
Asp Glu Phe Ser Pro Met Val Leu Pro Pro Gly Gln Ile Ser Asn
                365                 370                 375
Gln Ala Leu Ala Leu Ala Pro Ser Ser Ala Pro Val Leu Ala Gln
                380                 385                 390
Thr Met Val Pro Ser Ser Ala Met Val Pro Ser Leu Ala Gln Pro
                395                 400                 405
Pro Ala Pro Val Pro Val Leu Ala Pro Gly Pro Pro Gln Ser Leu
                410                 415                 420
Ser Ala Pro Val Pro Lys Ser Thr Gln Ala Gly Glu Gly Thr Leu
                425                 430                 435
Ser Glu Ala Leu Leu His Leu Gln Phe Asp Ala Asp Glu Asp Leu
                440                 445                 450
Gly Ala Leu Leu Gly Asn Asn Thr Asp Pro Gly Val Phe Thr Asp
                455                 460                 465
Leu Ala Ser Val Asp Asn Ser Glu Phe Gln Gln Leu Leu Asn Gln
                470                 475                 480
Gly Val Ala Met Ser His Ser Thr Ala Glu Pro Met Leu Met Glu
                485                 490                 495
Tyr Pro Glu Ala Ile Thr Arg Leu Val Thr Gly Ser Gln Arg Pro
                500                 505                 510
Pro Asp Pro Ala Pro Ala Thr Leu Gly Thr Ser Gly Leu Pro Asn
                515                 520                 525
Gly Leu Ser Gly Asp Glu Asp Phe Ser Ser Ile Ala Asp Met Asp
                530                 535                 540
Phe Ser Ala Leu Leu Ser Gln Ile Ser Ser
                545                 550

<210> 26
<211> 1043
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7515465CD1

<400> 26
Met Arg Arg Phe Lys Arg Lys His Leu Thr Ala Ile Asp Cys Gln
  1               5                  10                  15
His Leu Ala Arg Ser His Leu Ala Val Thr Gln Pro Phe Gly Gln
                 20                  25                  30
```

Arg Trp Thr Asn Arg Asp Pro Asn His Gly Leu Tyr Pro Lys Pro
35 40 45
Arg Thr Lys Arg Gly Ser Arg Gly Gln Gly Ser Gln Arg Cys Ile
50 55 60
Pro Glu Phe Phe Leu Ala Gly Lys Gln Pro Cys Thr Asn Asp Met
65 70 75
Ala Lys Ser Asn Ser Val Gly Gln Asp Ser Cys Gln Asp Ser Glu
80 85 90
Gly Asp Met Ile Phe Pro Ala Glu Ser Ser Cys Ala Leu Pro Gln
95 100 105
Glu Gly Ser Ala Gly Pro Gly Ser Pro Gly Ser Ala Pro Pro Ser
110 115 120
Arg Lys Arg Ser Trp Ser Ser Glu Glu Glu Ser Asn Gln Ala Thr
125 130 135
Gly Thr Ser Arg Trp Asp Gly Val Ser Lys Lys Ala Pro Arg His
140 145 150
His Leu Ser Val Pro Cys Thr Arg Pro Arg Glu Ala Arg Gln Glu
155 160 165
Ala Glu Asp Ser Thr Ser Arg Leu Ser Ala Glu Ser Gly Glu Thr
170 175 180
Asp Gln Asp Ala Gly Asp Val Gly Pro Asp Pro Ile Pro Asp Ser
185 190 195
Tyr Tyr Gly Leu Leu Gly Thr Leu Pro Cys Gln Glu Ala Leu Ser
200 205 210
His Ile Cys Ser Leu Pro Ser Glu Val Leu Arg His Val Phe Ala
215 220 225
Phe Leu Pro Val Glu Asp Leu Tyr Trp Asn Leu Ser Leu Val Cys
230 235 240
His Leu Trp Arg Glu Ile Ile Ser Asp Pro Leu Phe Ile Pro Trp
245 250 255
Lys Lys Leu Tyr His Arg Tyr Leu Met Asn Glu Glu Gln Ala Val
260 265 270
Ser Lys Val Asp Gly Ile Leu Ser Asn Cys Gly Ile Glu Lys Glu
275 280 285
Ser Asp Leu Cys Val Leu Asn Leu Ile Arg Tyr Thr Ala Thr Thr
290 295 300
Lys Cys Ser Pro Ser Val Asp Pro Glu Arg Val Leu Trp Ser Leu
305 310 315
Arg Asp His Pro Leu Leu Pro Glu Ala Glu Ala Cys Val Arg Gln
320 325 330
His Leu Pro Asp Leu Tyr Ala Ala Ala Gly Gly Val Asn Ile Trp
335 340 345
Ala Leu Val Ala Ala Val Val Leu Leu Ser Ser Ser Val Asn Asp
350 355 360
Ile Gln Arg Leu Leu Phe Cys Leu Arg Arg Pro Ser Ser Thr Val
365 370 375
Thr Met Pro Asp Val Thr Glu Thr Leu Tyr Cys Ile Ala Val Leu
380 385 390
Leu Tyr Ala Met Arg Glu Lys Gly Ile Asn Ile Ser Asn Arg Ile
395 400 405
His Tyr Asn Ile Phe Tyr Cys Leu Tyr Leu Gln Glu Asn Ser Cys
410 415 420
Thr Gln Ala Thr Lys Val Lys Glu Glu Pro Ser Val Trp Pro Gly
425 430 435
Lys Lys Thr Ile Gln Leu Thr His Glu Gln Gln Leu Ile Leu Asn
440 445 450
His Lys Met Glu Pro Leu Gln Val Val Lys Ile Met Ala Phe Ala
455 460 465
Gly Thr Gly Lys Thr Ser Thr Leu Val Lys Tyr Ala Glu Lys Trp
470 475 480
Ser Gln Ser Arg Phe Leu Tyr Val Thr Phe Asn Lys Ser Ile Ala
485 490 495
Lys Gln Ala Glu Arg Val Phe Pro Ser Asn Val Ile Cys Lys Thr

500 505 510
Phe His Ser Met Ala Tyr Gly His Ile Gly Arg Lys Tyr Gln Ser
515 520 525
Lys Lys Lys Leu Asn Leu Phe Lys Leu Thr Pro Phe Met Val Asn
530 535 540
Ser Val Leu Ala Glu Gly Lys Gly Gly Phe Ile Arg Ala Lys Leu
545 550 555
Val Cys Lys Thr Leu Glu Asn Phe Phe Ala Ser Ala Asp Glu Glu
560 565 570
Leu Thr Ile Asp His Val Pro Ile Trp Cys Lys Asn Ser Gln Gly
575 580 585
Gln Arg Val Met Val Glu Gln Ser Glu Lys Leu Asn Gly Val Leu
590 595 600
Glu Ala Ser Arg Leu Trp Asp Asn Met Arg Lys Leu Gly Glu Cys
605 610 615
Thr Glu Glu Ala His Gln Met Thr His Asp Gly Tyr Leu Lys Leu
620 625 630
Trp Gln Leu Ser Lys Pro Ser Leu Ala Ser Phe Asp Ala Ile Phe
635 640 645
Val Asp Glu Ala Gln Asp Cys Thr Pro Ala Ile Met Asn Ile Val
650 655 660
Leu Ser Gln Pro Cys Gly Lys Ile Phe Val Gly Asp Pro His Gln
665 670 675
Gln Ile Tyr Thr Phe Arg Gly Ala Val Asn Ala Leu Phe Thr Val
680 685 690
Pro His Thr His Val Phe Tyr Leu Thr Gln Ser Phe Arg Phe Gly
695 700 705
Val Glu Ile Ala Tyr Val Gly Ala Thr Ile Leu Asp Val Cys Lys
710 715 720
Arg Val Arg Lys Lys Thr Leu Val Gly Gly Asn His Gln Ser Gly
725 730 735
Ile Arg Gly Asp Ala Lys Gly Gln Val Ala Leu Leu Ser Arg Thr
740 745 750
Asn Ala Asn Val Phe Asp Glu Ala Val Arg Val Thr Glu Gly Glu
755 760 765
Phe Pro Ser Arg Ile His Leu Ile Gly Gly Ile Lys Ser Phe Gly
770 775 780
Leu Asp Arg Ile Ile Asp Ile Trp Ile Leu Leu Gln Pro Glu Glu
785 790 795
Glu Arg Arg Lys Gln Asn Leu Val Ile Lys Asp Lys Phe Ile Arg
800 805 810
Arg Trp Val His Lys Glu Gly Phe Ser Gly Phe Lys Arg Tyr Val
815 820 825
Thr Ala Ala Glu Asp Lys Glu Leu Glu Ala Lys Ile Ala Val Val
830 835 840
Glu Lys Tyr Asn Ile Arg Ile Pro Glu Leu Val Gln Arg Ile Glu
845 850 855
Lys Cys His Ile Glu Asp Leu Asp Phe Ala Glu Tyr Ile Leu Gly
860 865 870
Thr Val His Lys Ala Lys Gly Leu Glu Phe Asp Thr Val His Val
875 880 885
Leu Asp Asp Phe Val Lys Val Pro Cys Ala Arg His Asn Leu Pro
890 895 900
Gln Leu Pro His Phe Arg Val Glu Ser Phe Ser Glu Asp Glu Trp
905 910 915
Asn Leu Leu Tyr Val Ala Val Thr Arg Ala Lys Lys Arg Leu Ile
920 925 930
Met Thr Lys Ser Leu Glu Asn Ile Leu Thr Leu Ala Gly Glu Tyr
935 940 945
Phe Leu Gln Ala Glu Leu Thr Ser Asn Val Leu Lys Thr Gly Val
950 955 960
Val Arg Cys Cys Val Gly Gln Cys Asn Asn Ala Ile Pro Val Asp
965 970 975

Thr Val Leu Thr Met Lys Lys Leu Pro Ile Thr Tyr Ser Asn Arg
980 985 990
Lys Glu Asn Lys Gly Gly Tyr Leu Cys His Ser Cys Ala Glu Gln
995 1000 1005
Arg Ile Gly Pro Leu Ala Phe Leu Thr Ala Ser Pro Glu Gln Val
1010 1015 1020
Arg Ala Met Glu Arg Thr Val Glu Asn Ile Val Leu Pro Arg His
1025 1030 1035
Glu Ala Leu Leu Phe Leu Val Phe
1040

<210> 27
<211> 486
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 4912891CD1

<400> 27
Met Ala Ala Ile Tyr Leu Ser Arg Gly Phe Phe Ser Arg Glu Pro
1 5 10 15
Ile Cys Pro Phe Glu Glu Lys Thr Lys Val Glu Arg Met Val Glu
20 25 30
Asp Tyr Leu Ala Ser Gly Tyr Gln Asp Ser Val Thr Phe Asp Asp
35 40 45
Val Ala Val Asp Phe Thr Pro Glu Glu Trp Ala Leu Leu Asp Thr
50 55 60
Thr Glu Lys Tyr Leu Tyr Arg Asp Val Met Leu Glu Asn Tyr Met
65 70 75
Asn Leu Ala Ser Val Glu Trp Glu Ile Gln Pro Arg Thr Lys Arg
80 85 90
Ser Ser Leu Gln Gln Gly Phe Leu Lys Asn Gln Ile Phe Ser Gly
95 100 105
Ile Gln Met Thr Arg Gly Tyr Ser Gly Trp Lys Leu Cys Asp Cys
110 115 120
Lys Asn Cys Gly Glu Val Phe Arg Glu Gln Phe Cys Leu Lys Thr
125 130 135
His Met Arg Val Gln Asn Gly Gly Asn Thr Ser Glu Gly Asn Cys
140 145 150
Tyr Gly Lys Asp Thr Leu Ser Val His Lys Glu Ala Ser Thr Gly
155 160 165
Gln Glu Leu Ser Lys Phe Asn Pro Cys Gly Lys Val Phe Thr Leu
170 175 180
Thr Pro Gly Leu Ala Val His Leu Glu Val Leu Asn Ala Arg Gln
185 190 195
Pro Tyr Lys Cys Lys Glu Cys Gly Lys Gly Phe Lys Tyr Phe Ala
200 205 210
Ser Leu Asp Asn His Met Gly Ile His Thr Asp Glu Lys Leu Cys
215 220 225
Glu Phe Gln Glu Tyr Gly Arg Ala Val Thr Ala Ser Ser His Leu
230 235 240
Lys Gln Cys Val Ala Val His Thr Gly Lys Lys Ser Lys Lys Thr
245 250 255
Lys Lys Cys Gly Lys Ser Phe Thr Asn Phe Ser Gln Leu Tyr Ala
260 265 270
Pro Val Lys Thr His Lys Gly Glu Lys Ser Phe Glu Cys Lys Glu
275 280 285
Cys Gly Arg Ser Phe Arg Asn Ser Ser Cys Leu Asn Asp His Ile
290 295 300
Gln Ile His Thr Gly Ile Lys Pro His Lys Cys Thr Tyr Cys Gly
305 310 315

Lys Ala Phe Thr Arg Ser Thr Gln Leu Thr Glu His Val Arg Thr
320 325 330
His Thr Gly Ile Lys Pro Tyr Glu Cys Lys Glu Cys Gly Gln Ala
335 340 345
Phe Ala Gln Tyr Ser Gly Leu Ser Ile His Ile Arg Ser His Ser
350 355 360
Gly Lys Lys Pro Tyr Gln Cys Lys Glu Cys Gly Lys Ala Phe Thr
365 370 375
Thr Ser Thr Ser Leu Ile Gln His Thr Arg Ile His Thr Gly Glu
380 385 390
Lys Pro Tyr Glu Cys Val Glu Cys Gly Lys Thr Phe Ile Thr Ser
395 400 405
Ser Arg Arg Ser Lys His Leu Lys Thr His Ser Gly Glu Lys Pro
410 415 420
Phe Val Cys Lys Ile Cys Gly Lys Ala Phe Leu Tyr Ser Ser Arg
425 430 435
Leu Asn Val His Leu Arg Thr His Thr Gly Glu Lys Pro Phe Val
440 445 450
Cys Lys Glu Cys Gly Lys Ala Phe Ala Val Ser Ser Arg Leu Ser
455 460 465
Arg His Glu Arg Ile His Thr Gly Glu Lys Pro Tyr Glu Cys Lys
470 475 480
Asp Met Ser Val Thr Ile
485

<210> 28
<211> 364
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 70552759CD1

<400> 28
Met Trp Ala Pro Arg Glu Gln Leu Leu Gly Trp Thr Ala Glu Ala
1 5 10 15
Leu Pro Ala Lys Asp Ser Ala Trp Pro Trp Glu Glu Lys Pro Arg
20 25 30
Tyr Leu Gly Pro Val Thr Phe Glu Asp Val Ala Val Leu Phe Thr
35 40 45
Glu Ala Glu Trp Lys Arg Leu Ser Leu Glu Gln Arg Asn Leu Tyr
50 55 60
Lys Glu Val Met Leu Glu Asn Leu Arg Asn Leu Val Ser Leu Ala
65 70 75
Glu Ser Lys Pro Glu Val His Thr Cys Pro Ser Cys Pro Leu Ala
80 85 90
Phe Gly Ser Gln Gln Phe Leu Ser Gln Asp Glu Leu His Asn His
95 100 105
Pro Ile Pro Gly Phe His Ala Gly Asn Gln Leu His Pro Gly Asn
110 115 120
Pro Cys Pro Glu Asp Gln Pro Gln Ser Gln His Pro Ser Asp Lys
125 130 135
Asn His Arg Gly Ala Glu Ala Glu Asp Gln Arg Val Glu Gly Gly
140 145 150
Val Arg Pro Leu Phe Trp Ser Thr Asn Glu Arg Gly Ala Leu Val
155 160 165
Gly Phe Ser Ser Leu Phe Gln Arg Pro Pro Ile Ser Ser Trp Gly
170 175 180
Gly Asn Arg Ile Leu Glu Ile Gln Leu Ser Pro Ala Gln Asn Ala
185 190 195
Ser Ser Glu Glu Val Asp Arg Ile Ser Lys Arg Ala Glu Thr Pro
200 205 210

Gly Phe Gly Ala Val Thr Phe Gly Glu Cys Ala Leu Ala Phe Asn
215 220 225
Gln Lys Ser Asn Leu Phe Arg Gln Lys Ala Val Thr Ala Glu Lys
230 235 240
Ser Ser Asp Lys Arg Gln Ser Gln Val Cys Arg Glu Cys Gly Arg
245 250 255
Gly Phe Ser Arg Lys Ser Gln Leu Ile Ile His Gln Arg Thr His
260 265 270
Thr Gly Glu Lys Pro Tyr Val Cys Gly Glu Cys Gly Arg Gly Phe
275 280 285
Ile Val Glu Ser Val Leu Arg Asn His Leu Ser Thr His Ser Gly
290 295 300
Glu Lys Pro Tyr Val Cys Ser His Cys Gly Arg Gly Phe Ser Cys
305 310 315
Lys Pro Tyr Leu Ile Arg His Gln Arg Thr His Thr Arg Glu Lys
320 325 330
Ser Phe Met Cys Thr Val Cys Gly Arg Gly Phe Arg Glu Lys Ser
335 340 345
Glu Leu Ile Lys His Gln Arg Ile His Thr Gly Asp Lys Pro Tyr
350 355 360
Val Cys Arg Asp

<210> 29
<211> 697
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7524194CD1

<400> 29
Met Met Ala Ala Val Lys Ser Thr Glu Ala His Pro Ser Ser Asn
1 5 10 15
Lys Asp Pro Thr Gln Gly Gln Lys Ser Ala Leu Gln Gly Asn Ser
20 25 30
Pro Asp Ser Glu Ala Ser Arg Gln Arg Phe Arg Gln Phe Cys Tyr
35 40 45
Gln Glu Val Thr Gly Pro His Glu Ala Phe Ser Lys Leu Trp Glu
50 55 60
Leu Cys Cys Gln Trp Leu Arg Pro Lys Thr His Ser Lys Glu Glu
65 70 75
Ile Leu Glu Leu Leu Val Leu Glu Gln Phe Leu Thr Ile Leu Pro
80 85 90
Glu Glu Ile Gln Ala Trp Val Arg Glu Gln His Pro Glu Asn Gly
95 100 105
Glu Glu Ala Val Ala Leu Val Glu Asp Val Gln Arg Ala Pro Gly
110 115 120
Gln Gln Val Leu Asp Ser Glu Lys Asp Leu Lys Val Leu Met Lys
125 130 135
Glu Met Ala Pro Leu Gly Ala Thr Arg Glu Ser Leu Arg Ser Gln
140 145 150
Trp Lys Gln Glu Val Gln Pro Glu Glu Pro Thr Phe Lys Gly Ser
155 160 165
Gln Ser Ser His Gln Arg Pro Gly Glu Gln Ser Glu Ala Trp Leu
170 175 180
Ala Pro Gln Ala Pro Arg Asn Leu Pro Gln Asn Thr Gly Leu His
185 190 195
Asp Gln Glu Thr Gly Ala Val Val Trp Thr Ala Gly Ser Gln Gly
200 205 210
Pro Ala Met Arg Asp Asn Arg Ala Val Ser Leu Cys Gln Gln Glu
215 220 225

Trp Met Cys Pro Gly Pro Ala Gln Arg Ala Leu Tyr Arg Gly Ala
230 235 240
Thr Gln Arg Lys Asp Ser His Val Ser Leu Ala Thr Gly Val Pro
245 250 255
Trp Gly Tyr Glu Glu Thr Lys Thr Leu Leu Ala Ile Leu Ser Ser
260 265 270
Ser Gln Phe Tyr Gly Lys Leu Gln Thr Cys Gln Gln Asn Ser Gln
275 280 285
Ile Tyr Arg Ala Met Ala Glu Gly Leu Trp Glu Gln Gly Phe Leu
290 295 300
Arg Thr Pro Glu Gln Cys Arg Thr Lys Phe Lys Ser Leu Gln Leu
305 310 315
Ser Tyr Arg Lys Val Arg Arg Gly Arg Val Pro Glu Pro Cys Ile
320 325 330
Phe Tyr Glu Glu Met Asn Ala Leu Ser Gly Ser Trp Ala Ser Ala
335 340 345
Pro Pro Met Ala Ser Asp Ala Val Pro Gly Gln Glu Gly Ser Asp
350 355 360
Ile Glu Ala Gly Glu Leu Asn His Gln Asn Gly Glu Pro Thr Glu
365 370 375
Val Glu Asp Gly Thr Val Asp Gly Ala Asp Arg Asp Glu Lys Asp
380 385 390
Phe Arg Asn Pro Gly Gln Glu Val Arg Lys Leu Asp Leu Pro Val
395 400 405
Leu Phe Pro Asn Arg Leu Gly Phe Glu Phe Lys Asn Glu Ile Lys
410 415 420
Lys Glu Asn Leu Lys Trp Asp Asp Ser Glu Glu Val Glu Ile Asn
425 430 435
Lys Ala Leu Gln Arg Lys Ser Arg Gly Val Tyr Trp His Ser Glu
440 445 450
Leu Gln Lys Gly Leu Glu Ser Glu Pro Thr Ser Arg Arg Gln Cys
455 460 465
Arg Asn Ser Pro Gly Glu Ser Glu Glu Lys Thr Pro Ser Gln Glu
470 475 480
Lys Met Ser His Gln Ser Phe Cys Ala Arg Asp Lys Ala Cys Thr
485 490 495
His Ile Leu Cys Gly Lys Asn Cys Ser Gln Ser Val His Ser Pro
500 505 510
His Lys Pro Ala Leu Lys Leu Glu Lys Val Ser Gln Cys Pro Glu
515 520 525
Cys Gly Lys Thr Phe Ser Arg Ser Ser Tyr Leu Val Arg His Gln
530 535 540
Arg Ile His Thr Gly Glu Lys Pro His Lys Cys Ser Glu Cys Gly
545 550 555
Lys Gly Phe Ser Glu Arg Ser Asn Leu Thr Ala His Leu Arg Thr
560 565 570
His Thr Gly Glu Arg Pro Tyr Gln Cys Gly Gln Cys Gly Lys Ser
575 580 585
Phe Asn Gln Ser Ser Ser Leu Ile Val His Gln Arg Thr His Thr
590 595 600
Gly Glu Lys Pro Tyr Gln Cys Ile Val Cys Gly Lys Arg Phe Asn
605 610 615
Asn Ser Ser Gln Phe Ser Ala His Arg Arg Ile His Thr Gly Glu
620 625 630
Ser Pro Tyr Lys Cys Ala Val Cys Gly Lys Ile Phe Asn Asn Ser
635 640 645
Ser His Phe Ser Ala His Arg Lys Thr His Thr Gly Glu Lys Pro
650 655 660
Tyr Arg Cys Ser His Cys Glu Arg Gly Phe Thr Lys Asn Ser Ala
665 670 675
Leu Thr Arg His Gln Thr Val His Met Lys Ala Val Leu Ser Ser
680 685 690
Gln Glu Gly Arg Asp Ala Leu

695

<210> 30
<211> 802
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7523973CD1

<400> 30
Met Tyr Arg Ser Thr Lys Gly Ala Ser Lys Ala Arg Arg Asp Gln
1   5   10   15
Ile Asn Ala Glu Ile Arg Asn Leu Lys Glu Leu Leu Pro Leu Ala
                20   25   30
Glu Ala Asp Lys Val Arg Leu Ser Tyr Leu His Ile Met Ser Leu
                35   40   45
Ala Cys Ile Tyr Thr Arg Lys Gly Val Phe Phe Ala Gly Gly Thr
                50   55   60
Pro Leu Ala Gly Pro Thr Gly Leu Leu Ser Ala Gln Glu Leu Glu
                65   70   75
Asp Ile Val Ala Ala Leu Pro Gly Phe Leu Leu Val Phe Thr Ala
                80   85   90
Glu Gly Lys Leu Leu Tyr Leu Ser Lys Ser Val Ser Glu His Leu
                95   100   105
Gly His Ser Met Val Asp Leu Val Ala Gln Gly Asp Ser Ile Tyr
                110   115   120
Asp Ile Ile Asp Pro Ala Asp His Leu Thr Val Arg Gln Gln Leu
                125   130   135
Thr Leu Pro Ser Ala Leu Asp Thr Asp Arg Leu Phe Arg Cys Arg
                140   145   150
Phe Asn Thr Ser Lys Ser Leu Arg Arg Gln Ser Ala Gly Asn Lys
                155   160   165
Leu Val Leu Ile Arg Gly Arg Phe His Ala His Pro Pro Gly Ala
                170   175   180
Tyr Trp Ala Gly Asn Pro Val Phe Thr Ala Phe Cys Ala Pro Leu
                185   190   195
Glu Pro Arg Pro Arg Pro Gly Pro Gly Pro Gly Pro Gly Pro Ala
                200   205   210
Ser Leu Phe Leu Ala Met Phe Gln Ser Arg His Ala Lys Asp Leu
                215   220   225
Ala Leu Leu Asp Ile Ser Glu Ser Val Leu Ile Tyr Leu Gly Phe
                230   235   240
Glu Arg Ser Glu Leu Leu Cys Lys Ser Trp Tyr Gly Leu Leu His
                245   250   255
Pro Glu Asp Leu Ala His Ala Ser Ala Gln His Tyr Arg Leu Leu
                260   265   270
Ala Glu Ser Gly Asp Ile Gln Ala Glu Met Val Val Arg Leu Gln
                275   280   285
Ala Lys Thr Gly Gly Trp Ala Trp Ile Tyr Cys Leu Leu Tyr Ser
                290   295   300
Glu Gly Pro Glu Gly Pro Ile Thr Ala Asn Asn Tyr Pro Ile Ser
                305   310   315
Asp Met Glu Ala Trp Ser Leu Arg Gln Gln Leu Asn Ser Glu Asp
                320   325   330
Thr Gln Ala Ala Tyr Val Leu Gly Thr Pro Thr Met Leu Pro Ser
                335   340   345
Phe Pro Glu Asn Ile Leu Ser Gln Glu Glu Cys Ser Ser Thr Asn
                350   355   360
Pro Leu Phe Thr Ala Ala Leu Gly Ala Pro Arg Ser Thr Ser Phe
                365   370   375
Pro Ser Ala Pro Glu Leu Ser Val Val Ser Ala Ser Glu Glu Leu

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                380                     385                     390
Pro Arg Pro Ser Lys Glu Leu Asp Phe Ser Tyr Leu Thr Phe Pro
                395                     400                     405
Ser Gly Pro Glu Pro Ser Leu Gln Ala Glu Leu Ser Lys Asp Leu
                410                     415                     420
Val Cys Thr Pro Pro Tyr Thr Pro His Gln Pro Gly Gly Cys Ala
                425                     430                     435
Phe Leu Phe Ser Leu His Glu Pro Phe Gln Thr His Leu Pro Thr
                440                     445                     450
Pro Ser Ser Thr Leu Gln Glu Gln Leu Thr Pro Ser Thr Ala Thr
                455                     460                     465
Phe Ser Asp Gln Leu Thr Pro Ser Ser Ala Thr Phe Pro Asp Pro
                470                     475                     480
Leu Thr Ser Pro Leu Gln Gly Gln Leu Thr Glu Thr Ser Val Arg
                485                     490                     495
Ser Tyr Glu Asp Gln Leu Thr Pro Cys Thr Ser Thr Phe Pro Asp
                500                     505                     510
Gln Leu Leu Pro Ser Thr Ala Thr Phe Pro Glu Pro Leu Gly Ser
                515                     520                     525
Pro Ala His Glu Gln Leu Thr Pro Pro Ser Thr Ala Phe Gln Ala
                530                     535                     540
His Leu Asp Ser Pro Ser Gln Thr Phe Pro Glu Gln Leu Ser Pro
                545                     550                     555
Asn Pro Thr Lys Thr Tyr Phe Ala Gln Glu Gly Cys Ser Phe Leu
                560                     565                     570
Tyr Glu Lys Leu Pro Pro Ser Pro Ser Ser Pro Gly Asn Gly Asp
                575                     580                     585
Cys Thr Leu Leu Ala Leu Ala Gln Leu Arg Gly Pro Leu Ser Val
                590                     595                     600
Asp Val Pro Leu Val Pro Glu Gly Leu Leu Thr Pro Glu Ala Ser
                605                     610                     615
Pro Val Lys Gln Ser Phe Phe His Tyr Ser Glu Lys Glu Gln Asn
                620                     625                     630
Glu Ile Asp Arg Leu Ile Gln Gln Ile Ser Gln Leu Ala Gln Gly
                635                     640                     645
Met Asp Arg Pro Phe Ser Ala Glu Ala Gly Thr Gly Gly Leu Glu
                650                     655                     660
Pro Leu Gly Gly Leu Glu Pro Leu Asp Ser Asn Leu Ser Leu Ser
                665                     670                     675
Gly Ala Gly Pro Pro Val Leu Ser Leu Asp Leu Lys Pro Trp Lys
                680                     685                     690
Cys Gln Glu Leu Asp Phe Leu Ala Asp Pro Asp Asn Met Phe Leu
                695                     700                     705
Glu Glu Thr Pro Val Glu Asp Ile Phe Met Asp Leu Ser Thr Pro
                710                     715                     720
Asp Pro Ser Glu Glu Trp Gly Ser Gly Asp Pro Glu Ala Glu Gly
                725                     730                     735
Pro Gly Gly Ala Pro Ser Pro Cys Asn Asn Leu Ser Pro Glu Asp
                740                     745                     750
His Ser Phe Leu Glu Asp Leu Ala Thr Tyr Glu Thr Ala Phe Glu
                755                     760                     765
Thr Gly Val Ser Ala Phe Pro Tyr Asp Gly Phe Thr Asp Glu Leu
                770                     775                     780
His Gln Leu Gln Ser Gln Val Gln Asp Ser Phe His Glu Asp Gly
                785                     790                     795
Ser Gly Gly Glu Pro Thr Phe
                800
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<210> 31
<211> 580
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7524553CD1

<400> 31
Met Ala Gly Gly Gly Ser Asp Leu Ser Thr Arg Gly Leu Asn Gly
1 5 10 15
Gly Val Ser Gln Val Ala Asn Glu Met Asn His Leu Pro Ala His
20 25 30
Ser Gln Ser Leu Gln Arg Leu Phe Thr Glu Asp Gln Asp Val Asp
35 40 45
Glu Gly Leu Val Tyr Asp Thr Val Phe Lys His Phe Lys Arg His
50 55 60
Lys Leu Glu Ile Ser Asn Ala Ile Lys Lys Thr Phe Pro Phe Leu
65 70 75
Glu Gly Leu Arg Asp Arg Glu Leu Ile Thr Asn Lys Met Phe Glu
80 85 90
Asp Ser Glu Asp Ser Cys Arg Asn Leu Val Pro Val Gln Arg Val
95 100 105
Val Tyr Asn Val Leu Ser Glu Leu Glu Lys Thr Phe Asn Leu Ser
110 115 120
Val Leu Glu Ala Leu Phe Ser Glu Val Asn Met Gln Glu Tyr Pro
125 130 135
Asp Leu Ile His Ile Tyr Lys Ser Phe Lys Asn Ala Ile Gln Asp
140 145 150
Lys Leu Ser Phe Gln Glu Ser Asp Arg Lys Glu Arg Glu Glu Arg
155 160 165
Pro Asp Ile Lys Leu Ser Leu Lys Gln Gly Glu Val Pro Glu Ser
170 175 180
Pro Glu Ala Arg Lys Glu Ser Asp Gln Ala Cys Gly Lys Met Asp
185 190 195
Thr Val Asp Ile Ala Asn Asn Ser Thr Leu Gly Lys Pro Lys Arg
200 205 210
Lys Arg Arg Lys Lys Lys Gly His Gly Trp Ser Arg Met Gly Thr
215 220 225
Arg Thr Gln Lys Asn Asn Gln Gln Asn Asp Asn Ser Lys Ala Asp
230 235 240
Gly Gln Leu Val Ser Ser Glu Lys Lys Ala Asn Met Asn Leu Lys
245 250 255
Asp Leu Ser Lys Ile Arg Gly Arg Lys Arg Gly Lys Pro Gly Thr
260 265 270
His Phe Thr Gln Ser Asp Arg Ala Pro Gln Lys Arg Val Arg Ser
275 280 285
Arg Ala Ser Arg Lys His Lys Asp Glu Thr Val Asp Phe Gln Ala
290 295 300
Pro Leu Leu Pro Val Thr Cys Gly Gly Val Lys Gly Ile Leu His
305 310 315
Lys Glu Lys Leu Glu Gln Gly Thr Leu Ala Lys Cys Ile Gln Thr
320 325 330
Glu Asp Gly Lys Trp Phe Thr Pro Met Glu Phe Glu Ile Lys Gly
335 340 345
Gly Tyr Ala Arg Ser Lys Asn Trp Arg Leu Ser Val Arg Cys Gly
350 355 360
Gly Trp Pro Leu Arg Arg Leu Met Glu Glu Gly Ser Leu Pro Asn
365 370 375
Pro Ser Arg Ile Tyr Tyr Arg Asn Lys Lys Arg Ile Leu Lys Ser
380 385 390
Gln Asn Asn Ser Ser Val Asp Pro Cys Met Arg Asn Leu Asp Glu
395 400 405
Cys Glu Val Cys Arg Asp Gly Gly Glu Leu Phe Cys Cys Asp Thr
410 415 420
Cys Ser Arg Val Phe His Glu Asp Cys His Ile Pro Pro Val Glu
425 430 435

```
Ser Glu Lys Thr Pro Trp Asn Cys Ile Phe Cys Arg Met Lys Glu
                440                 445                 450
Ser Pro Gly Ser Gln Gln Cys Cys Gln Glu Ser Glu Val Leu Glu
                455                 460                 465
Arg Gln Met Cys Pro Glu Glu Gln Leu Lys Cys Glu Phe Leu Leu
                470                 475                 480
Leu Lys Val Tyr Cys Cys Ser Glu Ser Ser Phe Phe Ala Lys Ile
                485                 490                 495
Pro Tyr Tyr Tyr Tyr Ile Arg Glu Ala Cys Gln Gly Leu Lys Glu
                500                 505                 510
Pro Met Trp Leu Asp Lys Ile Lys Lys Arg Leu Asn Glu His Gly
                515                 520                 525
Tyr Pro Gln Val Glu Gly Phe Val Gln Asp Met Arg Leu Ile Phe
                530                 535                 540
Gln Asn His Arg Ala Ser Tyr Lys Tyr Lys Asp Phe Gly Gln Met
                545                 550                 555
Gly Leu Arg Leu Glu Ala Glu Phe Glu Lys Asp Phe Lys Glu Val
                560                 565                 570
Phe Ala Ile Gln Glu Thr Asn Gly Asn Ser
                575                 580
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<210> 32
<211> 183
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 2666675CD1

<400> 32

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Met Ala Leu Leu Thr Ala Glu Thr Phe Arg Leu Gln Phe Asn Asn
 1               5                  10                  15
Lys Arg Arg Leu Arg Arg Pro Tyr Tyr Pro Arg Lys Ala Leu Leu
                20                  25                  30
Cys Tyr Gln Leu Thr Pro Gln Asn Gly Ser Thr Pro Thr Arg Gly
                35                  40                  45
Tyr Phe Glu Asn Lys Lys Lys Cys His Ala Glu Ile Cys Phe Ile
                50                  55                  60
Asn Glu Ile Lys Ser Met Gly Leu Asp Glu Thr Gln Cys Tyr Gln
                65                  70                  75
Val Thr Cys Tyr Leu Thr Trp Ser Pro Cys Ser Ser Cys Ala Trp
                80                  85                  90
Glu Leu Val Asp Phe Ile Lys Ala His Asp His Leu Asn Leu Gly
                95                  100                 105
Ile Phe Ala Ser Arg Leu Tyr Tyr His Trp Cys Lys Pro Gln Gln
                110                 115                 120
Lys Gly Leu Arg Leu Leu Cys Gly Ser Gln Val Pro Val Glu Val
                125                 130                 135
Met Gly Phe Pro Glu Phe Ala Asp Cys Trp Glu Asn Phe Val Asp
                140                 145                 150
His Glu Lys Pro Leu Ser Phe Asn Pro Tyr Lys Met Leu Glu Glu
                155                 160                 165
Leu Asp Lys Asn Ser Arg Ala Ile Lys Arg Arg Leu Glu Arg Ile
                170                 175                 180
Lys Gln Ser
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<210> 33
<211> 456
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 4032169CD1

<400> 33
Met Asn Lys Phe Gln Gly Pro Val Thr Leu Lys Asp Val Ile Val
1 5 10 15
Glu Phe Thr Lys Glu Glu Trp Lys Leu Leu Thr Pro Ala Gln Arg
20 25 30
Thr Leu Tyr Lys Asp Val Met Leu Glu Asn Tyr Ser His Leu Val
35 40 45
Ser Val Gly Tyr His Val Asn Lys Pro Asn Ala Val Phe Lys Leu
50 55 60
Lys Gln Gly Lys Glu Pro Trp Ile Leu Glu Val Glu Phe Pro His
65 70 75
Arg Gly Phe Pro Glu Asp Leu Trp Ser Ile His Asp Leu Glu Ala
80 85 90
Arg Tyr Gln Glu Ser Gln Ala Gly Asn Ser Arg Asn Gly Glu Leu
95 100 105
Thr Lys His Gln Lys Thr His Thr Thr Glu Lys Ala Cys Glu Cys
110 115 120
Lys Glu Cys Gly Lys Phe Phe Cys Gln Lys Ser Ala Leu Ile Val
125 130 135
His Gln His Thr His Ser Lys Gly Lys Ser Tyr Asp Cys Asp Lys
140 145 150
Cys Gly Lys Ser Phe Ser Lys Asn Glu Asp Leu Ile Arg His Gln
155 160 165
Lys Ile His Thr Arg Asp Lys Thr Tyr Glu Cys Lys Glu Cys Lys
170 175 180
Lys Ile Phe Tyr His Leu Ser Ser Leu Ser Arg His Leu Arg Thr
185 190 195
His Ala Gly Glu Lys Pro Tyr Glu Cys Asn Gln Cys Glu Lys Ser
200 205 210
Phe Tyr Gln Lys Pro His Leu Thr Glu His Gln Lys Thr His Thr
215 220 225
Gly Glu Lys Pro Phe Glu Cys Thr Glu Cys Gly Lys Phe Phe Tyr
230 235 240
Val Lys Ala Tyr Leu Met Val His Gln Lys Thr His Thr Gly Glu
245 250 255
Lys Pro Tyr Glu Cys Lys Glu Cys Gly Lys Ala Phe Ser Gln Lys
260 265 270
Ser His Leu Thr Val His Gln Arg Met His Thr Gly Glu Lys Pro
275 280 285
Tyr Lys Cys Lys Glu Cys Gly Lys Phe Phe Ser Arg Asn Ser His
290 295 300
Leu Lys Thr His Gln Arg Ser His Thr Gly Glu Lys Pro Tyr Glu
305 310 315
Cys Lys Glu Cys Arg Lys Cys Phe Tyr Gln Lys Ser Ala Leu Thr
320 325 330
Val His Gln Arg Thr His Thr Gly Glu Lys Pro Phe Glu Cys Asn
335 340 345
Lys Cys Gly Lys Thr Phe Tyr Tyr Lys Ser Asp Leu Thr Lys His
350 355 360
Gln Arg Lys His Thr Gly Glu Lys Pro Tyr Glu Cys Thr Glu Cys
365 370 375
Gly Lys Ser Phe Ala Val Asn Ser Val Leu Arg Leu His Gln Arg
380 385 390
Thr His Thr Gly Glu Lys Pro Tyr Ala Cys Lys Glu Cys Gly Lys
395 400 405
Ser Phe Ser Gln Lys Ser His Phe Ile Ile His Gln Arg Lys His
410 415 420
Thr Gly Glu Lys Pro Tyr Glu Cys Gln Glu Cys Gly Glu Thr Phe
425 430 435

Ile Gln Lys Ser Gln Leu Thr Ala His Gln Lys Thr Arg Thr Lys
440 445 450
Lys Arg Asn Ala Glu Lys
455

<210> 34
<211> 410
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7524926CD1

<400> 34
Met Asp Gly Thr Ile Lys Glu Ala Leu Ser Val Val Ser Asp Asp
1 5 10 15
Gln Ser Leu Phe Asp Ser Ala Tyr Gly Ala Ala Ala His Leu Pro
20 25 30
Lys Ala Asp Met Thr Ala Ser Gly Ser Pro Asp Tyr Gly Gln Pro
35 40 45
His Lys Ile Asn Pro Leu Pro Pro Gln Gln Glu Trp Ile Asn Gln
50 55 60
Pro Val Arg Val Asn Val Lys Arg Glu Tyr Asp His Met Asn Gly
65 70 75
Ser Arg Glu Ser Pro Val Asp Cys Ser Val Ser Lys Cys Ser Lys
80 85 90
Leu Val Gly Gly Gly Glu Ser Asn Pro Met Asn Tyr Asn Ser Tyr
95 100 105
Met Asp Glu Lys Asn Gly Pro Pro Pro Pro Asn Met Thr Thr Asn
110 115 120
Glu Arg Arg Val Ile Val Pro Ala Asp Pro Thr Leu Trp Thr Gln
125 130 135
Glu His Val Arg Gln Trp Leu Glu Trp Ala Ile Lys Glu Tyr Ser
140 145 150
Leu Met Glu Ile Asp Thr Ser Phe Phe Gln Asn Met Asp Gly Lys
155 160 165
Glu Leu Cys Lys Met Asn Lys Glu Asp Phe Leu Arg Ala Thr Ala
170 175 180
Leu Tyr Asn Thr Glu Val Leu Leu Ser His Leu Ser Tyr Leu Arg
185 190 195
Glu Ser Ser Leu Leu Ala Tyr Asn Thr Thr Ser His Thr Asp Gln
200 205 210
Ser Ser Arg Leu Ser Val Lys Glu Asp Pro Tyr Gln Ile Leu Gly
215 220 225
Pro Thr Ser Ser Arg Leu Ala Asn Pro Gly Ser Gly Gln Ile Gln
230 235 240
Leu Trp Gln Phe Leu Leu Glu Leu Leu Ser Asp Ser Ala Asn Ala
245 250 255
Ser Cys Ile Thr Trp Glu Gly Thr Asn Gly Glu Phe Lys Met Thr
260 265 270
Asp Pro Asp Glu Val Ala Arg Arg Trp Gly Glu Arg Lys Ser Lys
275 280 285
Pro Asn Met Asn Tyr Asp Lys Leu Ser Arg Ala Leu Arg Tyr Tyr
290 295 300
Tyr Asp Lys Asn Ile Met Thr Lys Val His Gly Lys Arg Tyr Ala
305 310 315
Tyr Lys Phe Asp Phe His Gly Ile Ala Gln Ala Leu Gln Pro His
320 325 330
Pro Thr Glu Ser Ser Met Tyr Lys Tyr Pro Ser Asp Ile Ser Tyr
335 340 345
Met Pro Ser Tyr His Ala His Gln Gln Lys Val Asn Phe Val Pro
350 355 360

Pro His Pro Ser Ser Met Pro Val Thr Ser Ser Ser Phe Phe Gly
365 370 375
Ala Ala Ser Gln Tyr Trp Thr Ser Pro Thr Gly Gly Ile Tyr Pro
380 385 390
Asn Pro Asn Val Pro Arg His Pro Asn Thr His Val Pro Ser His
395 400 405
Leu Gly Ser Tyr Tyr
410

<210> 35
<211> 835
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7509493CD1

<400> 35
Met Val Ala Met Asp Leu Gly Ser Pro Ser Leu Pro Lys Lys Ser
1 5 10 15
Leu Pro Val Pro Gly Ala Leu Glu Gln Val Ala Ser Arg Leu Ser
20 25 30
Ser Lys Val Ala Ala Glu Val Pro His Gly Ser Lys Gln Glu Leu
35 40 45
Gln Asp Leu Lys Ala Gln Ser Leu Thr Thr Cys Glu Val Cys Gly
50 55 60
Ala Cys Phe Glu Thr Arg Lys Gly Leu Ser Ser His Ala Arg Ser
65 70 75
His Leu Arg Gln Leu Gly Val Ala Glu Ser Glu Ser Ser Gly Ala
80 85 90
Pro Ile Asp Leu Leu Tyr Glu Leu Val Lys Gln Lys Gly Leu Pro
95 100 105
Asp Ala His Leu Gly Leu Pro Pro Gly Leu Ala Lys Lys Ser Ser
110 115 120
Ser Leu Lys Glu Val Val Ala Gly Ala Pro Arg Pro Gly Leu Leu
125 130 135
Ser Leu Ala Lys Pro Leu Asp Ala Pro Ala Val Asn Lys Ala Ile
140 145 150
Lys Ser Pro Pro Gly Phe Ser Ala Lys Gly Leu Gly His Pro Pro
155 160 165
Ser Ser Pro Leu Leu Lys Lys Thr Pro Leu Ala Leu Ala Gly Ser
170 175 180
Pro Thr Pro Lys Asn Pro Glu Asp Lys Ser Pro Gln Leu Ser Leu
185 190 195
Ser Pro Arg Pro Ala Ser Pro Lys Ala Gln Trp Pro Gln Ser Glu
200 205 210
Asp Glu Gly Pro Leu Asn Leu Thr Ser Gly Pro Glu Pro Ala Arg
215 220 225
Asp Ile Arg Cys Glu Phe Cys Gly Glu Phe Phe Glu Asn Arg Lys
230 235 240
Gly Leu Ser Ser His Ala Arg Ser His Leu Arg Gln Met Gly Val
245 250 255
Thr Glu Trp Tyr Val Asn Gly Ser Pro Ile Asp Thr Leu Arg Glu
260 265 270
Ile Leu Lys Arg Arg Thr Gln Ser Arg Pro Gly Gly Pro Pro Asn
275 280 285
Pro Pro Gly Pro Ser Pro Lys Ala Leu Ala Lys Met Met Gly Gly
290 295 300
Ala Gly Pro Gly Ser Ser Leu Glu Ala Arg Ser Pro Ser Asp Leu
305 310 315
His Ile Ser Pro Leu Ala Lys Lys Leu Pro Pro Pro Pro Gly Ser
320 325 330

Pro Leu Gly His Ser Pro Thr Ala Ser Pro Pro Pro Thr Ala Arg
335 340 345
Lys Met Phe Pro Gly Leu Ala Ala Pro Ser Leu Pro Lys Lys Leu
350 355 360
Lys Pro Glu Gln Ile Arg Val Glu Ile Lys Arg Glu Met Leu Pro
365 370 375
Gly Ala Leu His Gly Glu Leu His Pro Ser Glu Gly Pro Trp Gly
380 385 390
Ala Pro Arg Glu Asp Met Thr Pro Leu Asn Leu Ser Ser Arg Ala
395 400 405
Glu Pro Val Arg Asp Ile Arg Cys Glu Phe Cys Gly Glu Phe Phe
410 415 420
Glu Asn Arg Lys Gly Leu Ser Ser His Ala Arg Ser His Leu Arg
425 430 435
Gln Met Gly Val Thr Glu Trp Ser Val Asn Gly Ser Pro Ile Asp
440 445 450
Thr Leu Arg Glu Ile Leu Lys Lys Lys Ser Lys Pro Cys Leu Ile
455 460 465
Lys Lys Glu Pro Pro Ala Gly Asp Leu Ala Pro Ala Leu Ala Glu
470 475 480
Asp Gly Pro Pro Thr Val Ala Pro Gly Pro Val Gln Ser Pro Leu
485 490 495
Pro Leu Ser Pro Leu Ala Gly Arg Pro Gly Lys Pro Gly Ala Gly
500 505 510
Pro Ala Gln Val Pro Arg Glu Leu Ser Leu Thr Pro Ile Thr Gly
515 520 525
Ala Lys Pro Ser Ala Thr Gly Tyr Leu Gly Ser Val Ala Ala Lys
530 535 540
Arg Pro Leu Gln Glu Asp Arg Leu Leu Pro Ala Glu Val Lys Ala
545 550 555
Lys Thr Tyr Ile Gln Thr Glu Leu Pro Phe Lys Ala Lys Thr Leu
560 565 570
His Glu Lys Thr Ser His Ser Ser Thr Glu Ala Cys Cys Glu Leu
575 580 585
Cys Gly Leu Tyr Phe Glu Asn Arg Lys Ala Leu Ala Ser His Ala
590 595 600
Arg Ala His Leu Arg Gln Phe Gly Val Thr Glu Trp Cys Val Asn
605 610 615
Gly Ser Pro Ile Glu Thr Leu Ser Glu Trp Ile Lys His Arg Pro
620 625 630
Gln Lys Val Gly Ala Tyr Arg Ser Tyr Ile Gln Gly Gly Arg Pro
635 640 645
Phe Thr Lys Lys Phe Arg Ser Ala Gly His Gly Arg Asp Ser Asp
650 655 660
Lys Arg Pro Ser Leu Gly Leu Ala Pro Gly Gly Leu Ala Val Val
665 670 675
Gly Arg Ser Ala Gly Gly Glu Pro Gly Pro Glu Ala Gly Arg Ala
680 685 690
Ala Asp Gly Gly Glu Arg Pro Leu Ala Ala Ser Pro Pro Gly Thr
695 700 705
Val Lys Ala Glu Glu His Gln Arg Gln Asn Ile Asn Lys Phe Glu
710 715 720
Arg Arg Gln Ala Arg Pro Pro Asp Ala Ser Ala Ala Arg Gly Gly
725 730 735
Glu Asp Thr Asn Asp Leu Gln Gln Lys Leu Glu Glu Val Arg Gln
740 745 750
Pro Pro Pro Arg Val Arg Pro Val Pro Ser Leu Val Pro Arg Pro
755 760 765
Pro Gln Thr Ser Leu Val Lys Phe Val Gly Asn Ile Tyr Thr Leu
770 775 780
Lys Cys Arg Phe Cys Glu Val Glu Phe Gln Gly Pro Leu Ser Ile
785 790 795
Gln Glu Glu Trp Val Arg His Leu Gln Arg His Ile Leu Glu Met

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                800                 805                 810
Asn Phe Ser Lys Ala Asp Pro Pro Pro Glu Glu Ser Gln Ala Pro
                815                 820                 825
Gln Ala Gln Thr Ala Ala Ala Glu Ala Pro
                830                 835
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<210> 36
<211> 1531
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7511189CB1

<400> 36
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gtcagcggcc tctgaggttt caggctcttg gcggtgaagg tcagagtgca gacctgagac 60
cactgctcac cgacttcaga ctccagttta tctgtgcccc agcttcctca cttagtctca 120
gaagacttag gctgaggcct cagaaggaga tctccatcct ttgtccagag cagaacagga 180
tggccatgtc ccaggaatca ttgaccttca aggacgtgtt tgtggacttc accctggagg 240
agtggcagca actggactct gcccagaaga acctctacag ggatgtcatg cttgagaact 300
acagccacct ggtgtccgtg gggtatctag ttgcgaagcc agatgtgatc ttcaggttgg 360
gaccaggtga agagtcctgg atggcagatg gggggacccc ggtacggacc tgtgcagatg 420
tatccatttt tgcctcatgt attttgaagt gctgttatta gaagtctggc aagttgatga 480
gcagatagat cactacaagg aaagccaaga caaacttcct tggcaagctg cattcatagg 540
caaggaaaca ctgaaggatg aaagcggtca agaatccaga acatgtagaa aaagcattta 600
tctgagcaca gaatttgatt ctgtaaggca aagactccct aaatattatt cgtgggaaaa 660
ggcattcaaa acatcattta aactttcttg gtcaaaatgg aagctatgta agaaagaaag 720
atgatggatg taaagcatat tggaaagtat gcttccatta taatcttcat aaagctcaac 780
ctgcagagag atttttgac cctaatcaac gagggaaagc cctccaccaa aagcaagccc 840
ttagaaaaag tcagagaagt caaactgggg agaaactcta caaatgtact gaatgtggaa 900
aagtgtttat ccagaaagca aacttagttg tacatcaaag aactcacacc ggagagaaac 960
cttatgaatg ctgcgaatgt gcaaaagcct tcagccagaa gtcaaccctc atagcacacc 1020
agagaactca cacaggggag aagccctatg aatgcagtga atgtggaaaa acctttatcc 1080
agaagtcaac tctgattaaa cataagaaaa ttcatatgga gagagaccct ataaatgcag 1140
tgtctgtgag aaagccttca gtaggaagtc aactctcatt aaacatcaga taattcatat 1200
gggagaaacc ttatgaatgt aataaatgtg ggaaatcttt tagtgttaaa atcaactctc 1260
attgtatgtc acagaacata aatgcataag ttgcaacaac cacaaaacaa aacaccaaca 1320
cccaacaaca acaaaacaaa tagcactcac caccccacaa cccacactac aatcagaaac 1380
caaaaaactt taccctcgac acagagcaca cacccaacgc atcattactc aacaacaaca 1440
cctaaataat acaaatatat ataaacccca aaaaaaatca caaatagaga caaccatatc 1500
caaaataaac cactaataga cagacacata t                                 1531
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<210> 37
<211> 3012
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7511811CB1

<400> 37
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cgcgaatgcc gtggcgcgaa cttgggactg cagaggcgcg cctggcggat ctgagtgtgt 60
tgcccgggca gcggcgcgcg ggaccaacgc aaggagcagc tgacagacga agaaaagtgc 120
tggacaggaa gggagaattc tgacgccaac atgcagcgaa gtatcatgtc atttttccac 180
cccaagaaag agggtaaagc aaagaagcct gagaaggagg catccaatag cagcagagag 240
acggagcccc ctccaaaggc ggcactgaag gagtggaatg gagtggtgtc cgagagtgac 300
tctccggtga agaggccagg gaggaaggcg gcccgggtcc tgggcagcga aggggaagag 360
gaggatgaag cccttagccc tgctaaaggc cagaagcctg ccctggactg ctcacaggtc 420
tccccgcccc gtcctgccac atctcctgag aacaatgctt ccctctctga cacctctccc 480
atggacagtt ccccatcagg gattccgaag cgtcgcacag ctcggaagca gctcccgaaa 540
cggaccattc aggaagtcct ggaagagcag agtgaggacg aggacagaga agccaagagg 600
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aagaaggagg aggaagaaga ggagaccccg aaagaaagcc tcacagaggc tgaagtggca 660
acagagaagg aaggagaaga cggggaccag cccaccacgc ctcccaagcc cctaaagacc 720
tccaaagcag agaccccgac ggaaagcgtt tcagagcctg aggtggccac gaagcaggaa 780
ctgcaggagg aggaagagca gaccaagcct ccccgcagag ctcccaagac gctcagcagc 840
ttcttcaccc cccggaagcc agcagtcaaa aaagaagtga aggaagagga gccaggggct 900
ccaggaaagg agggagctgc tgagggaccc ctggatccat ctggttacaa tcctgccaag 960
aacaactatc atcccgtgga agatgcctgc tggaaaccgg gccagaaggt tccttacctg 1020
gctgtggccc ggacgtttga gaagatcgag gaggtgtctg ctcggctccg gatggtggag 1080
acgctgagca acttgctgcg ctccgtggtg gccctgtcgc ctccagacct cctccctgtc 1140
ctctacctca gcctcaacca ccttgggcca ccccagcagg gcctggagct tggcgtgggt 1200
gatggtgtcc ttctcaaggc agtggcccag gccacaggtc ggcagctgga gtccgtccgg 1260
gctgaggcag ccgagaaagg cgacgtgggg ctggtggccg agaacagccg cagcacccag 1320
aggctcatgc tgccaccacc tccgctcact gcctccgggg tcttcagcaa gttccgcgac 1380
atcgccaggc tcactggcag tgcttccaca gccaagaaga tagacatcat caaaggcctc 1440
tttgtggcct gccgccactc agaagcccgg ttcatcgcta ggtccctgag cggacggctg 1500
cgccttgggc tggcagagca gtcggtgctg gctgccctct cccaggcagt gagcctcacg 1560
cccccgggcc aagaattccc accagccatg gtggatgctg ggaagggcaa gacagcagag 1620
gccagaaaga cgtggctgga ggagcaaggc atgatcctga agcagacgtt ctgcgaggtt 1680
cccgacctgg accgaattat ccccgtgctg ctggagcacg gcctggaacg tctcccggag 1740
cactgcaagc tgagcccaga tccacgccct ggaaggcggg gaggtgaaga tcttcagcag 1800
gaatcaggaa gacaacactg ggaagtaccc ggacatcatc agccgcatcc ccaagattaa 1860
actcccatcg gtcacatcct tcatcctgga caccgaagcc gtggcttggg accgggaaaa 1920
gaagcagatc cagccattcc aagtgctcac cacccgcaaa cgcaaggagg tggatgcgtc 1980
tgagatccag gtgcaggtgt gtttgtacgc cttcgacctc atctacctca atggagagtc 2040
cctggtacgt gagccccttt cccggcgccg gcagctgctc cgggagaact ttgtggagac 2100
agagggcgag tttgtcttcg ccacctccct ggacaccaag gacatcgagc agatcgccga 2160
gttcctggag cagtcagtga aagactcctg cgaggggctg atggtgaaga ccctggatgt 2220
tgatgccacc tacgagatcg ccaagagatc gcacaactgg ctcaagctga agaaggacta 2280
ccttgatggc gtgggtgaca ccctggacct ggtggtgatc ggcgcctacc tgggccgggg 2340
gaagcgggcc ggccggtacg gggcttcct gctggcctcc tacgacgagg acagtgagga 2400
gctgcaggcc atatgcaagc ttggaactgg cttcagtgac gaggagctgg aggagcatca 2460
ccagagcctc aaggccctgg tgctgcccag cccacgccct tacgtgcgga tagatggcgc 2520
tgtgattccc gaccactggc tggaccccag cgctgtgtgg gaggtgaagt gcgctgacct 2580
ctccctctct cccatctacc ctgctgcgcg gggcctggtg gatagtgaca agggcatctc 2640
ccttcgcttc cctcggttta ttcgagtccg tgaagacaag cagccggagc aggccaccac 2700
cagtgctcag gtggcctgtt tgtaccggaa gcaaagtcag attcagaacc aacaaggcga 2760
ggactcaggc tctgaccctg aagataccta ctaagccctc gccctcctag ggcctgggta 2820
cagggcatga gttggacgga ccccagggtt attattgcct ttgcttttta gcaaatctgc 2880
tgtggcaggc tgtggatttt gagagtcagg ggaggggtgt gtgtgtgagg gggtggctta 2940
ctccggagtc tgggattcat cccgtcattt ctttcaataa ataattatcg gataaaaaaa 3000
aaaaaaaaaa gg 3012

<210> 38
<211> 2280
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7511870CB1

<400> 38
taggagtggt tgcgcgtgtt gcggcccggg ttctgttcgt aggtttacaa ttgtcgcggg 60
tagtggggcg ccccagctgc caactgcacg tggtttcttt ctggcgtttg cctagtgtcc 120
tgtccattcc gccccgttgg gaagtgggaa gcggagaggt tctgtgggga cagagctggg 180
agtcctgaag gagaaagcgc gcagccgagg ggctctggag tgccttagag cagagctgcg 240
gccgcggaag gggagcagct gagaaaggag ggccgctgca ggcggggttc gaaccgtggg 300
gtctgggctg ctcccgcgga gggcctgggc ggacgcggga tgctgggggt ccgctgcctg 360
ctgcggtccg tgcgcttctg ttcctccgcc cccttcccca agcacaaacc ttcagccaaa 420
ctgagcgtgc gggacgctct cggggctcag aacgcgagtg gggagcgcat taagatccag 480
ggatggattc gttctgtccg atcccagaag gaagtcttgt tcctgcatgt aaatgatggg 540
tcatctttgg aaagccttca ggttgttgca gattcaggcc ttgacagtag agaattaact 600
tttgggagtt ctgtggaagt acaagggcag ctgataaaaa gtccatccaa aaggcaaaat 660

gtggaactga aggcagaaaa aattaaagtt attggaaatt gtgatgccaa ggatttcccc 720
atcaaatata aagagaggca tcctctggag tatctgcgac aatatcctca ctttaggtgt 780
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<220>
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<213> Homo sapiens

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<223> Incyte ID No: 7511072CB1

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<213> Homo sapiens

<220>
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<212> DNA
<213> Homo sapiens

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<223> Incyte ID No: 7512244CB1

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<213> Homo sapiens

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<211> 2010
<212> DNA
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<213> Homo sapiens

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